Bayesian Axiom Induction from Instances:
Template Priors, Derivation Likelihoods and Time Penalties

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Abstract

Kaarel Hänni asked whether a Solomonoff-style inducer over axiom systems, with a prior over templates, a preference for short derivations and a time penalty, would pass from instances $\varphi(t)$ to $\forall x\varphi$, and whether it would robustly recover the axioms we have, or a deductively equivalent system. We make such an inducer precise. Theories are finite sets of templates with a prefix-code prior; the likelihood is a derivation grammar with a data-independent normaliser, so theories predicting unobserved sentences lose a constant factor per datum. First, in the calculi analysed, closed instances never make $\forall x\varphi$ more probable than its instance schema $\varphi(z)$, except through non-logical weight effects under a selection-aware likelihood: the odds stay at the prior odds, fall by a constant factor per datum, or drop to zero. "All future data are instances" is confirmed, while "the axioms prove $\forall x\varphi$" tends to a prior share or to zero: a Bayesian ω-gap. Second, with data drawn from a theory's law and fixed weights, the posterior concentrates on theories with that law and identifies their theorems, not their axioms. Third, human-stated theorems are not such data; in finite classes the posterior goes to the best-fitting generator, which can be weaker than the data's axioms, or unsound. Fourth, time-penalising axiom checking or generation leaves Hänni's collapse intact from function to axiom induction; penalising consistent function induction, not deduction, makes axiom induction strictly stronger; written derivation length prices the collapse. AI referees checked first versions; no human has checked the mathematics.

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*Claude is an AI system made by Anthropic. Claude did the research and wrote this report in response to questions posed by Kaarel Hänni and under his direction. The first was his question about Solomonoff-style axiom induction (Hänni, 2026); the report is a follow-up to the earlier reports IL (Hänni and Claude (Anthropic), 2026b) and AS (Hänni and Claude (Anthropic), 2026a). The results were checked by separate Claude instances acting as referees and reviewers (Appendix H); no human has checked all of the mathematics, and nothing has been verified in a proof assistant. Code, research notes and referee reports: `axiom-induction/` in the accompanying repository.

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1 Introduction

Kaarel Hänni asked the following question on 8 October 2026 (verbatim):

> let's consider the case of going from a bunch of phi(x) statements to forall x phi(x). i wonder if this should happen basically like so:
>
> - forall x phi(x) is a simple law which implies all the other statements. so once we see all these other statements, we should up the probability on forall x phi(x). i guess we should still be entertaining other hypotheses as well
> - i guess this is some sort of solomonoff induction idea. see my writing on solomonoff axiom induction here: `https://github.com/kaarelh/notes`
> - could you investigate whether there is a good version of sth like this here? some further ideas:
>   - we might want to not allow just any enumerable axioms, but ones with templates
>   - we might want to have a derivation length prior (like, statements given without proof in our data set should be fairly easily derived from axioms)
>   - we might want to have a time complexity penalty. like, it should not take too long to generate axioms / to check whether a statement is an axiom
> - i'm interested in whether you can come up with something that robustly picks up the axioms we actually have, or at least something that is equivalent in terms of theorems derived. or if this will at least get a lot of posterior mass

Short answer.

1. **There is a coherent version.** The prior is $2^{-\text{code length}}$ over finite sets of templates, i.e. schemas such as $\varphi(z)$ or induction. The likelihood of a datum is the probability that a random derivation process outputs it. The process cites an axiom, fills in its schematic variables at random, and applies modus ponens, generalisation and $\forall$-elimination. The probability is divided by a normaliser that does not depend on the data (Section 2).
2. **The normaliser does the work.** A theory pays for the probability it spends on sentences that are never observed (the size principle). Your "prove the givens" and "do not contradict" scores have no normaliser, so a consistent strengthening never loses its prior odds under them (Proposition 2.22).
3. **From $\varphi(t)$ to $\forall x\varphi$.** Instances push mass off memorisers and off over-general schemas. They do not make $\forall x\varphi$ more probable than the instance schema $\varphi(z)$, which is just as simple and implies exactly the data (Theorem 3.4).
4. **A derivation likelihood even favours $\varphi(z)$.** Used as a generator, $\{\forall x\varphi\}$ mostly states its own axiom: at $\forall$-elimination rate 0.3, 77% of its output is $\forall x\varphi$ itself. So each observed instance multiplies its odds against $\varphi(z)$ by 0.23. (This is proved in the simplest calculus. With a background theory and a selection-aware likelihood the $\forall$-version can gain through its effective mixture weight, a non-logical effect. With $\wedge$-rules on quantified formulas the direction is open.)
5. **So prediction is confirmed, derivability is not.** "All future data are instances" gets posterior probability tending to 1. With data drawn from the model, "the axioms prove $\forall x\varphi$" tends to its prior share (3–48% for $0 + x = x$, depending on the prior code) or to 0. One instance at a free parameter, $\varphi(p)$, or one quantified datum that entails $\forall x\varphi$, does license it (Theorem 3.9).
6. **Data drawn from some theory's law with fixed weights:** the posterior concentrates on the theories with that law. Under citation, or a derivation likelihood with parameters, these prove exactly the right theorems: a deductively equivalent system is found. With learned weights this holds under citation for almost every weight vector. The posterior does not find the particular axioms: among theories with the same law, the prior decides (Section 4).
7. **Human-stated theorems are not such data.** In finite classes the posterior then goes to the best-fitting generator, which can be strictly weaker than the axioms behind the data, or unsound. For PA it finds the axioms that are used. PA-equivalents won on direct uses of the axioms under

broad usage, within the candidates scored. Narrow or atomic induction practice ended on a weaker theory (Section 7.7.1).

8. **"Statements given without proof should be easily derivable"** turns short theorems with long proofs, and frequently used lemmas, into axioms. Under a two-part code, deriving each of six library theorems of PA costs about 6 to 24 times as much as adopting it as an axiom (Table 3).
9. **A time penalty on checking or generating axioms leaves your collapse intact in the direction from function to axiom induction:** padded Craig sets turn every consistent assigner into axioms at constant cost (a Kt penalty: $\log_2(|f|+1)+O(1)$ bits). Penalising consistent function induction but not proof search makes axiom induction strictly stronger, as you objected, through free deduction rather than its axioms. Templates block the acceptance half of your schema, but over PA one reflection sentence still reproduces every Σ_n-sound assigner on Σ_n sentences. Charging deduction prices the collapse: if the axiom instances of each derivation, logical ones included, have bounded total size, axiom induction is function induction with short certificates (Remark 6.1 and Section 6.6).
10. **For the trichotomy, keep (belief, plausibility):** renormalising and 50/50 are incoherent in general. All positive guarantees need data drawn from the model. AI referees with independent code checked first versions of the research notes. No human has checked the mathematics.

This paper checks an idea within a precise model; it does not design a prover.

1.1 Hänni's note

Hänni's working notes (Hänni, 2026) weigh the axiom systems that prove the labelled givens and not their negations, or that "do not contradict the given statements", possibly by the givens proved and proof length (Section 2.6); each other statement gets "a p(true/false/independent)", and he asks whether to renormalise, split 50/50 or keep the triple (Section 2.7). He argues that axiom induction equals function induction over consistent *assigners*, programs labelling sentences accept or reject, as an assigner becomes the decidable schema "if the assigner accepts φ, then φ", and turns to substitution schemas (Section 6). His other ideas and his time-bounded note are taken up in Section 9.3.

1.2 The question as we read it, and the model

We read the question as asking for (1) *a good version*, a precise Bayesian inducer with a template prior, a likelihood favouring short derivations and a time penalty; (2) what it does with instances $\varphi(t)$ of $\forall x\varphi$; (3) *guarantees*: when the posterior concentrates on the actual axioms or a deductively equivalent system, or gives them much mass, and why it fails.

The model (Section 2) is that of short-answer items 1–2 and builds on two earlier reports, AS (Hänni and Claude (Anthropic), 2026a), on axiom schemas, and IL (Hänni and Claude (Anthropic), 2026b), on inferential learning. A *theory* is a finite set of *templates* (formulas with metavariables, in AS's class DT$^\circ$) with prior π and fixed or Dirichlet mixture weights; data are drawn independently from the law P_{T^*} of a generating theory T^*. L1 is the derivation likelihood. Under *citation*, L0, a datum is a template filled in by the grammar, so the support of P_T is the *instance union* of T; under L1 with *parameters admissible* (free names in instantiations) it is the theorem set.

1.3 Answers

A1. A good version exists, in a precise and limited sense: the model above, whose likelihood is a computable real with undecidable positivity. The recommended inducer (Section 9.2) adds a model of which theorems get reported and of mistakes, and reports belief and plausibility.

A2. For $\forall x\varphi$, prediction is confirmed and derivability is not. In every comparison analysed (Theorem 3.4) closed instances keep the odds of $\forall x\varphi$ against $\varphi(z)$ at the prior odds (citation with the closure reading, a likelihood conditioned on closed-instance data, Hänni's scores), lower them by a constant factor per datum (derivation likelihoods in the minimal calculus of citation and $\forall$-elimination) or set them to zero (strict citation), up to non-logical weight effects under a selection-aware likelihood with a background ($\wedge$-rules: open). With data from a model generator, "the axioms prove $\forall x\varphi$" tends to the prior share of provers in the *generator class* C^* of theories whose law equals the data's, or to 0 if C^* has none (Theorem 3.9): a Bayesian ω-gap, since all closed instances together do not entail $\forall x\varphi$. If the likelihood ignores how the data were selected, the limit depends on the class.

A3. Templates block Hänni's collapse schema, not the collapse; nor do they see a schema's boundary. No non-ground template fits the acceptance part of his schema, but over PA one ground reflection sentence reproduces every Σ_n-sound assigner on Σ_n sentences; for merely consistent assigners this is open (Remark 6.1). A schema and its *root split* (one template per root symbol of the substituted expression) tie at matched weights, differ by Occam terms with learned weights, and the split wins linearly when the instantiation grammar misfits the usage: as far as proved, a derivation likelihood does not change AS's MDL finding, and the posterior tracks usage, not the logical boundary of a schema (Remark 4.13).

A4. Read as "statements given without proof should be easily derivable", a derivation-length prior makes the data theorems, and a derivation likelihood then favours the theory that makes them cheap to state: under a two-part code, on a computed stream of six library theorems, Q plus the theorems, strictly weaker than PA, beats $\mathsf{Q} + T_{\mathrm{Ind}}$ (Robinson arithmetic with the induction template) within the candidates scored (Remark F.10). The generator wins on data from its own derivation process under the full-sum likelihood, or on direct uses of its schemas.

A5. A time penalty on checking or generating axioms leaves the collapse intact in the direction from function to axiom induction; with deduction free, penalising consistent function induction makes axiom induction strictly stronger; charging deduction prices the collapse (Remark 6.1 and Section 6.6). For each polynomial bound on membership time, a hard decidable assigner forces every theory within that bound and consistent with its labels to use, infinitely often, derivations of at least a fixed polynomial root of its nondeterministic time; the collapse constructions pay at most a polynomial of the deterministic time (proof sketch). Bounding each derivation's axiom material makes axiom induction function induction with certificates, bounding each axiom instance makes it exponential-time, bounding only the theory's own axioms sets no limit. For plain L1 only a logarithmic charge is proved, a polynomial one conjectured.

A6. His last bullet depends on the data. *Data drawn from a law of the model* are sampled from some theory's likelihood, with fixed weights unless stated. On human-stated theorems the posterior goes, in finite classes, to the best-fitting (KL-minimising) generators; in the full class even this can fail (Theorem 4.16).

	data drawn from a law of the model	human-stated theorems
the actual axioms	No: inside the generator class the posterior equals the prior, so the actual axioms keep $\pi(T^*)/\pi(C^*)$. Under L1, equivalent axiomatisations with different laws are told apart, and the generating one wins (Cor. 4.3 and Prop. 4.5)	No: it picks up the axioms that are used. Usage decides among equivalent forms, and theorem data are memorised (§ 7.7.1)
a deductively equivalent system	Yes with fixed weights: the theorem set under L1 with parameters admissible, the instance union under L0. With learned weights, under L0, for almost every weight vector (Cor. 4.4 and Thm. 4.7)	Not robustly: limits can be weaker or unsound (Tab. 18). PA-equivalents win only on broad direct-use practice, within the candidates scored (§ 7.7.1)
a lot of posterior mass	The actual axioms keep their prior share inside the generator class, which the data never move	Only within the hand-picked candidate sets scored (§ 7.7.1)

A7. Side results. The posterior, used as a proof checker, accepts a sentence when theories deriving it carry at least $1-\delta$ of the mass. It is guaranteed sound only for well-specified data: with fixed weights and data from the model no adaptive prover ever gets a non-theorem accepted, with probability at least $1-\delta/\pi(C^*_d)$ ($C^*_d \subseteq C^*$: T^*'s theorems up to size d), uniformly in time (Theorem 5.2); with learned weights, on average over the weight prior and, for likelihoods linear in the weights, with a threshold shrinking in n, otherwise only on average. Misspecified, there is no guarantee: in an example a waiting prover wins with probability 1; elsewhere the verifier is merely incomplete. Hänni's trichotomy is (Bel, Dis, Indep), the masses of the theories that prove s, refute it, or neither: keep Bel and $\mathrm{Pl} = 1 - \mathrm{Dis}$, a coherent lower and upper probability, as renormalising and the 50/50 split are incoherent in general (Section 2.7).

1.4 What is not achieved

(i) No posterior over the full template class is computed; candidate sets and pools are hand-picked, and winners changed with the pool (Remark 8.2). (ii) Misspecified limits are known only for finite classes and specific families. (iii) Conjectured, not proved (Appendix H.6): the Occam rate under L1, identification for every weight vector, a polynomial charge by plain L1, the over-Q limits of the split–memoriser races, the memoriser rate. (iv) No human-written theorem corpus is used, and (v) PA is never computed under the generative L1. (vi) Nothing is checked in a proof assistant or by a human; results added in revision were checked only by their tracks and the section writers; several references are unverified (Appendix H).

1.5 Earlier work, method and reader's guide

AS and IL are cited by their own numbering. AS treats the same questions without priors (templates, matching, the ω-gap, the MDL finding, DTRC, a learner for unlabelled mixtures that clusters the data by refuting merged templates), and its code is reused; IL supplies the Ville argument of Section 5. The results come from four independently refereed research notes and two refereed follow-ups on the time penalty (Appendix H). Each theorem-like statement carries a status in brackets (proved; computed, by a seeded script; known, with a reference; proof sketch; conjecture; refuted, kept with its counterexample) and, as a small grey note, its item in the notes (e.g. "model Thm 4.1"). The research brief had seven hypotheses, judged in Table 41. *Reader's guide:* §2: the model, Hänni's scores and trichotomy; §3: $\forall x\varphi$; §4: what the posterior identifies;

§5: the verifier; §6: the collapse and time penalties; §7: PA, ZF and robust failures; §8: exact posteriors over finite pools, experiments E1–E8; §9: the good version, Hänni's note, literature, open problems; appendices: proofs, tables, the record of verification.

2 The model

This section fixes theories, calculi, priors, the instantiation grammar, likelihoods, Hänni's scores and his trichotomy. Proofs are in Appendix A, with Tables 6 and 7, which list the calculi and the likelihoods (normalisers, size principle, names in the research notes).

2.1 Syntax and theories

We use the syntax of AS §2. The languages are $L_A = \{0, S, +, \cdot, <, =\}$ and $L_\in$. Formulas may contain *parameters* $p, p_1, \ldots$ (free names) and are read under their universal closure; leading universal quantifiers are not stripped, so $\forall x\,(x + 0 = x)$ and $p + 0 = p$ are different data with equivalent closures (closure-normal form, AS §2.1). $\mathcal{S}$ is the set of these formulas, called *sentences*.

A *template* (AS Def 2.1) may contain metavariables applied to metavariable-free arguments, $M(t_1, \ldots, t_n)$. Term metavariables are lower case (z, z' if 0-ary; f, g), formula metavariables capital (P, F). An occurrence is a *pattern occurrence* if its arguments are distinct bound variables; in the class DT° every metavariable has one. $\mathrm{FO} \subseteq \mathrm{PAT} \subseteq \mathrm{DT}^\circ \subseteq \mathrm{SO}^\circ$, and DT°_F (0-ary term metavariables) is the class implemented in code. A *ground* template is a single sentence. A substitution θ maps metavariables to bodies, λ-terms whose only free names are parameters, and $\mathrm{inst}(\tau) = \{\tau\theta\}$ (AS Def 2.2); whether parameters are admissible in bodies is said at each use. *Matching* (AS Thm 2.5): for $\tau \in \mathrm{DT}^\circ$ and $s \in \mathcal{S}$ at most one θ has $\tau\theta = s$, found in time $O(|\tau| + |s|)$. The chains of Sections 3 and 8 give each metavariable a *guard*: *closed* (no parameter in the body; the default) or *open*.

Robinson arithmetic Q has the axioms Q1 $\forall x\, \neg Sx = 0$; Q2 $\forall x \forall y (Sx = Sy \to x = y)$; Q3 $\forall x (\neg x = 0 \to \exists y\, x = Sy)$; Q4 $\forall x\,(x + 0 = x)$; Q5 $\forall x \forall y\,(x + Sy = S(x + y))$; Q6 $\forall x\,(x \cdot 0 = 0)$; Q7 $\forall x \forall y\,(x \cdot Sy = x \cdot y + x)$, numbered as in AS. Q has no axiom about $<$; it is Σ_1-complete for $<$-free Σ_1 sentences. Section 7 writes Q3 as $x = 0 \vee \exists y\, x = Sy$. Induction is the template $T_{\mathrm{Ind}} = (P(0) \wedge \forall x(P(x) \to P(Sx))) \to \forall x P(x) \in \mathrm{DT}^\circ \setminus \mathrm{PAT}$.

Definition 2.1 (Theories)**.** model Def 1.1 A *theory* is a finite set $T = \{\tau_1, \ldots, \tau_k\}$ of DT° templates, taken up to renaming of metavariables; $\bigcup \mathrm{inst}(T) := \bigcup_i \mathrm{inst}(\tau_i)$. $\mathcal{H}$ is the class of all theories. A *weighted theory* is a pair (T, w) with w in the open simplex over T.

2.2 Calculi

Definition 2.2 (Background calculus; theorems)**.**
model §1.2 The background calculus is Mendelson's first-order calculus with equality K (Mendelson, 1997, Ch. 2; chapter and numbering not checked): axiom schemas A1 $B \to (C \to B)$; A2 $(B \to (C \to D)) \to ((B \to C) \to (B \to D))$; A3 $(\neg C \to \neg B) \to ((\neg C \to B) \to C)$; A4 $\forall x B \to B[t/x]$, t free for x; A5 $\forall x(B \to C) \to (B \to \forall x C)$, x not free in B; reflexivity and substitutivity $x = y \to (B(x, x) \to B(x, y))$ of equality; rules MP and Gen. Mendelson's free variables are our parameters; Gen infers $\forall x\, \varphi[x/p]$ from φ. $\mathrm{Th}(T)$ is the set of formulas derivable in K from $\bigcup \mathrm{inst}(T)$, with Gen on every parameter; $\mathrm{Th}_d(T)$ the set of those with a derivation of symbol size at most d, axioms included.

Since parameters are read under closure, $\varphi(p)$ and $\forall x\varphi$ are interderivable (AS Prop 4.10), while a closed instance $\varphi(t)$ is not (AS Ex 4.9). K is complete (known). $\mathrm{Th}(T)$ and $\mathrm{Th}_d(T)$ depend on T only through $\bigcup\mathrm{inst}(T)$. The unit of d is stated at each use: symbol size here and in Section 5, chain steps in Section 8, derivation size in Section 7.

$\forall$-elimination is not a template (Lemma A.5, in Appendix A.3), so it belongs to the fixed background (compare AS Rem A.3); the other logical axioms are templates. Derivation grammars use $\forall$E as a primitive rule; it is derivable in K, so Th is unchanged.

Definition 2.3 (The restricted calculi)**.** universal §1.4; experiments §1.6; pa Def 0.3; model Ex 3.6 Later sections use derivation grammars in restricted calculi (rules in Appendix A.3, comparison in Table 6): the chains $\mathsf{C}_{\min}$ (cite with probability $1-c$, else $\forall$E) and $\mathsf{C}_{\mathrm{open}}(c,g)$ (plus Gen on the parameter, probability g; $p_{\mathrm{cite}} = 1-c-g$) and the trees C_{U3}, $\mathsf{C}_{\wedge}$ of Section 3; the forward chain Ch_J of Section 8 (stop probability c_{stop} while a rule applies; $c_{\mathrm{ch}} := 1 - c_{\mathrm{stop}} = \frac{1}{2}$); natural deduction ND with the two-part code $\mathsf{L1}^{\mathrm{sch}}$ (Section 7); and the equational grammar of $\mathsf{L1}^{\mathrm{eq}}$.

2.3 Prior

Definition 2.4 (Template code and prior)**.** model Def 1.3 Templates are coded by a prefix code on their preorder token sequences, one token per node (details in Appendix A.1). Renamings get the same code, argument permutations ($M(x,y)$, $M(y,x)$) different ones. A theory is coded as its number of templates followed by their codes in lexicographic order; $\ell(T)$ is the length in bits, and

$$\pi_\lambda(T) := 2^{-\lambda\ell(T)}/Z_\lambda, \qquad \lambda \geq 1, \qquad Z_\lambda := \textstyle\sum_{T\in\mathcal{H}} 2^{-\lambda\ell(T)}.$$

Mixture weights get the prior $\mathrm{Dir}(\alpha,\dots,\alpha)$, by default $\alpha = \frac{1}{2}$ (Krichevsky and Trofimov, 1981). Statements that need fixed weights say so.

Lemma 2.5 (Kraft)**.** [proved] model Lemma 1.4 *Theory codes are prefix-free, so* $\sum_{T\in\mathcal{H}} 2^{-\ell(T)} \leq 1$ *and* π_λ *is a probability for* $\lambda \geq 1$*; and* $\sum_T \pi_\lambda(T)^{1/2} < \infty$ *for* $\lambda \geq 2$.

Other priors (Remark A.1, in Appendix A.1): the experiments' prior is proper for $\lambda \geq 1$; those of Sections 3 and 7 are improper, used only on finite or summable families; a Levin-style time factor changes $\ln\pi$ only by $O(\ln\ell(T))$ inside DT°.

2.4 Instantiation grammar

Definition 2.6 (Instantiation grammar $\mathcal{Q}$)**.** model Def 1.5 $\mathcal{Q}$ is a probabilistic context-free grammar on metavariable bodies with one production law per *type* (s,k): the sort $s \in \{\iota, o\}$ of the node and the number k of variables available there (holes and bound indices alike). A body of type $\iota^n \to s$ starts at type (s,n); a node chooses a function or relation symbol or connective (children of type $(\cdot,k)$), a quantifier (child of type $(o,k+1)$), one of the k variables, a constant, or a parameter (from a geometric law). $\mathcal{Q}(\theta) := \prod_M \mathcal{Q}_{\mathrm{type}(M)}(\theta(M))$. We require *full support* (no production has probability 0) and *uniform subcriticality*: at most $b < \infty$ children per production, and type weights $h \in [h_{\min}, h_{\max}]$, $h_{\min} > 0$, and $\rho < 1$ with $\mathbb{E}_i\big[\sum_{\text{children } c} h(\mathrm{type}(c))\big] \leq \rho\, h(i)$ for all i.

Under these conditions bodies are finite a.s. with exponential tails, and $\mathcal{Q}_\tau(s) := \mathcal{Q}(\theta_s)$ if $\tau\theta_s = s$, else 0, is a probability on $\mathcal{S}$, with support $\mathrm{inst}(\tau)$ under full support (Lemma A.2, in Appendix A.2). The uniform condition replaces a per-type condition that does not give finiteness (Remark A.3); it holds for a grammar for L_A with $\rho = 0.75$ (Example A.4), for the term laws of Definition 3.1 and for the experiments' grammar. The code of Section 7 instead *learns* a positional grammar (a sequential KT model of the body), not a fixed PCFG, so results that assume a fixed $\mathcal{Q}$, such as the exact split ties of Proposition 4.6, do not apply to it verbatim (Proposition F.6).

2.5 Likelihoods

Definition 2.7 (L0: axiom citation)**.** model §1.5 For data $D = (X_1, \dots, X_n)$, a sequence of sentences, and fixed weights, $P_{T,w}(s) := \sum_{\tau \in T} w_\tau \mathcal{Q}_\tau(s)$ and $P_{T,w}(D) := \prod_i P_{T,w}(X_i)$. With Dirichlet weights, $P_T^{\mathrm{Dir}}(D) := \int \prod_i P_{T,w}(X_i) \,\mathrm{Dir}(dw; \alpha)$.

Lemma 2.8 (Dirichlet labelling sum)**.** [proved; computed]
model §1.5 (c8); experiments Prop X2 *With $A := \sum_\tau \alpha_\tau$, summing over labellings $\ell : \{1..n\} \to T$ with $X_i \in \mathrm{inst}(\ell_i)$, and $n_\tau(\ell)$ the number of data labelled τ,*

$$P_T^{\mathrm{Dir}}(D) = \sum_\ell \frac{\Gamma(A)}{\Gamma(A+n)} \prod_{\tau \in T} \frac{\Gamma(\alpha_\tau + n_\tau(\ell))}{\Gamma(\alpha_\tau)} \prod_{i=1}^{n} \mathcal{Q}_{\ell_i}(X_i).$$

P_T^{Dir} is exchangeable and consistent in n, and its normaliser does not depend on D. For disjoint instance sets the sum has one term, and $P(X_{n+1} = s \mid D) = \sum_\tau \frac{\alpha_\tau + n_\tau}{A+n} \mathcal{Q}_\tau(s)$; at $\alpha_\tau = \frac{1}{2}$ this is, per template, the Krichevsky–Trofimov (KT) estimator, the Dirichlet($\frac{1}{2}$) predictive rule.

Definition 2.9 (L1: derivation grammar)**.** model §1.5 Fix rule probabilities $\alpha_{\mathrm{ax}}, \alpha_{\mathrm{lg}}, \alpha_{\mathrm{MP}}, \alpha_{\mathrm{Gen}}, \alpha_{\forall \mathrm{E}}$ summing to 1, with $\alpha_{\mathrm{ax}} + \alpha_{\mathrm{lg}} > 0$; put $\alpha_{\mathrm{r}} := \alpha_{\mathrm{MP}} + \alpha_{\mathrm{Gen}} + \alpha_{\forall \mathrm{E}}$ and $m := 2\alpha_{\mathrm{MP}} + \alpha_{\mathrm{Gen}} + \alpha_{\forall \mathrm{E}}$. A tree is grown from the root; each node independently cites a theory axiom (α_{ax}: τ with probability w_τ, then $\theta \sim \mathcal{Q}$; conclusion $\tau\theta$), cites a logical axiom (α_{lg}: a schema of K uniformly, instantiated by $\mathcal{Q}$), or applies MP (two children), Gen (one child; a parameter from a geometric law is abstracted) or $\forall$E (one child; a term $t \sim \mathcal{Q}$). A tree is *valid* if at each MP node the right child concludes (left conclusion $\to B$), the node concluding B, and each $\forall$E node's child has root $\forall$. Then $\mu_T(s) := \Pr[\text{valid, conclusion } s]$, $Z_T := \sum_s \mu_T(s)$ and $P_T(s) := \mu_T(s)/Z_T$.

Subscripted $\alpha_\bullet$ are rule probabilities; the Dirichlet parameter is α or α_τ. The code length $-\log_2 \Pr(\text{tree})$ counts *grammar choices*, not the symbols of the formulas; the two can differ exponentially (Remark 6.11).

Lemma 2.10 (L1 is proper; support)**.** [proved; extinction criterion known; computed] model Lemma 1.7
(a) Let $\mathcal{Q}$ be uniformly subcritical. The tree is finite a.s. iff $m \le 1$ (Athreya and Ney, 1972, Ch. I; numbering not checked); for $m < 1$ it has $1/(1-m)$ nodes in expectation; for $m > 1$ it is infinite with probability $1 - q > 0$, q the least root of $q = (\alpha_{\mathrm{ax}} + \alpha_{\mathrm{lg}}) + (\alpha_{\mathrm{Gen}} + \alpha_{\forall \mathrm{E}})q + \alpha_{\mathrm{MP}} q^2$. (b) $Z_T \ge 1 - \alpha_{\mathrm{r}} > 0$ for $T \ne \emptyset$, and $Z_\emptyset \ge \alpha_{\mathrm{lg}}$; so, when $Z_T > 0$, P_T is a probability whose normaliser does not depend on the data. (c) If all rule probabilities and weights are positive, $\mathcal{Q}$ has full support and parameters are admissible *(in logical-axiom instances, $\forall E$ terms and Gen), then* $\mathrm{supp}\, P_T = \mathrm{Th}(T)$. *Without parameters Gen is vacuous; e.g. $\forall x(R(x) \to R(x))$ then has no tree.*

Definition 2.11 (Variants)**.** model §1.5, Ex 3.6; universal §1.5; pa Def 0.3; experiments §1.7 L2: L1 with citation forced at depth d; support $\mathrm{Th}_{\le d}(T)$, the conclusions of valid trees of depth $\le d$. L1$^\sigma$: $\mu_T^\sigma(s) := \sum_{\pi \text{ valid, concluding } s} \Pr(\pi) e^{-\kappa|\pi|}$, $\kappa > 0$, $|\pi|$ the total symbol size of the tree's conclusions; $P_T^\sigma := \mu_T^\sigma / Z_T^\sigma$, a probability with a data-independent normaliser and the support of L1 (Lemma A.7). L1$^{\max}$ (two-part, Viterbi): $V_T(s) :=$ the largest probability of a single valid tree concluding s (under L0: of a single citation). L1^{sch}, L1^{naive} (Definition 7.1): two-part codes over ND derivations with β bits per *written symbol*, a two-part form of L1$^\sigma$, not of plain L1. L0^{cl}: as L0, but a universal sentence $\forall \bar{x} \chi$ is cited through its instances $\chi(\bar{t})$. L1^{sel} (stream filter): $P_T(s \mid S) = P_T(s)/P_T(S)$ for $s \in S$, a set known to contain every datum. L1$^{\mathrm{sel}}_{\mathrm{cit}}$ (per-citation rejection): $\sum_i w_i a_i(s)/c_i$, with $a_i(s)$ the probability that a chain from component i ends in s and c_i that it ends in S; it equals L1^{sel} for one-component theories only (Remark A.6). Noise mixtures: $P_T^\eta := (1-\eta)P_T + \eta N$, N

a fixed full-support law, $\eta \in (0,1)$. L_ε: $(1-\varepsilon)P_T(s) + \varepsilon\mu_0(s)[T \nvdash_k \neg s]$, μ_0 fixed, $\nvdash_k$ a bounded refutation search; a sub-probability whose mass depends on T, not on the data. $\mathsf{L1}^{\mathrm{eq}}$: μ_T/Z_T for an equational grammar (Appendix A.3), terms of size $\leq N$.

2.6 Hänni's scores

Hänni's notes (Hänni, 2026)[1] keep axiom systems that prove the givens "and not their negations", or those "that do not contradict the given statements", and let the weight "depend [...] on how many of the given statements can be proven. (one could also penalize longer proofs i guess?)".

Definition 2.12 (Hänni's scores)**.** model §1.5; universal §1.5; pa Def 0.3 For labelled data $D = D_+ \cup D_-$ (true, false): $S_{\mathrm{prove}}(T;D) := 1[T \text{ consistent}] \prod_{s\in D_+} 1[T \vdash s] \prod_{s\in D_-} 1[T \vdash \neg s]$; $S_{\mathrm{nc}}(T;D) := 1[T \cup D_+ \cup \neg D_- \text{ consistent}]$; $S_g(T;D) := S_{\mathrm{nc}}(T;D) \prod_{s\in D_+} g(\ell_T(s))$, $\ell_T(s)$ the least derivation size (∞ if none), g non-increasing and $g_\infty := g(\infty) \in [0,1)$. The posterior is $\pi(T)S(T;D)$. These are scores, not likelihoods: $\sum_D S(T;D) \neq 1$ and depends on T. For $\beta > 1$, the score $S_{\mathrm{nc},\beta}$ of Section 3 equals $\beta^n S_g$ with $g \equiv 1$ on provable data and $g_\infty = 1/\beta$; $S_{\mathrm{nc},1} = S_{\mathrm{nc}}$. The noisy likelihood L_ε (Definition 2.11) is related but not equal: at $\varepsilon = 1$ it is $\mu_0(D)\prod_{s\in D}[T \nvdash_k \neg s]$, a per-datum, bounded-search relaxation of S_{nc} on positive data, which equals S_{nc} up to the common factor $\mu_0(D)$ only when joint consistency of $T \cup D$ reduces to non-refutation of each datum within k steps; as $\varepsilon \to 0$ it tends to the generative P_T, whose support ($\mathsf{L1}$, parameters admissible) is $\mathrm{Th}(T)$.

2.7 Hänni's trichotomy

His notes also give each sentence "a p(true/false/independent)" and ask three questions about it.

Definition 2.13 (Belief, disbelief, independence)**.**
model §7; universal §1.6 For a posterior π_D on a countable class, a sentence s and $\vdash$ derivability in K without a bound: $\mathrm{Bel}(s) := \pi_D(T \vdash s)$, $\mathrm{Dis}(s) := \pi_D(T \vdash \neg s)$, $\mathrm{Indep}(s) := \pi_D(T \nvdash s \text{ and } T \nvdash \neg s)$, $\mathrm{Inc} := \pi_D(T \text{ inconsistent})$, $\mathrm{Pl}(s) := 1 - \mathrm{Dis}(s)$ and $\mathrm{Bel}_{\mathrm{cons}}(s) := \pi_D(T \text{ consistent and } T \vdash s)$.

Bounded $\mathrm{Th}_d(T)$ is not closed under consequence, so Proposition 2.14(a), (c), (d) and Proposition 2.18 need $d = \infty$; Proposition 2.19 holds for every d (proofs in Appendix A.7).

Proposition 2.14 (Basic laws)**.** [proved] model Prop 7.1 *(a)* $\mathrm{Bel}(s) + \mathrm{Dis}(s) + \mathrm{Indep}(s) = 1 + \mathrm{Inc}$*; from now on the posterior is on consistent theories, so the three sum to 1. (b)* $\mathrm{Dis}(s) = \mathrm{Bel}(\neg s)$*;* $\mathrm{Bel}(\top) = 1$, $\mathrm{Bel}(\bot) = 0$*;* $\mathrm{Pl}(s) = \mathrm{Bel}(s) + \mathrm{Indep}(s)$*. (c) If* $s \vdash s'$ *then* $\mathrm{Bel}(s) \leq \mathrm{Bel}(s')$*. (d) If* $\vdash \neg(s \wedge s')$ *then* $\mathrm{Bel}(s \vee s') \geq \mathrm{Bel}(s) + \mathrm{Bel}(s')$*.*

Example 2.15 (Renormalising is incoherent)**.** [proved; computed] model Ex 7.4 Hänni's first question proposes to "get true/false by either just removing the independent fraction and renormalizing"; this is incoherent in general, as he later suspects. Let $R(s) := \mathrm{Bel}(s)/(\mathrm{Bel}(s) + \mathrm{Dis}(s))$. With atoms a, b and theories $\{a\}$, $\{b\}$, $\{\neg(a \wedge b)\}$ of posterior $\frac{1}{3}$ each, $R(a) = R(b) = 1$ but $R(a \wedge b) = 0$; no probability does this.

Example 2.16 (The 50/50 rule is incoherent)**.** [proved; computed]
model Ex 7.5 His second proposes to "turn the independent fraction into 50/50" and expects "the downside of not giving coherent probabilities usually"; confirmed, with an exact criterion in Proposition 2.17. Let $F_{1/2}(s) := \mathrm{Bel}(s) + \frac{1}{2}\mathrm{Indep}(s)$. Already for the empty theory over atoms b, c: $F_{1/2}(b) = F_{1/2}(b \wedge c) = F_{1/2}(b \wedge \neg c) = \frac{1}{2}$, while coherence needs $P(b \wedge c) + P(b \wedge \neg c) = P(b)$.

[1] File `research/prior/hanni-solomonoff-axiom-induction.md` of the accompanying repository.

Proposition 2.17 (When the 50/50 rule is coherent)**.** [proved; computed] model Prop 7.6 *Over finitely many propositional atoms, $F_{1/2}$ is coherent (equal to some probability on valuations, on every sentence) iff every theory of positive posterior mass has at most two models.*

Proposition 2.18 (Bel is a belief function)**.** [proved; framework known] model Prop 7.2 *For all sentences $s_1, \dots, s_k$, $\mathrm{Bel}(s_1 \vee \cdots \vee s_k) \geq \sum_{\emptyset \neq I \subseteq [k]} (-1)^{|I|+1} \mathrm{Bel}(\bigwedge_{i \in I} s_i)$. So* Bel *is a totally monotone capacity on the Lindenbaum algebra, with focal elements* $\mathrm{Mod}(T)$*, and* Pl *is its plausibility (Shafer, 1976).*

Proposition 2.19 (Coherent bracketing)**.** [proved] model Prop 7.3 *Some probability P on complete consistent theories has $\mathrm{Bel}(s) \leq P(s) \leq \mathrm{Pl}(s)$ for all s.*

So his third, "hmm maybe this motivates not removing the independent fraction?", gets a yes: $(\mathrm{Bel}, \mathrm{Pl})$ is a coherent lower and upper probability. Like the semimeasure he mentions, it keeps the mass on "undecided" (the convention under which Theorem 6.3 holds with a constant); a coherent point value would need a distribution over the completions of each theory.

Remark 2.20 (How the likelihood shapes Indep)*.* [proved for ground ψ; proof sketch when ψ is generated by a schema] model §7 Let ψ be independent of the data. Under Hänni's 0/1 scores the posterior odds of $T^* \cup \{\psi\}$ against T^* stay at their prior value (Proposition 2.22(b)). Under L0 and L1 a theory that proves *and generates* ψ loses mass while ψ never appears: polynomially with Dirichlet weights (Proposition 4.14), exponentially with fixed weights (Lemma 2.21). So that part of $\mathrm{Bel}(\psi)$ falls (all of it under L1, parameters admissible), and $\mathrm{Indep}(\psi)$ grows if the mass moves to theories that neither prove nor refute ψ.

2.8 The size principle

Lemma 2.21 (Size principle, exact form)**.** [proved; computed] model Lemma 1.8 (c1); universal Thm U4(1) *Let P^*, P be probabilities on a countable $\mathcal{S}$, $\varepsilon := P(\mathcal{S} \setminus \operatorname{supp} P^*) < 1$ and $\tilde{P} := P(\cdot \mid \operatorname{supp} P^*)$. (a) $\mathrm{KL}(P^* \| P) = \mathrm{KL}(P^* \| \tilde{P}) + \ln \frac{1}{1-\varepsilon} \geq \ln \frac{1}{1-\varepsilon}$. (b) On data in $\operatorname{supp} P^*$, $P(D) = (1-\varepsilon)^n \tilde{P}(D)$, and $\mathbb{E}_{P^*}[\tilde{P}(D)/P^*(D)] = 1$. Computed checks: Appendix A.5.*

So a hypothesis that puts mass ε outside the data's support loses at least $\ln \frac{1}{1-\varepsilon}$ nats per datum: the size principle (Tenenbaum and Griffiths, 2001) (for sub-probabilities: Theorem B.4).

Proposition 2.22 (Which likelihoods have the size principle)**.** [proved] model Prop 1.9 *(a)* L0*,* L1*,* L2 *and* $\mathsf{L1}^\sigma$ *have it: each P_T is a probability on $\mathcal{S}$ with a data-independent normaliser, so Lemma 2.21 applies to every T that puts mass outside the data law's support (under* L0 *with full-support $\mathcal{Q}$: every T with $\bigcup \mathrm{inst}(T) \not\subseteq \operatorname{supp} P^*$). (b) Hänni's scores do not. Let $\mathrm{Th}(T) \subseteq \mathrm{Th}(T')$. Then $S_{\mathrm{nc}}(T') \leq S_{\mathrm{nc}}(T)$. If T' is consistent with the labelled data, $S_{\mathrm{prove}}(T') \geq S_{\mathrm{prove}}(T)$; if moreover $\bigcup \mathrm{inst}(T) \subseteq \bigcup \mathrm{inst}(T')$, $S_g(T') \geq S_g(T)$. So if $\psi \notin \mathrm{Th}(T^*)$ and $T^* \cup \{\psi\} \cup D_+ \cup \neg D_-$ is consistent, the posterior odds of $T^* \cup \{\psi\}$ against T^* never fall below the prior odds under S_{prove} or S_g, and equal them under S_{nc}. The consistency hypothesis is needed ($\psi = \forall x R(x)$, $T^* = \emptyset$, $D_+ = \{\neg R(0)\}$). (c) Generative likelihoods penalise that strengthening (ψ ground): exactly by $(1-w_\psi)^n$ under* L0 *with fixed weights (T^*'s weights scaled by $1-w_\psi$), on data not containing ψ; at rate at least $\ln \frac{1}{1-\alpha_{\mathrm{ax}} w_\psi}$ nats per datum under* L1*; like $n^{-\alpha}$ with Dirichlet weights (Proposition 4.14).*

Remark 2.23 (Hänni's graded score, normalised)*.* [proved; refuted (old sentence); computed] model Rem 1.10; referee m3 (r6) Let $\ell_T(s)$ be the least length in bits of an explicit derivation of s from T in a prefix-free code that writes every formula out, $g(\ell) := 2^{-\kappa \ell}$, $\kappa \geq 1$, $g_\infty = 0$. Then $Z_T := \sum_s 2^{-\kappa \ell_T(s)} \leq 1$ and $P_T^g := 2^{-\kappa \ell_T}/Z_T$ is a probability; with $g_\infty > 0$ the normaliser is infinite. Like $\mathsf{L1}^\sigma$, P_T^g charges

the *written* length of the derivation (Proposition E.5). If $\bigcup \mathrm{inst}(T) \subseteq \bigcup \mathrm{inst}(T')$, then $Z_{T'} \geq Z_T$, so $P^g_{T'}(s) \leq P^g_T(s)$ whenever $\ell_{T'}(s) = \ell_T(s)$. Hänni's remark that one can "think of this as there being some model for how the statements get generated" holds once the score is divided by Z_T and $g_\infty = 0$. *Refuted* (Appendix H.2): the pre-referee sentence "a stronger theory has more short theorems, hence a larger Z_T, hence a smaller $P_T(s)$ on each datum". For $T = \{a, a \to b\}$ and $T' = T \cup \{b\}$ (same theorems, nested instances) b has a one-citation derivation from T' and needs MP from T, so $P^g_{T'}(b)/P^g_T(b) = 2^{\kappa(\ell_T(b) - \ell_{T'}(b))} Z_T/Z_{T'} \geq 2^{\kappa(\ell_T(b) - \ell_{T'}(b))} Z_T$, which exceeds 1 once $\kappa(\ell_T(b) - \ell_{T'}(b)) > \log_2(1/Z_T)$: a datum whose shortest derivation gets shorter can gain.

2.9 Computing the likelihood

Proposition 2.24 (What can be computed)**.** [proved; (b), (c) given Church's theorem (known)] model Prop 2.7 *Assume* L1 *with $m < 1$, $\mathcal{Q}$ uniformly subcritical, computable grammar probabilities and weights, and the hypotheses of Lemma 2.10(c). (a) $\mu_T(s)$, Z_T and $P_T(s)$ are computable reals, uniformly in (T, s). (b) $\{(T, s) : P_T(s) > 0\} = \{(T, s) : s \in \mathrm{Th}(T)\}$ is recursively enumerable and, for a language with a binary relation symbol, undecidable already for $T = \emptyset$ (Church, 1936; Turing, 1936). (c) Whether a theory has likelihood 0 on a datum is undecidable. For a finite class, $\pi_n(T)$ is a computable real whenever defined, but whether $\pi_n(T) = 0$ is undecidable.*

The proof enumerates trees, exponentially many in $1/\epsilon$ for precision ϵ, and nothing better is known (Appendix A.6); later sections restrict the calculus (Ch_J, $J \leq 2$) or score explicit derivations ($\mathsf{L1}^{\mathrm{sch}}$). The two-part $\mathsf{L1}^{\max}$ is not the sum: the maximum can break a tie inside the generator class, with two-part odds $(\pi(T_2)/\pi(T_1))\, 2^{-n} \to 0$ between theories of equal L0 law and equal theorems (Proposition A.8, in Appendix A.6); whether it can change a *deductive* limit is open. The codes of Section 7 are also two-part.

3 The case $\forall x \varphi$: what instances support

Hänni's question starts from statements $\varphi(t)$ and the law $\forall x \varphi$, which "implies all the other statements" (Section 1). In the model of Section 2 the data favour the generator that emits exactly the instances. In every comparison analysed (Theorem 3.4) they never favour $\forall x \varphi$ over its instance schema σ_φ, except through non-logical weight effects under a selection-aware likelihood (Proposition 3.6); with $\wedge$-rules on quantified formulas the question is open. "All future data are instances" is confirmed; "the axioms prove $\forall x \varphi$" is not, a Bayesian form of the ω-gap of AS §4.5: all closed instances $\varphi(0), \varphi(S0), \ldots$ together still do not entail $\forall x \varphi$.

So the intuition is right against hypotheses that fit worse (over-general templates, memorisers), for prediction, and for truth under a Gaifman-type credence, safely in $\mathbb{N}$. It needs three refinements: σ_φ is a second simple law that implies exactly the data, brought into the class by templates; "the axioms prove $\forall x \varphi$" gains from closed instances only through non-logical differences between the laws, as under $\mathsf{L1}^{\mathrm{sel}}$ with a background (Proposition 3.6), and with data from a model generator it tends to the prior share of provers in the generator class (Theorem 3.9); and a derivation likelihood rewards the most specific generator that fits, and frequently used lemmas, not the logically strongest law. $\forall x \varphi$ is licensed by an instance at a parameter, a quantified consequence, or a selection-aware likelihood without a guard (Proposition 3.10(a), Proposition B.14(a), Proposition 3.12(c)), soundly in $\mathbb{N}$ and unsoundly in $\mathbb{R}$; data at parameters are common in proof-assistant practice. Proofs and further tables: Appendix B; logarithms are natural.

3.1 Setup

Definition 3.1 (Instances, schema, term laws)**.** universal §1.1–1.3 $\varphi(x)$ is an L_A-formula with exactly one free variable x, which occurs in it; the examples ($0 + x = x$, $x + 0 = x$, $x \cdot 0 = 0$, $\neg Sx = 0$, $x < Sx$) are quantifier-free unless stated. $I_{\mathrm{c}}(\varphi) := \{\varphi(t) : t \text{ closed}\}$; $I_{\mathrm{o}}(\varphi)$ also allows parameters in t. The instance schema is $\sigma_\varphi := \varphi[z/x]$, z a 0-ary term metavariable, closed guard unless stated. Term laws: $\mathcal{Q}_{\mathrm{num}}(S^j 0) = (1-q)q^j$, $q = \frac{1}{2}$ by default; $\mathcal{Q}_{\mathrm{GW}}$, each node picking its root $0, S, +, \cdot$ with probabilities $.4, .3, .2, .1$; $\mathcal{Q}_{\mathrm{open}} := \rho[p] + (1-\rho)\mathcal{Q}_{\mathrm{num}}$ for a parameter p. The first two are root-generated: $\mathcal{Q}(f(t_1, \ldots, t_r)) = p_f \prod_i \mathcal{Q}(t_i)$. One law supplies all bodies and $\forall$E terms.

The hypotheses universal §1.3 (Table 8 in Appendix B) are $H_\forall = \{\forall x\varphi\}$; $H_{\mathrm{sch}} = \{\sigma_\varphi\}$; H_{open}, σ_φ with an open guard ($z \sim \mathcal{Q}_{\mathrm{open}}$); $H_{\mathrm{both}} = \{\forall x\varphi, \sigma_\varphi\}$ with weight $w_\forall$ on $\forall x\varphi$; memorisers, finite sets of closed instances; over-specific templates such as $\sigma_\varphi[z := Sz]$; over-general ones, with a subterm or subformula of σ_φ made a fresh metavariable (down to the inconsistent bare F_0); and root splits $\{\sigma_\varphi[z := f(z_1, \ldots, z_r)] : f \text{ a root of } \mathcal{Q}\}$. For $\varphi = 0 + x = x$, $H_\forall$, H_{open} and H_{both} prove $\forall x\varphi$; H_{sch}, memorisers, over-specific templates and root splits do not, even over Q (AS Ex 4.9). $B \oplus_w A$ adds A with weight w to B, scaling B's weights by $1 - w$. Calculi and likelihoods are as in Definitions 2.3 and 2.11. In $\mathsf{C}_{\mathrm{min}}$ each chain node cites (probability $1 - c$) or applies $\forall$E with $t \sim \mathcal{Q}$ (probability c); $\mathsf{C}_{\mathrm{open}}(c, g)$ adds Gen on the parameter (g; cite $p_{\mathrm{cite}} = 1 - c - g$); μ_T is the unnormalised derivation mass and $\mathsf{L1} = \mu_T / Z_T$. Hänni's scores are S_{prove}, S_{nc} and S_g with $g \equiv 1$ on provable data (Definition 2.12); for $\beta > 1$, $S_{\mathrm{nc},\beta} := \beta^n S_g$ with $g_\infty = 1/\beta$ rewards each proved datum by a factor β, and $S_{\mathrm{nc},1} := S_{\mathrm{nc}}$. Theorems about countable classes assume a proper prior (Definition 2.4); the computations use finite or summable families with the prior $2^{-(\sum_A |A| + \#\mathrm{axioms})}$. Defaults: $c = 0.3$; $(c, g, \rho, q) = (0.3, 0.2, 0.1, \frac{1}{2})$ in $\mathsf{C}_{\mathrm{open}}$; tables state α.

3.2 Instance data do not favour $\forall x\varphi$ over its schema

Proposition 3.2 (Factorisation)**.** [proved; computed] universal Prop U1 *In $\mathsf{C}_{\mathrm{min}}$, let φ have one free variable and m leading universal quantifiers. Then $\mu_{H_\forall}(s) = c\,\mu_{H_{\mathrm{sch}}}(s)$ for every sentence $s \neq \forall x\varphi$; $\mu_{H_\forall}(\forall x\varphi) = 1 - c$, $\mu_{H_{\mathrm{sch}}}(\forall x\varphi) = 0$; $Z_{H_{\mathrm{sch}}} = (1-c)\sum_{k \le m} c^k$, $Z_{H_\forall} = (1-c) + cZ_{H_{\mathrm{sch}}}$. With a background, $\mu_{B \oplus_w \forall x\varphi} = (1-w)\mu_B + wc\,\mu_{H_{\mathrm{sch}}}$ and $\mu_{B \oplus_w \sigma_\varphi} = (1-w)\mu_B + w\,\mu_{H_{\mathrm{sch}}}$ on $s \neq \forall x\varphi$. For k variables the factor is c^k on each full instance.*

Theorem 3.3 (Odds on instance data)**.** [proved; computed]
universal Thm U2 *In $\mathsf{C}_{\mathrm{min}}$, after n closed instances the odds $O_n := \pi_n(H_\forall)/\pi_n(H_{\mathrm{sch}})$ satisfy: O_n/O_0 is 1 under $\mathsf{L0}^{\mathrm{cl}}$, under $\mathsf{L1}^{\mathrm{sel}}(S)$ with $\forall x\varphi \notin S$, and under the scores ($\forall x\varphi$ consistent); c^n under μ_T and $\mathsf{L1}^{\mathrm{max}}$; $\left(cZ_{H_{\mathrm{sch}}}/((1-c) + cZ_{H_{\mathrm{sch}}})\right)^n$ under $\mathsf{L1}$, which is $(c/(1+c))^n$ for quantifier-free φ; 0 for $n \ge 1$ under strict $\mathsf{L0}$; and, with a background B under μ_T, $\prod_i r(X_i)$ with $r(s) \in [c, 1]$.*

So at $c = 0.3$, under $\mathsf{L1}$ with quantifier-free φ, a generator whose axiom is $\forall x\varphi$ outputs $\forall x\varphi$ itself with probability $1/(1+c) = 0.77$, and each observed instance multiplies its odds against σ_φ by $c/(1+c) = 0.23$ (Table 2; checks in Table 9).

Theorem 3.4 (Instance data do not favour $\forall x\varphi$)**.** [proved] universal Thm B *On data not containing $\forall x\varphi$, the posterior odds of the $\forall$-version against the schema version are at most the prior odds:* (B1) *in $\mathsf{C}_{\mathrm{min}}$, single axioms, closed instances, under $\mathsf{L0}^{\mathrm{cl}}$, μ_T, $\mathsf{L1}$, $\mathsf{L1}^{\mathrm{sel}}$, $\mathsf{L1}^{\mathrm{max}}$ and the scores (equality under $\mathsf{L0}^{\mathrm{cl}}$, under $\mathsf{L1}^{\mathrm{sel}}$ when $\forall x\varphi \notin S$, and under the scores when $\forall x\varphi$ is consistent);* (B2) *in $\mathsf{C}_{\mathrm{min}}$, $B \oplus_w \forall x\varphi$ against $B \oplus_w \sigma_\varphi$ for any background B and any data, under μ_T and $\mathsf{L1}$, with a fixed w or a common weight prior (equality under $\mathsf{L0}^{\mathrm{cl}}$);* (B3) *in C_{U3} calculi, quantifier-free*

likelihood	hypothesis	$n=0$	1	5	20	200
$\mathsf{L0}^{\mathrm{cl}}$ ($=\mathsf{L1}^{\mathrm{sel}}$ here)	$H_\forall$; H_{sch}	.021; .042	.194; .387	.303; .605	.332; .665	.332; .665
	over-general; over-specific	.926; .011	.307; .110	.022; .067	.000; .000	.000; .000
	root split, weights p_f	.000	.002	.002	.003	.003
μ_T	$H_\forall$; H_{sch}	.021; .042	.067; .448	.001; .891	.000; .996	.000; .996
	root split, weights p_f	.000	.002	.003	.004	.004
$\mathsf{L1}$	$H_\forall$; H_{sch}	.021; .042	.053; .455	.000; .892	.000; .996	.000; .996

Table 2: Running example (c2): mean posterior over 50 seeds, $\varphi = 0 + x = x$, $\mathcal{Q}_{\mathrm{num}}$, $\mathsf{C}_{\min}$, prior $2^{-(\text{symbols+1 per axiom})}$. Class: $H_\forall$, H_{sch}, four over-general templates (among them the inconsistent bare F_0, 0.674 at $n=0$), $\varphi(Sz)$, and the root split with fixed and with learned ($\alpha=1$) weights; the fixed-weight split has exactly H_{sch}'s law (Proposition 4.6). Four other formulas and $\mathcal{Q}_{\mathrm{GW}}$ behave alike.

φ, any background (Proposition B.2), under μ_T and $\mathsf{L1}^{\max}$; and, for single axioms and quantifier-free data, under the scores, with equality (by Herbrand's theorem, $\{\forall x\varphi\} \vdash s$ iff $I_{\mathrm{c}}(\varphi) \vdash s$ for quantifier-free s).

Proof idea. The mechanism is the size principle, through the extra $\forall$E step that $\forall x\varphi$ needs to reach an instance (Lemma 2.21). (B2) holds at every w, and $Z_{H_\forall} \geq Z_{H_{\mathrm{sch}}}$; (B3) rests on a simulation inequality (Proposition B.2 in Appendix B.1). □

Not covered: $\mathsf{L1}^{\mathrm{sel}}$ with a background. There $B \oplus_w \forall x\varphi$ has the selected law of $B \oplus_{w_e} \sigma_\varphi$ with the effective weight $w_e = wc/(1-w+wc)$ (proof of Proposition 3.6): with a common fixed w the two theories have different laws and whichever matches the data's mixture proportion wins at a linear rate, and with a learned weight the $\forall$-version can gain a bounded factor (Proposition 3.6). Neither effect is logical, and under the per-citation $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ both vanish (Remark 3.7). Also not covered: $\wedge$-detours (Remark 3.5), normalised likelihoods in C_{U3}, and the scores with a background or with quantified data, where the $\forall$-version can be favoured (the datum $\neg\exists x\neg\varphi$ is proved by $H_\forall$ and not by H_{sch}).

The direction is robust (Proposition B.1): in $\mathsf{C}_{\min}$, at every $s \neq \forall x\varphi$, a full-support noise component leaves the ratio of the laws of $H_\forall$ and H_{sch} in $[c,1]$ under μ_T, ≤ 1 under $\mathsf{L1}$ and $= 1$ under $\mathsf{L1}^{\mathrm{sel}}$ with noise after selection; chain weights non-increasing in the number of eliminations keep $P^f_{H_\forall}(s) \leq P^f_{H_{\mathrm{sch}}}(s)$; and under $\mathsf{L1}$, quantifier-free φ and any prior on c, $O_n/O_0 \leq 2^{-n}$.

Remark 3.5 ($\wedge$-detours). [refuted (conjecture); counterexample: proof sketch; computed] universal §2.3 The factor c of Proposition B.2 depends on the calculus (an earlier version conjectured otherwise; Appendix H.2): in $\mathsf{C}_\wedge$ at $(p_{\mathrm{cite}}, c, a, e) = (0.35, 0.15, 0.25, 0.25)$, $\varphi = 0 + x = x$, the ratio $\mu_{H_\forall}(\varphi(t))/\mu_{H_{\mathrm{sch}}}(\varphi(t))$ is $0.15553 > c$ (Appendix B.1); on a 9139-point grid it stays ≤ 0.925 (normalised ≤ 0.48). Whether detours can reverse the direction is open.

Proposition 3.6 (Learned weights under $\mathsf{L1}^{\mathrm{sel}}$)**.** [proved; computed] universal Prop B2 *In $\mathsf{C}_{\min}$ let $B = \{\sigma_\psi\}$, ψ quantifier-free, $\mathrm{inst}(\sigma_\psi) \cap \mathrm{inst}(\sigma_\varphi) = \emptyset$, and compare $B \oplus_w \forall x\varphi$ with $B \oplus_w \sigma_\varphi$ (uniform prior on w, equal prior masses) under $\mathsf{L1}^{\mathrm{sel}}$, S the closed quantifier-free sentences. On data i.i.d. from $(1-w^*)\mathcal{Q}_{\sigma_\psi} + w^*\mathcal{Q}_{\sigma_\varphi}$ the Bayes factor tends a.s. to $h(w^*)$, $h(v) := c/(c+v(1-c))^2 \in [c, 1/c]$, which exceeds 1 iff $w^* < \sqrt{c}/(1+\sqrt{c})$.*

The cause is the prior induced on the effective weight w_e (computed to $4 \cdot 10^{-4}$ at $n = 10^5$). With learned weights the effect is bounded; it is not evidence that the axioms prove $\forall x\varphi$.

Remark 3.7 (Per-citation selection). [proved] editor, from experiments §1.7, Prop X6(b) and universal Prop B2 Under the per-citation $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ the selected law of the component $\forall x\varphi$ is $\mathcal{Q}_{\sigma_\varphi}$, so both theories of Proposition 3.6 have law $(1-w)\mathcal{Q}_{\sigma_\psi} + w\mathcal{Q}_{\sigma_\varphi}$ at every w: the Bayes factor is 1 (Remark A.6).

A spare-slot prover dies only polynomially: with a uniform prior on $w_\forall$, under $\mathsf{L1}$ and quantifier-free φ, H_{both}'s mass falls like $\Theta(1/n)$, as $\forall x\varphi$ is a spare slot, a template beside the generator's own that the data never or rarely use (Proposition B.3).

The experiments reproduce Theorem 3.3 in their chain Ch_1 (factor $\frac{1}{2}$: one bit per datum, to $1.1 \cdot 10^{-11}$) and, under $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$, an exact tie with $P(T \vdash \forall x\varphi \mid D_n) \to 0.484$ (0.269 for $\neg Sx = 0$), the prior share in their code (E1, Table 30).

3.3 Memorisers, over-general and over-specific templates

Over-general templates lose exponentially (Theorem B.4 in Appendix B.2): a law with mass $p < 1$ on $I_{\mathrm{c}}(\varphi)$ loses at least $\ln \frac{1}{p}$ nats per datum, a.s. and in expectation ($\mathbb{E}\, P'(D_n)/P^*(D_n) \le p^n$), and under the citation law ($\mathsf{L0}$, $\mathsf{L0}^{\mathrm{cl}}$) a first-order template strictly more general than σ_φ, with metavariables i.i.d. from $\mathcal{Q}$ and a formula law F, has $p \le \max(\max_f p_f, \max_g F(\mathrm{root} = g))$. Over-specific templates $\sigma_\varphi[z := f(z_1, \dots, z_r)]$ fail too: under a root-generated $\mathcal{Q}$ their odds against σ_φ are a martingale that grows like p_f^{-n} while all data have root f (probability p_f^n) and then vanishes (Proposition B.5). The memoriser class, however, is not exponentially small in general (Proposition B.6): with uniform weights and $\mathcal{Q}$ of infinite support and finite entropy its log-odds against H_{sch} fall faster than linearly, but with Laplace weights under $\mathcal{Q}_{\mathrm{num}}$ they stay $\ge -C \ln^2 n$ a.s. for all large n (conjecturally $-\Theta(\ln^2 n)$); under $\mathcal{Q}_{\mathrm{GW}}$ the computed rate is exponential, as distinct data grow linearly. E1 tests only the single memoriser $\mathrm{Mem}(D_n)$, whose axioms are the distinct data seen so far; every fixed memoriser dies under well-specification (Remark C.4).

3.4 Predictive confirmation

Theorem 3.8 (Predictive confirmation)**.** [proved; computed] universal Thm U8 *Assume setting W (Definition 4.1). Let* $M(\cdot \mid D_n) = \sum_T \pi_n(T) P_T$, $G := \{T : P_T(I) = 1\}$, *and* $T^* \in G$ *the generator.* (1) $\sum_{n \ge 0} \mathbb{E}[1 - M(I \mid D_n)] \le \ln \frac{1}{\pi(T^*)}$. (2) *The probability that all future data lie in* I *is* $\pi_n(G)$, *and* $\pi_n(G) \to 1$ *a.s.* (3) $\mathbb{E}[1 - \pi_n(G)] \le \min\big(1, \sum_{T \notin G} \frac{\pi(T)}{\pi(T^*)} P_T(\mathrm{supp}\, P_{T^*})^n\big)$. (4) *If* $T \in G$ *have law* $\hat{P}$ *and* $T \notin G$ *have law* $(1-\varepsilon_T)\hat{P} + \varepsilon_T N$, $N(I) = 0$, *then on instance sequences* $1 - \pi_n(G) = \sum_{T \notin G} \pi(T)(1-\varepsilon_T)^n / \big(\pi(G) + \sum_{T \notin G} \pi(T)(1-\varepsilon_T)^n\big)$.

The rate depends on near-universal competitors: $\approx (1-\pi_0)/(\pi_0 n)$ for Laplace's rule with a point mass π_0, but $1 - \pi_n(G) = 0.37$ at $n = 10^{50}$ for a dyadic family ($\varepsilon_j = 2^{-j}$ with prior $(1-\pi_0)/(j(j+1))$ on $j \ge 1$, $\pi_0 = 0.01$). Part (2) is the countable-class analogue of Hutter's result for the universal prior, but G contains H_{sch}, which does not prove $\forall x\varphi$; and confirmation is not stepwise monotone (Remark B.7).

3.5 The Bayesian ω-gap

Theorem 3.9 answers "do the axioms prove $\forall x\varphi$?". Its headline is (a) with (b): the posterior probability of a prover tends to the prior share of provers in the generator class $C^* := \{T : P_T = P_{T^*}\}$, which no data move. A likelihood is *faithful* for a calculus C if $\mathrm{supp}\, P_T = \mathrm{Th}_{\mathsf{C}}(T) \cap X$ for all T (X the data space).

Theorem 3.9 (Bayesian ω-gap)**.** [(a)–(c), (d1), (e) proved; computed]
universal Thm U10 *Let $\mathcal{H}$ be countable with a proper prior $\pi > 0$, $\mathrm{Prov}_\forall := \{T \in \mathcal{H} : T \vdash \forall x\varphi\}$ (first-order; inconsistent theories included), and the data i.i.d. from P_{T^*}, $T^* \in \mathcal{H}$, unless stated.*

(a) $P(T \vdash \forall x\varphi \mid D_n) \to \pi(\mathrm{Prov}_\forall \cap C^*)/\pi(C^*)$ *a.s.; also for* $\vdash_{\mathsf{C}}$, $\mathrm{Bel}, \mathrm{Dis}, \mathrm{Indep}, \mathrm{Inc}$ *(Definition 2.13).*

(b) Prior share. *If C^* has a prover and a non-prover, the limit is in $(0,1)$: $H_\forall, H_{\mathrm{sch}}$ under $\mathsf{L0}^{\mathrm{cl}}$ or $\mathsf{L1}^{\mathrm{sel}}$; $B \oplus \forall x\varphi$, $B \oplus \sigma_\varphi$ under $\mathsf{L0}^{\mathrm{cl}}$ with a common weight or weight prior. For the two theories of Proposition 3.6 under $\mathsf{L1}^{\mathrm{sel}}$ the laws differ, but that proposition gives such a limit.*

(c) Faithful regime. *(c1) If $\forall x\varphi \in X$, $P(T \vdash_{\mathsf{C}} \forall x\varphi \mid D_n) \to 1[T^* \vdash_{\mathsf{C}} \forall x\varphi]$. (c2) If also C is sound and all axiom instances lie in X, members of C^* have T^*'s first-order theorems and $P(T \vdash \forall x\varphi \mid D_n) \to 1[T^* \vdash \forall x\varphi]$. (c3) In $\mathsf{C}_{\min}$, and $\mathsf{C}_{\mathrm{open}}$ with a guard, $\mathsf{L1}$ with full-support $\mathcal{Q}$ is faithful and $\operatorname{supp} P_{H_{\mathrm{sch}}} = I_{\mathrm{c}}(\varphi)$ for quantifier-free φ, so for $T^* = H_{\mathrm{sch}}$ the limit in (c1) is 0. If C has a rule that maps an instance to a non-instance with positive probability ($\wedge I$, $\vee I$, symmetry of $=$, $\to I$, MP with cited logical axioms), every T with $P_T(I_{\mathrm{c}}) > 0$ emits non-instances: instance-only data are then misspecified for every hypothesis, and the limit is set by the best-fitting generators of the class (Theorem 4.16).*

(d) Filtered data *from $H_\forall$, seen only in $S = I_{\mathrm{c}}(\varphi)$. (d1) For quantifier-free φ in $\mathsf{C}_{\min}$ the filtered law is H_{sch}'s: limit 0 under $\mathsf{L1}$, the prior share under $\mathsf{L1}^{\mathrm{sel}}$. (d2)* [proved for the pair; proof sketch in general] *For quantified φ, H_{sch} is the KL-minimiser in $\{H_\forall, H_{\mathrm{sch}}\}$ and the limit is 0; for larger classes the minimiser was not determined.*

(e) Unknown filter. *With a countable family $\mathcal{F}$ of filters S with $P_T(S) > 0$ for every $T \in \mathcal{H}$, a prior ν on $\mathcal{F}$ independent of π, and pair laws $P_T(s)1[s \in S]/P_T(S)$, (a) applies to pairs. For $\{H_\forall, H_{\mathrm{sch}}\}$, $\mathsf{C}_{\min}$, quantifier-free φ and data law $\mathcal{Q}_{\sigma_\varphi}$ the limit is $\frac{\pi(H_\forall)\nu(\mathcal{F}_1)}{\pi(H_\forall)\nu(\mathcal{F}_1)+\pi(H_{\mathrm{sch}})\nu(\mathcal{F}_0)}$, $\mathcal{F}_0 = \{S \supseteq I_{\mathrm{c}}\}$, $\mathcal{F}_1 = \{S \in \mathcal{F}_0 : \forall x\varphi \notin S\}$.*

In the prior-share regime the limit lies in $(0,1)$, robustly, but its value depends on the code: for $\varphi = 0 + x = x$ the share of $H_\forall$ in $\{H_\forall, H_{\mathrm{sch}}\}$ is 0.333 in the symbol-count prior above and in π_1 of Definition 2.4, 0.200 in π_2, 0.484 in the experiments' code and 0.030–0.042 in the per-symbol codes of Section 7 (Table 11).

Hänni's own scores leave the question open (Proposition B.8 in Appendix B.4): for $\varphi = 0 + x = x$ under S_{prove} the posterior tends a.s. in total variation to the prior on the consistent provers of $I_{\mathrm{c}}(\varphi)$, which include independent, true and false theories (Q, $\mathsf{Q} + \forall x\varphi$, $\mathsf{Q} + \neg\forall x\varphi$), so all three trichotomy values keep positive mass and over-generalisation is never penalised; under S_{nc} and S_g ($g \equiv 1$ on provable data) the provers of all the data share one factor, with the same conclusion, and under S_{nc} the limit is the prior on the larger set $\{T : T \cup I_{\mathrm{c}}(\varphi) \text{ consistent}\}$, which contains non-provers such as $\emptyset$. Inconsistent theories inflate Bel and Dis alike, so the verifier should use $\mathrm{Bel}_{\mathrm{cons}}$ (Remark 5.9). Closed instances cannot tell Q4 from its closed schema $z+0 = z$, as ($\mathsf{Q}-$Q4) plus all closed instances of $x + 0 = x$ does not prove $\forall x(x + 0 = x)$ (Proposition B.9). For truth, a credence on structures believes $\forall x\varphi$ in the limit of its true closed instances if $P(\forall x\varphi) > 0$ and almost every structure is an elementary extension of its closed-term substructure (Gaifman's condition, cited from memory): safe for $\mathbb{N}$, unsafe for credences that weigh $\mathbb{R}$, V, models of $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ or some models of Q (Proposition B.10).

3.6 Open instances and the closedness guard

Proposition 3.10 (Parameter data and the guard)**.** [proved; computed] universal Prop U11(a)–(d) *Assume the closure reading, Gen over parameters, theories without parameters, and a likelihood supported on theorems.* (a) *Once D contains some $\varphi(p)$, every T with $\pi(T \mid D) > 0$ proves $\forall x\varphi$.* (b) *If $P_{T^*}(\varphi(p)) = \beta > 0$, then $P(P(T \vdash \forall x\varphi \mid D_n) < 1) \le (1-\beta)^n$.* (c) *In $\mathsf{C}_{\mathrm{open}}(c, g)$ with $\mathcal{Q}_{\mathrm{open}}$, per closed datum under $\mathsf{L1}$: $H_{\mathrm{open}} : H_{\mathrm{sch}} = \frac{1-\rho}{1+g\rho}$, $H_\forall : H_{\mathrm{sch}} = \frac{c(1-\rho)}{1+c}$, $H_\forall : H_{\mathrm{open}} = \frac{c(1+g\rho)}{1+c}$.* (d) *If the class is finite and contains the guarded H_{sch}, closed data give $P(T \vdash \forall x\varphi \mid D_n) \to 0$: the guard is learned at rate $\mathcal{W}_{\mathrm{o}} := \ln \frac{1+g\rho}{1-\rho}$ (0.1252 nats per datum at the defaults); the spare-slot prover H_{both} decays only polynomially (like $1/n$ in $\mathsf{C}_{\mathrm{min}}$, Proposition B.3).*

For countable classes with learned-weight provers the limit 0 is not proved. Here ρ is fixed; in the two-part code of Section 7, where it is learned, a head-symbol split of $x + 0 = x$ closes in on the schema at only about 5 bits per doubling of n (Remark F.25).

Without a guard (DT$^\circ$, λ-convention, parameters admissible) the outcome depends on which competitors the class contains; with a selection-aware likelihood the ω-step is taken. The competitors are the unguarded splits $R_k := \{\varphi(0), \dots, \varphi(S^{k-1}0), \varphi(S^k z)\}$ with weights $(v_0, \dots, v_{k-1}, u)$, provers of $\forall x\varphi$ over Q but not in pure logic; on closed numeral data under $\mathsf{L1}$ they lose only $q^k\mathcal{W}_{\mathrm{o}}$ nats per datum at their best weights, against $\mathcal{W}_{\mathrm{o}}$ (H_{open}) and $\ln \frac{1+c}{c(1-\rho)}$ ($H_\forall$; Proposition B.13).

Remark 3.11 (Without a guard). [refuted; computed] universal §7.2 Without a guard the posterior need not take the ω-step on closed data (contrary to an earlier version; Appendix H.2): in $\{H_\forall, H_{\mathrm{open}}, R_1\}$, where $R_1 = \{\varphi(0), \varphi(Sz)\}$ does not prove $\forall x\varphi$ in pure logic, $P(T \vdash \forall x\varphi \mid D_n)$ is 0.697 at $n = 100$ and 0.000 from $n = 1000$. The claim survives for $\{H_\forall, H_{\mathrm{open}}\}$ and under $\mathsf{L1}^{\mathrm{sel}}$ (cf. AS Prop 4.5(b), (c)).

Proposition 3.12 (Who wins without a guard)**.** [proved; computed] universal Prop N3 *Closed data i.i.d. from $\mathcal{Q}_{\mathrm{num}}$; the class contains $H_\forall$, H_{open}, R_k for k in a nonempty $\mathcal{K}$ (learned or optimal weights), possibly memorisers. In pure logic $H_\forall$, H_{open} prove $\forall x\varphi$, and R_k and memorisers do not; over Q all but the memorisers do.* (a) $\mathsf{L1}$*, pure logic, provers only $H_\forall, H_{\mathrm{open}}$: $P(T \vdash \forall x\varphi \mid D_n) \to 0$ a.s., exponentially, at rate $\ge (1-q^k)\mathcal{W}_{\mathrm{o}}$ for each $k \in \mathcal{K}$.* (b) $\mathsf{L1}$*, over Q, $\mathcal{K}$ finite: $\to 0$ with Laplace-weighted memorisers, $= 1$ at every n without them.* (c) [proof sketch] $\mathsf{L1}^{\mathrm{sel}}$*, finite $\mathcal{K}$: $\to 1$ in pure logic.* (d) [computed; limit conjecture] $\mathsf{L1}$*, $\mathcal{K} = \{1, \dots, 40\}$, memorisers:* 1.000 *over Q from $n = 10$ to 10^{12}, with ever deeper splits.*

(b) is proved for Laplace-weighted memorisers; with the Dirichlet($\frac{1}{2}$) weights of Table 13 the over-Q value also falls to 0 with $R_{1..4}$. By (d), a likelihood that ignores the selection approximates the missing guard by ever deeper splits, whose waste $q^k\mathcal{W}_{\mathrm{o}}$ tends to 0; over Q the answer is a race between the split family (provers) and the memoriser family (non-provers), a property of the class. A misspecified term law causes a related drift (Remark 4.20).

3.7 Quantified data, lemmas and the equational fragment

Quantified data decide (Proposition B.14 in Appendix B.6). A datum s with $B \cup I_{\mathrm{c}}(\varphi) \nvdash s$ gives $B \oplus \sigma_\varphi$ posterior 0 under any likelihood supported on theorems; over Q, one datum $\forall y(0{+}Sy = Sy)$ leaves only provers of $\forall x(0 + x = x)$. In $\mathsf{C}_{\mathrm{min}}$, between provers that differ in cost ($\forall x\varphi$ behind L vacuous quantifiers against $\forall x\varphi$, both beside σ_φ, uniform weight priors, equal prior masses) the log Bayes factor tends to $\ln(1 + c + \cdots + c^L)$ on instance data and grows linearly, a little above $f_q L \ln \frac{1}{c}$, when a fraction $f_q > 0$ of the data is $\forall x\varphi$ (proof sketch). In $\mathsf{L1}^{\max}$ form a derived sentence becomes an axiom when the derivation steps it saves, at $\ge \ln \frac{1}{c}$ each, exceed its prior cost

and an $O(\ln n)$ dilution (proof sketch): frequently used lemmas become axioms. So the posterior identifies the data producer's working set of primitives, not a canonical axiomatisation (compare Proposition 7.4). On the equational fragment $E = \{\mathrm{Q4}, \mathrm{Q5}\}$ every derivation of $0 + S^k 0 = S^k 0$ has at least $k+1$ axiom leaves, so the two-part ($\mathsf{L1}^{\max}$) comparison favours the schema $0 + z = z$ linearly in k at $q = \frac{1}{2}$, and the full sum does so at least when $q > 0.61$ (Proposition B.15); full Q, and the full sum at $q = \frac{1}{2}$, are open.

3.8 Without templates

Without templates σ_φ is unavailable and hypotheses are finite sets of sentences. Setting: $\mathsf{C}_{\min}$, $c = 0.3$, numeral data, $q = \frac{1}{2}$, $\varphi = 0 + x = x$, learned weights; besides $H_\forall$ and memorisers the class has $\mathrm{Split}_k := \{\varphi(0), \ldots, \varphi(S^{k-1}0), \forall y \varphi(S^k y)\}$ and $\mathrm{Both0} := \{\varphi(0), \forall x \varphi\}$.

Under $\mathsf{L0}^{\mathrm{cl}}$, $H_\forall$ has the true law and the probability of a prover tends to 1 (proof sketch). Under $\mathsf{L1}$, Split_k loses only $q^k \ln \frac{1+c}{c}$ nats per datum at its best weights, against $\ln \frac{1+c}{c} = 1.466$ for $H_\forall$; it proves $\forall x \varphi$ over Q but not in pure logic, and memorisers prove it in neither. So in pure logic $P(T \vdash \forall x \varphi \mid D_n) \to 0$ in every class computed, and over Q the limit is a race between families: with depth ≤ 4 memorisers win (limit 0), with depth ≤ 40 splits win to $n = 10^{12}$ under four prior variants, and with all depths the limit is conjecturally 1 (Proposition B.16 in Appendix B.7). Every finite sentence set that emits infinitely many numeral instances (numeral $\forall$E terms) proves $\forall x \varphi$ over Q (Lemma B.17). Split_1 shows that "all instances have short derivations, so $\forall x \varphi$ is derivable", a naive form of Kreisel's conjecture, fails without axioms such as Q3 (Remark B.18).

3.9 A good version for this case, and its soundness

For $\forall x \varphi$ the good version of Section 9.2 models the selection of reported data by a known filter ($\mathsf{L1}^{\mathrm{sel}}$, or the per-citation $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$, under which the background effect of Proposition 3.6 vanishes, Remark 3.7) or by a prior over filters (Theorem 3.9(e)), and its verifier accepts s iff $\mathrm{Bel}_{\mathrm{cons}}(s) \geq 1 - \delta$. On closed instances it accepts $\forall x \varphi$ only if, eventually, the prior share of provers over theories and filters is $\geq 1 - \delta$ (for finite n see Corollary 3.13); one parameter instance, or one quantified datum entailing it over the background, suffices.

Corollary 3.13 (Soundness for $\forall x \varphi$)**.** [proved] universal §11 *Let the data be i.i.d. from P_{T^*} with* fixed *weights, $T^* \in \mathcal{H}$, $T^* \nvdash \forall x \varphi$, and $C^*_\forall := \{T \in C^* : T \vdash \forall x \varphi \Leftrightarrow T^* \vdash \forall x \varphi\}$. The probability that the verifier (on* Bel *or* $\mathrm{Bel}_{\mathrm{cons}}$*) ever accepts $\forall x \varphi$ is at most $\delta / \pi(C^*_\forall) \leq \delta / \pi(T^*)$, uniformly in time: Theorem 5.2 read at the sentence $\forall x \varphi$ (proof in Appendix B.4).*

Without well-specification there is no guarantee. In the misspecified settings of Theorem 3.9(d) and Propositions 3.12 and B.16 the limit theories (σ_φ, R_k, Split_k, memorisers) are true in $\mathbb{N}$, so in pure logic the verifier errs towards incompleteness; misspecification can also make it accept a falsehood with probability 1 (Example 4.19 and Proposition 5.4). The learned-weight computations here are outside the corollary (see Theorems 5.6 and 5.7).

Remark 3.14 (Splits of $\forall x \varphi$ and the MDL finding). [proved (from the cited results)] universal §10 Well specified, a root split with weights p_f ties with the schema under $\mathsf{L0}$ and $\mathsf{L1}$ (Proposition 4.6) and with learned weights pays a polynomial Occam factor (Proposition 4.10). An unmodelled selection creates linear split wins, at rates $(1 - q^k)\mathcal{W}_{\mathrm{o}}$ for R_k and $(1 - q^k) \ln \frac{1+c}{c}$ for Split_k (AS Prop E.16, split by numeral depth); modelling the selection removes them (Proposition 3.12(c)). This is Remark 4.13 for $\forall x \varphi$.

4 What the posterior identifies

Well specified, with fixed weights, the posterior concentrates on the theories with the data's law and is the prior among them: it identifies the theorem set (under L1, parameters admissible) or the instance union (under L0), not the axioms. With learned (Dirichlet) weights the exact generator class has mass 0, and under L0 the instance union is identified for almost every weight vector. Splits of a schema are decided by Occam terms or by usage statistics. With Dirichlet weights, *spare templates* (templates beside the generator's own that the data never or rarely use) are never refuted and decay at best polynomially (exponentially with fixed weights). Misspecified, in finite classes the posterior goes to the best-fitting generator, which can be weaker than the axioms behind the data, or unsound. Logarithms are natural, KL in nats; the experiments report bits.

4.1 Well-specified consistency

Definition 4.1 (Setting W)**.**
model §2 $\mathcal{H}$ is a countable class with a prior π, $\sum_T \pi(T) = 1$; each T carries a *fixed* law P_T on the countable set $\mathcal{S}$ (fixed weights). The data $D_n = (X_1, \ldots, X_n)$ are i.i.d. from P_{T^*}, $\pi(T^*) > 0$. The posterior is $\pi_n(T) := \pi(T)P_T(D_n)/M(D_n)$, $M(D_n) := \sum_T \pi(T)P_T(D_n)$; the predictive law is $M_n := \sum_T \pi_n(T)P_T$; the *generator class* is $C^* := \{T : P_T = P_{T^*}\}$.

Theorem 4.2 (Concentration on the generator class)**.** [proved; known (Doob 1949)] model Thm 2.1; universal Lemma U7 *Under W, $\pi_n(C^*) \to 1$ a.s.; moreover $\pi_n(T) \to \pi(T)1[T \in C^*]/\pi(C^*)$ for every T, and in total variation.*

Corollary 4.3 (Inside C^* the posterior is the prior)**.** [proved; computed] model Cor 2.2 *For $T, T' \in C^*$ and every n, $\pi_n(T)/\pi_n(T') = \pi(T)/\pi(T')$.*

Corollary 4.4 (Limits of deductive beliefs)**.** [proved] model Cor 2.3 *Under W, a.s., for all s and $d \le \infty$ at once, $\pi_n(\{T : s \in \mathrm{Th}_d(T)\}) \to \pi(\{T \in C^* : s \in \mathrm{Th}_d(T)\})/\pi(C^*)$.*

The proof is Doob's argument (Doob, 1949) (Appendix C.1). The data decide a deductive question only through C^*; inside it the prior decides. The limit is $1[s \in \mathrm{Th}_d(T^*)]$ under L0 (full-support $\mathcal{Q}$, positive weights) for every d ($\mathrm{supp}\, P_T = \bigcup \mathrm{inst}(T)$) and under L1 with parameters admissible for $d = \infty$ ($\mathrm{supp}\, P_T = \mathrm{Th}(T)$; Lemmas 2.10 and A.2); under L2 only $\mathrm{Th}_{\le d}$ is fixed, and for Hänni's scores nothing of this kind holds.
The rates (Proposition C.1, in Appendix C.1) are the usual ones: $\mathrm{KL}(P_{T^*} \| P_T)$ per datum against each $T \notin C^*$, cumulative expected predictive KL at most $\ln \frac{1}{\pi(C^*)}$, and decay through the Hellinger affinity, which degenerates for spare templates (Proposition 4.14).

4.2 Identification is of the generator

Proposition 4.5 (L1 separates deductively equivalent theories)**.** [proved (open parameter region); computed] model Prop 2.5 *Let $a := \neg(S0 = 0)$, $b := \neg(SS0 = 0)$, $T_1 := \{a, b\}$, $T_2 := \{a, a \to b\}$ (ground, weights $\frac{1}{2}$); $\mathrm{Th}(T_1) = \mathrm{Th}(T_2)$. Under L1 or L2, $P_{T_1}(b) \ge \alpha_{\mathrm{ax}}/2$ and $P_{T_2}(b) \le \alpha_{\mathrm{r}}/(1-\alpha_{\mathrm{r}})$; so $P_{T_1} \ne P_{T_2}$ if $\alpha_{\mathrm{ax}}/2 > \alpha_{\mathrm{r}}/(1-\alpha_{\mathrm{r}})$.*

Outside that region the laws differ in every computed setting (Appendix C.2); in general it is open. On data from P_{T_1} or P_{T_2} the posterior picks the generating axiomatisation, not a logically privileged one; from a third generator the drift, a difference of two KL divergences, can vanish.

Proposition 4.6 (Equal laws; splits are exact ties)**.** [proved; computed] model Prop 2.6 and remark; pa Prop 2.1 *Let $\mathcal{Q}$ have full support and all weights be positive. (a) Under* L0*, $P_{T,w} = P_{T',w'}$ implies $\bigcup \mathrm{inst}(T) = \bigcup \mathrm{inst}(T')$, hence $\mathrm{Th}_d(T) = \mathrm{Th}_d(T')$ for all d. (b) Let $\tau \in T$ have a metavariable M whose law factors at the root over finitely many productions, $\mathcal{Q}_M(\lambda \bar{z}.\, r(\beta_1, \ldots, \beta_a)) = p_r \prod_j \mathcal{Q}(\beta_j)$, and let τ_r be τ with M's body fixed to root r and fresh metavariables for its children. The* root split *T', with τ replaced by the τ_r at weights $w'_{\tau_r} := w_\tau p_r$, consists of DT° templates, and $P_{T',w'} = P_{T,w}$ under* L0*; $\mu_{T'} = \mu_T$, $Z_{T'} = Z_T$ under* L1, L2*. (c) A renamed or argument-permuted copy (AS Lemma A.2) has the same law if $\mathcal{Q}$ is symmetric in holes. Equal laws give equal instance sets, hence equivalence for first-order templates with only 0-ary term metavariables (AS Prop C.1, under its richness condition); in general (e.g. A1, or DT°) this converse is open.*

So with weights matched to $\mathcal{Q}$ no data tell a schema (e.g. T_{Ind}) from its root split; only the prior does. The tie needs $\mathcal{Q}$ to factor at the root, which the learned grammar of Section 7 does not (Proposition F.6). In the experiments' chain Ch_J the tie holds for a 0-ary closed-guard term metavariable of a component that is not an MP major premise, also with latent citations (Remark C.2, in Appendix C.2; E6: Remark 7.7).

4.3 Positive data and Gold

Theorem 4.7 (Identification in the limit with probability one)**.** [(a), (b) proved; (b′) known, proof sketch; (c) conjecture] model Thm 5.1 *Let $\mathcal{H}$ be all theories and $\pi(T^*) > 0$. (a) For fixed generators (*L0 *with fixed positive weights; or* L1 *with fixed positive parameters, $m \leq 1$, parameters admissible; full-support $\mathcal{Q}$) and data i.i.d. from P_{T^*}, the learner $g_n := L$ if $\pi_n(\{T : \mathrm{supp}\, P_T = L\}) > \frac{1}{2}$ is eventually $\mathrm{supp}\, P_{T^*}$, a.s.: $\bigcup \mathrm{inst}(T^*)$ under* L0*, $\mathrm{Th}(T^*)$ under* L1*. No bound on the number of templates is needed. (b) With Dirichlet weights (prior $\pi(T)\mathrm{Dir}_T(dw; \alpha)$,* L0*, full-support $\mathcal{Q}$, data i.i.d. from P_{T^*,w^*}), for Lebesgue-a.e. w^*, $\pi_n(\{T : \bigcup \mathrm{inst}(T) = \bigcup \mathrm{inst}(T^*)\}) \to 1$ a.s. (b′) Under (b), $\pi_n(\{(T, w) : \mathrm{TV}(P_{T,w}, P_{T^*,w^*}) < \varepsilon\}) \to 1$ a.s. for all $\varepsilon > 0$ (Ghosal and van der Vaart, 2017, Thm 6.9; numbering checked only via citing papers). (c) Conjecture: (b) for all interior w^*.*

(b) runs Doob's argument with the support functional $\bigcup \mathrm{inst}(T)$, which takes countably many values and a.s. equals the set of sentences seen (Appendix C.3). Of the computed posteriors, these theorems cover one-component generators in fixed hand pools; data-dependent pool members are covered only by the soundness bounds, and samples at one fixed weight vector are a yardstick, not an instance of a theorem (Remark C.3, in Appendix C.3).

Remark 4.8 (The exact class has posterior mass zero)*.* [refuted; counter-statement proved] model Thm 5.1(b″) Under the hypotheses of (b), if $|T^*| \geq 2$ with linearly independent component laws, the exact class $\{(T, w) : P_{T,w} = P_{T^*,w^*}\}$ has posterior mass 0 at every n for all w^* off a countable set, since each $\{w : P_{T,w} = P_{T^*,w^*}\}$ is Lebesgue-null. So only the instance union is identified (an earlier version claimed more; Appendix H.2).

Proposition 4.9 (Gold's limit point $\mathcal{L}_\infty$ against $\mathcal{L}_5$)**.** [proved] model Prop 5.2 *Let $\mathcal{L}_1 \subsetneq \mathcal{L}_2 \subsetneq \cdots$, $\mathcal{L}_\infty := \bigcup_i \mathcal{L}_i$, each $\mathcal{L}_i$ ($i \leq \infty$) with a generator P_i, $\mathrm{supp}\, P_i = \mathcal{L}_i$, whose class has prior mass $\pi_i > 0$ (setting W). (a) On data i.i.d. from P_i, $g_n = \mathcal{L}_i$ eventually, a.s. (b) Some text for $\mathcal{L}_\infty$ has $g_n \neq \mathcal{L}_\infty$ infinitely often (Gold, 1967). (c) Such texts have P_∞^∞-probability 0.*

The learner has no fixed preference between the ends of the limit point; the size principle decides (on P_5-data the ratio of $\mathcal{L}_\infty$'s generator to $\mathcal{L}_5$'s is $P_\infty(\mathcal{L}_5)^n$ times a factor of mean at most 1; Lemma 2.21). E5 shows both ends identified and the MAP changing at each of 14 stages of Gold's

text (Table 37). For the $\mathrm{I}\Sigma_n$ chain both directions separate, after presumably astronomical waiting times (a heuristic; Appendix C.3, Proposition 7.13). Fixed memorisers die at the first datum outside their set, so the Bayesian needs no bound on the number of templates, unlike the cautious learner; the memoriser class can keep mass $\exp(-O(\ln^2 n))$ (Laplace-weighted memorisers, geometric numerals; Remark C.4, in Appendix C.3).

4.4 Splits and the MDL finding

AS §6.10 found that MDL "tracks the statistics of usage, not the logical boundaries of schemas" (a naive code splits T_{Ind} by a linear margin, a well-specified one gains at most $O(\ln n)$; AS Props E.16, E.17). Does a derivation likelihood change this? With learned weights a root split must learn the root law that the schema takes from $\mathcal{Q}$.

Proposition 4.10 (Schema against root split, L0)**.** [proved; known (the χ^2 limit); computed] model Prop 5.4 *Let $T = \{\tau\}$, T' its root split over K productions with $\mathrm{Dir}(\alpha)$ weights, the data n instances of τ with root law r, root counts n_f, $\hat{r} := n_f/n$, and p the root law of $\mathcal{Q}$. (a) $\Delta_n := \ln P^{\mathrm{Dir}}_{T'}(D_n) - \ln P_T(D_n) = \ln \mathrm{DirMult}(n_f;\alpha) - \sum_f n_f \ln p_f$, $\mathrm{DirMult}(n_f;\alpha) := \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_f \frac{\Gamma(\alpha+n_f)}{\Gamma(\alpha)}$. (b) If all $n_f \to \infty$: $\Delta_n = n\mathrm{KL}(\hat{r}\|p) - \frac{K-1}{2}\ln n + c_{K,\alpha} + (\alpha - \frac{1}{2})\sum_f \ln \hat{r}_f + o(1)$, $c_{K,\alpha} := \ln \frac{\Gamma(K\alpha)}{\Gamma(\alpha)^K} + \frac{K-1}{2}\ln 2\pi$. (c) If $r = p$: $\Delta_n = -\frac{K-1}{2}\ln n + O_P(1)$ ($2n\mathrm{KL}(\hat{r}\|p) \to \chi^2_{K-1}$); the schema wins at rate $n^{-(K-1)/2}$. (d) If $r \neq p$: $\Delta_n/n \to \mathrm{KL}(r\|p) > 0$ a.s.; the split wins linearly. (e) The prior adds $\lambda(\ell(T') - \ell(T)) > 0$ bits for the schema.*

Proposition 4.11 (Schema against root split, derivation likelihoods)**.** [(a)–(c) proved; (d) conjecture; computed] model Prop 5.6 *Let $T = \{\tau\}$ root-factor as in Proposition 4.6(b), $|R| = K$, root law p interior, T' the split with $\mathrm{Dir}(\alpha)$ weights, under L0, L1 ($m < 1$) or L2 with other parameters fixed and shared; $P^{(p')}$ is the law of $\{\tau\}$ with root law p', and $\mathrm{BF}_n := P^{\mathrm{Dir}}_{T'}(D_n)/P_T(D_n)$. (a) $P_{T',w'} = P^{(w')}$, so $P^{\mathrm{Dir}}_{T'}(D) = \int \prod_i P^{(p')}(X_i)\,\mathrm{Dir}(dp')$: the split with learned weights is the schema with a learned root law. (b) If $\mathcal{Q}$ fits the usage (data i.i.d. from P_T), BF_n is a nonnegative supermartingale from 1: $P(\sup_n \mathrm{BF}_n \geq 1/\delta') \leq \delta'$. (c) If not (data i.i.d. from $P^{(r)} \neq P^{(p)}$, r interior), $\frac{1}{n}\ln \mathrm{BF}_n \to \mathrm{KL}(P^{(r)}\|P^{(p)}) \in (0,\infty]$ a.s. (d) Conjecture: under (b), with $p' \mapsto P^{(p')}$ identifiable and nonsingular Fisher information, $\ln \mathrm{BF}_n = -\frac{K-1}{2}\ln n + O_P(1)$.*

(b) is Ville's inequality (IL Lemma C.4); (d) would need a Laplace theorem for a model with latent citations. Computed: Tables 15 and 16 (slope -0.507 for (d)).

Remark 4.12 (E8: latent citations). [computed] experiments E8 $\{\varphi(z), \psi(z), \varphi(z) \to \psi(z)\}$ against the split of $\varphi(z)$ by its term's root ($K = 4$; chain L1 data, 10 seeds; prior excluded): well-specified usage makes the split lose 1.43 bits per doubling of n under L1 and 1.51 under L0 ($n = 256 \ldots 4096$; conjectured 1.5); skewed usage makes it win 0.202 against 0.074 bits per datum ($n = 1024 \ldots 4096$), since a ψ-datum may be φ followed by MP, so under L1 its term informs the split too (Table 40). E8 does not test Proposition 4.11(b), whose setting differs.

Remark 4.13 (What this says about the MDL finding). [proved; computed (as cited)] model §5.3; pa §2; universal §10; experiments E8 When the instantiation model is misspecified, the posterior tracks usage, not the logical boundary of a schema; as far as proved, a derivation likelihood does not change the MDL finding. A schema and its root split tie at matched weights. With learned weights and usage that $\mathcal{Q}$ fits, the split loses under L0 and gains at most $\ln(1/\delta')$ nats with probability $1-\delta'$ under L1 (that it loses there is Proposition 4.11(d), a conjecture); with usage that $\mathcal{Q}$ misfits, it wins linearly under both. Three qualifications. (i) Partial splits that only L1 keeps alive are not covered: a

split of T_{Ind} with a non-atomic connective derives every induction instance, an unseen-connective datum costing 369–489 bits in the two-part code instead of being impossible (Proposition 7.8); a general comparison is open. (ii) A learned grammar can reverse the Occam drift: under the shared positional grammar of Section 7 the split into roots F costs $\Delta\mathrm{prior} + \frac{|F|-9}{2}\log_2 n + O(1)$ bits more than the schema (Proposition F.6); with seven roots it gains 1 bit per doubling yet is 762 bits behind at $n = 256000$. (iii) Unmodelled selection of the reported data creates linear split wins; for $\forall x\varphi$ at rates $(1-q^k)\mathcal{W}_{\mathrm{o}}$ for R_k and $(1-q^k)\ln\frac{1+c}{c}$ for Split_k, which modelling the selection removes (Proposition 3.12(c), Remark 3.14).

4.5 Spare templates

Proposition 4.14 (Cost of a spare template)**.** [(a), (d1) proved; (b), (c), (d2) proof sketch; computed] model Prop 5.5; experiments Prop X7; pa Prop 5.1 *Let the truth be (T^*, w^*) under* L0 *with* $\mathrm{Dir}(\alpha_\tau)$ *priors, $A := \sum_{\tau\in T^*}\alpha_\tau$, σ a spare with parameter α_σ, $P^* := P_{T^*,w^*}$, and $\mathrm{BF}_n^\sigma := P^{\mathrm{Dir}}_{T^*\cup\{\sigma\}}(D_n)/P^{\mathrm{Dir}}_{T^*}(D_n)$. (a) If no datum is in* $\mathrm{inst}(\sigma)$*, exactly $\mathrm{BF}_n^\sigma = \frac{\Gamma(A+\alpha_\sigma)\Gamma(A+n)}{\Gamma(A)\Gamma(A+\alpha_\sigma+n)} \sim \frac{\Gamma(A+\alpha_\sigma)}{\Gamma(A)}n^{-\alpha_\sigma}$; at $\alpha_\sigma = \frac12$, $\frac12\log_2 n$ bits plus a constant and the prior bits. (b) If σ covers data but $\mathcal{Q}_\sigma(\mathrm{supp}\,P^*) = 1-c < 1$: $\ln\mathrm{BF}_n^\sigma = -\alpha_\sigma\ln n + O_P(1)$. (c) If $\mathrm{inst}(\sigma)\subseteq\mathrm{supp}\,P^*$, $\mathcal{Q}_\sigma$ is outside the span of T^*'s components, and $\sum_s \mathcal{Q}_\sigma(s)^2/P^*(s) < \infty$: $\ln\mathrm{BF}_n^\sigma = -\frac{\alpha_\sigma}{2}\ln n + O_P(1)$. (d1) If $\mathcal{Q}_\sigma = \mathcal{Q}_{\tau_1}$, $\tau_1\in T^*$ (e.g. an argument-permuted copy), and T^*'s instance sets are disjoint, $\mathrm{BF}_n^\sigma \to \frac{\Gamma(A+\alpha_\sigma)\Gamma(\alpha_1)}{\Gamma(A)\Gamma(\alpha_1+\alpha_\sigma)}(w_1^*)^{\alpha_\sigma} > 0$ a.s.: the spare never vanishes. (d2) If $\mathcal{Q}_\sigma$ is in the convex hull of T^*'s components but equals none, and the induced prior on $P_{T^*\cup\{\sigma\},w}$ has a continuous positive density at P^*, BF_n^σ tends to a positive constant. The case of a $\mathcal{Q}_\sigma$ in the span of T^*'s components but outside their convex hull is not covered.*

(b) and (c) need a boundary Laplace approximation (not written out) of the kind studied for overfitted mixtures by Rousseau and Mengersen (2011). In E5 unused spares follow (a) exactly and a nested spare, case (c), has fitted exponent -0.251 ± 0.002 on $n = 10^2 \ldots 10^8$; without the density hypothesis of (d2) a spare can gain slowly (Table 17).

Remark 4.15 (All spares; what spares cost). [proved (the sum); waiting time in case (b) from a sketch; computed] model §5.4; experiments E2, E5 With a common $\alpha_\sigma = \alpha$, BF_n^σ in (a) does not depend on σ; so the posterior mass of all $T^*\cup\{\sigma\}$ with σ covering no datum is at most $\mathrm{BF}_n^\sigma/\pi(T^*) = O(n^{-\alpha})$. With fixed weights the factor is $(1-w_\sigma)^n$. Spares inside the support do not threaten identification of the instance union. What spares cost is slow rejection: about $(\mathrm{prior\ ratio}/\delta_r)^{1/\alpha}$ data to fall below δ_r in case (b) (from the sketch of Proposition 4.14(b)). A spare sentence that T^* refutes makes $T^*\cup\{\sigma\}$ inconsistent, and positive data remove it only like $n^{-1/2}$ (E2, $0 = S0$ beside Q1: posterior 0.004 at $n = 512$); the size principle does not see inconsistency (Proposition 7.19).

4.6 Misspecification

Human data are theorems of the actual axioms, but are not drawn from any P_T of the model.

Theorem 4.16 (Finite class: concentration on the KL-minimisers)**.** [proved; known (general form)] model Thm 3.1 *Let $\mathcal{H}$ be finite, $\pi > 0$, the data i.i.d. from any P_H, $K_T := \mathrm{KL}(P_H\|P_T)$, $\min_T K_T < \infty$, $\mathbb{M} := \arg\min_T K_T$. Then $\pi_n(\mathbb{M}) \to 1$ a.s., with $\frac1n\ln\frac{\pi_n(T)}{\pi_n(T_0)} \to K_{T_0} - K_T < 0$ for $T\notin\mathbb{M}\ni T_0$; inside $\mathbb{M}$ the posterior need not converge.*

The general form is known (Berk, 1966; Kleijn and van der Vaart, 2006) (abstracts read, conditions not). In the full class with Dirichlet weights even this picture can fail: hybrids of memorisation

and an unsound escape approach KL 0, and unsound mass $\to 1$ is conjectured, with a computed illustration (Remark C.5, in Appendix C.6).

Example 4.17 (The axiom's own theorems lead to a weaker theory)**.** [proved] model Ex 3.2 In L_A, parameters not admissible, under L0 with fixed full-support $\mathcal{Q}$ and $\pi(\{0+z=z\})>0$, let the human's axiom be $\forall x(0+x=x)$, the data its theorems $0+t=t$, $t\sim\mathcal{Q}$. They are well specified for $\{0+z=z\}$, whose generator class has instance union $I_{\mathrm{c}}(0+x=x)$ (Proposition 4.6), and none proves $\forall x(0+x=x)$ (take $\mathbb{N}\cup\{a\}$ with $0+a:=0$). So $P(T\vdash\forall x(0+x=x)\mid D_n)\to 0$ a.s.: the ω-gap of AS §4.5 (all closed instances together do not entail $\forall x\varphi$), forced by the likelihood.

Under L1 without parameters $\operatorname{supp} P_T=\operatorname{Th}(T)$ fails; Theorem 3.9 treats that case.

Lemma 4.18 (Sound templates cover at most one zero-sum equation)**.** [proved] model Lemma 3.3 *Let $Z:=\{t=0: t$ a closed term over $0,+\}$. A DT° template all of whose instances are true in $\mathbb{N}$ (parameters, if admissible, read universally) covers at most one member of Z.*

Example 4.19 (An unsound limit)**.** [proved; computed] model Ex 3.4 Let the human's axioms be Q and the data true equations $t=0$, $t\in Z$, from a law of infinite support; the class is finite, made of sound theories $T_1,\dots,T_m$ and $T_{\mathrm{esc}}:=\{z=0\}$, all with positive prior; L0 with full-support $\mathcal{Q}$. By Lemma 4.18 the T_j cover finitely many members of Z, so a.s. $\pi_n(T_{\mathrm{esc}})=1$ eventually. But $T_{\mathrm{esc}}\vdash S0=0$ and $\mathsf{Q}\vdash\neg S0=0$; Q itself, if in the class, dies at the first datum, as its axioms are not in Z. Computed: against twenty best-possible sound theories at prior 0.99, $S0=0$ reached posterior ≥ 0.99 in 2000/2000 runs (median 17 data).

In Example 4.19 Q fails because the data are not axiom instances. A derivation likelihood does not reliably rescue the axioms: in a truncated equational grammar the best fit is unsound in all 16 settings with $p_+=0.25$ (Example C.6, in Appendix C.6; untruncated: Conjecture C.7). Table 18 (in Appendix C.6) collects the cases: misspecified usage of axiom instances gives the same or weaker theorems; derived theorems, absent counterexamples, mistakes and small samples can give unsound ones.

Remark 4.20 (E3(a): misspecified usage of instances). [computed]
experiments E3(a) On instances of $x+0=x$ with a term law not the model's, the posterior moves at a linear rate to a best-fitting family member: under heavy tails a depth-2 nesting with the same instances ($\{\varphi(z),\varphi(Sz),\varphi(SSz)\}$ in 3 seeds, $\{\varphi(z),\varphi(SSz)\}$ in 2); under numerals closed-guard splits R_m, m growing in n (9–10 at $n=512$, 11–12 at 1024; $\varphi(2+3)$ underived); on a finite selected support $\mathrm{Mem}(D_n)$. No unsound mass at large n; completeness is lost (Table 33).

4.7 Predictive versus deductive

Theorem 4.21 (Predictive versus deductive)**.** [proved]
model Thm 8.1 *Assume W. (a) $\sum_{n\geq 0}\mathbb{E}\,\mathrm{KL}(P_{T^*}\|M_n)\leq\ln\frac{1}{\pi(C^*)}$. (b) $\pi_n(\{T: T\vdash s\})\to\pi(\{T\in C^*: T\vdash s\})/\pi(C^*)$ a.s., which is $1[T^*\vdash s]$ iff the members of C^* agree on s up to prior-null sets; they do when the generator determines the theorem set (L0 with full support; L1 with parameters admissible), and need not under L2 or when $\operatorname{supp} P_T\subsetneq\operatorname{Th}(T)$. (c) If $T_1\vdash s$, $T_2\nvdash s$ and the data come from P_{T_1}, the expected log-odds of T_1 against T_2 grow by exactly $\mathrm{KL}(P_{T_1}\|P_{T_2})$ per datum, while the per-datum predictions differ in total variation by at most $(\mathrm{KL}(P_{T_1}\|P_{T_2})/2)^{1/2}$.*

So when two theories differ in what they prove but hardly in what they generate, prediction is settled early and the deductive question stays near its prior odds for about 1/KL data. Misspecified, predictions go to the KL-projection and beliefs to its theorems, so predictive success certifies

nothing deductive. For $\forall x\varphi$ this is the gap between Theorem 3.8 and Theorem 3.9 (compare Hutter, 2007). Data that come with their proofs identify the instance union of the generator (Proposition C.8, in Appendix C.7). Hänni's "grade near misses", read as a corruption channel that is part of the well-specified model, keeps Theorems 4.2 and 5.2 and Corollary 4.4; with a full-support channel the size principle is vacuous, and how much the generator class grows is open (Remark C.9, in Appendix C.7).

5 Soundness of the thresholded verifier

A posterior over theories can serve as a proof checker: it accepts a sentence when theories deriving it carry at least $1-\delta$ of the mass. With fixed weights and data drawn from the model, no adaptive prover ever gets a non-theorem accepted, with probability at least $1-\delta/\pi(C_d^*)$, uniformly in time (Theorem 5.2). With learned (Dirichlet) weights this holds on average over the weight prior (Theorem 5.6), and, for likelihoods linear in the weights, at each weight vector with a threshold that shrinks with n (Theorem 5.7); a constant threshold can fail (Example 5.8). For a general derivation-grammar likelihood with learned weights only the averaged guarantee is proved. Misspecified, there is no guarantee (Proposition 5.4).

5.1 Protocol

Definition 5.1 (Protocol; thresholded verifier)**.** model §4 Fix a countable class $\mathcal{H}$ with prior π, a truth $T^* \in \mathcal{H}$, a derivability bound $d \le \infty$ and an infinite data sequence $X_1, X_2, \ldots$. In each round $t = 1, 2, \ldots$ either the next datum arrives or a *prover* submits a query q_t; the prover chooses which. $n(t)$ counts the data that have arrived by round t. The verifier $V_{\delta,d}$ answers ACCEPT iff $\pi_t(\{T : q_t \notin \mathrm{Th}_d(T)\}) \le \delta$. Otherwise it may ESCALATE to a deterministic oracle telling whether $q_t \in \mathrm{Th}_d(T^*)$; answer j becomes the constraint $c_j(T) := 1[q \in \mathrm{Th}_d(T) \Leftrightarrow q \in \mathrm{Th}_d(T^*)]$. The posterior at round t is $\pi_t(T) \propto \pi(T) \prod_{i \le n(t)} P_T(X_i) \prod_j c_j(T)$ with fixed weights; with Dirichlet weights the marginal $P_T^{\mathrm{Dir}}(X_1, \ldots, X_{n(t)})$ replaces the product over data. The verifier is *sound* on a run if it never accepts a sentence outside $\mathrm{Th}_d(T^*)$. $C_d^* := \{T : P_T = P_{T^*},\ \mathrm{Th}_d(T) = \mathrm{Th}_d(T^*)\}$ and $W_d^* := \pi(C_d^*) \ge \pi(T^*)$.

The prover may be randomised, adaptive and computationally unbounded, may know T^*, the verifier and all future data. Soundness is relative to $\mathrm{Th}_d(T^*)$, not to truth; d is symbol size (Definition 2.2), chain steps ($\vdash_J$) in the experiments, and 0 (being an instance) in Remark F.23. This is the protocol of AS Def 2.6 with $V_{\delta,d}$ in place of the version-space verifier.

5.2 Fixed weights

Theorem 5.2 (Time-uniform soundness, fixed weights)**.** [proved; computed] model Thm 4.1 *Assume setting W (Definition 4.1): every $T \in \mathcal{H}$ carries a fixed law P_T (fixed mixture weights), the data are i.i.d. from P_{T^*} with $\pi(T^*) > 0$, the prior does not depend on n, and the oracle is truthful. If $\delta \le W_d^*\,\delta'$, there is an event G with $P(G) \ge 1-\delta'$, depending only on the data sequence, on which no prover ever gets a sentence outside $\mathrm{Th}_d(T^*)$ accepted, in any round.*

Proof idea. As in IL Thms 4.15(b) and 10.9: members of C_d^* share T^*'s likelihood and pass every constraint, so $\pi_t(C_d^*) = W_d^*/Z_t$ with $Z_t \le M(X^{n(t)})/P_{T^*}(X^{n(t)})$ (M the Bayes mixture), a nonnegative supermartingale from 1 that by Ville's inequality (IL Lemma C.4; Ville, 1939) reaches $1/\delta'$ with probability at most δ'. □

One event covers all queries and rounds, so no union bound over the prover's attempts is needed. The constant improves from $\pi(T^*)$ to W_d^*, and it cannot be improved further: a pair of ground theories with fixed weights attains IL's acceptance probability, which tends to δ' (Remark D.1, in Appendix D.1). The guarantee is per target: under π_λ it covers the T^* with $\lambda\ell(T^*) \le \log_2(\delta'/\delta)$, so for long axiom systems it is weak. IL's escalation bound (IL Thm 4.16) transfers to escalations of valid queries; those of invalid queries are not bounded, as the protocol has no reject threshold (Appendix D.1). Read at the single sentence $\forall x\varphi$, with the members of C^* that agree with T^* on it, the theorem gives Corollary 3.13.

Remark 5.3 (Hypotheses). [proved] model §4.1 (referee m8) What matters: (i) well-specification of the *data law*, not only of the theorem set (Proposition 5.4; selected data in Remark 5.10(A)); (ii) fixed weights, or Section 5.4; (iii) a proper prior that does not depend on n (a penalty λ_n growing with n breaks the supermartingale step; no replacement is known); (iv) truthful constraints. A finite truncation $\mathcal{H}_N \ni T^*$ keeps the theorem with W_d^* replaced by $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N) \ge \pi(T^*)$, and so does a computable verifier for a finite class and $d < \infty$ (Appendix D.1; an earlier version claimed more, Appendix H.2).

Proposition 5.4 (Misspecification: a waiting prover wins)**.** [proved; computed] model Prop 4.2 *In the setting of Example 4.19 (human axioms* Q*; data* $t = 0$ *for closed* $\{0, +\}$*-terms* t *from a law with infinite support; a finite class of sound theories, possibly with* Q*, plus* $T_{\mathrm{esc}} = \{z = 0\}$*;* $\mathsf{L0}$*), let the prover wait until* $\pi_t(\{T : T \nvdash S0 = 0\}) \le \delta$*, then submit* $S0 = 0$*. For every* $\delta > 0$*, a.s. the verifier* $V_{\delta,\infty}$ *eventually accepts* $S0 = 0$*, whether or not it escalates; yet* $\mathsf{Q} \vdash \neg\, S0 = 0$*.*

By Example 4.19, almost surely $\pi_n(T_{\mathrm{esc}}) = 1$ for all large n, and $T_{\mathrm{esc}} \vdash S0 = 0$; the prover's only query is made when it is accepted. Against an escalating verifier the waiting prover won 400/400 runs, a prover asking in every round 11/400 (Appendix D.1). Misspecification can also make the verifier incomplete: in Example 4.17, where the best-fitting generator is weaker than T^*, it accepts $\forall x(0 + x = x)$ at most finitely often.

5.3 The cautious limit and completeness

In the limit $\delta \to 0$ the verifier is the cautious verifier of AS, read on theorems: $V_{0,d}$ accepts exactly the common Th_d of the theories of positive prior mass whose support contains the data, and is sound deterministically and without well-specification (Proposition D.2, in Appendix D.2).

Proposition 5.5 (Completeness without anchors)**.** [proved] model Prop 4.4 *Assume setting* W *with* $\mathsf{L0}$ *(full-support* $\mathcal{Q}$*, any* d*) or with* $\mathsf{L1}$ *(*$d = \infty$*, parameters admissible). For every* $s \in \mathrm{Th}_d(T^*)$ *and every* $\delta > 0$*, almost surely* $V_{\delta,d}$ *accepts* s *at all sufficiently large* n*. With Dirichlet weights under* $\mathsf{L0}$ *the same holds for Lebesgue-almost every* w^**.*

By Corollary 4.4 the non-deriving mass of s tends to $\pi(C^* \cap \{T : s \notin \mathrm{Th}_d(T)\})/\pi(C^*)$, which is 0 because under these likelihoods every member of C^* has the theorems of T^*; with Dirichlet weights use Theorem 4.7(b). Without a bound on the number of templates the cautious verifier accepts only $\mathrm{Th}_d(D_n)$ (under $\mathsf{L0}$ with a prior that charges every finite theory; AS Prop 6.4), and with a bound it becomes exact only once the data contain an *anchor*, a set of data that only the target template covers among its competitors (AS Def 2.6, Lemma 2.8). The Bayesian verifier needs neither, because the likelihood moves mass away from memorisers and splits; it gives up deterministic soundness (Table 21).

5.4 Dirichlet weights

Read with parameter (T, w) and prior $\pi(T)\mathrm{Dir}_T(dw)$, Theorem 5.2 is vacuous for Dirichlet weights under L0 and likelihoods affine in w: if $|T^*| \geq 2$ with linearly independent component laws, its class has prior mass 0 for all w^* off a countable set, which forces $\delta = 0$; under L1 this is plausible but not proved (Remark D.3, in Appendix D.3).

At a fixed w and for likelihoods linear in w, the Dirichlet marginal loses at most a regret factor: $P_T^{\mathrm{Dir}} \geq R_\alpha(n, K)\, P_{T,w}$ with $K = |T|$, where $-\ln R_\alpha(n, K) = \frac{K-1}{2} \ln n + O(1)$ for $\alpha_\tau \equiv \frac{1}{2}$, and $R_\alpha(n, K) \geq \frac{\Gamma(A)}{e\,(K-1)! \prod_\tau \Gamma(\alpha_\tau)} (n+1)^{-(K-1)}$ if all $\alpha_\tau \leq 1$, $A := \sum_\tau \alpha_\tau$ (Lemma D.4, in Appendix D.3).

Theorem 5.6 (Soundness on average over w^*)**.** [proved] model Thm 4.7; experiments Prop X8(a), (b) *Let $\mathcal{H}$ be countable with prior π and Dirichlet weights, each T with its marginal law P_T^{Dir} on data sequences (under L0, L1 or L2), and truthful constraints. Let the data come from $P_{T^*}^{\mathrm{Dir}}$: first $w^* \sim \mathrm{Dir}_{T^*}$, then i.i.d. from P_{T^*,w^*}. Put $C_d^{\mathrm{Dir}} := \{T : P_T^{\mathrm{Dir}} = P_{T^*}^{\mathrm{Dir}}$ on $\mathcal{S}^n$ for all n, $\mathrm{Th}_d(T) = \mathrm{Th}_d(T^*)\} \ni T^*$. If $\delta \leq \pi(C_d^{\mathrm{Dir}})\, \delta'$, there is a data event G with $P_{T^*}^{\mathrm{Dir}}(G) \geq 1 - \delta'$ on which no prover ever gets a sentence outside $\mathrm{Th}_d(T^*)$ accepted. Hence $\mathrm{Dir}_{T^*}\{w^* : P^\infty_{T^*,w^*}(G^c) > \delta'/\gamma\} \leq \gamma$ for every $\gamma \in (0, 1]$.* Pool version *(experiments): with the experiments' prior weights $2^{-\mathrm{bits}(T)}$ and $\vdash_J$ for Th_d, the verifier may use the posterior restricted to any pool $R_n \ni T^*$, even one chosen from the data, and the conclusion holds for $\delta \leq 2^{-\mathrm{bits}(T^*)}\delta'$. If the family is a fixed pool H plus all memorisers and the memorisers' codes lie outside the template code, the prior weights may sum to more than 1, and the threshold must be $\delta \leq 2^{-\mathrm{bits}(T^*)}\delta'/(\sum_{T \in H} 2^{-\mathrm{bits}(T)} + 1)$.*

Theorem 5.7 (Soundness at a fixed w^*)**.** [proved]
model Thm 4.8; experiments Prop X8(c) *As in Theorem 5.6, but with data i.i.d. from P_{T^*,w^*} for a* fixed *w^* in the closed simplex, a likelihood linear in w (L0; also the experiments' chain Ch_J), $K := |T^*|$, and a verifier that accepts at round t iff the non-deriving mass is at most $\delta_{n(t)}$. If $\delta_n \leq \pi(C_d^{\mathrm{Dir}})\, \delta'\, R_\alpha(n, K)$ for every n, then with probability at least $1 - \delta'$ no prover ever gets a sentence outside $\mathrm{Th}_d(T^*)$ accepted. For $\alpha \leq 1$, Lemma D.4(b) gives an explicit $\delta_n \propto (n+1)^{-(K-1)}$; with $\alpha = \frac{1}{2}$ and the exact R_α, $\delta_n \asymp n^{-(K-1)/2}$. To cover every target with at most k templates, use $\min_{K \leq k} R_\alpha(n, K)$.* Pool version *(experiments): with the experiments' prior weights $2^{-\mathrm{bits}(T)}$ and $\vdash_J$ for Th_d, the conclusion holds for the posterior restricted to any pool $R_n \ni T^*$, even one chosen from the data, if $\delta_n \leq 2^{-\mathrm{bits}(T^*)}\, R_\alpha(n, K)\, \delta'$ for every n.*

The proofs (Appendix D.3) run that of Theorem 5.2 under $P_{T^*}^{\mathrm{Dir}}$, or under P_{T^*,w^*} with the regret bound. Theorem 5.7 is time-uniform, but its threshold slows acceptance; whether completeness (Proposition 5.5) survives is open. Theorem 5.6 covers a fixed w^* only outside a set of small prior mass. The shrinking threshold is needed:

Example 5.8 (Constant threshold)**.** [computed; mechanism proof sketch]
model Ex 4.9; experiments check_fixed_weight L0; root law of $\mathcal{Q}$ $(p_0, p_S, p_+, p_\cdot) = (0.5,\, 0.5 - \varepsilon,\, \varepsilon/2,\, \varepsilon/2)$; truth $T^* = \{0 + 0 = 0,\ (Sz) + 0 = Sz\}$ with $\mathrm{Dir}(\frac{1}{2}, \frac{1}{2})$ weights, data from P_{T^*,w^*} with the interior $w^* = (p_0, p_S)/(1 - \varepsilon)$. The competitor $T' = \{z + 0 = z\}$ proves $q := (0 \cdot 0) + 0 = 0 \cdot 0$, which T^* does not. Prior $\frac{1}{2}$ each; the verifier accepts q iff $\pi_n(T^*) \leq \delta = 0.01$, so Theorem 5.6's bound is 0.02. In three implementations (Table 22) q was accepted with probability 0.977–0.997 ($\varepsilon = 10^{-5}$) and 1.000 ($\varepsilon = 10^{-6}$) at the fixed w^*, 0.003–0.010 with w^* drawn from the prior, and in 0/300 runs at the fixed w^* with the threshold of Theorem 5.7; in runs at $\varepsilon = 10^{-3}$ and 10^{-4}, never. The mechanism (Appendix D.3): on the data's support $P_{T'} = (1 - \varepsilon) P_{T^*,w^*}$, so T' gains the regret $\frac{1}{2} \ln n + O(1)$ against $P_{T^*}^{\mathrm{Dir}}$ and loses only $n\varepsilon$; the constant threshold fails when T' is so close to

T^*'s law that $1 \ll n \ll 1/\varepsilon$ is a long window. With fixed weights the ratio is $(1-\varepsilon)^n \leq 1$ and q is never accepted.

Remark 5.9 (The $\mathrm{Bel}_{\mathrm{cons}}$ verifier). [computed; transfer proved]
universal §6.2, §11; editor, from model §4 Inconsistent theories inflate Bel and Dis alike (Definition 2.13): in the class of Table 2 at $n = 0$, $\mathrm{Bel}(\forall x \varphi) \approx 0.69$ is almost all Inc, and generative likelihoods remove such theories within a few data (Table 12). Section 3.9 therefore recommends "accept s iff $\mathrm{Bel}_{\mathrm{cons}}(s) \geq 1-\delta$". When this verifier accepts, so would $V_{\delta,d}$ at the same posterior. The proofs of Theorems 5.2, 5.6 and 5.7 bound $\pi_t(C_d^*)$ (or $\pi_t(C_d^{\mathrm{Dir}})$) from below for all truthful constraints, so all three transfer to it ($\mathrm{Bel}_{\mathrm{cons}}$ read at level d).

5.5 What the experiments show

Remark 5.10 (E4: adaptive provers in the code). [computed] experiments E4 Verifier on $\vdash_1$, $\mathrm{Dir}(\frac{1}{2})$ weights, one-component T^*, so Theorem 5.6 applies (Table 36). (A) The tight construction of IL Prop C.5 with a Dirichlet decoy $T' = \{\varphi(0), 0 = S0\}$: win probability exactly $(1-u)^{t^*}$, 9–46 times below δ' in five settings. With *selected* data (only $\varphi(0)$, each a theorem of T^*) the prover wins 20/20 in every setting. (B) Well specified ($\varphi = x + 0 = x$, $T^* = H_{\mathrm{sch}}$, bound 0.05, every $n \leq 256$): 0/100 wins under L0 and L1; the mass on theories deriving an invalid query peaks at 0.45 and 0.53 at $n \leq 6$, always on the true but underivable $\forall x(x + 0 = x)$. (C1) 10% false near misses $t + 0 = St$, with the mistake schema in the pool: 100/100 wins, first acceptance at the first mistaken datum in 90/100 streams. (C2) Heavy-tailed terms: 0/100. (C3) Selection with the decoy and a clean memoriser in the pool: 0/100.

In (C1) the data's actual generator is a two-schema theory with fixed weights $(0.9, 0.1)$, and the queries first accepted, $0 + 0 = 1$ and $2 + 0 = 3$, are theorems of it: the verifier is unsound only relative to the intended T^*, since nothing forces the best-fitting generator to be the actual axioms. Theorem 5.7 would cover a verifier with its shrinking threshold on that generator, not the constant-threshold verifier used in E4. Comparing (C3) with (A): under misspecification soundness depends on what else is in the pool, and no theorem covers this.

Remark 5.11 (Early unsound lumps). [computed] experiments E2; pa §5.4 E2 (L0, $T^* = \mathrm{Q1..Q7} + T_{\mathrm{Ind}}$ at fixed weights, 25 seeds, verifier on citation; causal pools with $\mathrm{SeenQ}(D_n)$, the theory of T_{Ind} and the Q axioms cited so far, Section 8.1): a $\delta = 0.05$ verifier accepts the false $\forall x(x + 0 = 0)$, $\forall x\,(Sx = x)$, $\forall x \forall y\,(y = x)$ in 5 of 25 seeds, at $n = 8$ or 16, never from $n = 32$ (Table 31). The two-part code of Remark F.23 behaves alike. No contradiction with the pool version of Theorem 5.7: for E2 it needs $\delta_n \leq 2^{-226} R_\alpha(n, 8)\, \delta' \asymp 2^{-226} n^{-3.5}$ (T^* costs 226.2 bits in the experiments' code).

6 Time penalties and Hänni's collapse

Hänni's working notes (Hänni, 2026)[2] compare axiom induction, which weighs axiom systems that prove the labelled data, with Solomonoff function induction, which weighs programs that label statements directly. He argues that the two are equivalent for consistent assigners: an assigner T becomes the schema "`T(quoted-phi) = accept => phi`", where "being an axiom of the right form is in fact decidable", and the result is consistent because "a model of the phis alone would be a model of the added sentences also". So unrestricted axiom induction collapses into function induction, and he considers restricting axioms to "second-order substitution schemas of a specific

[2] File `research/prior/hanni-solomonoff-axiom-induction.md`; working notes, quoted verbatim.

kind". The question adds a penalty on the time to generate axioms or to check axiomhood. The result (proofs in Appendix E):

Remark 6.1 (What the time penalty does). [(1) proved, under a self-simulation assumption on U, given the recursion and MRDP theorems (known); (2), (4) proved; (3) lower bound proved, upper bound proof sketch; (5) as in Section 6.6] model §6.5; time-followup §11; short-derivations §13 (1) Penalties on the time to check or generate axioms leave intact the direction of Theorem 6.3 from function to axiom induction: padded Craig sets have polynomial membership and polynomial-delay enumeration, so axiom induction stays within a constant of unpenalised $\mathrm{FI}_{\mathrm{cons}}$, and a Kt penalty costs $\log_2(|f|+1)+O(1)$ bits (Proposition 6.8, Theorem 6.14(a), Proposition E.13). They leave free the proof search of the other direction: once consistent function induction is clocked ($\mathrm{FI}^{\tau}_{\mathrm{cons}}$), axiom induction dominates it and beats it by $n - O(1)$ bits on a computable sequence (Theorem 6.14(b), Theorem 6.15), as Hänni objected. (2) Templates block Hänni's schema (its acceptance part, Proposition 6.6) but, over PA, not the collapse for Σ_n-sound assigners on Σ_n sentences: one ground reflection sentence reproduces them at a cost linear in $|f|$ (Proposition 6.7); for merely consistent assigners that sentence can be inconsistent with PA, and whether templates block the collapse for them is open. (3) Derivation size in written symbols prices the assigner: one decidable X for all finite DT° theories, and one for each e for all axiom sets with membership in time $O(n^e)$, forces every such theory consistent with Γ_{f_X} to use, infinitely often, derivations of at least a fixed polynomial root of the assigner's nondeterministic time (Theorem 6.9 and Corollary 6.10), which $\mathsf{L1}^{\sigma}$ and the graded score charge (Proposition 6.12). The collapse constructions through A^C_f and the two-sorted A_f pay at most a polynomial of the deterministic running time (proof sketch, Remark E.3); for $\rho_{f,n}$ no upper bound is established. (4) Plain L1 is proved to charge at least about the logarithm of the assigner's nondeterministic time (Corollary E.7); a polynomial charge is Conjecture 6.13. Function induction's total log loss on f_X's labels is at most $|f_X|$ bits (when $B \cup \Gamma_{f_X}$ is consistent). (5) Which computation is free decides (Section 6.6): with deduction free the gain is free proof search, which unpenalised $\mathrm{FI}_{\mathrm{cons}}$ also has, and a slow assigner keeps its weight; with deduction charged, axiom induction is certificate function induction ($\mathrm{NP} \cap \mathrm{coNP}$ at polynomial budgets), exponential-time, or not time-limited, as each derivation's axiom material, each axiom instance, or only the theory's own axioms are bounded (Theorem 6.16 and Corollary 6.17).

6.1 The two inductors

Fix a countable first-order language L with sentences Sent_L, a universal prefix machine U (program p has weight $2^{-|p|}$), and a decidable background $B \subseteq \mathrm{Sent}_L$, possibly empty, that is part of every axiom hypothesis. Data are labelled sentences $D = ((\varphi_1, b_1), \ldots, (\varphi_n, b_n))$, $b_i \in \{1, 0\}$ (true, false).

Definition 6.2 (Axiom and function induction). model §6.1

- AI: a hypothesis is a program p that halts on every input and decides a set $A_p \subseteq \mathrm{Sent}_L$. It is *compatible* with D if $B \cup A_p$ is consistent and proves φ_i when $b_i = 1$ and $\neg\varphi_i$ when $b_i = 0$. $W_{\mathrm{AI}}(D) := \sum_{p \text{ compatible}} 2^{-|p|}$.
- $\mathrm{FI}_{\mathrm{cons}}$: a hypothesis is a program f computing a partial map $\mathrm{Sent}_L \to \{\mathrm{acc}, \mathrm{rej}\}$ with $B \cup \Gamma_f$ consistent, where $\Gamma_f := \{\varphi : f(\varphi) = \mathrm{acc}\} \cup \{\neg\varphi : f(\varphi) = \mathrm{rej}\}$; it is compatible if it gives each φ_i the label b_i; $W_{\mathrm{FI}}(D) := \sum_{f \text{ compatible}} 2^{-|f|}$. $\mathrm{FI}_{\mathrm{all}}$ is the same over all assigners.
- Semimeasure convention: $P_X(b_1 \ldots b_n \mid \varphi_1 \ldots \varphi_n) := W_X(D_n)/W_X(\emptyset)$, and log loss is $-\log_2$ of it. Hypotheses that leave φ undecided drop out, so $W(D + (\varphi, 1)) + W(D + (\varphi, 0)) \leq W(D)$ (Hänni: "maybe we should speak of a semimeasure here instead of renormalizing"). $W_X(D)$ depends only on the set D.

Function induction over all assigners is not the fair comparison: on a consistent labelling of fair coin flips $\mathrm{FI}_{\mathrm{all}}$ has bounded log loss while AI and $\mathrm{FI}_{\mathrm{cons}}$ lose at least n bits in expectation (Proposition E.1), as Hänni argued informally.

Theorem 6.3 (Hänni's equivalence, repaired)**.** [proved; Craig's theorem known] model Thm 6.1 *There is a constant c, depending only on U, the coding of syntax and the decider for B, such that $2^{-c}W_{\mathrm{AI}}(D) \le W_{\mathrm{FI}}(D) \le 2^{c}W_{\mathrm{AI}}(D)$ for every finite labelled D. Hence for every reveal order and labelling the semimeasure log losses of* AI *and* $\mathrm{FI}_{\mathrm{cons}}$ *differ by at most $2c$ bits.*

Proof idea. AI $\to$ FI is Hänni's argument: map p to "search for proofs of φ and of $\neg\varphi$ from $B \cup A_p$". For FI $\to$ AI his schema is replaced by Craig's trick (Craig, 1953): f becomes a decider of $A_f^C := \{\varphi^{\wedge(k+1)} : f(\varphi) = \text{acc in exactly } k \text{ steps}\} \cup \{(\neg\varphi)^{\wedge(k+1)} : f(\varphi) = \text{rej in exactly } k \text{ steps}\}$, $\psi^{\wedge m}$ the right-nested m-fold conjunction. A_f^C is decidable and has the consequences of Γ_f; its decider is f plus a fixed wrapper (Appendix E.1). □

The constant equivalence is proved only in the semimeasure convention; renormalising at each step, the argument gives only $2c$ bits per step (Remark E.2).

6.2 Hänni's schema

Let $\mathrm{Accept}_f(x)$ be a Σ_1 formula of arithmetic for "f halts on input x with output acc" (via Kleene's T predicate), and $\mathrm{Reject}_f(x)$ likewise. Both are written without $<$ (bounded quantifiers as $\exists z\,(z{+}x = y)$), so that Σ_1-completeness of Q, which in L_A holds for $<$-free Σ_1 sentences, applies. Hänni's schema is $A_f := \{\mathrm{Accept}_f(\ulcorner\varphi\urcorner) \to \varphi : \varphi \in \mathrm{Sent}_L\} \cup \{\mathrm{Reject}_f(\ulcorner\varphi\urcorner) \to \neg\varphi : \varphi \in \mathrm{Sent}_L\}$.

Proposition 6.4 (Two sorts: his argument works)**.** [proved; Σ_1-completeness of Q for $<$-free sentences known] model Prop 6.2 *Let L-sentences live in one sort and arithmetic in another, with $B := \mathsf{Q}$ on the second. If Γ_f has a model, then $B \cup A_f$ is consistent and $\mathrm{Cn}(B \cup A_f) \cap \mathrm{Sent}_L = \mathrm{Cn}(\Gamma_f)$.*

Proposition 6.5 (One sort over PA)**.** [proved; Hänni's argument refuted in this reading] model Prop 6.3 *Assume* $\mathrm{Con}(\mathsf{PA})$ *and the standard formalisation of computations. Let $L = L_A$, $B = \mathsf{PA}$, and let f accept $\ulcorner\neg\mathrm{Con}(\mathsf{PA})\urcorner$, search on input $\ulcorner\mathrm{Con}(\mathsf{PA})\urcorner$ for a* PA*-proof of $\bot$ and accept if it finds one, and loop otherwise. Then $\Gamma_f = \{\neg\mathrm{Con}(\mathsf{PA})\}$ is consistent with* PA*;* $\mathsf{PA} \cup A_f$ *is not.*

His consistency argument assumes that the antecedents are evaluated correctly; in a model of $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, $\mathrm{Accept}_f(\ulcorner\mathrm{Con}(\mathsf{PA})\urcorner)$ holds via a nonstandard proof of $\bot$. His *conclusion*, the equivalence, stands (Theorem 6.3); his *argument* needs two sorts or a sound f.

6.3 What templates do

Proposition 6.6 (No non-ground template in the collapse schema)**.** [proved] model Prop 6.4; pa Prop 4.3 *Write Gödel numbers as unary numerals S^k0 (or any numerals in which some term symbol, here $+$, never occurs), and let $C_f := \{\mathrm{Accept}_f(\ulcorner\varphi\urcorner) \to \varphi : \varphi \in \mathrm{Sent}_L\}$, the acceptance part of A_f, single- or two-sorted. Every* DT° *template τ with* $\mathrm{inst}(\tau) \subseteq C_f$ *is ground. So C_f is not* $\bigcup \mathrm{inst}(T)$ *for any finite set T of* DT° *templates, and such a set with instances in C_f covers finitely many of its members. Likewise for the reflection schema* $\mathrm{Refl}_T = \{\mathrm{Prv}_T(\ulcorner\varphi\urcorner) \to \varphi\}$ *(Proposition F.11).*

The rejection part, and templates whose instances mix the two parts, are not treated here (Appendix H.6). For the acceptance part, Hänni's restriction blocks his construction: inside DT° it costs one ground template per sentence, which is memorisation. But one ground sentence can carry the collapse over PA.

Proposition 6.7 (One reflection sentence carries the collapse)**.** [proved, given partial truth predicates (known)] model Prop 6.5 *For $n \geq 1$ let $\mathrm{Tr}_n(x)$ be a Σ_n partial truth predicate, $\mathsf{PA} \vdash \varphi \leftrightarrow \mathrm{Tr}_n(\ulcorner\varphi\urcorner)$ for every Σ_n sentence φ (known; Hájek and Pudlák, 1993, Ch. I; Kaye, 1991, Ch. 9; chapter numbers not checked), and $\mathrm{Sent}_{\Sigma_n}(x)$ a Δ_1 formula for "x codes a Σ_n sentence". The single sentence (ground DT° template)*

$$\rho_{f,n} := \forall x\big[(\mathrm{Sent}_{\Sigma_n}(x) \wedge \mathrm{Accept}_f(x)) \to \mathrm{Tr}_n(x)\big] \wedge \forall x\big[(\mathrm{Sent}_{\Sigma_n}(x) \wedge \mathrm{Reject}_f(x)) \to \neg\mathrm{Tr}_n(x)\big]$$

has $\ell(\rho_{f,n}) \leq a|f| + b_n$ for constants a, b_n (binary numeral for f's index), and: (a) $\mathsf{PA} + \rho_{f,n}$ proves every Σ_n sentence that f accepts and the negation of every one it rejects; (b) if f is Σ_n-sound (it accepts only true and rejects only false Σ_n sentences), $\mathsf{PA} + \rho_{f,n}$ is true in $\mathbb{N}$.

The overhead, $\ell(\mathsf{PA}) + a|f| + b_n$, is linear in $|f|$ rather than $|f| + c$ as for Craig's sets (Theorem 6.3).

6.4 Time penalties

Proposition 6.8 (Membership is cheap in both collapse sets)**.** [proved] model Prop 6.6 *(a) With binary numerals, membership of χ in A_f is decidable in time polynomial in $|\chi|$ without running f: parse χ as $\mathrm{Accept}_f(t) \to \varphi$ or $\mathrm{Reject}_f(t) \to \neg\varphi$, compute $\ulcorner\varphi\urcorner$, compare with t. With unary numerals likewise, since then $|\chi| \geq \ulcorner\varphi\urcorner$. (b) Membership in A_f^C is decidable in time polynomial in $|\chi|$: at most $|\chi|$ decompositions $\chi = \psi^{\wedge m}$, each running f for $m - 1 < |\chi|$ steps.*

So a penalty in the style of Levin's Kt or the speed prior on the time to *check* axiomhood puts these sets in the same polynomial class as DT° templates (Remark A.1(iii)), and a penalty on the time to *generate* axioms is met by a Craig set padded by the enumerator's own work (Theorem 6.14(a)): neither prices the direction of the collapse from function to axiom induction, and neither touches the proof search of the other direction (Section 6.6).

The computation of f moves into the proofs (A_f: a derivation must prove $\mathrm{Accept}_f(\ulcorner\varphi\urcorner)$) or into the axioms' length (A_f^C), both counted by derivation size in symbols; no theory with polynomial-time membership escapes this. Let $L \supseteq \{0, S, +, \cdot, R\}$, R unary. For $X \subseteq \{0,1\}^*$ put $\varphi_w := R(\underline{w})$, $\underline{w}$ a binary numeral term, and let f_X accept φ_w iff $w \in X$ and reject it otherwise; Γ_{f_X} is consistent (interpret R as X).

Theorem 6.9 (Derivation size bounds the assigner's nondeterministic time)**.** [proved] model Thm 6.7 *Let T be any axiom set with membership decidable in time $O(n^e)$ and $T \cup \Gamma_{f_X}$ consistent. If every φ_w, $w \in X$, has a K-derivation from T of symbol size at most $\ell(|w|)$, with $\ell(m) \geq m$ time-constructible, then $X \in \mathrm{NTIME}(\ell(m)^{c_e})$, c_e depending only on e and K's checking overhead.*

Proof idea. Guess a derivation of size at most $\ell(|w|)$ and check it. If $w \notin X$, then $\neg\varphi_w \in \Gamma_{f_X}$, so consistency rules out a derivation (Appendix E.4). □

Corollary 6.10 (Hard assigners force long derivations, for every theory)**.** [proved, given the NTIME hierarchy theorem (known; not checked)] model Cor 6.8 (referee m1) *For finite DT° theories membership takes time $O(|T| \cdot |\chi|)$ (AS Thm 2.5), so $e = 1$ and c_1 is one constant for all of them. (a) Let $s(m) \geq m$ be time-constructible with $s(m+1)^{c_1} = o(s(m)^{c_1+1})$ (every polynomial). There is a decidable X such that every finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent has, for infinitely many m, some $w \in X$ of length m with no K-derivation of φ_w from T of symbol size $\leq s(m)$. (b) For each e the same holds, with X depending on e, for every axiom set with membership in time $O(n^e)$; a single X for all e exists under a growth condition* [proof sketch] *(Appendix E.4).*

6.5 What the derivation likelihoods charge

Theorem 6.9 bounds the *symbol* size of derivations. The code length $-\ln \Pr(\pi)$ of an L1 tree counts its grammar choices instead (Definition 2.9), and the two can differ exponentially.

Remark 6.11 (Refuted: plain L1 charges symbol size). [refuted; counterexample proved and computed] model Prop 6.9 Under L1 with subcritical parameters, $-\ln P_T(\varphi) \geq \kappa\, \ell_{\min}(\varphi) - \ln(C/Z_T)$, $\ell_{\min}$ the least symbol size of a derivation, fails (an earlier sketch claimed it; Appendix H.2). From $\forall x\,(x = x)$, repeat k times "$\forall$E with $t = p + p$, then Gen on p", and close with one $\forall$E: the conclusion has size $2^{k+3} - 1$, while $-\ln \Pr(\text{tree}) = 12.77 + 12.21k$ nats (12.206 per round) and $-\ln P_T \leq -\ln \Pr(\text{tree})$ (Appendix E.5, Table 23).

$\forall$E and Gen substitute into every occurrence of a variable at the price of one choice, so Theorem 6.9 and Corollary 6.10 transfer to $\mathsf{L1}^\sigma$ and to Hänni's graded score, not to plain L1. Plain L1 charges at least linearly in the least number of grammar nodes of a tree, in which conclusions grow at most exponentially; so a hard assigner makes it pay at least $\zeta s(m)$ nats infinitely often, about the logarithm of its nondeterministic time [proved, given the hierarchy theorem] (Lemma E.4, Proposition E.5, Theorem E.6, and Corollary E.7 in Appendix E.5).

Proposition 6.12 (Symbol-size penalties charge polynomially). [(a) proved; (b), (c) proved, given the hierarchy theorem] model Prop 6.14 *(a) Under* $\mathsf{L1}^\sigma$ *(Definition 2.11),* $P_T^\sigma(\varphi) \leq e^{-\kappa\, \ell^\sigma_{\min}(\varphi)} Z_T / Z_T^\sigma$*, with* $\ell^\sigma_{\min}(\varphi)$ *the least total symbol size of a valid* L1 *tree concluding* φ*. (b) A valid* L1 *tree of total symbol size* ℓ *yields a* K*-derivation of symbol size at most* 4ℓ*. So with* X *from Corollary 6.10(a), for every finite* DT° *theory* T *consistent with* Γ_{f_X}*, infinitely often* $-\ln P_T^\sigma(\varphi_w) \geq \kappa s(m)/4 - \ln(Z_T/Z_T^\sigma)$*. (c) Hänni's graded score with* $g(\ell) = 2^{-\kappa\ell}$ *on explicit* K*-derivations, in a code with at least one bit per written symbol (Remark 2.23), charges at least* $\kappa s(m)$ *bits on the same data (*$\kappa s(m) + \log_2 Z_T$ *normalised).*

Conjecture 6.13 (Plain L1 charges polynomially too). [conjecture] model Conj 6.15 *Theorem E.6 holds with* $\mathrm{NTIME}(\ell(m)^c)$*, for a constant* c*, in place of* $\mathrm{NTIME}(2^{(d+1)\ell(m)})$.

The missing step is Gen with de Bruijn indices on shared representations (Appendix E.5). Until it is proved, the question's derivation-length prior should measure written symbols ($\mathsf{L1}^\sigma$, or the graded score normalised as in Remark 2.23); with a hard bound on written size, axiom induction is certificate function induction (Theorem 6.16).

6.6 Which computation is free

Hänni objected to Claude's summary "The time penalty doesn't block your collapse argument": with time penalised "on the function induction side and in the axiom generator but crucially NOT in the proof search from axioms", the penalty should help. He then proposed requiring "that given statements have derivations from short axioms". Two follow-up tracks, each refereed once by two referees, answer both (Appendix H.4; proofs in Appendix E.6); that summary had conflated the two directions of Theorem 6.3.

Function induction charged, deduction free. $\mathrm{AI}_t^{\mathrm{chk}}$ admits only deciders running in time $O(t(|\chi|))$, $\mathrm{AI}_t^{\mathrm{gen}}$ only enumerators that write each axiom a within $O(t(|a|))$ steps of the last; $\mathrm{FI}^\tau_{\mathrm{cons}}$ clocks each assigner at $\tau(|\varphi|)$ (Definition E.10).

Theorem 6.14 (The equivalence becomes one-sided). [proved, under a self-simulation assumption on U] time-followup Lemma 2.1, Thms 2.2, 3.1 *Let* $t(m) \geq m^{a^*}$*,* a^* *fixed by* U*, and* τ *be time-constructible.*

(a) On every finite D the weights of $\mathrm{AI}_t^{\mathrm{chk}}$ and $\mathrm{AI}_t^{\mathrm{gen}}$ are within a constant factor of $W_{\mathrm{FI}}(D)$, unpenalised. (b) On every D the losses of AI, $\mathrm{AI}_t^{\mathrm{chk}}$ *and* $\mathrm{AI}_t^{\mathrm{gen}}$ *exceed that of* $\mathrm{FI}^{\tau}_{\mathrm{cons}}$ *by at most a constant c_τ.*

Theorem 6.15 (Diagonal separation)**.** [proved, given the recursion and MRDP theorems (known)] time-followup Thm 3.2, Cor 3.3, Prop 3.4 *Let $L \supseteq L_A$, B true in an expansion of $\mathbb{N}$. For every computable predictor P there is a computable sequence of sentences "e outputs 1 on n", each decided by* Q *plus one true Π_1 sentence, on which P loses at least $n - 1/(4 \ln 2)$ bits while* AI, $\mathrm{AI}_t^{\mathrm{chk}}$, $\mathrm{AI}_t^{\mathrm{gen}}$ *lose a constant independent of P. P can be Hänni's polynomial-time predictor S;* $\mathrm{FI}^{\tau}_{\mathrm{all}}$ *and* $\mathrm{FI}^{\tau}_{\mathrm{cons}}$ *lose at least $n - 1.37$ and $n - 1.37 - c_\perp$. Yet S and* $\mathrm{FI}^{\tau}_{\mathrm{all}}$ *beat* AI *by $n - O(1)$ expected bits on the fair-coin literals of Proposition E.1.*

Theorem 6.14(a) pads Craig's set by the enumerator's own work, writing $\chi^{\wedge(T+1)}$ on detecting a decision on χ at work T; in Theorem 6.15, e runs P on its own sentences and outputs the less probable label. With deduction free, axiom induction is thus strictly stronger than time-penalised consistent function induction, and incomparable with time-bounded predictors that may be inconsistent. The gain is free deduction, not axioms: a constant-size consistent evaluator gives unpenalised $\mathrm{FI}_{\mathrm{cons}}$ the same gain [proved]; under a time bound the defeating sequence is computable within a slightly larger bound, a time-hierarchy effect [proof sketch] (Proposition E.15). Nor do these penalties favour genuine axiom systems: posterior odds of classes of theories move by at most a constant factor, and a slow assigner keeps weight $2^{-|f|-c}$ (Proposition E.16) [proved, as Theorem 6.14].

Deduction charged. Let hypotheses decide in time $O(t)$ axiom sets closed under renaming parameters (Proposition E.27 shows why), and let each datum need a K-derivation in which (a1) every axiom instance used, logical ones included, has at most $h(|\varphi|)$ symbols, or (b) the *material*, the total size of the distinct instances used, is at most $g(|\varphi|)$; (a2), (b2) count nonlogical instances only (Definition E.18). $\mathrm{FI}_{\mathrm{cert}}$ weighs consistent assigners with certificates of length d' checkable in time t'.

Theorem 6.16 (Subformula lemma; sandwich)**.** [(a) proved; (b) proved, for renaming-closed axiom sets; Cook–Reckhow (known)] short-derivations §2, Thm 3.1; time-followup Thm 4.2 *(a) In a derivation of φ with distinct lines, each an ancestor of the last, each line sits at its own node of an axiom instance used or of φ, up to abstracting parameters of a chain of Gen steps; so the derivation's size is at most $(M + |\varphi|)(M + |\varphi| + 1)/2$, M its material; the $\frac{1}{2}$ is attained. (b) Hence reading (b) lies between derivation-size budgets g and $O((g + m)^2)$, and is equivalent to* $\mathrm{FI}_{\mathrm{cert}}$ *up to a constant in description length and polynomial changes of the budgets;* $\mathrm{FI}_{\mathrm{cert}}$ *also embeds into reading (a1) (Lemma E.19 and Proposition E.21). The free resource of reading (b) is guessing the proof.*

Corollary 6.17 (Separations, and their conditions)**.** short-derivations Cor 3.2, Prop 3.4, Cor 4.4, Props 4.5, 4.12; time-followup Cor 4.3 *On literal sequences (data $R(t_w)$, t_w spelling w, labelled $[w \in X]$, for all words w), with standard simulation overheads; polynomial budgets charge their degrees, and that of membership time, in the prior: (a)* [proved, for renaming-closed axiom sets; conditional where stated; the last clause under a self-simulation assumption on U; hierarchy theorem (known)] *at polynomial budgets reading (b) fits exactly the $X \in \mathrm{NP} \cap \mathrm{coNP}$. So per sequence it beats polynomial-time* $\mathrm{FI}_{\mathrm{cons}}$ *if* $\mathrm{NP} \cap \mathrm{coNP} \neq \mathrm{P}$*, and fits the same sequences if* $\mathrm{P} = \mathrm{NP}$*; with degrees uncharged, polynomial-time* $\mathrm{FI}_{\mathrm{cons}}$ *fails to dominate it; and for each τ, at a budget a hierarchy gap above τ, it beats* $\mathrm{FI}^{\tau}_{\mathrm{cons}}$*. (b)* [proved, for renaming-closed axiom sets; hierarchy theorem and APSPACE = EXPTIME (known)] *At polynomial budgets reading (a1) fits exactly the $X \in$* EXPTIME*; at any computable budgets some*

decidable X defeats it; and every predictor computable in time polynomial in the number of data, S among them (given Hänni's account of its time), loses at least $n - 0.361$ bits where (a1) at a linear budget loses $O(1)$. (c) [proved] *Readings (a2), (b2) fit every decidable X, by a finite set of universally closed Horn sentences.*

Soft charges [proved]: a factor $2^{-\kappa\ell}$ per datum, ℓ the bit length of a shortest derivation, matches certificate function induction only up to a constant factor in the rate; no pair of rates gives two-sided constant regret (Proposition E.26). *Templates* [proved, given Book–Greibach–Wegbreit (known)]: on literal data finite Horn template theories simulate any verifier with derivations quadratic in its time, so Corollary 6.10 is tight up to polynomials there (Proposition E.24 and Corollary E.25); non-literal data are open.

6.7 Links to the other sections

Links to Sections 3 and 7 are in Remark E.9: no penalty monotone in the number of derivation steps reverses Theorem 3.4; a Kt-style penalty charges only $O(\ln k)$ for $0 + S^k 0 = S^k 0$ on the equational fragment; and a time-bounded likelihood favours making theorems with long proofs axioms.

Conjecture 6.18 (A time-bounded analogue)**.** [conjecture; not written out] model §9.3 *Hänni's note on time-bounded Solomonoff induction*[3] *gives, for each polynomial p, a predictor whose log loss exceeds that of any "prediction algorithm P with runtime $O(p(n))$" by "at most a constant c_P". By his argument, the derivation-bounded* L2 *inducer with a growing depth budget plausibly has constant regret against theories whose data have derivations within the budget.*

7 The axioms we actually have: PA and ZF

Which axioms does the posterior pick up for arithmetic and set theory? Three caveats hold throughout. (i) The computations use the two-part code $\mathsf{L1}^{\mathrm{sch}}$ or L0 with a learned grammar, never the generative L1 of Section 2; their numbers are not L1 results. (ii) Candidate sets are hand-picked and finite. (iii) Derivation costs are upper bounds (Definition 7.1). Units are bits; "proved; computed" means a proof plus a derivation checked line by line (Appendix F.1); proofs and full tables are in Appendix F (for Proposition 7.6 in Appendix G.3).

7.1 Setting

Definition 7.1 (The two-part derivation code $\mathsf{L1}^{\mathrm{sch}}$)**.** pa Defs 0.2, 0.3 Derivations are in a sound and complete natural-deduction calculus ND (Table 6). An instance $f(\varphi)$ of a schema $f(P)$ is derived schematically in P and costs $\mathrm{bits}(\pi) + L_{\mathcal{Q}}(\varphi)$, the *motive* φ (the formula substituted for P) being coded once by the grammar $\mathcal{Q}$. Per line, $\mathrm{bits}(\pi)$ charges $\log_2 10$ for the rule, $\log_2 |T|$ per axiom citation, $\log_2(\text{line index})$ per premise reference and β per *written symbol*. On a finite candidate set the prior is $\pi(T) \propto 2^{-\beta_{\mathrm{ax}} \sum_{\tau \in T} |\tau|}$, with β_{ax} bits per axiom symbol; by default $\beta_{\mathrm{ax}} = \beta$ ($\log_2 23 \approx 4.52$ in the derivation computations, 5 in those built on AS's code; Appendix F.1). $\mathrm{CL}(T; D)$ is the total code length, ℓ_{cite} the cost of a one-line citation. Costs of explicit derivations are upper bounds on the shortest ones.

[3] File `research/prior/hanni-polytime-solomonoff.md`; working notes, in which he says he has not "checked all the technical claims completely carefully".

With at least $\log_2$ of the alphabet size bits per written symbol a decodable variant of $\mathsf{L1}^{\mathrm{sch}}$ is a prefix code, so Kraft holds [proved for $\beta \geq \log_2$(alphabet size)] and $\mathsf{L1}^{\mathrm{sch}}$ is the two-part form of a sub-probability likelihood of the $\mathsf{L1}^{\sigma}$ kind, not of plain $\mathsf{L1}$ (Proposition E.5); $\beta = \log_2 23$ under-charges symbols by about 3%, which changes no conclusion. $\mathcal{Q}$ is *learned*: sequential KT, a Dirichlet($\frac{1}{2}$) marginal (Krichevsky and Trofimov, 1981), per context (sort, depth, parent, child index) over 9 formula and 14 term symbols; with Dirichlet($\frac{1}{2}$) weights, CL differences under $\mathsf{L0}$ are exact $\log_2$ posterior odds, except where templates overlap (Remark F.24, in Appendix F.7).

Definition 7.2 (Refutation rule R_d)**.** pa Def 0.4 With data D and negative data N, R_d sets the likelihood of T to 0 if $T \vdash_{\leq d} \neg s'$ for some $s' \in D$ or $T \vdash_{\leq d} s$ for some $s \in N$ (Th_d in derivation size; under $\mathsf{L0}$, $d = 0$ and "derives" means "has as an instance"). R_∞ is not computable.

7.2 Equivalent axiomatisations: usage decides

Let $\mathrm{Dlt} := \forall u \forall v\,(u < v \leftrightarrow \exists z\,(u + Sz = v))$. For a motive φ in x (parameters allowed), $\mathrm{Ind}(\varphi) := \varphi(0) \wedge \forall x(\varphi \to \varphi(Sx)) \to \forall x\varphi$; $\mathrm{CVI}(\varphi) := \forall x(\theta_\varphi \to \varphi) \to \forall x\varphi$ with $\theta_\varphi := \forall y(y < x \to \varphi(y))$; $\mathrm{LNP}(\varphi) := \exists x\varphi \to \exists x(\varphi \wedge \forall y(y < x \to \neg\varphi(y)))$. For nonempty $S \subseteq \{\mathrm{Ind}, \mathrm{CVI}, \mathrm{LNP}\}$ let $\mathsf{PA}_S := \mathsf{Q} + \mathrm{Dlt} + \{f(P) : f \in S\}$, $\mathsf{PA}_f := \mathsf{PA}_{\{f\}}$, and $\mathsf{PA}_{\mathrm{all}}$ with all three.

Proposition 7.3 (Three forms of induction)**.** [proved; computed]
pa Prop 1.1 *With* $B := \{\mathrm{Q1}, \ldots, \mathrm{Q5}, \mathrm{Dlt}\}$*, for every motive* φ*: (A)* $B \vdash \mathrm{CVI}(\varphi) \to \mathrm{Ind}(\varphi)$*; (B)* $B \vdash \mathrm{Ind}(\theta_\varphi) \to \mathrm{CVI}(\varphi)$*; (C1)* $\vdash \mathrm{CVI}(\neg\varphi) \to \mathrm{LNP}(\varphi)$*; (C2)* $\vdash \mathrm{LNP}(\neg\varphi) \to \mathrm{CVI}(\varphi)$*. So* $\mathsf{PA}_{\mathrm{Ind}}$*,* $\mathsf{PA}_{\mathrm{CVI}}$*,* $\mathsf{PA}_{\mathrm{LNP}}$ *have the same theorems.*

The derivations were checked schematically in P, on 200 random motives, and by independent code (Appendix F.2). So PA is Q, Dlt and one higher-order pattern, CVI(P) or LNP(P); T_{Ind} is not a pattern (Remark F.1, in Appendix F.2).

Usage. Let the data be i.i.d. uses: a form f with probability p_f and a motive $\varphi \sim \mathcal{Q}$, the datum being $f(\varphi)$. In PA_S a use costs $c_S(f) + L_{\mathcal{Q}}(\varphi)$, $c_S(f)$ the cost of the cheapest derivation of $f(P)$ in a fixed library of checked derivations (a citation if $f \in S$). In the library a derivation costs 646–2234 bits more than a citation (Table 24); any derivation of CVI(P) from $\mathsf{PA}_{\mathrm{Ind}}$, or of Ind($P$) from $\mathsf{PA}_{\mathrm{CVI}}$, provably costs at least 53 or 44 bits more (Proposition F.2; both in Appendix F.2).

Proposition 7.4 (Usage limit)**.** [proved, given the library constants]
pa Prop 1.3 $\frac{1}{n}(\mathrm{CL}(\mathsf{PA}_S; D_n) - \mathrm{CL}(\mathsf{PA}_{S'}; D_n)) \to \sum_f p_f(c_S(f) - c_{S'}(f))$ *a.s.; the posterior concentrates on the* S *minimising* $\mathbb{E}_p[c_S(f)]$ *if it is unique.*

Remark 7.5 (The usage-closed axiomatisation). [computed] pa §1.2 For $\mathsf{L1}^{\mathrm{sch}}$ with this library: (i) $\mathsf{PA}_{\mathrm{Ind}}$ beats $\mathsf{PA}_{\mathrm{CVI}}$ iff $p_{\mathrm{CVI}}/p_{\mathrm{Ind}} < 0.600$, and $\mathsf{PA}_{\mathrm{LNP}}$ iff $p_{\mathrm{LNP}}/p_{\mathrm{Ind}} < 0.757$; these ratios are code-specific (Appendix F.2). (ii) $\mathsf{PA}_{\mathrm{all}}$ pays $81 + 86$ prior bits and 0.29 bits per citation more than $\mathsf{PA}_{\mathrm{Ind}}$, and is ahead after about one use of a non-primitive form (in expectation after 1–12 schema uses for the mixes tried). (iii) So the posterior concentrates on the *usage-closed* axiomatisation: for this library every form used at a rate above about $5 \cdot 10^{-4}$ per use is primitive (with only the lower bounds of Proposition F.2, the threshold is at most about $7 \cdot 10^{-3}$). All candidates are PA-equivalent, so the theorems are right whichever wins (compare Proposition 4.5).

The same holds for the two recursion conventions for $+$ (Q4, Q5 against $0 + x = x$, $Sx + y = S(x+y)$): each proves the other's laws with one induction, and the posterior adopts the convention the data use (Proposition F.3, in Appendix F.2).

In ZF, Separation from Replacement, Replacement from Collection and Foundation from $\in$-induction cost 574–1432 bits per use (template priors 72–185 bits); the reverse overheads, not formalised, are conjectured to be much larger. Given them, the textbook ZF axioms win when the data are direct uses of them and of nothing they derive only at a cost (Remark F.4, in Appendix F.2).

Proposition 7.6 (Forms of commutativity)**.** [proved; computed]
experiments Prop X12 *Let* $A_{xy} := \{\forall x\forall y\,(x + y = y + x)\}$, $A_{yx} := \{\forall y\forall x\,(x + y = y + x)\}$, $S_{ab} := \{z + z' = z' + z\}$, $M_x := \{\forall x\,(x + z = z + x)\}$, $M_y := \{\forall y\,(z + y = y + z)\}$, *with closed guards, and* S_{ab}^{open}, M_x^{open}, M_y^{open} *the same with open guards (Section 8.1).* (a) A_{xy}, A_{yx}, M_x^{open} *and* M_y^{open} *are logically equivalent.* (b) $S_{ab} \cup M_x \cup M_y \cup S_{ab}^{\text{open}} \nvdash \forall x\forall y\,(x + y = y + x)$. *So* S_{ab}, M_x, M_y *and* S_{ab}^{open} *are strictly weaker than* A_{xy}, *although* S_{ab}, M_x *and* M_y *yield exactly its closed instances* $t + t' = t' + t$.

Remark 7.7 (E6: commutativity). [computed] experiments E6 Data from the chain Ch_2 on A_{xy} and on M_x, and $\mathsf{L0}$ citations of S_{ab} (5 seeds, $n \le 256$; Appendix G.4). The chain $\mathsf{L1}$ identifies the generating form by $n = 32$ in every run, among logically equivalent and among instance-equivalent weaker forms (compare Proposition 4.5), through partial instances such as $\forall y\,(2 + y = y + 2)$. Under $\mathsf{L1}_{\text{cit}}^{\text{sel}}$ the five closed-guard forms have equal likelihoods on every datum, so the mass on provers of $\forall x\forall y\,(x+y = y+x)$ tends to their prior share 0.088, not to 1: the ω-gap of Theorem 3.9, not a tie among equivalents. A redundant second form is a spare slot (mass below 10^{-9}; Table 38).

7.3 The MDL finding in PA

AS's MDL finding (usage, not the logical boundary of a schema, decides whether it is split; Remark 4.13) concerns sequential KT codes, whose comparisons are $\log_2$ posterior odds. We re-ran AS's code on $T^* := \mathsf{Q} + T_{\text{Ind}}$ against its splits H_F, one T_f per motive root $f \in F$ (H_{root}: all 7 roots; H_{used}: those that occur), under several grammars and usage laws, among them AS's natural law G1 (motives of depth 2, all seven roots used; Appendix F.3). With fixed parameters and a grammar that factors at the root, H_F has exactly the law of T^* (Proposition F.5, in Appendix F.3). With learned parameters Occam terms decide: under SDPC, a learned positional grammar shared by all templates, $\text{CL}(H_F) - \text{CL}(T^*) = \Delta\text{prior} + \frac{|F|-9}{2}\log_2 n + O(1)$ when all nonzero counts grow linearly (Proposition F.6, in Appendix F.3), to 0.1 bit of the computed code length; so the SDPC margin is not constant (Remark F.7). With a fixed $\mathcal{Q}$ only the split pays Occam terms (Proposition 4.10); a learned grammar can reverse the drift (Remark 4.13(ii)).

On G1 at $n = 256000$ the split wins by 17242 to 274656 bits, a margin linear in n, under a fixed symbol code, per-template learned codes and a shared grammar with one context per sort; under SDPC it stays 762 bits behind (Table 26, in Appendix F.3). So the finding is about the grammar, not the likelihood family: a split pays off linearly whenever $\mathcal{Q}$ ignores a feature that the split conditions on and that usage depends on. A derivation likelihood changes the consequences:

Proposition 7.8 (Detours)**.** [(a) proved; computed; (b) known] pa Prop 2.2 *(a) A split* H_F *derives every induction instance if* F *contains a non-atomic connective, through the wrappers* $\neg\neg P$, $P\wedge P$, $P\vee P$, $(0 = 0) \to P$, $\forall z P$, $\exists z P$ *(six checked derivations; under* $\mathsf{L1}^{\text{sch}}$ *they cost 369.2–488.9 bits more than a citation). (b) If* $F = \{=\}$, $\mathsf{Q} + H_F \subseteq \text{IOpen}$, *which does not prove the irrationality of* $\sqrt{2}$ *(Shepherdson, 1964) (secondary sources only), while* PA *does.*

So under $\mathsf{L1}$ MDL's incompleteness for schemas (AS App. E.7) disappears once a split has a non-atomic root; the usage effect and the unsoundness error remain (Remark F.23; Remark 4.13(i)).

theorem s	$\lvert s\rvert$	lines	$D_T(s)$	$\beta\lvert s\rvert + \ell_{\text{cite}}$	ratio	$\beta^*(s)$
$\forall x\, \neg Sx = x$	7	11	224.6	38.2	5.9	31.2
$\forall x\, 0 \cdot x = 0$	7	14	281.3	38.2	7.4	39.3
$\forall x\, 0 + x = x$	7	11	220.1	38.2	5.8	30.5
$\forall a \forall v\, Sa + v = S(a + v)$	13	23	576.0	65.3	8.8	43.8
$\forall q \forall x\, x + q = q + x$	11	55	1358.6	56.3	24.2	122.9
$\forall a \forall b \forall x\, (a + b) + x = a + (b + x)$	17	27	793.4	83.4	9.5	46.3

Table 3: Deriving versus memorising in $\mathsf{Q} + T_{\text{Ind}}$ ($\mathsf{L1}^{\text{sch}}$, $\beta = \log_2 23 \approx 4.52$; bits; $\beta|s| + \ell_{\text{cite}}$ memorises at $\beta_{\text{ax}} = \beta$): memorising wins at the first occurrence for every theorem at the default $\beta_{\text{ax}} = \beta$. Computed; checked derivations; upper bounds (pa §3.2; Appendix F.4).

7.4 Theorem data: statements given without proof

Hänni proposed that "statements given without proof in our data set should be fairly easily derived from axioms". Then the data are theorems. A theory T can derive a datum s, at cost $D_T(s)$ (cheapest library derivation under $\mathsf{L1}^{\text{sch}}$), or adopt it as a ground axiom.

Proposition 7.9 (Memorisation threshold)**.** [proved]
pa Prop 3.1 *If s occurs r times and the other data have the same best derivations under T and $T \cup \{s\}$, then $\mathrm{CL}(T \cup \{s\}) - \mathrm{CL}(T) = \beta_{\text{ax}}|s| + \Delta I - r(D_T(s) - \ell_{\text{cite}})$, with $D_T(s)$ and ℓ_{cite} computed at the proof-text rate β and ΔI the index cost change ($\log_2 \frac{k+1}{k}$ per citation over k axioms). Up to ΔI, memorising at the first occurrence is cheaper iff $\beta_{\text{ax}} < \beta^*(s) := (D_T(s) - \ell_{\text{cite}})/|s|$.*

Frequent lemmas become axioms, and a schema compresses its direct instances about fourfold; on a stream of theorems from a library fixed in advance the MAP is Q plus the library, or, with steady direct induction uses, $\mathsf{Q} + T_{\text{Ind}}$ plus the library from $n = 10$ (Remark F.10, in Appendix F.4). On data from P_{T^*} the generator's code length for a new datum is in expectation at most the prior cost of memorising it, when that prior is a prefix code ($\beta_{\text{ax}} \geq \log_2$ of the alphabet size; Gibbs' inequality, Proposition F.8, in Appendix F.4).

Proposition 7.10 (Well specified, the generator wins)**.** [proved, given Theorem 4.2 and known facts; proof sketch (Dirichlet weights)] pa Prop 3.3 *Let the data be i.i.d. from PA's $\mathsf{L1}$ generator in setting W (Definition 4.1), weights positive and parameters admissible ($\operatorname{supp} P_{\mathsf{PA}} = \mathrm{Th}(\mathsf{PA})$, Lemma 2.10), in a countable class with $\pi(\mathsf{PA}) > 0$. Then a.s. every $\mathsf{Q} \cup S$ (S a finite set of PA-theorems) has posterior 0 eventually, and $\pi_n(C^*) \to 1$. Under W a theory $\mathsf{PA} \cup S$ with generic fixed weights has a law different from P_{PA} and loses exponentially, at rate $\mathrm{KL}(P_{\mathsf{PA}} \| P_{\mathsf{PA} \cup S})$ (Proposition C.1(a)); with Dirichlet weights (outside W) such theories behave like spare slots and should decay only polynomially (proof sketch, Proposition 4.14).*

Under the 0/1 score S_{prove} with R_d, $\mathsf{Q}+T_{\text{Ind}}$ beats Q plus memorised theorems once they cost more prior bits than T_{Ind} (72 bits at $\beta_{\text{ax}} = \beta$; associativity alone suffices), if d covers its derivations; but any unrefuted theory with a smaller prior that derives every datum within d beats both, e.g. an inconsistent theory that R_d does not refute (Proposition F.9, in Appendix F.4).

Answer. A derivation-length likelihood finds $\mathsf{Q}+T_{\text{Ind}}$ from theorem data (1) under the full-sum generative $\mathsf{L1}$, when the data come from its own derivation process (Propositions 7.10 and F.8), which human theorems, selected for short statements and interest, do not (under the two-part $\mathsf{L1}^{\text{sch}}$ this case was not studied); and, under $\mathsf{L1}^{\text{sch}}$, (2) when the data contain direct uses of the

schema; (3) when the axiom prior is much steeper than the proof code ($\beta_{\text{ax}} \geq \beta^*(s)$, here 30–123 bits per symbol against $\beta = 4.52$); (4) under the 0/1 constraint with refutation, at the price of the size principle. Otherwise $\mathsf{L1}^{\text{sch}}$ memorises short theorems, and on a computed stream of six library theorems the better of two candidates is Q plus the theorems as axioms, strictly weaker than PA (Remark F.10). $\mathsf{L1}^{\text{eq}}$ (Example C.6) and the fragment $\{\text{Q4}, \text{Q5}\}$ (Proposition B.15) show the same pattern: a derivation likelihood favours the theory that makes the data cheap to state. Not computed: theorem data under plain $\mathsf{L1}$ or the full sum. A time bound on computing the likelihood sums over fewer derivations and so pushes further towards memorisation (Remark E.9).

7.5 Data from $\mathrm{Th}(\mathbb{N})$, the $\mathrm{I}\Sigma_n$ chain and narrow practice

Proposition 7.11 (No consistent r.e. theory survives)**.** [proved] pa Prop 4.1 *If $P_T(s) > 0$ iff $T \vdash s$ (e.g. $\mathsf{L1}$, weights positive, parameters admissible) and the data are i.i.d. from a law with full support on $\mathrm{Th}(\mathbb{N})$, every consistent r.e. theory has posterior 0 eventually, a.s.*

Proposition 7.12 (Refutation)**.** [proved; known] pa Prop 4.2 *Without a refutation rule inconsistent theories are never refuted; R_∞ refutes them at once but is not computable; R_{d_n}, $d_n \to \infty$, refutes each eventually; no computable rule does better uniformly, as consistency of r.e. theories is Π_1-complete.*

Along PA, $\mathsf{PA} + \mathrm{Con}(\mathsf{PA})$, ... the posterior drifts: in a toy model the MAP is the largest level seen and the regret grows by about 50 bits, the prior cost of one level, per level (Remark F.14). In DT° each step must add a ground sentence, since every DT° template whose instances lie in the reflection schema Refl_T is ground (Proposition F.11; both in Appendix F.5).
Pointwise limits are trivial: under 0/1 support memorisation alone eventually decides every sentence correctly in every surviving consistent theory (Remark F.13). So for $\mathrm{Th}(\mathbb{N})$ one can promise regret against every theory plus its exceptions (under L_ε, within $-\log_2 \pi(T)$ bits of T with its exceptions coded as noise; Proposition F.12; both in Appendix F.5) and depth-relative soundness, not identification. In Gold's model the chain $\mathrm{I}\Sigma_1 \subset \mathrm{I}\Sigma_2 \subset \cdots$ with union PA is a limit point.

Proposition 7.13 (The $\mathrm{I}\Sigma_n$ chain)**.** [(a) proved, given known facts; (b) proof sketch; (c) refuted] pa Prop 4.6; referee M4, m7 *(a) For i.i.d. theorems from PA's $\mathsf{L1}$ generator (setting W, parameters admissible), each $\mathrm{I}\Sigma_n = \mathsf{Q} + \sigma_n$ has posterior 0 eventually and $\pi_n(C^*_{\mathsf{PA}}) \to 1$ a.s. in any countable class with $\pi(\mathsf{PA}) > 0$; the waiting times are presumably astronomical (a heuristic). (b) Before that, σ_n pays at each induction use for a Tarski biconditional that grows with the motive, so the schema wins linearly on induction-use data. (c)* Refuted: *"if the community uses only Σ_n motives, every DT° theory covering its practice contains PA-strength induction, so the posterior lands on the $\mathcal{L}_\infty$ side".*

(c) fails because AS Thm 5.16 concerns a single covering template. What is true: a read-once induction template $T_{\text{Ind}}[P := \lambda x.C]$, each metavariable occurring once in C, gives full induction over Q1 if C has a formula metavariable, and otherwise yields only instances with C's formula skeleton, so that Q plus its instances lies in some $\mathrm{I}\Sigma_k \subsetneq \mathsf{PA}$ (Proposition F.16; equivalent motives give equivalent induction instances, Lemma F.15). The condition is sufficient, not necessary: the non-read-once $T_{\text{Ind}}[P := \lambda x.\, x = 0 \vee F(x)]$ still gives full induction over Q1, Q2 (Remark F.17; all in Appendix F.5); which other templates give full induction is open.

Example 7.14 (Narrow practice ends on the weaker side)**.** [computed] pa §4.3 (1); referee r8 Practice: Q plus induction on motives $\exists z(s = t)$ and $\neg(s = t)$. $H_{\text{narrow}} := \mathsf{Q} + T_E + T_N$, with the two induction templates $T_E := \mathrm{Ind}(\lambda x.\exists z\, A(x,z) = B(x,z))$ and $T_N := \mathrm{Ind}(\lambda x.\neg C(x) = D(x))$ (term

metavariables), covers it and lies in $\mathrm{I}\Sigma_1 \subsetneq \mathsf{PA}$. $\mathrm{CL}(H_{\text{narrow}}) - \mathrm{CL}(T^*) = +230.8$ bits at $n = 100$ and $+174.6$ at 3000, matching Proposition F.6(a) to 0.1 bit; slope $\frac{9-8}{2} - 3 \cdot \frac{9-1}{2} = -11.5$ bits per doubling (three fixed formula contexts), crossover $n \approx 2^{26.7}$.

On G1 coverage does not force PA either: the term-only theory of all 157 motive skeletons lies inside $\mathrm{I}\Sigma_2$ and has slope $+2.0$ bits per doubling against T^* (proof sketch; code length not computed), while the theory of the 112 skeletons seen by $n = 3000$ dies at the first unseen one (Example F.18, in Appendix F.5). So strength beyond usage is set by an Occam balance, a term-only theory gaining $\frac{|A_c|-1}{2}$ bits per doubling per context it fixes and paying $\frac{1}{2}$ per extra template, and narrow practice lands on the weaker side: the Bayesian $\mathcal{L}_\infty$-versus-$\mathcal{L}_5$ choice (Proposition 4.9).

Conjecture 7.15 (Broad practice)**.** [conjecture] pa Conj 4.10 *If the motives have unbounded depth and use every connective at every depth with positive probability, the mass on theories deriving all of* PA *tends to* 1*. (Open because covering theories may use non-read-once templates.)*

The experiments of Section 8 (fixed PCFG, L0, causal pools; $<$ in the language, nine motive roots) find the narrow effect with atomic motives. Let T_f be T_{Ind} with motive root f; for a connective root f and any set Q^- of axioms among Q1–Q7, $\mathsf{Q}^- + T_f$ has the same consequences as $\mathsf{Q}^- + T_{\text{Ind}}$ (Proposition F.19), and Q3 is redundant given T_{Ind} (Proposition F.20; both in Appendix F.6).

Proposition 7.16 (Atomic fragments are weaker)**.** [proved, given two known facts] experiments Prop X10 $T^* :=$ Q1..Q7 $+ T_{\text{Ind}}$ *contains* PA *and proves* $\sigma := \neg\exists x \exists y(\neg y = 0 \wedge x \cdot x = (SS0 \cdot y) \cdot y)$*;* $\mathsf{Q} + \{T_=, T_<\} \nvdash \sigma$ *(known facts:* $\mathsf{PA} \vdash \sigma$*; Shepherdson's model, Shepherdson, 1964, secondary sources only).*

Remark 7.17 (E3(b)). [computed] experiments E3(b) L0 data from T^* with misspecified motive laws (5 seeds, $n \leq 2048$). Three connective-rich laws keep the theorems (mass 0.998–1.000 on T^*-equivalents at $n = 2048$); under two of them the MAP leaves T^* for $T^* + T_\exists$ or an observed-root split. Atomic motives: T^* has mass 0.988 at $n = 64$ in 4 of 5 seeds, then frag-atoms $= \mathsf{Q} + \{T_=, T_<\}$ has mass 1.000 in 5 of 5 seeds from $n = 1024$ (mass on T^*-equivalents *in this pool*: $2 \cdot 10^{-178}$ at $n = 2048$, carried by frag-complete, Q plus all nine T_f, whose seven never-used fragments put it about 590 bits behind frag-atoms; a T^*-equivalent that only adds T_{Ind} to frag-atoms was not in the pool and would lose only polynomially, as a spare slot, Proposition 4.14): the posterior moves to a strictly weaker theory (Table 35).

7.6 Unlabelled mixtures: a Bayesian DTRC

AS's DTRC is a learner for unlabelled mixtures that clusters the data by refuting merged templates (AS §7). Its Bayesian form: finite sets of DT$^\circ$ templates, prior 5 bits per symbol, Dirichlet($\frac{1}{2}$) weights, one shared positional grammar, negatives under R_0; data from G1 practice (Q1–Q7 as ground sentences, induction with probability $\frac{1}{2}$), one seed; candidates: the root split, a spare $F \to F$, memorisation, AS's over-general template $T_{\mathcal{L}_\infty}$, the lump $F_0 + T_{\text{Ind}}$ and the bare F_0 (code lengths in Table 27, in Appendix F.7).

Without negatives $\mathsf{Q} + T_{\text{Ind}}$ has mass $\geq 1 - 6 \cdot 10^{-6}$ from $n = 100$; with two negatives ($0 = S0$ and a false instance of $T_{\mathcal{L}_\infty}$), $\geq 1 - 1.6 \cdot 10^{-5}$ from $n = 10$. The root split and the spare are PA-equivalent, so the mass on PA-equivalents is $\geq 1 - 10^{-96}$ from $n = 100$, *within this candidate set.* At $n = 10, 30$, without negatives, the unsound lump $F_0 + T_{\text{Ind}}$ leads, and $V_{\delta,0}$ accepts the false $0 = S0$ for every $\delta \geq 3 \cdot 10^{-13}$ until a negative datum refutes the lump: AS's unsoundness for rarely used ground axioms in Bayesian form (Remark F.23, in Appendix F.7; frequencies over

	what fails	evidence	fixable?
F1	usage over logic	Prop. 7.4 and Rem. 7.5	change the target
F2	misspecified grammar: linear fragmentation	Tab. 26	shared positional grammar
F3	never-used parts split off (ω-gap)	Prop. F.6 and Rem. F.25	harmless for theorems only for splits with a non-atomic formula root (Prop. 7.8(a)); term-root splits lose $\forall x\varphi$ (Rem. F.25); atomic-root splits are weaker (Prop. 7.8(b))
F4	false generalisation, no counterexamples	Ex. F.26	only by refutation
F5	unsound lumps at small n	Rems. 7.18 and F.23	a negative datum, more data
F6	inconsistency unseen by the size principle	Prop. 7.19	only R_{d_n}, not certifiable
F7	no well-specified theory for Th($\mathbb{N}$)	Prop. 7.11 and Rem. F.14	as regret (Prop. F.12)
F8	strength beyond usage set by Occam terms	Ex. 7.14 and Rem. 7.17	broad practice (Conj. 7.15)
F9	bounded search memorises long proofs	§ 7.4	not in general (Prop. 2.24)
F10	theorem data memorised, MAP $\subsetneq$ PA	Prop. 7.9 and Rem. F.10	schema uses, steep prior

Table 4: Robust failures (pa §6); statuses as in the cited items.

seeds: Remark 5.11). The split gains 1 bit per doubling (Proposition F.6); the spare loses $\frac{1}{2}$, as a never-used template costs its prior plus $\frac{1}{2}\log_2 n$ bits and a constant (Proposition F.21, in Appendix F.7); the rest lose linearly.

Remark 7.18 (E2). [computed] experiments E2 L0 citations from T^* at fixed weights, 25 seeds, $n \leq 512$, causal pools. T^* takes the mass once Q1, Q2, Q4–Q7 have each been cited (median datum 55; Q3 is redundant): mass on its equivalents ≥ 0.95 from $n = 32$, 64 or 128 (5, 15, 5 seeds); mean mass of T^* 0.751 at $n = 64$, 0.996 at 512. Before that the sound but weaker SeenQ(D_n), T_{Ind} plus the Q axioms cited so far (Section 8.1), leads; the DTRC candidates above (other data) lack it. The false spare $T^* + \{0 = S0\}$ keeps 0.004 at $n = 512$ (Table 31); false acceptances: Remark 5.11.

On data $t+0 = t$ the schema $z+0 = z$ wins; the sound split by the head of t, which does not entail $\forall x(x + 0 = x)$, closes in at 5 bits per doubling (a Bayesian ω-gap), and $\forall x(x + 0 = x)$ loses at a rate set by the rule code, from $\log_2 10$ bits per datum to $\frac{R-1}{2}\log_2 n$ for R rules (Remark F.25). "Does not contradict" alone learns nothing: the posterior is the prior restricted to compatible theories (Proposition F.22; both in Appendix F.7).

Proposition 7.19 (Inconsistency is invisible to the size principle)**.** [proved] pa Prop 5.3 *Under* $\mathsf{L1}^{\max}$ *with citation weights, adding* σ *with fixed weight fraction* w *gives* $P_{T\cup\{\sigma\}}(s) \geq (1-w)^{c(s)}P_T(s)$, $c(s)$ *the citations in* s*'s best* T*-derivation; with Dirichlet weights an unused* σ *costs its prior plus about* $\frac{1}{2}\log_2 n$ *bits. Consistency must be enforced by* R_d.

7.7 Robust failures and the overall answer

F4: on true data Prime($k \cdot k + k + 41$) the false schema Prime($z \cdot z + z + 41$) beats memorisation at every computed $n \leq 10^6$, no finite unguarded covering union is true without memorising, and a refutation removes the schema; memorisation should eventually win under a flat grammar, not under a depth-indexed one (proof sketches; Example F.26, in Appendix F.7).

7.7.1 The overall answer

The inducer does *not* robustly pick up the textbook axioms; it picks up the axioms that are *used*. Within the hand-picked candidate sets and pools scored, for data that are direct uses of axioms under the broad usage laws scored (usage mixes of equivalent forms, G1 practice, connective-rich motives) and once every needed axiom has been cited, it puts nearly all mass on theories deductively equivalent to PA; narrow or atomic practice, also direct use, ends on a strictly weaker theory (Example 7.14 and Proposition 7.16). In three of the four hand-picked cases this is automatic, since every candidate is PA-equivalent (usage mixes, recursion conventions, root splits), and in the DTRC run PA-equivalents get mass $\geq 1 - 10^{-96}$ from $n = 100$. It fails robustly on theorem data, narrow practice, absent counterexamples, small n and inconsistency (Table 4). Not scored: all finite template unions, non-read-once templates, stronger true theories such as $\mathsf{PA} + \mathrm{Con}(\mathsf{PA})$ (spare slots), further ZF candidates.

8 Experiments: a small exact laboratory

The package `bai` (`code/bai`) computes exact posteriors over finite pools of theories made of DT$^\circ$ templates, reusing the code of AS (Hänni and Claude (Anthropic), 2026a) for syntax, matching, Min and DTRC; eight experiments, E1–E8, use it as a small exact laboratory for the claims of §§3–7. Each posterior is exact *over its pool*, i.e. conditioned on "the theory is in the pool" (Remark 8.2); code lengths are in bits, $\alpha = \frac{1}{2}$, every run is seeded (Appendix G.6), and all findings are computed unless a proposition is cited.

8.1 What is implemented

A *component* is a DT$^\circ$ template with a closed or open *guard* on each metavariable (may its body contain the parameter?); a theory is a finite set of components. The language is L_A; no axiom of Q mentions $<$. There is one parameter p: a datum $\varphi(Sp)$ means $\forall y\, \varphi(Sy)$, and data with two parameters have probability 0. The instantiation grammar $\mathcal{Q}$ is a fixed PCFG over bodies (Appendix G.1) that satisfies Definition 2.6 and counts holes and bound variables alike, so splitting a schema by the root of a body is an exact reparametrisation (Proposition 4.6(b), Remark C.2). The prior is $\pi(T) \propto 2^{-\lambda \,\mathrm{bits}(T)}(1 + |T|)^{-\tau}$, where $\mathrm{bits}(T)$ is the length of T in a prefix code that satisfies Kraft's inequality, $|T|$ is the template size and τ is the time exponent of Remark A.1; by default $\lambda = 1$ and $\tau = 0$. For instance $\{z + 0 = z\}$ costs 17.1 bits, $\{\forall x\, (x + 0 = x)\}$ 17.2 bits and $T^* := \mathrm{Q1..Q7} + T_{\mathrm{Ind}}$ 226.2 bits.

The mixture weights get a $\mathrm{Dir}(\frac{1}{2})$ prior and are integrated out. L0 is citation: $P(s \mid T, w) = \sum_i w_i a_i(s)$ with $a_i(s) = \prod_M \mathcal{Q}(\theta_s(M))$ for the unique matcher θ_s. L1 is the forward chain Ch_J (Definition 2.3; rules in Appendix A.3): cite component i with probability w_i, then take at most $J \in \{1, 2\}$ steps of ∀E (term from the elimination-term law $\mathcal{Q}_{\mathrm{e}}$), Gen on p, or MP with a *cited* major premise, stopping with probability $c_{\mathrm{stop}} = \frac{1}{2}$ while a rule applies; $c_{\mathrm{ch}} := 1 - c_{\mathrm{stop}}$. The chain never fails, so it needs no normaliser, and its law is linear in w; induction followed by MP and ∀E is outside it, so the PA runs use L0. $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ observes the chain through "closed quantifier-free sentences only" by per-citation rejection, a_i/c_i with c_i the probability that a chain from component i ends there (Remark A.6). The marginal $P^{\mathrm{Dir}}_T(D_n)$ is computed exactly by a dynamic programme over count vectors (Lemma 2.8). Above 20000 states rigorous bounds replace it; the largest posterior mass a theory so bounded could have had never exceeded $8 \cdot 10^{-14}$ in E1–E4, where it is recorded for every run, and was 0 in the one E6 seed checked (Appendix G.5).

	setup	result tested	finding	where
E1	data from H_{sch}, $H_{\forall}$, H_{open} (L0 or chain) on three φ; 5 seeds, $n \leq 256$	Thms. 3.3, 3.9 and B.4	in the chain, one bit per datum against $H_{\forall}$, exactly; under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ an exact tie, so $P(T \vdash \forall x\varphi)$ tends to a prior share; memorisers and over-general templates die fast	§ 3.2
E2	L0 citations of $T^* = \text{Q1..Q7} + T_{\text{Ind}}$ at fixed weights; causal pools; 25 seeds, $n \leq 512$	Prop. 4.14; pool version of Thm. 5.7	T^* wins once every needed axiom is cited, SeenQ(D_n) before; early unsound lumps; spares decay polynomially	Rems. 5.11 and 7.18
E3(a)	instances of $x + 0 = x$ under four other term laws; $n \leq 1024$	Thm. 4.16	linear drift to a best-fitting family member; completeness lost, nothing unsound	Rem. 4.20
E3(b)	E2's T^* under four other motive laws; $n \leq 2048$	Props. 7.16 and F.19	connective-rich laws keep the theorems, not the axioms; atomic motives end on a strictly weaker theory	Rem. 7.17
E4	an adaptive prover against $V_{\delta,1}$: IL's tight construction, and pools on $x + 0 = x$	Thm. 5.6	well specified, wins stay below δ'; with false near misses in the data, the prover always wins	Rem. 5.10
E5	spare slots; $\mathcal{L}_k$ against $\mathcal{L}_\infty$; Gold's text; 10 seeds	Props. 4.9 and 4.14	unused spares follow (a) exactly; both ends of Gold's limit point identified; 14 stages of Gold's text end	after Props. 4.9 and 4.14
E6	eight forms of commutativity; chain and L0 data; 5 seeds, $n \leq 256$	Props. 4.5 and 7.6 and Thm. 3.9	the chain identifies the generating form; under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ the provers keep their prior share	Rem. 7.7
E7	E2 with a time factor or a steeper prior	Rem. A.1	the time factor slightly favours small unsound lumps; steepness trades early for late soundness	Rem. E.8
E8	a schema against its root split, chain data with latent citations; 10 seeds, $n \leq 4096$	Prop. 4.11	a derivation likelihood does not change the MDL finding	Rem. 4.12

Table 5: The experiments. Pool names used in the tables of App. G.4: skel$k$@b is a skeleton clustering of D_b stopped at k clusters, DTRC@b is DTRC with a PA refuter on D_b, frag-complete, frag-atoms and frag-observed@b are Q plus the root fragments T_f of all nine roots, of $=$ and $<$, and of the roots seen in D_b, and IndSwap is Q plus an equivalent, swapped induction template. Computed.

The pool at sample size n holds hand-picked theories; theories built *causally* from the data D_b seen at build points $b \leq n$ (Min of random subsets, skeleton clusters and, in PA runs, DTRC and the observed-root split; Appendix G.1); in PA runs, for every pool theory T, $\text{Trim}(T, D_n)$, the components of T that could have produced a seen datum, so that $\text{SeenQ}(D_n) := \text{Trim}(T^*, D_n)$ is T_{Ind} plus the Q axioms cited so far; and $\text{Mem}(D_n)$, the set of data seen. So every pool member uses only data already seen. $T \vdash_J s$ iff some chain of at most J steps from T, with all rules enabled, ends in s: a sound under-approximation of derivability, Th_d of Definition 2.2 with d in chain steps. $V_{\delta,J}$, the verifier $V_{\delta,d}$ of Definition 5.1 on the pool, accepts s iff the pool posterior of $\{T : T \vdash_J s\}$ is at least $1 - \delta$; $J = 1$ in E1, E3(a), E4 and $J = 0$ in E2, E3(b), E7.

Proposition 8.1 (The computed likelihoods are exact)**.** [(a)–(c) proved; (d) proved under the hypothesis of (d3), computed otherwise] experiments Props X1–X4 (a) *For a component τ and a sentence s with at most the parameter p, citing τ with independent bodies from $\mathcal{Q}$ yields s with probability*

$\prod_M \mathcal{Q}(\theta_s(M))$ *if* τ *matches* s *by a matcher* θ_s *that respects the guards, and* 0 *otherwise.* (b) *For* $a_{ij} \geq 0$, $\mathbb{E}_{w\sim\mathrm{Dir}(\alpha)} \prod_{j=1}^{n} \sum_{i=1}^{K} w_i a_{ij} = \sum_z \prod_j a_{z_j j} \cdot \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_i \frac{\Gamma(\alpha+n_i(z))}{\Gamma(\alpha)}$, *summed over assignments* z *of the data to the* K *components, with counts* $n_i(z)$. *The dynamic programme computes this sum, and the bounds bracket it.* (c) *For every theory, weight vector and* J, *the law of the chain's output is a probability on sentences.* (d) *The backward recursion over predecessors computes the* L1 *coefficients* $a_i(s)$ *exactly whenever the predecessor sets it enumerates are complete. They are complete for the component shapes of Appendix G.2, and the code admits into* L1 *pools only theories of the shapes it checks (this left out 65 and 43 data-derived theories in two E1 configurations, none elsewhere); the hypothesis of (d3) in Appendix G.2, no rigid p inside a metavariable argument, is not checked by the code. It holds for every hand theory of E1 and E6; for data-derived theories at* $J = 2$ *(E1 allq2) exactness is computed (brute force to* 10^{-9} *on 10 theories; the referee's 904 triples), not proved.*

The proofs are in Appendix G.2. The code agrees with brute force on 500 random Dirichlet sums ($1.4 \cdot 10^{-14}$) and with an independent reimplementation, written from the notes, on 400 marginals ($1.4 \cdot 10^{-14}$) and 904 L1 coefficients at $J = 1, 2$ ($2.1 \cdot 10^{-14}$); a forward sampler passes χ^2 tests; 21 unit tests pass.

8.2 Results

Table 5 lists the eight experiments, the result each one tests and where its finding is stated; setups, tables and further findings are in Appendix G.4.

For a one-component generator in a fixed hand pool (E1, E4, E6) the generator is an atom of the prior, and Doob's argument in the (T, w) form gives concentration on it (Remark C.3, in Appendix C.3); the data-dependent pool members ($\mathrm{Mem}(D_n)$, causal theories) are covered only by the pool versions of Theorems 5.6 and 5.7. E2, E3(b), E5(b) and E8 sample at one fixed weight vector; there the runs are a yardstick, not an instance of a theorem.

8.3 What the experiments do not show

Remark 8.2 (Limitations). [(i), (v) computed; (ii)–(iv) properties of the code] experiments §§11, 12, 16.5 (i) *Pool, not class.* The winners, and even the frequency of unsound acceptance, changed with the pool: E2's acceptances fell from 19 to 5 of 25 seeds with causal pools and trimmed theories, and E3(a)'s winners moved when the families R_m, C_k were extended. The pool version of Theorem 5.6 keeps soundness under pool restriction; nothing comparable is shown for identification. Search beyond a pool is open. (ii) *Chains,* $J \leq 2$. MP needs a cited major premise, so no run computes PA under a derivation likelihood. (iii) *One parameter; fixed* $\mathcal{Q}$. A misfit of usage statistics is absorbed by extra components (E3, E8); a learned grammar can change the sign of the Occam drift (Proposition F.6, in Appendix F.3). (iv) *Bounded derivability.* $\vdash_J$ under-approximates provability, so the $P(T \vdash s)$ columns are lower bounds. (v) *Small runs.* 25 seeds in E2 and E7, 10 in E5 and E8, 5 in E1, E3 and E6; E4 100–400 streams; $n \leq 4096$ (quadrature to 10^8). The exact Dirichlet sum is exponential in the number of overlapping components, which also stops E3(a)'s nesting at C_2. No run uses human-written theorem lists: every misspecified generator perturbs or selects from the model's own term or motive law. Neither Hänni's collapse (Section 6) nor his "do not contradict" variant is implemented.

9 Discussion

The positive answers of Section 1.3 need data drawn from a law of the model; the theorems humans state are not such data. We interpret the three refinements, state the good version, and go through Hänni's note, related work and open problems.

9.1 The three refinements

Templates. Templates, which Hänni proposed against his collapse, block the acceptance part of his schema and the reflection schema (Propositions 6.6 and F.11), not the collapse itself: over PA one ground reflection sentence reproduces every Σ_n-sound assigner on Σ_n sentences (Proposition 6.7). Inside the class membership is matching, so the citation likelihood, its Dirichlet sum and the experiments' two-step chain can be computed exactly over a pool (Proposition 8.1); the general derivation likelihood is only a computable real whose positivity is undecidable (Proposition 2.24). Unexpectedly, templates bring in the instance schema $\varphi(z)$, as simple as $\forall x\varphi$ and implying exactly the instances, which $\forall x\varphi$ never overtakes on closed instances in the comparisons analysed (Theorem 3.4); and usage, not the logical boundary of a schema, decides whether it is split (Remark 4.13).

A derivation-length prior. As a generative derivation grammar with a data-independent normaliser it supplies the size principle that Hänni's scores lack (Proposition 2.22). Its code length counts grammar choices, which can be exponentially fewer than the written symbols (Remark 6.11); to price computation, length should be measured in written symbols ($\mathsf{L1}^{\sigma}$, or the graded score divided by its normaliser; Remark 2.23). Read as "statements given without proof in our data set should be fairly easily derived from axioms", it makes the data theorems and identifies the working set of primitives of whoever produced them, frequently used lemmas included, not a canonical axiomatisation (Proposition 7.9, Proposition B.14(c)).

A time penalty. The penalty the question names, on the time to generate axioms or check axiomhood, leaves the direction from function to axiom induction intact: padded Craig sets have polynomial membership and polynomial-delay enumeration (Proposition 6.8 and Theorem 6.14). It leaves proof search free, so, as Hänni objected, penalising consistent function induction but not deduction makes axiom induction strictly stronger, through free deduction rather than its axioms (Section 6.6). Derivation size in written symbols prices the collapse (Remark 6.1); a bound on each derivation's axiom material, logical instances included, one reading of his proposal of short axioms, makes axiom induction certificate function induction (Theorem 6.16). For plain $\mathsf{L1}$ only a logarithmic charge is proved, and a polynomial one is conjectured (Conjecture 6.13).

9.2 A good version, and what it cannot promise

Collecting the pieces, a good version of the inducer has:

- *Hypotheses and prior*: finite sets of DT° templates, a proper prefix-code prior and Dirichlet weights (Definition 2.4 and Lemma 2.5); inconsistent theories removed by refutation at growing depth (Proposition 7.12) or reported as a fourth value Inc (Definition 2.13).
- *Likelihood*: a derivation grammar with a data-independent normaliser and a uniformly subcritical instantiation grammar (Definitions 2.6 and 2.9), in written symbols ($\mathsf{L1}^{\sigma}$) if it is to price computation; a model of which theorems get reported, by a known filter ($\mathsf{L1}^{\mathrm{sel}}$) or a prior over

filters (Theorem 3.9(e); for $\forall x\varphi$, Section 3.9); and a noise component or corruption channel, so that one false datum does not refute the true theory (Proposition B.1, Remark C.9).

- *Output*: the triple $(\mathrm{Bel}, \mathrm{Dis}, \mathrm{Indep})$, a belief function with its plausibility (Section 2.7); and a verifier that accepts s when $\mathrm{Bel}_{\mathrm{cons}}(s) \geq 1-\delta$ (Remark 5.9), with a threshold that shrinks with n when the weights are learned and the likelihood is linear in them (Theorem 5.7); otherwise with the averaged guarantee (Theorem 5.6).

What it can promise, when the data are drawn from a law of the model: concentration on the generator class and convergence of predictions (Theorems 4.2 and 4.21); with plain L1 (parameters admissible) identification of the theorem set, with L0 of the instance union, and with a selection model only of the reported part $\mathrm{Th}(T) \cap S$, deductive questions beyond S keeping their prior share (Corollary 4.4 and Theorem 3.9); no bound on the number of templates and a way around Gold's theorem (Theorem 4.7 and Proposition 4.9); soundness of the verifier with probability $1-\delta/\pi(C_d^*)$ for fixed weights, on average over learned weights, and with a shrinking threshold at each weight vector for likelihoods linear in the weights; for a derivation-grammar likelihood with learned weights only the averaged version is proved (Theorems 5.2, 5.6 and 5.7).

What it cannot promise. It cannot single out the actual axioms: inside the generator class the posterior is the prior (Corollary 4.3). It cannot certify $\forall x\varphi$ from closed instances (Theorem 3.9). With learned weights it removes a false spare sentence σ, and with it, when T^* refutes σ, the inconsistent theory $T^* \cup \{\sigma\}$, only polynomially (Proposition 4.14 and Remark 4.15): the likelihood charges a theory for the mass it spends off the data, not for being inconsistent (Proposition 7.19), so certifying consistency needs refutation at growing depth (Proposition 7.12). Above all, it promises nothing when the data are not drawn from a law of the model, the case the question cares about: in finite classes the posterior then goes to the best-fitting generator, weaker or unsound by turns (Theorem 4.16, Table 18), in the full class even this can fail (Remark C.5), and in an example a verifier on it accepts a falsehood with probability 1 (Proposition 5.4). The main lever is to model how the data were produced (which theorems are reported, how often each axiom form is used, which mistakes occur), a choice positive data cannot check.

9.3 Hänni's note, point by point

His two variants are scores without the size principle; "does not contradict" alone learns nothing beyond compatibility (Proposition 2.22(b), Proposition F.22). His graded score, normalised, is a two-part code (Remark 2.23), which makes good his reading of the scores as a model of how the statements get generated. The trichotomy is best kept as a triple (Section 2.7). His equivalence holds with the constant he expects, via Craig's trick, in the semimeasure convention he also considers; his consistency argument needs two sorts or a sound assigner (Theorem 6.3 and Proposition 6.5); his "easy" separation is Proposition E.1. Generating "only a very small subset of observations" is what a likelihood with a data-independent normaliser does. Allowing for "making some mistakes" is harmless for the general results if the mistakes are a noise component or corruption channel of the well-specified generator (Proposition B.1 and Remark C.9); modelled as axioms, as in E4(C1), they are accepted as theorems (Remark 5.10). Grading near misses is a corruption channel; his time-bounded note plausibly has an analogue for derivation-bounded inducers (Conjecture 6.18), and his predictor S is incomparable with axiom induction (Theorem 6.15).

9.4 Relation to the literature

We cite and do not re-prove (unchecked references: Appendix H.7). Theorem 3.8 is the countable-class analogue of the confirmation of universal hypotheses by the universal prior (Hutter, 2007), though what is confirmed is "all future data are instances", not derivability; confirmation is not stepwise monotone (Leike and Hutter, 2015), and for truth Gaifman's condition applies (Proposition B.10; Gaifman, 1964). Theorem 4.2 is Doob's argument (Doob, 1949), extended to larger models by Schwartz (1965) (see Ghosal and van der Vaart, 2017); misspecified posteriors concentrate on KL-minimisers under conditions (Berk, 1966; Kleijn and van der Vaart, 2006) not checked here. Gold's theorem limits learning from text (Gold, 1967; Angluin, 1980), not from stochastic positive data (Horning, 1969; Angluin, 1988). The size principle is that of Tenenbaum and Griffiths (2001), the code-length reading is MDL (Rissanen, 1978) with the KT estimator (Krichevsky and Trofimov, 1981), spare slots are overfitted mixture components (Rousseau and Mengersen, 2011), and Bel, Pl are a belief function and its plausibility (Shafer, 1976). The collapse results use Craig's trick (Craig, 1953), the nondeterministic time hierarchy (Cook, 1972; Seiferas et al., 1978; Žák, 1983) and Pudlák (1998); the diagonal separation is a log-loss form of an argument of Legg (2006), and certificate function induction rests on Cook and Reckhow (1979); Kreisel's conjecture is related (Remark B.18). AS is the non-Bayesian counterpart of this paper, close to inductive logic programming (Muggleton, 1991), and IL supplies its Ville argument (Ville, 1939).

9.5 Open problems

Main open problems: identification for every weight vector, and rigorous spare-slot rates, also for a spare in the span of T^*'s components but outside their hull (Theorem 4.7(c), Proposition 4.14); whether equal laws or instance sets imply equivalence for single DT° templates (Proposition 4.6(c)); unsound mass in the full class (Remark C.5); completeness under the shrinking threshold, and soundness at a fixed weight vector for derivation-grammar likelihoods with learned weights; reversal through $\wedge$-detours, the split–memoriser races over Q, and the limit of Proposition 3.10(d) for countable classes with learned-weight provers; Conjecture 6.13; whether templates block the collapse for merely consistent assigners, and the rejection part of Hänni's schema; templates for non-literal data, soft certificate correspondences with additive overhead, and the case $\mathrm{P} \neq \mathrm{NP}$, $\mathrm{NP} \cap \mathrm{coNP} = \mathrm{P}$ (Section 6.6); for PA, the posterior over all template unions, non-read-once induction templates (Conjecture 7.15) and unrefuted inconsistent theories; search beyond a pool; PA under $\mathsf{L1}$ in code. Further items and sketches: Appendix H.6.

Closing. This was a check of an idea, not the design of a prover. A template prior with a generative derivation likelihood is a coherent inducer: it confirms predictions and, on data drawn from the model, finds a system with the right theorems and a verifier that adaptive provers rarely fool. It does not turn human-stated theorems into their axioms, because mathematicians do not select what they write down the way the model generates data.

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A The model: proofs and details

Scripts cited here are in `research/tracks/model/checks/` (the track's) and `.../referee_code/` (its referee's); each is seeded and writes its output next to itself.

A.1 Prior

The template code (Definition 2.4). Templates are coded in preorder, one token per node from a fixed alphabet (the symbols of the language, connectives, quantifiers, and escapes IDX, PAR, MV) at $\lceil \log_2(\text{alphabet size}) \rceil$ bits. The escapes are followed by Elias-γ codes: $\gamma(k+1)$ for a de Bruijn index k, $\gamma(j+1)$ for p_j, and for a metavariable a new/old bit, then its arity ($\gamma(n+1)$) and sort bit if new or its index if old, then its arguments. A theory $\{\tau_1, \dots, \tau_k\}$ is coded as $\gamma(k+1)$ and the k template codes in lexicographic order.

Proof of Lemma 2.5. The template code is prefix-free: each token fixes the number of children that follow, the escapes are followed by self-delimiting Elias-γ codes and fixed bits, and an old metavariable's arity is already known. $\gamma(k+1)$ fixes how many template codes follow, so theory codes are prefix-free. The code determines a template up to renaming of metavariables, which is how Definition 2.1 identifies templates, and sorting makes the code of a set canonical; so distinct theories have distinct codes, and Kraft's inequality gives $\sum_T 2^{-\ell(T)} \le 1$. Hence $0 < Z_\lambda \le 1$ for $\lambda \ge 1$. For $\lambda \ge 2$, $\sum_T \pi_\lambda(T)^{1/2} = Z_\lambda^{-1/2} \sum_T 2^{-\lambda\ell(T)/2} \le Z_\lambda^{-1/2} \sum_T 2^{-\ell(T)} < \infty$. □

Implicit in the notes. (1) Theories must be taken up to renaming of metavariables: renamings share a code, so otherwise infinitely many theories would share one code and the sum would diverge. (2) The notes code the number of templates as $\gamma(k)$; Elias-γ codes positive integers, and the empty theory (used in Proposition 2.24(b)) needs a code, so we write $\gamma(k+1)$. Neither changes a statement.

Remark A.1 (Other priors; time factors). [proved] experiments §1.3; universal §1.6; pa Def 0.2; model §1.3 (i) The experiments' prior $\propto 2^{-\lambda\,\mathrm{bits}(T)}(1+|T|)^{-\tau}$ uses a stochastic template code that satisfies Kraft, so it is proper for $\lambda \ge 1$ ($\lambda = \frac{1}{2}$ only on finite pools); $\mathsf{Q} \cup \{T_{\mathrm{Ind}}\}$ costs 226.2 bits. Its factor $(1+|T|)^{-\tau}$ (time exponent τ, default 0) prices membership time. (ii) The priors $2^{-\lambda(\sum_A |A| + \#\text{axioms})}$ of Section 3 and $2^{-\beta_{\mathrm{ax}} \sum_\tau |\tau|}$ of Section 7 ($\beta_{\mathrm{ax}} = 5$ or $\log_2 23$ bits per axiom symbol) are improper over all theories and are used only on finite or summable families. Prior

shares depend on the code (Table 11). (iii) A Levin-style time factor for a decider of the axiom set is polynomial in $\ell(T)$ inside DT°, since membership is matching; it changes $\ln \pi$ by $O(\ln \ell(T))$ and barely bites. Time matters in Section 6.

Proof of Remark A.1. (i) The experiments' template code is $-\log_2$ of a subcritical stochastic grammar over templates whose arities fix the tree shape, so $\sum_\tau 2^{-\mathrm{bits}(\tau)} \le 1$. A set of m distinct components costs $|\gamma(m)| + \sum_i \mathrm{bits}(\tau_i) - \log_2 m!$, so the sum over such sets is the sum over ordered tuples of distinct components times $2^{-|\gamma(m)|}$, at most $2^{-|\gamma(m)|}(\sum_\tau 2^{-\mathrm{bits}(\tau)})^m \le 2^{-|\gamma(m)|}$; summing over m gives at most 1, and $(1+|T|)^{-\tau} \le 1$. An independent implementation of the code gives the same lengths to four decimals (experiments §1.3). (iii) Membership in a finite DT° theory is decided by matching against each template, in time $O(\sum_{\tau\in T}(|\tau|+|s|))$ (AS Thm 2.5); so the time constant is polynomial in $\ell(T)$ and its logarithm is $O(\ln \ell(T))$. □

A.2 Instantiation grammar

Lemma A.2 ($\mathcal{Q}_\tau$ is a law on $\mathrm{inst}(\tau)$)**.** [proved] model Lemma 1.6 *Let $\mathcal{Q}$ be uniformly subcritical, $\tau \in \mathrm{DT}^\circ$, and N the size of a body drawn from type i. (a) $\mathbb{E}_i N \le h(i)/(h_{\min}(1-\rho))$; bodies are finite a.s. (b) With $\Lambda := 2/((1-\rho)h_{\min})$ and $0 < \zeta \le (\Lambda b h_{\max})^{-2}$: $\mathbb{E}_i[e^{\zeta N}] \le e^{\zeta\Lambda h(i)}$, so $P_i(N \ge \ell) \le e^{\zeta(\Lambda h_{\max} - \ell)}$. (c) $\mathcal{Q}_\tau(s) := \mathcal{Q}(\theta_s)$ if $\tau\theta_s = s$, else 0, is a probability on $\mathcal{S}$; with full support its support is $\mathrm{inst}(\tau)$. (d) (a) and (b) hold for every multi-type branching process with independent children given the type, at most b children and the uniform condition.*

Proof of Lemma A.2. (a) Let G_j be the sum of h over generation j. Given the first j generations, $\mathbb{E}[G_{j+1}] \le \rho G_j$ by the uniform condition and linearity, so $\mathbb{E}G_j \le \rho^j h(i)$. Generation j has at most $G_j/h_{\min}$ nodes; sum over j.

(b) Let N_D count the nodes of depth at most D. We show $\mathbb{E}_j[e^{\zeta N_D}] \le e^{\zeta\Lambda h(j)}$ for all types j by induction on D. For $D=0$, $N_0 = 1 \le \Lambda h_{\min} = 2/(1-\rho)$. For the step, let the root's children have types $c_1,\dots,c_r$ ($r \le b$) and $H := \sum_l h(c_l) \in [0, b h_{\max}]$. The subtrees are independent given the types, so $\mathbb{E}_j[e^{\zeta N_{D+1}}] \le e^\zeta \mathbb{E}_j[e^{\zeta\Lambda H}]$. Put $x := \zeta\Lambda H \in [0,1]$ (as $\zeta \le (\Lambda b h_{\max})^{-2}$ and $\Lambda b h_{\max} \ge 2$). For $x \in [0,1]$, $e^x \le 1+x+x^2$, since $e^x - 1 - x \le x^2(e-2)$. So $\mathbb{E}_j[e^{\zeta\Lambda H}] \le 1 + \zeta\Lambda\rho h(j) + \zeta^2\Lambda^2 b^2 h_{\max}^2 \le \exp(\zeta\Lambda\rho h(j) + \zeta^2\Lambda^2 b^2 h_{\max}^2)$, and $\mathbb{E}_j[e^{\zeta N_{D+1}}] \le e^{\zeta\Lambda h(j)}$ iff $1 + \zeta\Lambda^2 b^2 h_{\max}^2 \le \Lambda(1-\rho)h(j)$. The right side is at least 2, the left at most 2. Monotone convergence gives the bound for N; the tail is Markov's inequality.

(c) By AS Thm 2.5(a), $\theta \mapsto \tau\theta$ is injective, so $\sum_s \mathcal{Q}_\tau(s) = \sum_\theta \mathcal{Q}(\theta) = \prod_M 1 = 1$, each body law being a probability on finite bodies by (a). With full support $\mathcal{Q}(\theta) > 0$ for every θ.

(d) The proofs of (a) and (b) use only independence given the type, the bound b and the uniform condition. □

Remark A.3 (The earlier condition). [refuted (old condition); exclusion proved; computed] model §1.4; referee M5 (r1); c10 The pre-referee condition "every type has mean offspring below 1" is refuted (referee M5) as a usable hypothesis. It dominated the process by one Galton–Watson process with the largest mean. (1) No formula type of L_A or $L_\in$ satisfies it: every formula production has a child. (2) With infinitely many types it does not give finiteness: with a nullary predicate $\top$, let a formula node with k variables choose $\forall$ (child with $k+1$ variables) with probability $1 - 1/(k+2)^2$ and $\top$ otherwise; the chain never stops with probability $\prod_{m\ge 2}(1-1/m^2) = \frac{1}{2}$ (`r1_subcritical.out`: product 0.500000, simulated 0.4989). (3) Means do not give domination: $\{0\colon 0.55,\ 2\colon 0.45\}$ and $\{0\colon 0.1,\ 1\colon 0.9\}$ both have mean 0.9, and a law dominating both has mean ≥ 1.35. *The uniform condition excludes the chain of (2)* [proved]: it requires $h(k+1)(1-1/(k+2)^2) \le \rho h(k)$, so

$h(k) \le h(0)\rho^k \prod_{j<k} \frac{(j+2)^2}{(j+1)(j+3)} = h(0)\rho^k \frac{2(k+1)}{k+2} \to 0$, against $h \ge h_{\min} > 0$ (`c10_subcritical.out`, Part C: e.g. $6.1 \cdot 10^{-92}$ at $k = 2000$ for $\rho = 0.9$). The uniform condition holds for a grammar for L_A with $\rho = 0.75$ (Example A.4).

Example A.4 (Grammars that satisfy Definition 2.6)**.** [computed] model §1.4 (c10); experiments §1.2 $\mathcal{Q}_A$ for L_A: term node 0 .4, S .2, + .1, · .1, a variable .2 (uniform over the k variables; renormalised away when $k = 0$); formula node = .3, < .2, $\neg, \wedge, \to, \forall, \exists$ .1 each. Term nodes have mean 0.75 ($k = 0$) or 0.6 term children; formula nodes 0.7 formula and 1.0 term children. With $h(o) = 1$, $h(\iota) = 0.05$ the weighted ratio is at most 0.75 at every type, so $\rho = 0.75$. From $2 \cdot 10^5$ bodies: mean size 14.62 (s.e. 0.04) from a formula root, against the bound 80; 4.016 (s.e. 0.015) from a term root at $k = 0$, against the bound 4, which is the exact mean $1/(1 - 0.75)$. Both tails decay exponentially. Track experiments' grammar has mean 0.72 same-sort children at formula nodes and at most 0.75 at term nodes, and satisfies the condition with $h(o) = 1$ and $h(\iota)$ small.

A.3 Calculi

One law $\mathcal{Q}$ instantiates metavariables and, unless said otherwise, supplies the ∀E terms (universal §1.2).

$\mathsf{C}_{\min}$. Chains. A node cites (probability $1 - c$: an axiom by weight, metavariables from $\mathcal{Q}$) or applies ∀E (probability c) to a recursively generated premise with $t \sim \mathcal{Q}$, failing if the premise has no leading ∀. For $\{\sigma_\varphi\}$, $Z = (1 - c)\sum_{k \le m'} c^k$ with m' the number of leading ∀ of φ (Proposition 3.2). $\mathsf{C}_{\text{open}}(c, g)$. Adds Gen (probability g), which abstracts the parameter of the premise and fails if there is none; $p_{\text{cite}} = 1 - c - g$; terms from $\mathcal{Q}_{\text{open}} = \rho[p] + (1 - \rho)\mathcal{Q}_{\text{num}}$ (ρ here is the parameter probability). Per closed datum, normalised: $H_\forall / H_{\text{sch}} = c(1 - \rho)/(1 + c)$, $H_{\text{open}}/H_{\text{sch}} = (1 - \rho)/(1 + g\rho)$ (Proposition 3.10). C_{U3}. Each node picks cite, ∀E, or a rule whose premises and conclusion are quantifier-free (equality rules; propositional rules and MP on quantifier-free formulas), with fixed probabilities; trees. $\mathsf{C}_\wedge$. Cite (p_{cite}), ∀E (c), ∧I (a), ∧E (e; side left or right with probability $\frac{1}{2}$), the ∧-rules on all formulas; trees. Used for Remark 3.5.

Ch_J *(experiments §1.6).* A chain $s_0, \ldots, s_k$, $k \le J$. s_0 cites component i with probability w_i (bodies from $\mathcal{Q}$, guards respected). At step $j < J$, if some rule applies, the chain stops with probability c_{stop}, otherwise it applies an applicable rule r with probability $\propto c_r$: *elim* ($s_j = \forall x \psi \mapsto \psi[t/x]$, t from $\mathcal{Q}_{\text{e}}$, closed terms, with p iff the run is open); *gen* ($s_j \mapsto \forall x\, s_j[x/p]$ if s_j contains p); *mp* (a uniformly chosen component $A \to B$, $A, B \in \mathrm{DT}^\circ$ with the metavariables of A in B, whose A matches s_j; $s_{j+1} = B\theta$, θ read off s_j and drawn from $\mathcal{Q}$ on the rest). If no rule applies, or $j = J$, it stops. Defaults $c_{\text{stop}} = \frac{1}{2}$, all $c_r = 1$, $J = 1$ ($J = 2$ in parts of E1 and in E6); one parameter p. The chain never fails, so its law is a probability (experiments Prop X3), and $P(s \mid T, w) = \sum_i w_i a_i(s)$. It is the special case of $\mathsf{L1}$ with cited major premises and depth $\le J$, so induction followed by MP and ∀E is outside it (Remark 8.2).

ND *and* $\mathsf{L1}^{\text{sch}}$ *(pa Def 0.3).* Sequents $\Gamma \vdash A$ with rules hyp, ax (an axiom, or a schema instance at a motive), tc (from $\Gamma_j \vdash A_j$ infer $\bigcup_j \Gamma_j \vdash B$ if $\bigwedge_j A_j \to B$ is a tautology), →I, ∀E, ∀I, ∃I, ∃E, refl, subst, with the usual side conditions; a standard sound and complete calculus, checked by `nd.py`. The derivation is written schematically in a predicate metavariable P and the motive is coded once by the grammar. Per line it costs $\log_2 10$ bits for the rule, $\log_2 |T|$ per axiom citation, $\log_2$(line index) per premise reference and $\beta = \log_2 23 \approx 4.52$ bits per written symbol. The decodable variant also codes the number of lines, the premise count of each tc line and the variables named by subst and ∃I; it adds 4.8–7.3% to the costs of pa §1's derivations. With at least $\log_2$ of the alphabet size bits per written symbol (about $\log_2 25$ with the variable names used) it is a prefix code, so Kraft holds and $\mathsf{L1}^{\text{sch}}$ is the two-part form of a sub-probability

likelihood [proved for $\beta \geq \log_2(\text{alphabet size})$]. The reported $\beta = \log_2 23$ under-charges symbols by about 3%, which changes no conclusion (pa §0).

The grammar of $\mathsf{L1}^{\mathrm{eq}}$ *(model Ex 3.6).* A node cites a template (α_{ax}; metavariables from $\mathcal{Q}$) or applies refl ($t \sim \mathcal{Q}$), sym, trans (valid iff the middle terms agree), congruence for S, or congruence for $+$ (side term $u \sim \mathcal{Q}$, left or right). A tree is valid iff its steps are and all its terms are closed of size $\leq N$. μ_T is the exact least fixpoint; Monte Carlo agrees ($Z = 0.54909$ against 0.54965 ± 0.00091). The rules are complete for closed equational consequence (known; Birkhoff, 1935, not checked).

Table 6 compares the calculi.

calculus	rules	normalisation	factor $P_{H_\forall}/P_{H_{\mathrm{sch}}}$ per closed instance	used in
K	cite theory or logical axiom, MP, Gen, $\forall$E	μ_T/Z_T; finite a.s. iff $m \leq 1$	not computed; with parameters admissible no hypothesis emits only instances (Thm. 3.9(c3))	§§4–6
$\mathsf{C}_{\mathrm{min}}$	cite $(1-c)$, $\forall$E (c)	μ_T or μ_T/Z_T	c unnormalised; $c/(1+c)$ normalised (φ quantifier-free)	§3
$\mathsf{C}_{\mathrm{open}}$	$\mathsf{C}_{\mathrm{min}}$ + Gen on p (g)	same	$c(1-\rho)/(1+c)$ normalised, against guarded H_{sch} (ρ: parameter probability of $\mathcal{Q}_{\mathrm{open}}$)	§3
C_{U3}	cite, $\forall$E, quantifier-free rules	trees, μ_T	$\leq c$ on the mass that cites the axiom (Prop. B.2)	§3
$\mathsf{C}_{\wedge}$	cite, $\forall$E, $\wedge$I, $\wedge$E	trees, μ_T	$> c$ (refuted conjecture, Rem. 3.5); ≤ 0.925 on the computed grid	§3
Ch_J	cite, then $\leq J$ steps ($\forall$E, Gen, MP)	proper, no normaliser	$c_{\mathrm{ch}} = 1 - c_{\mathrm{stop}}$	§8
ND	hyp, ax, tc, $\to$I, $\forall$E, $\forall$I, $\exists$I, $\exists$E, refl, subst	two-part code; sub-probability	1/10 per $\forall$E step (fixed rule code; learned codes: Tab. 29)	§7
$\mathsf{L1}^{\mathrm{eq}}$	cite, refl, sym, trans, cong_S, cong_+	μ_T/Z_T, least fixpoint	— (closed equations only)	§4

Table 6: The calculi of Defs. 2.2 and 2.3. "Factor" is the likelihood ratio of $H_\forall = \{\forall x \varphi\}$ to its instance schema H_{sch} (Tab. 8) on one closed instance $\varphi(t)$; m is the mean offspring of Def. 2.9. Rates are comparable only within one calculus and one code: Ch_J is proper, yet its factor equals the *unnormalised* $\mathsf{C}_{\mathrm{min}}$ factor with $c = c_{\mathrm{ch}}$. The direction (factor at most 1) is proved for $\mathsf{C}_{\mathrm{min}}$ (all listed likelihoods) and, unnormalised, for C_{U3} (Thm. 3.4); it follows from the exact laws for $\mathsf{C}_{\mathrm{open}}$ (Prop. 3.10), and holds in Ch_J (experiments Prop X6) and under pa's codes (pa §5.5); in $\mathsf{C}_{\wedge}$ it holds on the computed grid and is open in general.

Lemma A.5 (A4 is not a template)**.** [(a) proved; (b) proof sketch] model Lemma 1.2 *(a) A4, written as* $\forall x P(x) \to P(f)$ *with* f *a* 0*-ary term metavariable, is not in* SO°*, so not in* DT°*. (b) No finite set of* DT° *templates has the instances of A4 as its instance union.*

(a) AS Def 2.1 requires metavariable-free arguments. The other logical axioms are templates: A1–A3 in FO, A5 in PAT as $\forall x(P \to F(x)) \to (P \to \forall x F(x))$, substitutivity in DT°. (b) is sketched next: a template inside A4 covers $\forall x B_m \to B_m[t/x]$, $B_m := (x + \cdots + x = x)$, for one m or one t.

Proof sketch of Lemma A.5(b) Let $s_{m,t} := \forall x B_m \to B_m[t/x]$, $B_m := (x + \cdots + x = x)$ with m summands. Suppose $\tau \in \mathrm{DT}^\circ$ with $\mathrm{inst}(\tau) \subseteq \mathrm{inst}(\mathrm{A4})$ covers $s_{m,t}$ and $s_{m',t'}$, $m \neq m'$. The antecedents differ under $\forall x$ at positions whose subterms contain the bound x, so τ has an occurrence there with x among its arguments, say $f(x)$. The matching position of the consequent

carries $B_m[t/x]$; in τ it is either $f(c)$ with a fixed metavariable-free c, forcing $t = t' = c$ there, or a metavariable independent of f, which gives instances outside A4. So each template covers $s_{m,t}$ for one m or one t, and a finite set misses some $s_{m,t}$. *Not written out:* turning "the matching position in the consequent" into a statement about every DT° shape (model §10, open problem 9).

A.4 Likelihoods

Table 7 lists the likelihoods of Definitions 2.7, 2.9 and 2.11 and the scores of Definition 2.12.

name	definition	normaliser	size pr.	model / universal / pa / experiments
$\mathsf{L0}$	$\sum_\tau w_\tau \mathcal{Q}_\tau(s)$, or P_T^{Dir}	probability	yes	L0 / L0-strict / L0 / L0
$\mathsf{L0}^{\mathrm{cl}}$	$\mathsf{L0}$; $\forall \bar{x}\chi$ cited via instances	probability	yes†	— / L0-closure / — / —
μ_T	Pr[valid tree concluding s]	mass Z_T	yes‡	μ_T / L1(c) / — / —
$\mathsf{L1}$	μ_T/Z_T	$Z_T \geq 1-\alpha_{\mathrm{r}}$	yes	L1 / L1-norm / L1 (not computed) / L1 (Ch_J)
$\mathsf{L2}$	citation forced at depth d	$Z_T^{(d)}$	yes	L2 / — / "bounded" / —
$\mathsf{L1}^\sigma$	$\mathsf{L1} \cdot e^{-\kappa\lvert\pi\rvert}$, renormalised	Z_T^σ	yes	$\mathrm{L1}^\sigma$ / — / — / —
$\mathsf{L1}^{\max}$	best single tree	sub-prob.	yes‡	two-part / L1-max / L1-max / —
$\mathsf{L1}^{\mathrm{sch}}$	two-part ND code	sub-prob.	yes‡	— / — / L1-sch / —
$\mathsf{L1}^{\mathrm{sel}}$	$P_T(s)/P_T(S)$ on S	$P_T(S)$	yes†	— / L1-sel(S) / — / —
$\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$	$\sum_i w_i a_i(s)/c_i$	c_i	yes†	— / — / — / L1sel
P_T^η	$(1-\eta)P_T + \eta N$	as P_T	vacuous¶	(channel P_T^K) / P_T^η / — / —
L_ε	$(1-\varepsilon)P_T + \varepsilon\mu_0[T \nvdash_k \neg s]$	mass dep. on T	partly§	— / — / L_ε / —
$\mathsf{L1}^{\mathrm{eq}}$	equational grammar	Z_T	yes†	L1-eq / — / — / —
P_T^g	$2^{-\kappa\ell_T(s)}/Z_T$	$Z_T \leq 1$	restricted	P_T^g / — / — / —
scores	Def. 2.12	none	no	$S_{\mathrm{prove}}, S_{\mathrm{nc}}, S_g$ / $S_{\mathrm{nc},\beta}$ / (L_ε, a relaxation) / —

Table 7: Likelihoods and scores. Normalisers are data-independent. "yes": Prop. 2.22(a); "restricted": Rem. 2.23; "no": Prop. 2.22(b). †A fixed probability on $\mathcal{S}$, so Lem. 2.21 applies as in the proof of Prop. 2.22(a). ‡A fixed sub-probability, so Thm. B.4 applies. §Only $(1-\varepsilon)P_T$ separates theories that refute no datum; at $\varepsilon = 1$ nothing is learned (Prop. F.22). ¶With full-support noise (or a full-support channel, Rem. C.9) $\operatorname{supp} P_{T^*}^\eta = \mathcal{S}$, so no competitor puts mass outside the data's support and Lem. 2.21 says nothing; theories are separated only through KL. The marks say that mass outside the data's support is lost, not how strongly two theories are separated.

Proof of Lemma 2.8. $\prod_i P_{T,w}(X_i) = \sum_\ell \prod_\tau w_\tau^{n_\tau(\ell)} \prod_i \mathcal{Q}_{\ell_i}(X_i)$, a finite sum over the labellings with $X_i \in \mathrm{inst}(\ell_i)$; integrate term by term with the Dirichlet moments (known)

$$\mathbb{E}\prod_\tau w_\tau^{n_\tau} = \tfrac{\Gamma(A)}{\Gamma(A+n)} \prod_\tau \tfrac{\Gamma(\alpha_\tau+n_\tau)}{\Gamma(\alpha_\tau)}.$$

The result is symmetric in the data. P_T^{Dir} is a mixture of the i.i.d. laws $P_{T,w}^{\otimes n}$, each a probability on $\mathcal{S}^n$ consistent in n (Lemma A.2(c)), so its normaliser is 1 for every D. With disjoint instance sets each datum has one label, and $P_T^{\mathrm{Dir}}(D,s)/P_T^{\mathrm{Dir}}(D) = \frac{\Gamma(A+n)}{\Gamma(A+n+1)} \frac{\Gamma(\alpha_\tau+n_\tau+1)}{\Gamma(\alpha_\tau+n_\tau)} \mathcal{Q}_\tau(s) = \frac{\alpha_\tau+n_\tau}{A+n} \mathcal{Q}_\tau(s)$ for $s \in \mathrm{inst}(\tau)$. □

Computed. `c8_formulas.out` (1): closed form $9.908294678 \cdot 10^{-6}$ against quadrature over $w_1 \sim \mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$ $9.908294680 \cdot 10^{-6}$ (asserted to 10^{-12}). Experiments Prop X2: the exact dynamic

programme agrees with brute force on 500 random instances and with the referee's independent count-vector sum on 400 cases to $1.4 \cdot 10^{-14}$; above 20000 states it returns rigorous bounds (Appendix G).

Proof of Lemma 2.10. (a) A derivation node has 0, 1 or 2 premises with probabilities $\alpha_{\mathrm{ax}}+\alpha_{\mathrm{lg}} > 0$, $\alpha_{\mathrm{Gen}} + \alpha_{\forall\mathrm{E}}$, α_{MP}, independently: a Galton–Watson tree with mean m. The extinction theorem (known) gives finiteness iff $m \le 1$, given positive mass at 0 (without it the tree is infinite even at $m = 1$; referee m11), and extinction probability q for $m > 1$. $\mathbb{E}N = 1 + m\mathbb{E}N$, with $\mathbb{E}N < \infty$ for $m < 1$ since generation k has mean m^k. Bodies and terms are finite a.s. by Lemma A.2(a). (b) A one-node citation tree is valid, so $Z_T \ge \alpha_{\mathrm{ax}} + \alpha_{\mathrm{lg}} = 1 - \alpha_{\mathrm{r}}$ for $T \ne \emptyset$; for $T = \emptyset$ only logical-axiom citations are valid, so $Z_\emptyset \ge \alpha_{\mathrm{lg}}$. (c) $\subseteq$: a valid tree is a K-derivation from $\bigcup \mathrm{inst}(T)$, each $\forall$E node replaced by an A4 instance and MP. $\supseteq$: a K-derivation from $\bigcup \mathrm{inst}(T)$ and logical axioms is a grammar tree of positive probability: citations have $w_\tau \mathcal{Q}(\theta) > 0$, logical-axiom instances (A4's term included) have positive $\mathcal{Q}$-probability with parameters where used, and each Gen has positive parameter probability. Without parameters all formulas in a tree are closed and Gen adds only vacuous quantifiers (referee m12). □

Implicit in the notes. For $T = \emptyset$ no theory axiom can be cited; reading such a node as invalid gives the bound $Z_\emptyset \ge \alpha_{\mathrm{lg}}$ of (b), positive under the hypotheses of (c), which is what Proposition 2.24(b) uses.

Computed (`c8_formulas.out` (2), derivation nodes): $(\alpha_{\mathrm{MP}}, \alpha_{\mathrm{Gen}}+\alpha_{\forall\mathrm{E}}) = (0.15, 0.2)$, $m = 0.5$: mean 2.010 against 2; $(0.3, 0.3)$, $m = 0.9$: 9.965 against 10; $m = 1$: heavy tail; $(0.4, 0.3)$, $m = 1.1$: $P(\text{finite},\ < 20000 \text{ nodes}) = 0.7489$ against $q = 0.75$.

Remark A.6 (Two selection-aware likelihoods). [proved]
experiments §1.7 (referee m8) $\mathsf{L1}^{\mathrm{sel}}$ and $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ coincide for one-component theories and differ for mixtures ($\sum_i w_i a_i / \sum_i w_i c_i$ against $\sum_i w_i a_i / c_i$). Proposition 3.6 uses $\mathsf{L1}^{\mathrm{sel}}$; experiments E1 and E6 use $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$, for which multi-component pool members are a modelling choice (Remark 3.7).

Lemma A.7 ($\mathsf{L1}^\sigma$ is proper)**.** [proved] model §1.5 *$0 < Z^\sigma_T \le Z_T \le 1$; so $\mathsf{L1}^\sigma$ is a probability with a data-independent normaliser and the support of* $\mathsf{L1}$.

Proof of Lemma A.7 and Remark A.6. A one-node citation tree π has $\Pr(\pi)\, e^{-\kappa|\pi|} > 0$, and $e^{-\kappa|\pi|} \le 1$ gives $\mu^\sigma_T \le \mu_T$; $\mu^\sigma_T(s) > 0$ iff some valid tree concludes s with positive probability, iff $\mu_T(s) > 0$. For Remark A.6: with one component $P_T = a_1$ and $P_T(S) = c_1$; with several, the stream filter divides by $\sum_i w_i c_i$ and the per-citation form each term by its own c_i. □

A.5 The size principle

Proof of Lemma 2.21. On $\mathrm{supp}\, P^*$, $P = (1-\varepsilon)\tilde{P}$; substitute into

$$\mathrm{KL}(P^* \| P) = \textstyle\sum_{s \in \mathrm{supp}\, P^*} P^*(s) \ln(P^*(s)/P(s)).$$

For (b), $\mathbb{E}_{P^*}[\tilde{P}(X)/P^*(X)] = \sum_{s \in \mathrm{supp}\, P^*} \tilde{P}(s) = 1$; extend to n data by independence. □

Computed (`c1_size_principle.out`): root law $(0, S, +, \cdot) = (.5, .3, .1, .1)$; P^* the law of $\{z + 0 = z\}$, P that of $\{z_1 + 0 = z_2\}$ on the 30901 terms of size ≤ 11. (a) holds to 10^{-9} on this truncated grammar (asserted; there KL = 2.618 nats). Separately, the simulated per-datum log-ratio of the untruncated grammar is 3.81, 4.19, 3.94 nats at $n = 10^2, 10^3, 10^4$ (no standard error reported), against its $\mathrm{KL} = H(\mathcal{Q}) = 3.894$ nats.

Proof of Proposition 2.22. (a) Each P_T is a fixed probability on $\mathcal{S}$ (Lemmas 2.10, A.2 and A.7; under L2 all trees are finite and one-node citations valid); apply Lemma 2.21. Under L0 with full support and positive weights, $\operatorname{supp} P_{T,w} = \bigcup \operatorname{inst}(T)$. (b) A model of $T' \cup D_+ \cup \neg D_-$ is a model of $\operatorname{Th}(T') \supseteq \bigcup \operatorname{inst}(T)$, so $S_{\mathrm{nc}}(T') \le S_{\mathrm{nc}}(T)$. If $S_{\mathrm{prove}}(T) = 1$, T' proves what T proves and is consistent by hypothesis, so $S_{\mathrm{prove}}(T') = 1$. For S_g: $S_{\mathrm{nc}}(T') = 1$ by hypothesis, and instance inclusion makes every derivation from T one from T', so $\ell_{T'} \le \ell_T$ and $g(\ell_{T'}) \ge g(\ell_T)$. The consequence is the case $T = T^*$, $T' = T^* \cup \{\psi\}$, where the instances are nested; if T^* proves the labelled data, the consistency hypothesis reduces to that of $T^* \cup \{\psi\}$. (c) Under L0, $P_{T'}(s) = (1 - w_\psi)P_{T^*}(s) + w_\psi[s = \psi]$. Under L1, the one-node tree citing ψ gives $P_{T'}(\psi) \ge \alpha_{\mathrm{ax}} w_\psi$ (as $Z_{T'} \le 1$), while $\psi \notin \operatorname{Th}(T^*) \supseteq \operatorname{supp} P_{T^*}$ by the inclusion $\subseteq$ of Lemma 2.10(c), which needs no hypothesis. So $\varepsilon \ge \alpha_{\mathrm{ax}} w_\psi$ in Lemma 2.21, and by the strong law $n^{-1} \ln \frac{P_{T'}(D)}{P_{T^*}(D)} \to -\mathrm{KL}(P_{T^*} \| P_{T'}) \le -\ln \frac{1}{1 - \alpha_{\mathrm{ax}} w_\psi}$ a.s. The Dirichlet case is Proposition 4.14(a). □

Proof of Remark 2.23. A derivation determines its conclusion, so distinct sentences have distinct shortest derivations with codes in one prefix-free set; with $\kappa \ge 1$, Kraft gives $Z_T \le 1$. With $g_\infty > 0$ the infinitely many sentences without derivation make the sum infinite. The code writes axioms as formulas, so instance inclusion makes every derivation from T one from T' with the same code: $\ell_{T'} \le \ell_T$, $Z_{T'} \ge Z_T$. The refuted sentence fails where the numerator grows too: $P^g_{T'}(b)/P^g_T(b) = 2^{\kappa(\ell_T(b) - \ell_{T'}(b))} Z_T / Z_{T'}$, and $Z_{T'} \le 1$.

Computed by the referee (`r6_stronger_theory.out`), in a different setting: ℓ is the symbol size of the least MP tree, normalised over a finite universe of 570 formulas of size ≤ 7, where the normaliser need not be ≤ 1. There $\ell_T(b) = 5$ and $\ell_{T'}(b) = 1$; $\kappa = 1$ gives $P_T(b) = 0.0321$, $P_{T'}(b) = 0.3456$ ($\kappa = \frac{1}{2}$: 0.0432, 0.1491), 10 sentences gaining and 36 losing. Under L2 (depth 2) the probability of b is 0.0134, 0.0156, 0.0168 under T and 0.263, 0.240, 0.196 under T', in three rule settings. □

A.6 Computing the likelihood

Proof of Proposition 2.24. (a) Let $\nu(\pi)$ count the grammar nodes of a tree: derivation nodes, $\mathcal{Q}$ nodes of bodies and terms, and the nodes of the geometric chains that pick parameters. Weight them 1, $\eta h_{\mathcal{Q}}$ and η. A derivation node then has expected weighted offspring at most $m + \eta(v_T h_{\mathcal{Q},\max} + 1)$, with v_T the largest number of metavariable slots of a template of T or of a logical schema; a $\mathcal{Q}$ node has ratio at most $\rho_{\mathcal{Q}}$; a chain node continues with a fixed probability below 1. As $m < 1$, a small η makes all ratios less than 1, so the whole tree satisfies Lemma A.2(d) (as in the proof of Proposition E.5) and $\mathbb{E}\nu \le C_\nu$, computable from the parameters and T. Finitely many trees have $\nu \le N$ (each node chooses from a finite list: rule, template, logical schema, production, one of at most the arity plus N variables, chain step); validity is decidable and each probability is a computable real. So $\mu_T^{\le N}(s) \le \mu_T(s) \le \mu_T^{\le N}(s) + C_\nu/N$ by Markov's inequality, likewise for Z_T, and $Z_T \ge 1 - \alpha_{\mathrm{r}} > 0$ ($Z_\emptyset \ge \alpha_{\mathrm{lg}} > 0$; Lemma 2.10(b)). (b) The identity is Lemma 2.10(c). $\bigcup \operatorname{inst}(T)$ is decidable by matching, so $\operatorname{Th}(T)$ is recursively enumerable. $\operatorname{Th}(\emptyset)$ is the set of valid formulas, undecidable for a language with a binary relation symbol (Church, Turing; known). (c) Deciding $P_T(s) = 0$ would decide $s \notin \operatorname{Th}(T)$. For a finite class, $\pi_n(T)$ is a quotient of computable reals whose denominator is positive when defined. □

Cost. Precision ϵ needs all trees with up to $N \approx C_\nu/\epsilon$ grammar nodes (C_ν bounds their mean number), exponentially many in N; a likelihood ratio needs ϵ below likelihoods that are exponentially small in the datum's least tree size (Proposition E.5). No better general algorithm and no lower bound are known.

Proposition A.8 (The maximum is not the sum)**.** [proved] model Prop 2.8 *Let* $\tau := \forall x \forall y\, F(x,y)$ *(*$F : \iota^2 \to o$*),* $\tau' := \forall x \forall y\, F(y,x)$*,* $T_1 := \{\tau\}$*,* $T_2 := \{\tau, \tau'\}$ *with weights* $\frac{1}{2}, \frac{1}{2}$*, and* $\mathcal{Q}$ *symmetric in the holes. Under* $\mathsf{L0}$*,* $P_{T_2} = P_{T_1}$*, so the posterior odds equal the prior odds at every* n*. Under* $\mathsf{L1}^{\max}$*,* $V_{T_2}(s) = \frac{1}{2} V_{T_1}(s)$ *on every instance, so the two-part odds are* $(\pi(T_2)/\pi(T_1))\, 2^{-n} \to 0$*: the maximum breaks a tie inside the generator class.*

Proof of Proposition A.8. For a body β of type $\iota^2 \to o$ let β^{sw} swap its holes. Then $\tau'[F := \beta] = \tau[F := \beta^{\mathrm{sw}}]$, so $\mathrm{inst}(\tau) = \mathrm{inst}(\tau')$, and hole symmetry gives $\mathcal{Q}_\tau = \mathcal{Q}_{\tau'}$. Under $\mathsf{L0}$, $P_{T_2} = \frac{1}{2}\mathcal{Q}_\tau + \frac{1}{2}\mathcal{Q}_{\tau'} = P_{T_1}$, and the posterior odds are the prior odds (Corollary 4.3). The codes of τ and τ' differ, so $T_1 \neq T_2$. Under the maximum each instance has exactly one matcher per template (AS Thm 2.5), so two citations of probability $\frac{1}{2}\mathcal{Q}_\tau(s)$ each, and $V_{T_2}(s) = \frac{1}{2} V_{T_1}(s)$. □

A.7 Hänni's trichotomy

Derivability is unbounded here (Definition 2.13), except in Proposition 2.19.

Proof of Proposition 2.14. (a) A consistent theory proves at most one of s, $\neg s$, so it is counted once in $\mathrm{Bel}(s) + \mathrm{Dis}(s) + \mathrm{Indep}(s)$; an inconsistent one proves both and is counted twice. The sum is $(1 - \mathrm{Inc}) + 2\mathrm{Inc}$. (b) $\mathrm{Dis}(s) = \mathrm{Bel}(\neg s)$ by definition; classically $T \vdash \neg\neg s$ iff $T \vdash s$. On consistent theories $\mathrm{Bel}(\top) = 1$, $\mathrm{Bel}(\bot) = 0$, and $1 - \mathrm{Dis} = \mathrm{Bel} + \mathrm{Indep}$ by (a). (c) $\mathrm{Th}(T)$ is closed under consequence. (d) A consistent T proves at most one of two incompatible sentences, and proves $s \vee s'$ if it proves either. □

Proof of Proposition 2.18. For one consistent T put $u(s) := 1[T \vdash s]$. Closure under $\wedge$-introduction and elimination gives $u(\bigwedge_{i \in I} s_i) = \prod_{i \in I} u(s_i)$. So the right side equals $\sum_{\emptyset \neq I} (-1)^{|I|+1} \prod_{i \in I} u(s_i) = 1 - \prod_i (1 - u(s_i)) = 1[T \text{ proves some } s_i] \leq u(s_1 \vee \cdots \vee s_k)$. The inequality is linear in u, so it survives the mixture $\mathrm{Bel} = \sum_T \pi_D(T) u_T$. By completeness, $u_T(s) = 1[\mathrm{Mod}(T) \subseteq \mathrm{Mod}(s)]$, which identifies the focal elements. □

Proof of Proposition 2.19. For each T in the posterior's (countable) support choose a complete consistent extension $\bar{T}$ (Lindenbaum), and put $P(s) := \sum_T \pi_D(T)\, 1[s \in \bar{T}]$, a mixture of point masses on complete theories. Then $1[T \vdash s] \leq 1[s \in \bar{T}] \leq 1 - 1[T \vdash \neg s]$, the second because $\bar{T}$ is consistent and contains $\neg s$ when $T \vdash \neg s$. Average over T. The argument uses only $\mathrm{Th}_d(T) \subseteq \mathrm{Th}(T) \subseteq \bar{T}$, so it holds for bounded derivability as well. □

Proof of Examples 2.15 and 2.16. Example 2.15: $\mathrm{Bel}(a) = \frac{1}{3}$ (from $\{a\}$) and $\mathrm{Dis}(a) = 0$, since no theory proves $\neg a$; so $R(a) = 1$, and likewise $R(b) = 1$. No theory proves $a \wedge b$, and $\{\neg(a \wedge b)\}$ refutes it, so $R(a \wedge b) = 0$. A probability with $P(a) = P(b) = 1$ has $P(a \wedge b) = 1$. Example 2.16: the empty theory proves none of b, $b \wedge c$, $b \wedge \neg c$ or their negations, so each gets $F_{1/2} = \frac{1}{2}$, and $\frac{1}{2} + \frac{1}{2} \neq \frac{1}{2}$. □

Proof of Proposition 2.17. Identify a consistent theory with its nonempty set of models $\mathrm{Mod}(T)$ among the valuations, and a sentence with its set of models; over finitely many atoms every set of valuations is a sentence. For one theory

$$F_T(s) = 1[\mathrm{Mod}(T) \subseteq s] + \tfrac{1}{2}\, 1[\mathrm{Mod}(T) \text{ meets both } s \text{ and its complement}],$$

and $F_{1/2} = \sum_T \pi_D(T) F_T$. *If.* With $|\mathrm{Mod}(T)| = 1$, F_T is the point mass on the model. With $\mathrm{Mod}(T) = \{v, v'\}$, $F_T(s)$ is 1, $\frac{1}{2}$ or 0 as s contains both, one or none of v, v', i.e. $F_T = \frac{1}{2}(\delta_v + \delta_{v'})$. A mixture of probabilities is a probability. *Only if.* For valuations $v \neq v'$ put $D_T := F_T(\{v, v'\}) -$

$F_T(\{v\}) - F_T(\{v'\})$. If $v, v' \in \mathrm{Mod}(T)$ and $|\mathrm{Mod}(T)| \geq 3$, all three sets meet $\mathrm{Mod}(T)$ and its complement, so $D_T = \frac{1}{2} - \frac{1}{2} - \frac{1}{2} = -\frac{1}{2}$. In every other case $D_T = 0$: if $\mathrm{Mod}(T) = \{v, v'\}$, $D_T = 1 - \frac{1}{2} - \frac{1}{2}$; if exactly one of them, say v, is a model, then $F_T(\{v'\}) = 0$ and $F_T(\{v, v'\}) = F_T(\{v\})$, because $\{v, v'\}$ and $\{v\}$ meet $\mathrm{Mod}(T)$ and its complement in the same way; if neither is a model, all three terms are 0. So if some T_0 with $\pi_D(T_0) > 0$ has three or more models and v, v' are two of them, $\sum_T \pi_D(T) D_T \leq -\frac{1}{2}\pi_D(T_0) < 0$, whereas a probability gives 0. □

Linear-programming checks. `c5`: Examples 2.15 and 2.16 are infeasible as probabilities; 300 random posteriors on three atoms gave no violation of 2- or 3-monotonicity and no bracketing failure; renormalisation was incoherent in 90/300, the 50/50 rule in 290/300. `r5` (independent generator): no violation of 2-, 3- or 4-monotonicity on two atoms, no bracketing failure with other completions; the 50/50 rule was coherent in 12/300 trials, exactly those in which every theory had at most two models. `c14`: the criterion of Proposition 2.17 agrees with LP coherence in 3225/3225 posteriors on two atoms (all one- and two-theory supports plus 3000 random) and 600/600 on three. (Model §7 also credits c5 with 4-monotonicity; c5's output records only 2 and 3, and 4 was checked by the referee on two atoms.)

Remark A.9 (Renormalisation and complete theories). [refuted; computed] model §7 (referee m4) The model track's first version said: "the coherent cases [of renormalisation] are essentially those where every theory is complete". **Refuted**: in the referee's trials renormalisation was coherent in 199 of 300 posteriors, only 2 of them with all theories complete; in `c14` it was coherent in 1761 cases on two atoms (218 with all theories complete) and 306 on three atoms (42). For the 50/50 rule the exact condition is Proposition 2.17; for renormalisation no characterisation is given.

B Proofs and computations for Section 3

Proofs follow the universal track's final notes; steps they leave out are named. Checks are the seeded scripts c1–c10 in `research/tracks/universal/checks/`; the referee's independent scripts r1–r7 (which do not import the track's code) are in `referee_code/`. A second run of all scripts reproduced every output byte for byte. Dirichlet $\alpha = 1$ in c2–c7, $\alpha = \frac{1}{2}$ in c8–c10.

B.1 Factorisation, odds and robustness

name	axioms	$\vdash \forall x\varphi$: pure / over Q
$H_\forall$; H_{sch}	$\{\forall x\varphi\}$; $\{\sigma_\varphi\}$, closed guard	yes/yes; no/no
H_{open}; H_{both}	$\{\sigma_\varphi\}$, open guard ($z \sim \mathcal{Q}_{\mathrm{open}}$); $\{\forall x\varphi, \sigma_\varphi\}$, weight $w_\forall$ on $\forall x\varphi$	yes/yes; yes/yes
memorisers; over-specific	finite sets of closed instances; e.g. $\sigma_\varphi[z := Sz]$, closed guard	no/no; no/no
over-general	a subterm or subformula of σ_φ made a fresh metavariable; the bare F_0 (inconsistent)	varies
root split	$\{\sigma_\varphi[z := f(z_1, \ldots, z_r)] : f$ a root of $\mathcal{Q}\}$, closed guards	no/no

Table 8: Hypotheses of Section 3; provability for $\varphi = 0 + x = x$ (closed instances do not prove $\forall x\varphi$ over Q: AS Ex 4.9). $B \oplus_w A$ adds A with weight w to B, scaling B's weights by $1 - w$. R_k, Split_k, Both0: Sections 3.6 and 3.8.

Proof of Proposition 3.2. A $\mathsf{C}_{\min}$-derivation from $\{\forall x\varphi\}$ is a chain: cite $\forall x\varphi$, then $k \geq 0$ eliminations with terms $t_1, \ldots, t_k$, with probability $c^k(1-c)\prod_j \mathcal{Q}(t_j)$. With $k = 0$ it yields

$\forall x\varphi$. For $k \geq 1$ map it to the σ_φ-chain "cite σ_φ with $z := t_1$, then eliminate with $t_2, \ldots, t_k$", of probability $c^{k-1}(1-c)\prod_j \mathcal{Q}(t_j)$. The conclusions coincide, since $\forall$E with t_1 gives $\varphi(t_1) = \sigma_\varphi[z := t_1]$ (t_1 has no bound variable) and the remaining steps agree; failing chains map to failing chains. The map is a bijection onto all σ_φ-chains, and no σ_φ-chain yields $\forall x\varphi$ (its conclusions have fewer quantifiers). Summing over chains with conclusion s gives the factor c. Citation factorises over axioms, $\mu_T = \sum_A w_A \mu_A$, which gives the background form. The leading universal quantifiers of φ are rigid, so exactly $m+1$ chain lengths succeed from σ_φ and $m+2$ from $\forall x\varphi$, which gives the normalisers. For k variables the first k eliminations are absorbed into one citation. □

c1 checks the identity on every output of $2 \cdot 10^5$ sampled derivations ($0 + x = x$, $\neg Sx = 0$, $\forall y(x + y = y + x)$; $\mathcal{Q}_{\text{num}}$, $\mathcal{Q}_{\text{GW}}$) to $5 \cdot 10^{-14}$; referee r1 to $1.6 \cdot 10^{-15}$.

Proof of Theorem 3.3. Each case follows from Proposition 3.2. L0: $H_\forall$ cites only $\forall x\varphi$. $\mathsf{L0}^{\text{cl}}$: identical laws by definition. μ_T: factor c per datum. L1: divide by the normalisers; for quantifier-free φ, $Z_{H_{\text{sch}}} = 1 - c$ and the factor is $c(1-c)/((1-c) + c(1-c)) = c/(1+c)$. $\mathsf{L1}^{\text{sel}}$: $\mu_{H_\forall}(s)/\mu_{H_\forall}(S) = c\mu_{H_{\text{sch}}}(s)/(c\mu_{H_{\text{sch}}}(S))$ as $\forall x\varphi \notin S$. $\mathsf{L1}^{\max}$: best chains map to best chains, each scaled by c. Scores: both theories prove every instance and, when $\forall x\varphi$ is consistent, are consistent with the data (if it is not, $S_{\text{prove}}(H_\forall) = 0$, which still agrees with Theorem 3.4). Background: $r(s) = \frac{(1-w)\mu_B(s) + wc\mu_{H_{\text{sch}}}(s)}{(1-w)\mu_B(s) + w\mu_{H_{\text{sch}}}(s)} \in [c, 1]$. □

likelihood	O_n/O_0
L0 (strict citation)	0 for $n \geq 1$
$\mathsf{L0}^{\text{cl}}$; $\mathsf{L1}^{\text{sel}}(S)$ with $\forall x\varphi \notin S$; S_{prove}, S_{nc}, S_g ($\forall x\varphi$ consistent)	1
μ_T (unnormalised); $\mathsf{L1}^{\max}$	c^n
L1	$\big(cZ_{H_{\text{sch}}}/((1-c)+cZ_{H_{\text{sch}}})\big)^n$; $(c/(1+c))^n$ for quantifier-free φ
background B, under μ_T	$\prod_i r(X_i)$, $r(s) \in [c, 1]$; $r(s) = c$ if $\mu_B(s) = 0$

Table 9: Odds of $H_\forall$ against H_{sch} after n closed instances, relative to the prior odds (Theorem 3.3); asserted on every sampled path (5 formulas, 2 term laws, 50 seeds, $n \leq 200$), referee's code to $7 \cdot 10^{-13}$. The consistency condition in the score row is ours; the notes leave it implicit.

Proof of Theorem 3.4. (B1) is Theorem 3.3. (B3) under μ_T and $\mathsf{L1}^{\max}$ is Proposition B.2, whose map acts derivation by derivation. Under the scores, for single axioms and quantifier-free data: by Herbrand's theorem a quantifier-free sentence s follows from $\{\forall x\varphi\}$ iff it follows from finitely many closed instances of φ, i.e. from $I_{\text{c}}(\varphi) = \bigcup \text{inst}(H_{\text{sch}})$; in particular $\{\forall x\varphi\}$ is consistent iff $I_{\text{c}}(\varphi)$ is. So $H_\forall$ and H_{sch} prove, and are consistent with, the same quantifier-free data, and S_{prove}, S_{nc}, S_g give them equal scores. (B2): fix w and $s \neq \forall x\varphi$. Under μ_T, $\mu_{B \oplus_w \forall}(s) = (1-w)\mu_B(s) + wc\mu_{H_{\text{sch}}}(s) \leq (1-w)\mu_B(s) + w\mu_{H_{\text{sch}}}(s) = \mu_{B \oplus_w \sigma}(s)$. Under L1, $Z_{B \oplus_w \forall} = (1-w)Z_B + w((1-c) + cZ_{H_{\text{sch}}}) \geq (1-w)Z_B + wZ_{H_{\text{sch}}} = Z_{B \oplus_w \sigma}$, since $(1-c) + cZ_{H_{\text{sch}}} - Z_{H_{\text{sch}}} = (1-c)(1 - Z_{H_{\text{sch}}}) \geq 0$; so the normalised likelihoods obey the same inequality datum by datum. A pointwise inequality at every w survives integration against a common weight prior. Under $\mathsf{L0}^{\text{cl}}$ the laws coincide at every w. □

Proposition B.1 (Robustness of the direction)**.** [proved; computed]
universal Props U2n, U2t, U2c *In* $\mathsf{C}_{\min}$*, for every* $s \neq \forall x\varphi$*:* (U2n) *with a full-support noise law* N *and* $\eta \in (0, 1)$*, the ratio of the noisy laws of* $H_\forall$ *and* H_{sch} *at* s *is in* $[c, 1]$ *under* μ_T*,* ≤ 1 *under* L1*,*

and 1 *under* $\mathsf{L1}^{\mathrm{sel}}$ *with noise after selection; a false datum no longer refutes.* (U2t) *If chains are weighted by* $f(\ell)w_A\prod_j \mathcal{Q}(t_j)$, f *non-increasing in the number* ℓ *of eliminations (e.g. Levin-style* $f(\ell)\propto(\ell+1)^{-2}$*), then* $P^f_{H_\forall}(s)\le \sup_\ell \frac{f(\ell+1)}{f(\ell)} P^f_{H_{\mathrm{sch}}}(s)\le P^f_{H_{\mathrm{sch}}}(s)$. (U2c) *Under* $\mathsf{L1}$, *quantifier-free* φ *and any prior on* c, $O_n/O_0\le 2^{-n}$.

Proof of Proposition B.1. (U2n) The unnormalised claim is the background case of Theorem 3.3 with $\mu_B := N$. For $\mathsf{L1}$, $c\mu_{H_{\mathrm{sch}}}(s)/Z_{H_\forall}\le \mu_{H_{\mathrm{sch}}}(s)/Z_{H_{\mathrm{sch}}}$ because $Z_{H_\forall}=(1-c)+cZ_{H_{\mathrm{sch}}}\ge Z_{H_{\mathrm{sch}}}$ (as $Z_{H_{\mathrm{sch}}}\le 1$); mixing both with the same ηN keeps the inequality. Under $\mathsf{L1}^{\mathrm{sel}}$ the selected laws coincide. A false datum has likelihood $\ge \eta N(s)>0$. (U2t) The bijection maps a $\forall x\varphi$-chain with $\ell+1$ eliminations to a σ_φ-chain with ℓ, with the same term product; $f(\ell+1)$ becomes $f(\ell)$. (U2c) H_{sch}'s normalised law is $\mathcal{Q}_{\sigma_\varphi}$ for every c, and $c/(1+c)$ increases in c to $\frac12$ at $c=1$; a bound valid for every c survives integration. □c10(h) checked (U2n) at $2\cdot 10^5$ random parameter points.

Proposition B.2 (Simulation inequality)**.** [proved] universal Prop U3 *In a* C_{U3} *calculus, for quantifier-free* φ *and* s, $\mu_{B\oplus\forall x\varphi}(s)-P^0(s)\le c\,(\mu_{B\oplus\sigma_\varphi}(s)-P^0(s))$, *with* $P^0(s)$ *the mass of derivations of* s *that never cite the added axiom.*

Proof of Proposition B.2. In a derivation of a quantifier-free s from $B\oplus\forall x\varphi$, a leaf citing $\forall x\varphi$ is not the root, and its parent takes a premise that is not quantifier-free; in a C_{U3} calculus only ∀E does. So each such leaf sits in a pattern "∀E(cite $\forall x\varphi$, t)" concluding $\varphi(t)$. Replace each pattern by "cite σ_φ, $z:=t$": probability $c\,\mathcal{Q}(t)p_{\mathrm{cite}}w$ becomes $p_{\mathrm{cite}}w\mathcal{Q}(t)$, and conclusions are kept. The map is injective, because cited axioms are labelled and every σ_φ-leaf in the image came from a pattern. Derivations with $m\ge 1$ patterns contribute $\sum c^m\Pr(f(\pi))\le c\sum_{\pi'\text{ citing }\sigma_\varphi}\Pr(\pi')$; those with $m=0$ never cite the added axiom and have the same probability in both theories. □

The ∧-detour counterexample (Remark 3.5; proof sketch). $\mathsf{C}_\wedge$ with probabilities p_c (cite), c (∀E), a (∧I), e (∧E, side chosen with probability $\frac12$); $B=\emptyset$; $\varphi=0+x=x$. A formula G made at a base node (cite or ∀E) reaches the root through a balanced word of ∧I (into a conjunction with a free partner) and ∧E (take G's side), each pair of weight eaZ_T with Z_T the success probability. So $P_T(G)=\mathrm{base}_T(G)D(eaZ_T)$ with $D(x)=\sum_j \mathrm{Cat}_j x^j=(1-\sqrt{1-4x})/(2x)$; the ansatz solves $P(G)=\mathrm{base}(G)+e\sum_W P(G\wedge W)$ exactly when $D=1+xD^2$. For $H_\forall$, $\mathrm{base}(\varphi(t))=cP(\forall x\varphi)\mathcal{Q}(t)$ and $P(\forall x\varphi)=p_cD_\forall$, so $P_{H_\forall}(\varphi(t))=cp_cD_\forall^2\mathcal{Q}(t)$; for H_{sch}, $P(\varphi(t))=p_cD_\sigma\mathcal{Q}(t)$. Z_T is the least fixed point of $Z=D(eaZ)(p_c+cP_U+aZ^2)$, $P_U=p_cD$ for $H_\forall$, 0 for H_{sch}; the first map dominates, so $D_\forall\ge D_\sigma\ge 1$, with $D_\forall>1$ if $a,e>0$, and $R=cD_\forall^2/D_\sigma\ge cD_\forall>c$. Not written out: that the least fixed point is the success probability. At $(0.35,0.15,0.25,0.25)$, $D_\forall=1.03160$, $D_\sigma=1.02636$, $R=0.15553$ (c10(a), referee r2); two samplers give 0.15528 ± 0.00067 and 0.1562 ± 0.0005.

Proof of Proposition 3.6. Under $\mathsf{L1}^{\mathrm{sel}}$, T_σ gives $\varphi(t)$ probability $w\mathcal{Q}(t)$ and $\psi(t)$ probability $(1-w)\mathcal{Q}(t)$. $T_\forall$ reaches S through $\forall x\varphi$ only after one elimination, so it gives $w_e\mathcal{Q}(t)$ and $(1-w_e)\mathcal{Q}(t)$ with $w_e=wc/(1-w+wc)$. The likelihoods are $K_Dw_e^{n_\varphi}(1-w_e)^{n_\psi}$ and $K_Dw^{n_\varphi}(1-w)^{n_\psi}$ with the same K_D. $w\mapsto w_e$ is a C^1 bijection of $[0,1]$ with inverse $w=v/(c+v(1-c))$ and $dw/dv=h(v)$, so

$$\mathrm{BF}_n=\int_0^1 v^{n_\varphi}(1-v)^{n_\psi}h(v)\,dv\Big/\int_0^1 v^{n_\varphi}(1-v)^{n_\psi}dv=\mathbb{E}[h(V)],\quad V\sim\mathrm{Beta}(n_\varphi+1,n_\psi+1).$$

$n_\varphi/n \to w^*$ a.s. and the Beta variance is $\le 1/(4(n+3))$, so $V \to w^*$ in law; h is bounded and continuous. $h(v) > 1$ iff $\sqrt{c} > c + v(1-c)$ iff $v < \sqrt{c}/(1+\sqrt{c})$. □

c10(b) ($c = 0.3$, 10 seeds, $n = 10^5$): 2.1918, 1.1537, 0.7102, 0.4056 at $w^* = 0.1, 0.3, 0.5, 0.8$, against $h(w^*) = 2.1914$, 1.1534, 0.7101, 0.4056.

Proof of Remark 3.7. Under $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ each citation is repeated until its chain lands in S. The component $\forall x\varphi$ lands only after one elimination, so its selected law is $\mathcal{Q}_{\sigma_\varphi}$, as is that of σ_φ; σ_ψ has $\mathcal{Q}_{\sigma_\psi}$. Both theories therefore have law $(1-w)\mathcal{Q}_{\sigma_\psi} + w\mathcal{Q}_{\sigma_\varphi}$ at every w. □

Proposition B.3 (Spare-slot prover)**.** [proved; computed]
universal §4 *In $\{H_{\mathrm{sch}}, H_\forall, H_{\mathrm{both}}\}$ with a uniform prior on $w_\forall$, under $\mathsf{L1}$ with quantifier-free φ, the likelihood ratio $H_{\mathrm{both}} : H_{\mathrm{sch}}$ is $I_n = \int_0^1 (\frac{1-w(1-c)}{1+wc})^n dw \in [\frac{1}{n+1}, \frac{1}{n(1-c)}]$, so $P(T \vdash \forall x\varphi \mid D_n) \to 0$ like $\Theta(1/n)$: $\forall x\varphi$ is a spare slot of H_{both}, a template beside the generator's own that the data never or rarely use (Proposition 4.14).*

Proof of Proposition B.3. H_{both} with weight w on $\forall x\varphi$ has $\mu(\varphi(t)) = (1-w+wc)(1-c)\mathcal{Q}(t)$ and $Z = (1-c)(1+wc)$, so its ratio to H_{sch} is $(1-w(1-c))/(1+wc)$. Since $1-w+wc-(1-w)(1+wc) = w^2c \ge 0$, $I_n \ge \int_0^1 (1-w)^n dw = \frac{1}{n+1}$; and the integrand is $\le e^{-nw(1-c)}$, so $I_n \le \frac{1}{n(1-c)}$. □Computed (c5 Part C): $nI_n = 0.958$, 0.996, 1.000 at $n = 10, 10^2, 10^5$.

B.2 Size principle, over-specific templates and memorisers

Theorem B.4 (Size principle on instances)**.** [proved; computed] universal Thm U4 *Let the data be i.i.d. from P^*, $P^*(I) = 1$ for $I = I_{\mathrm{c}}(\varphi)$, and P' a sub-probability with $P'(I) = p < 1$. Then $\mathrm{KL}(P^* \| P') \ge \ln \frac{1}{p}$, $\frac{1}{n} \ln \frac{P'(D_n)}{P^*(D_n)} \to -\mathrm{KL}(P^* \| P')$ a.s., and $\mathbb{E}\frac{P'(D_n)}{P^*(D_n)} \le p^n$. Under the citation law ($\mathsf{L0}$, $\mathsf{L0}^{\mathrm{cl}}$), a first-order template strictly more general than σ_φ, with metavariables i.i.d. from $\mathcal{Q}$ and a formula law F, has $P_\tau(I) \le \max(\max_f p_f, \max_g F(\mathrm{root} = g))$.*

Proof of Theorem B.4. (1) Log-sum inequality: $\sum_{s \in I} P^*(s) \ln \frac{P^*(s)}{P'(s)} \ge P^*(I) \ln \frac{P^*(I)}{P'(I)} = \ln \frac{1}{p}$. (2) With $Y_i = \ln \frac{P'(X_i)}{P^*(X_i)}$, $y^+ \le e^y$ gives $\mathbb{E}Y^+ \le \mathbb{E}e^Y = P'(\mathrm{supp}\, P^*) \le 1$, so $\mathbb{E}Y = -\mathrm{KL} \in [-\infty, 0)$ and the strong law applies (it allows $\mathbb{E}Y = -\infty$ when $\mathbb{E}Y^+ < \infty$). (3) Tonelli: $\mathbb{E}\prod_i P'(X_i)/P^*(X_i) = P'(\mathrm{supp}\, P^*)^n$. (4) Write $\sigma_\varphi = \tau\rho$, ρ not a renaming. Either (a) some metavariable u of τ has $\rho(u)$ with a rigid root f, or (b) ρ identifies two metavariables $u \ne u'$; otherwise ρ would be a renaming. If $\tau\theta \in \mathrm{inst}(\sigma_\varphi)$, then $\tau\theta = \tau\rho\theta'$ and unique matching gives $\theta = \rho\theta'$ on τ's metavariables. In case (a) $\theta(u)$ has root f (probability $\mathcal{Q}(\mathrm{root} = f)$, or $F(\mathrm{root} = f)$); in case (b) $\theta(u) = \theta(u')$, of probability $\sum_t \mathcal{Q}(t)^2 \le \max_t \mathcal{Q}(t) \le \max_f \mathcal{Q}(\mathrm{root} = f)$. □

c2: under $\mathcal{Q}_{\mathrm{num}}$, $?y + ?z = ?z$ has $P_\tau(I) = \frac{1}{2}$ and KL 0.693, $0 + ?y = ?z$ has $\frac{1}{3}$ and 1.386; (4) held in 38 of 38 cases and in the referee's 130 random generalisations (r7).

Proposition B.5 (Over-specific templates)**.** [proved; computed]
universal Prop U5 *For $\tau = \sigma_\varphi[z := f(z_1, \ldots, z_r)]$ and root-generated $\mathcal{Q}$, $P_\tau(\varphi(t))/P_{H_{\mathrm{sch}}}(\varphi(t))$ is $1/p_f$ if $\mathrm{root}(t) = f$ and 0 otherwise. So the odds $\tau : \sigma_\varphi$ grow like p_f^{-n} while all data have root f (probability p_f^n), then vanish; they form a martingale. Some single-root template survives with probability $\sum_f p_f^n$, the failure probability of AS Thm 4.4(a).*

Proof of Proposition B.5. $\mathcal{Q}(f(\bar{t})) = p_f \prod_i \mathcal{Q}(t_i)$ and τ matches $\varphi(t)$ iff $\mathrm{root}(t) = f$, with $z_i := t_i$; so $P_\tau(\varphi(t)) = \mathcal{Q}(t)/p_f$ there and 0 elsewhere. Under H_{sch}'s law the ratio has mean $\sum_{\mathrm{root}(t)=f} \mathcal{Q}(t)/p_f = 1$. □c2: gain per datum $\ln 2$ under $\mathcal{Q}_{\mathrm{num}}$ and $\ln(1/0.3) = 1.204$ under $\mathcal{Q}_{\mathrm{GW}}$; over-specific mass 0.110 at $n = 1$, 0 at $n = 10$ (Table 2).

Proposition B.6 (Memorisers)**.** [(i), (ii), (ii′) proved; (iii) conjecture; computed] universal Prop U6 *Give the memorisers a product-Bernoulli prior over a named universe of sentences s, with $r_s = 2^{-(|s|+1)}$ and $\sum r_s < \infty$. Let $Z_0 := \prod_s (1 - r_s) > 0$, S_n the distinct data, $m_n = |S_n|$, μ the prior of the memoriser class.* (i) *The class marginal lies in $[Z_0, 1] \cdot \prod_{s \in S_n} r_s\, m_n^{-n}$ for uniform weights, with the Dirichlet(α)-multinomial probability $\mathrm{DirMult}_\alpha(D_n)$ of the counts (formula in Proposition 4.10) in place of m_n^{-n} for Dirichlet weights.* (ii) *Uniform weights, $\mathcal{Q}$ of infinite support and finite entropy: the log-odds against H_{sch} divided by n tend to $-\infty$ a.s.* (ii′) *Laplace weights: deterministically the log-odds are $\geq \ln \frac{\mu Z_0}{\pi(H_{\mathrm{sch}})} + \sum_{s \in S_n} \ln r_s - (m_n - 1) \ln(n + m_n - 1)$, which under $\mathcal{Q}_{\mathrm{num}}$ (numeral universe) is $\geq -C \ln^2 n$ a.s. for all large n: the class is not exponentially small.* (iii) *Conjecture: $-\Theta(\ln^2 n)$ under $\mathcal{Q}_{\mathrm{num}}$.*

Proof of Proposition B.6. (i) $\pi(F) = Z_0 \prod_{s \in F} r_s/(1 - r_s)$. $F = S_n$ gives the lower bound ($r_s/(1 - r_s) \geq r_s$). For the upper, $\sum_{F \supseteq S_n} \pi(F) = \prod_{s \in S_n} r_s$, and each likelihood is at most that of S_n: $|F|^{-n} \leq m_n^{-n}$, and j unused components multiply $\mathrm{DirMult}_\alpha$ by $\frac{\Gamma((m+j)\alpha)\Gamma(n+m\alpha)}{\Gamma(m\alpha)\Gamma(n+(m+j)\alpha)} \leq 1$. (ii) The log-odds are at most a constant $-n \ln m_n + \sum_i \ln \frac{1}{\mathcal{Q}(t_i)}$, the sum is $n(H(\mathcal{Q}) + o(1))$ a.s., and $m_n \to \infty$ a.s. (ii′) $\mathrm{DirMult}_1(D_n) = 1/[\binom{n}{(n_s)}\binom{n+m-1}{m-1}]$; the multinomial coefficient is $\leq 1/\mathrm{ML}(D_n)$ (a type class has probability ≤ 1 under the maximum-likelihood law), $\binom{n+m-1}{m-1} \leq (n+m-1)^{m-1}$, and $\mathrm{ML}(D_n) \geq P_{H_{\mathrm{sch}}}(D_n)$; with the lower bracket this is the deterministic bound. For numerals $|\varphi(S^k 0)| = |\sigma_\varphi| + \mathrm{occ} \cdot k$. With K_n the largest k seen, $P(K_n \geq 3 \log_{1/q} n) \leq n \cdot n^{-3} = n^{-2}$, summable, so by Borel–Cantelli a.s. $K_n < 3 \log_{1/q} n$ eventually; then $m_n \leq K_n + 1$ and $\sum_{s \in S_n} (|s| + 1) = O(\ln^2 n)$. □

(The first version had probability $1 - 1/n$ at threshold $2 \log_{1/q} n$; c10(g) checks the n^{-2} tail exactly.) The universe must be named: over all closed sentences $\sum r_s = \infty$ at $\lambda = 1$ (referee r6). *c3* ($\alpha = 1$, 10 seeds, log-odds up to a constant in $[\ln Z_0, 0]$): under $\mathcal{Q}_{\mathrm{num}}$, $q = \frac{1}{2}$, $m_n = 10.4$, 16.4 at $n = 10^3, 10^5$ and the Laplace log-odds are -133.8, -327.7 ($-2.80 \ln^2 n$, $-2.47 \ln^2 n$; uniform weights -1062, $-1.4 \cdot 10^5$); at $q = 0.9$, $D_l / \ln^2 n$ moves from -35 to -53 (pre-asymptotic); under $\mathcal{Q}_{\mathrm{GW}}$, $m_n = 2.2 \cdot 10^4$ at $n = 10^5$ and $D_l/n \approx -4.0$: exponential.

B.3 Predictive confirmation

Proof of Theorem 3.8. (1) By the chain rule, $\sum_{n<N} \mathbb{E}\mathrm{KL}(P^* \| M(\cdot \mid D_n)) = \mathbb{E} \ln \frac{P^*(D_N)}{M(D_N)} \leq \ln \frac{1}{\pi(T^*)}$, as $M \geq \pi(T^*) P^*$. Pushing forward to the indicator of I, $\mathrm{KL}(P^* \| M(\cdot \mid D_n)) \geq \ln \frac{1}{M(I \mid D_n)} \geq 1 - M(I \mid D_n)$; a nonnegative series with finite mean is finite a.s. (2) $P_T(I)^\infty = 1[T \in G]$, and $C^* \subseteq G$, so Theorem 4.2 applies. (3) $1 - \pi_n(G) \leq \min(1, \sum_{T \notin G} \pi_n(T)/\pi_n(T^*))$ and Theorem B.4(3). (4) $P_T(D_n) = (1 - \varepsilon_T)^n \hat{P}(D_n)$ on instance sequences. □

Remark B.7 (Relation to Hutter). [known; computed] universal §5 In the Bayes–Laplace model the universal hypothesis, even its observable form, has posterior 0, while the universal prior confirms it (Hutter, 2007) (abstract checked; full text not). Theorem 3.8(2) is the analogue for countable classes, by the same mechanism: prior mass on instance-only hypotheses. But G contains H_{sch}, which does not prove $\forall x \varphi$. Nor is confirmation stepwise monotone: with prior $\frac{1}{2}$ on each of σ_φ, $\varphi(Sz)$, the datum $\varphi(S0)$ lowers σ_φ to $\frac{1}{3}$, a Nicod-type effect (Leike and Hutter, 2015).

c4: with a point mass $\pi_0 = 0.01$ and Uniform$(0,1)$ on ε, $P(\text{all future} \mid n) = \pi_0/(\pi_0 + \frac{1-\pi_0}{n+1})$; with $\pi_0 = 0$ it is 0 while $P(\text{next } m) = \frac{n+1}{n+m+1}$. Summed misses 4.22 (Laplace) and 2.70 (dyadic) $\leq \ln 100 = 4.61$.

B.4 The Bayesian ω-gap

Proof of Theorem 3.9. (a) Sum the limit of Theorem 4.2 over $\mathrm{Prov}_\forall$, or over the theories counted by Bel, Dis, Indep or Inc. (b) Both theories lie in C^*, one in $\mathrm{Prov}_\forall$. For the learned-weight pair the marginal laws are exchangeable, not equal, so (a) does not apply; Proposition 3.6 gives the limit $\pi(T_\forall)h/(\pi(T_\forall)h + \pi(T_\sigma)) \in (0,1)$ directly. (Under $\mathsf{L1}^{\mathrm{sel}}$, $\{\sigma_\varphi, \exists x \neg(0 + x = x)\}$ with S excluding the second sentence is a consistent refuter in C^*, so all three trichotomy values can keep mass.) (c1) $T \in C^*$ implies $\mathrm{Th}_{\mathsf{C}}(T) \cap X = \mathrm{Th}_{\mathsf{C}}(T^*) \cap X$. (c2) Each axiom instance of $T \in C^*$ lies in $\mathrm{Th}_{\mathsf{C}}(T^*) \cap X$, hence follows from T^* by soundness; symmetrically. (In $\mathsf{C}_{\min}$, $T^* = \{\forall x(\varphi \wedge \psi)\}$ has limit 0 in (c1) and 1 in (c2): different questions, conflated in the first version.) (c3) In $\mathsf{C}_{\min}$ every chain has positive probability under full-support $\mathcal{Q}$, so $\mathsf{L1}$ is faithful, and a σ_φ-chain with quantifier-free φ can only cite. For the second part apply the rule to a derivation of an instance. (d1) $\mu_{H_\forall}(\cdot \mid S) = c\mu_{H_{\mathrm{sch}}}(\cdot \cap S)/(c\mu_{H_{\mathrm{sch}}}(S)) = P_{H_{\mathrm{sch}}}$ since $\operatorname{supp} \mu_{H_{\mathrm{sch}}} = I_{\mathrm{c}} \subseteq S$; then (c3) with $T^* := H_{\mathrm{sch}}$, or (b) under $\mathsf{L1}^{\mathrm{sel}}$. (d2) For quantified φ, $\tilde{P} := \mu_{H_{\mathrm{sch}}}(\cdot \mid S)$ is the law of neither hypothesis. On S, $\mu_{H_\forall} = c\mu_{H_{\mathrm{sch}}}$, so both log-ratios are constant there and $\mathrm{KL}(\tilde{P} \| P_{H_{\mathrm{sch}}}) = \ln \frac{Z_{H_{\mathrm{sch}}}}{\mu_{H_{\mathrm{sch}}}(S)} < \ln \frac{Z_{H_\forall}}{c\mu_{H_{\mathrm{sch}}}(S)} = \mathrm{KL}(\tilde{P} \| P_{H_\forall})$, because $cZ_{H_{\mathrm{sch}}} < Z_{H_\forall}$; the posterior concentrates on H_{sch} (Theorem 4.16). For a general class this is a sketch. (e) Pairs form a countable class of i.i.d. laws. (H_{sch}, S) has the data law iff $S \supseteq I_{\mathrm{c}}$; $(H_\forall, S)$ has it iff $S \supseteq I_{\mathrm{c}}$ and $\forall x\varphi \notin S$ (Theorem 3.3); otherwise some instance gets probability 0 or $\forall x\varphi$ positive probability. Summing prior masses over C^* gives the limit. □

c10(e): filters {all sentences, closed quantifier-free, closed instances, closed instances with S-rooted argument}, $\pi = (\frac{1}{3}, \frac{2}{3})$, ν uniform: 0.333, 0.284, 0.252, 0.250 at $n = 0, 1, 5, 10$, predicted 0.2500.

Proof of Corollary 3.13. Members of $C^*_\forall$ do not prove $\forall x\varphi$ and share T^*'s likelihood, so

$$\pi_n(C^*_\forall) = \pi(C^*_\forall)\, P_{T^*}(D_n)/M(D_n).$$

Acceptance needs $\pi_n(C^*_\forall) \leq \delta$, that is, $M(D_n)/P_{T^*}(D_n) \geq \pi(C^*_\forall)/\delta$. This ratio is a nonnegative supermartingale started at 1; by Ville's inequality it ever reaches that level with probability at most $\delta/\pi(C^*_\forall)$. □

stream	likelihood	class	$n = 0$	10	50	100
closed instances	$\mathsf{L1}$	full	0.750	0.364	0.004	0.000
closed instances	$\mathsf{L1}$	$\{H_\forall, H_{\mathrm{open}}\}$	1.000	1.000	1.000	1.000
closed instances	$\mathsf{L1}^{\mathrm{sel}}$	full	0.750	0.750	0.750	0.750
output of H_{open}	$\mathsf{L1}$	full	0.750	0.860	1.000	1.000
output of $H_\forall$	$\mathsf{L1}$	full	0.750	1.000	1.000	1.000

Table 10: $P(T \vdash \forall x\varphi \mid D_n)$ in $\mathsf{C}_{\mathrm{open}}$ (c5 Part A); closed instances are also $H_\forall$'s output filtered to closed instances; $\varphi = 0 + x = x$, defaults, prior $2^{-(\text{symbols+1 per axiom})}$; full class $\{H_\forall, H_{\mathrm{open}}, H_{\mathrm{sch}}\ (\text{guarded}), H_{\mathrm{both}}\}$, with a uniform prior on H_{both}'s weight, prior share of provers 0.750. On the full output of $H_\forall$ (itself reported), $H_\forall$ has 0.993 at $n = 10$.

prior code	bits($H_\forall$)	bits(H_{sch})	share of $H_\forall$
universal: dtrc symbols +1 per axiom, $\lambda = 1$	7	6	0.333
model π_1 (Definition 2.4; 4 bits per token)	27	26	0.333
model π_2			0.200
experiments' template code, closed guard	17.18	17.09	0.484
pa, $\beta = 5$ bits per symbol	30	25	0.030
pa, $\beta = \log_2 23$	27.14	22.62	0.042

Table 11: Prior share of $H_\forall$ in $\{H_\forall, H_{\text{sch}}\}$ for $\varphi = 0 + x = x$ under the four tracks' codes (c10(f); the model rows by hand from Definition 2.4). The experiments' value equals their E1 limit under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ universal §6.6.

Proposition B.8 (Hänni's variants on $\forall x\varphi$)**.** [proved; computed] universal §6.2 *Let $\varphi = 0 + x = x$, $\mathcal{Q}$ of full support. Under S_{prove} the posterior converges a.s. in total variation to the prior restricted to $\{T$ consistent $: T \vdash I_{\text{c}}(\varphi)\}$, which contains Q (independent), $\mathsf{Q} + \forall x\varphi$ (true), $\mathsf{Q} + \exists x\neg\varphi$ (false), H_{sch}, $H_\forall$ and consistent over-general templates such as $\{0 + z_1 = z_2\}$: all three trichotomy values keep positive mass, and over-generalisation is never penalised. Under S_{nc} and S_g ($g \equiv 1$ on provable data) the theories that prove all the data share one factor, so their mutual odds keep the prior odds and again all three trichotomy values keep positive mass; under S_{nc} the limit is the prior restricted to the larger set $\{T : T \cup I_{\text{c}}(\varphi)$ consistent$\}$, which also contains non-provers of the instances such as $\emptyset$. Under generative likelihoods inconsistent theories die within a few data (Table 12).*

Proof of Proposition B.8. Under S_{prove} the posterior is the prior restricted to $\text{Adm}_n = \{T \text{ consistent} : T \vdash D_n\}$; every closed instance eventually appears a.s., so $\text{Adm}_n \downarrow \{T \text{ consistent} : T \vdash I_{\text{c}}(\varphi)\}$, and continuity of measure gives convergence in total variation. The model of AS Ex 4.9 shows that Q, $\mathsf{Q} + \forall x\varphi$ and $\mathsf{Q} + \exists x\neg\varphi$ are consistent with all closed instances, each of which Q proves; $\{0 + z_1 = z_2\}$ has a one-element model. Under S_{nc} and S_g ($g \equiv 1$ on provable data) all theories proving the data get the same factor, so their mutual odds stay at the prior odds. Under S_{nc} the score is $1[T \cup D_n$ consistent]; as every closed instance eventually appears a.s., compactness and continuity of measure give the limit: the prior restricted to $\{T : T \cup I_{\text{c}}(\varphi)$ consistent$\}$, which contains $\emptyset$. Inc $= 0$ under the scores by definition. □

φ, law, likelihood	$n = 0$	1	2	5	10	20
$0 + x = x$, $\mathcal{Q}_{\text{num}}$; $\mathsf{L0}^{\text{cl}}$, $\mathsf{L1}$, $\mathsf{L1}^{\text{sel}}$	0.674	0.000	0.000	0.000	0.000	0.000
$\neg Sx = 0$, $\mathcal{Q}_{\text{num}}$; $\mathsf{L0}^{\text{cl}}$ / $\mathsf{L1}$	0.931	.585/.635	.319/.390	.056/.081	.002/.003	0.000
$x < Sx$, $\mathcal{Q}_{\text{num}}$; $\mathsf{L0}^{\text{cl}}$	0.627	0.080	0.004	0.000	0.000	0.000
$\neg Sx = 0$, $\mathcal{Q}_{\text{GW}}$; $\mathsf{L0}^{\text{cl}}$	0.931	0.454	0.191	0.010	0.000	0.000

Table 12: Mass Inc of inconsistent theories in the class of Table 2 without its learned-weight root split, so all members have fixed weights and no Dirichlet α enters (c10(d), 20 seeds): the bare formula metavariable and, for $\neg Sx = 0$, three templates with instances such as $\neg(0 = 0)$. $\text{Bel}_{\text{cons}}(\forall x\varphi)$ tends to the prior share ≈ 0.332 under $\mathsf{L0}^{\text{cl}}$, $\mathsf{L1}^{\text{sel}}$ and to 0 under $\mathsf{L1}$.

Proposition B.9 (An ω-gap for Q4)**.** [proved; computed] universal Prop U13 *($\mathsf{Q} - Q4$) plus all closed instances of $x + 0 = x$ does not prove $\forall x(x + 0 = x)$ (a model on $\mathbb{N} \cup \{a, b\}$). So closed instances*

cannot tell the axiom Q4 from its closed schema $z + 0 = z$, which differ in their theorems over $\mathsf{Q} - Q4$.

Proof of Proposition B.9. On $\mathbb{N} \cup \{a, b\}$, standard on $\mathbb{N}$, let $Sa = a$, $Sb = b$; $a + m = m$, $b + m = b$ ($m \in \mathbb{N}$); $x + a = a$ for $x \in \mathbb{N} \cup \{a\}$, $b + a = b$, $x + b = b$ for all x; $x \cdot 0 = 0$; $a \cdot m = a$, $b \cdot m = b$ ($m \geq 1$); $n \cdot a = n \cdot b = b$ ($n \geq 1$); $0 \cdot a = 0 \cdot b = 0$; $a \cdot a = a \cdot b = a$; $b \cdot a = b \cdot b = b$. Q1–Q3: S is the identity on $\{a, b\}$ and $n \mapsto n + 1$ on $\mathbb{N}$. Q5: for $y = m$, $a + (m + 1) = m + 1 = S(a + m)$, both sides are b for $x = b$, and $x \in \mathbb{N}$ is standard; for $y \in \{a, b\}$, $Sy = y$ and $x + y \in \{a, b\}$ is fixed by S. Q6 holds by definition. Q7: for $y = m$, $a \cdot (m + 1) = a = a \cdot m + a$ (as $0 + a = a + a = a$) and $b \cdot (m + 1) = b = b \cdot m + b$; for $y \in \{a, b\}$ we need $z + x = z$ with $z = x \cdot y$, which holds for $x = 0$ ($0 + 0 = 0$), $x = n \geq 1$ ($b + n = b$), $x = a$ ($a + a = a$) and $x = b$ ($b + b = b$). Closed terms denote standard numbers, so every closed instance holds, but $a + 0 = 0 \neq a$. □

Each case was re-checked for this paper; c6 and referee r5 find no violation on $\{0..40\} \cup \{a, b\}$ and $\{0..80\} \cup \{a, b\}$.

Proposition B.10 (The Gaifman condition)**.** [proved; known] universal Prop U15 *Let μ be a probability on L-structures, $P(\psi) := \mu\{M : M \models \psi\}$, and $t_1, t_2, \ldots$ the closed terms. Then*

$$P\Big(\forall x\varphi \;\Big|\; \textstyle\bigwedge_{i \leq n} \varphi(t_i)\Big) \to P(\forall x\varphi)/\mu\{M : M \models \varphi(t) \text{ for all closed } t\}.$$

The Gaifman condition (Gaifman, 1964) $P(\forall x\varphi) = \inf_n P(\bigwedge_{i \leq n} \varphi(t_i))$ holds for all φ iff $M_0 \preccurlyeq M$ for μ-a.e. M, with M_0 the closed-term substructure (AS Prop 4.7); the limit is then 1 if $P(\forall x\varphi) > 0$ and 0 if $P(\forall x\varphi) = 0$.

So believing $\forall x\varphi$ from true closed instances is safe for $\mathbb{N}$ and unsafe for credences that weigh $\mathbb{R}$, V, models of $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$ (AS Ex 4.8) or some models of Q (AS Ex 4.9). The condition is cited from memory by the track; secondary sources state it with constants rather than closed terms.

Proof of Proposition B.10. $\forall x\varphi$ entails each instance, so $P(\forall x\varphi \wedge \bigwedge_{i \leq n} \varphi(t_i)) = P(\forall x\varphi)$, and the conjunctions decrease to "all closed instances hold"; continuity of μ gives the limit. The gap in the Gaifman condition is $\mu\{M : \text{all closed instances hold}, \ M \not\models \forall x\varphi\}$, which vanishes for every φ (countably many) iff μ-a.e. M satisfies (i) of AS Prop 4.7, i.e. $M_0 \preccurlyeq M$. □

B.5 Open instances, guards and root splits

Proof of Proposition 3.10. (a) $P_T(D) > 0$ gives $T \vdash \varphi(p)$, and p occurs in no axiom, so Gen gives $T \vdash \forall x\varphi$. (b) follows from (a). (c) With weights $(w_\forall, w_{\text{open}}, w_{\text{sch}})$ on $\forall x\varphi$ and the open- and closed-guarded σ_φ, the only outputs are $\forall x\varphi$, $\varphi(p)$ and closed $\varphi(t)$ (Gen fails on closed formulas, $\forall$E on quantifier-free ones). With $a := \mu(\forall x\varphi)$, $b := \mu(\varphi(p))$: $a = p_{\text{cite}} w_\forall + gb$ and $b = \rho(p_{\text{cite}} w_{\text{open}} + ca)$, so

$$a = \frac{p_{\text{cite}}(w_\forall + g\rho w_{\text{open}})}{1 - gc\rho}, \qquad Z = a(1 + c) + p_{\text{cite}}(w_{\text{open}} + w_{\text{sch}}),$$
$$\mu(\varphi(t)) = \mathcal{Q}_{\text{num}}(t)\big[(1 - \rho)(p_{\text{cite}} w_{\text{open}} + ca) + p_{\text{cite}} w_{\text{sch}}\big] \quad (t \text{ closed}).$$

Hence $Z_{H_{\text{open}}} = p_{\text{cite}} \frac{1+g\rho}{1-gc\rho}$, $Z_{H_\forall} = p_{\text{cite}} \frac{1+c}{1-gc\rho}$, $Z_{H_{\text{sch}}} = p_{\text{cite}}$; unnormalised, $\mu_{H_\forall}/\mu_{H_{\text{open}}} = c$ at instances and $1/(g\rho)$ at $\forall x\varphi$, and $\mu_{H_{\text{open}}}/\mu_{H_{\text{sch}}} = \frac{1-\rho}{1-gc\rho}$ at closed instances; divide by the normalisers. (d) follows, for the finitely many hypotheses compared, from (c), Theorem B.4(2) and

Proposition B.3; the rate $1/n$ for H_{both} is computed in $\mathsf{C}_{\min}$, and in C_{open} its per-datum ratio to H_{sch} (the formulas above with $w_{\text{open}} = 0$) is still below 1 at every $w_\forall > 0$. □

c1 (sampler, max $|z| = 1.60$), c5 and referee r7 (2.15) confirm (c).

Lemma B.11 (Waste)**.** [proved] universal Lemma N0 *Let P^* be a law on a countable set, $H \cup T$ a partition of* $\operatorname{supp} P^*$ *with $\tau := P^*(T) \in (0,1]$, N_W a law on a set disjoint from* $\operatorname{supp} P^*$*, and $r_{\text{w}} > 0$. Over the laws $p_h P^*(\cdot \mid H) + p_t P^*(\cdot \mid T) + r_{\text{w}} p_t N_W$ with $p_h + (1+r_{\text{w}})p_t = 1$, the minimum of* $\mathrm{KL}(P^* \| \cdot)$ *is $\tau \ln(1 + r_{\text{w}})$, attained at $p_t = \tau/(1 + r_{\text{w}})$, $p_h = 1 - \tau$.*

Proof. The conditionals are exact and N_W lives off the support, so $\mathrm{KL} = (1-\tau)\ln\frac{1-\tau}{p_h} + \tau \ln\frac{\tau}{p_t}$, convex in p_t once $p_h = 1 - (1 + r_{\text{w}})p_t$. The derivative $(1-\tau)(1+r_{\text{w}})/p_h - \tau/p_t$ vanishes at $p_t = \tau/(1+r_{\text{w}})$, $p_h = 1 - \tau$, where the value is $\tau \ln(1 + r_{\text{w}})$. □

Proposition B.12 (Exact law of R_k)**.** [proved; computed]
universal Prop N1 *In C_{open}, let $A_\rho := \frac{1-\rho}{1-gc\rho}$ and $B_\rho := \frac{1+g\rho}{1-gc\rho}$, and give R_k sentence weights $v_0, \ldots, v_{k-1}$ and template weight u. Its unnormalised law is $p_{\text{cite}} v_j$ at $\varphi(S^j 0)$ $(j < k)$; $p_{\text{cite}} u A_\rho \mathcal{Q}_{\text{num}}(S^m 0)$ at $\varphi(S^{k+m}0)$; $p_{\text{cite}} u\rho/(1 - gc\rho)$ at $\varphi(S^k p)$; $p_{\text{cite}} ug\rho/(1 - gc\rho)$ at $\forall y \varphi(S^k y)$; and 0 elsewhere. $Z = p_{\text{cite}}(1 - u + uB_\rho)$. H_{open} is the case $k = 0$, $u = 1$.*

Proof. Citing a sentence yields it, and Gen and ∀E then fail. Citing the template yields $\varphi(S^k t)$, $t \sim \mathcal{Q}_{\text{open}}$; Gen applies only to $\varphi(S^k p)$, giving $U := \forall y \varphi(S^k y)$, and ∀E only to U. With $a := \mu(U)$, $b := \mu(\varphi(S^k p))$: $a = gb$, $b = \rho(p_{\text{cite}} u + ca)$, so $a = g\rho p_{\text{cite}} u/(1 - gc\rho)$, $b = \rho p_{\text{cite}} u/(1 - gc\rho)$, and a closed tail instance has mass $(1-\rho)\mathcal{Q}_{\text{num}}(S^m 0)(p_{\text{cite}} u + ca) = p_{\text{cite}} u A_\rho \mathcal{Q}_{\text{num}}(S^m 0)$. Summing gives $Z = p_{\text{cite}}(1-u) + p_{\text{cite}} u(1 + g\rho)/(1 - gc\rho)$. □

c8 Part 1: sampler for $R_1..R_3$, max $|z| = 2.07$ (referee r3: 2.19).

Proposition B.13 (Rates on closed data)**.** [proved; computed] universal Prop N2 *On closed numeral data from $\mathcal{Q}_{\text{num}}$, under* L1*: $\mathrm{KL}(P^* \| P_{H_\forall}) = \ln\frac{1+c}{c(1-\rho)}$, $\mathrm{KL}(P^* \| P_{H_{\text{open}}}) = \mathcal{W}_{\text{o}}$, $\inf_w \mathrm{KL}(P^* \| P_{R_k}) = q^k \mathcal{W}_{\text{o}}$. Each split level halves the loss at $q = \frac{1}{2}$ (0.0626, 0.0313, 0.0156 for R_1, R_2, R_3).*

The proof uses the exact law of R_k (Proposition B.12) and the waste lemma (Lemma B.11): a law that reproduces P^* on a tail of mass τ and wastes r_{w} times the tail's mass loses at best $\tau \ln(1 + r_{\text{w}})$; for R_k, $\tau = q^k$ and $\ln(1 + r_{\text{w}}) = \mathcal{W}_{\text{o}}$.

Proof of Proposition B.13. Conditioned on closed outputs, $H_\forall$ and H_{open} have law P^* (Proposition 3.10(c)), so their KL is $-\ln$ of their closed mass, $c(1-\rho)/(1+c)$ and $A_\rho/B_\rho = (1-\rho)/(1+g\rho)$. For R_k (Proposition B.12) the normalised law has head mass $(1-u)/Z_R$, $Z_R = 1 - u + uB_\rho$, tail mass $p_t = A_\rho u/Z_R$ with the exact tail shape (because $\mathcal{Q}_{\text{num}}(S^{k+m}0) = q^k \mathcal{Q}_{\text{num}}(S^m 0)$), and waste $(B_\rho - A_\rho)u/Z_R = r_{\text{w}} p_t$ with $r_{\text{w}} = B_\rho/A_\rho - 1$. For fixed u the best head is $v_j \propto \mathcal{Q}_{\text{num}}(S^j 0)$ (Gibbs), and Lemma B.11 applies with $\tau = q^k$: p_t ranges over $(0, A_\rho/B_\rho] = (0, 1/(1+r_{\text{w}})]$, which contains the optimum, attained at $u = q^k/(B_\rho(1-q^k) + q^k)$. The value is $q^k \ln(B_\rho/A_\rho) = q^k \mathcal{W}_{\text{o}}$. □

c8 Part 2: numerical minimisation agrees to $1.8 \cdot 10^{-15}$ ($k \le 6$, four settings). For Remark 3.11, the referee's independent run (r3) of $\{H_\forall, H_{\text{open}}, R_1\}$ gives 0.880 at $n = 100$, then 0.000 at $n = 300$.

Proof of Proposition 3.12. *Provability.* On $\mathbb{N} \cup \{e, e'\}$, standard on $\mathbb{N}$, with $Se = Se' = e'$ and $0 + e = 0 + e' = e'$, every $S^k x$ ($k \geq 1$) is a standard number $\geq k$ or e', so every instance of $\varphi(S^k z)$, closed or under closure, and $\varphi(0), \ldots, \varphi(S^{k-1}0)$ hold, while $0 + e \neq e$. Over Q: Q proves each $0 + S^j 0 = S^j 0$ (Q4, Q5, induction on j outside the theory), and k uses of Q3 show that every x is one of $0, \ldots, S^{k-1}0$ or $S^k y$, where $\varphi(S^k p)$, read as $\forall y \varphi(S^k y)$, applies. Memorisers consist of closed instances, which Q proves, and $\mathsf{Q} \nvdash \forall x \varphi$ (AS Ex 4.9). (a) $H_\forall$, H_{open} have fixed laws with KL $\geq \mathcal{W}_{\text{o}}$ (note $\ln \frac{1+c}{c(1-\rho)} \geq \mathcal{W}_{\text{o}}$), so their normalised log-likelihood ratio to P^* is $-n\text{KL} + o(n)$ a.s. For R_k with learned weights, take the interior optimum w_0 and an ε-neighbourhood N_ε on which the weights are bounded away from 0; there the per-datum log-likelihood ($\ln v_j - \ln Z_R$, or $\ln(A_\rho u) + \ln \mathcal{Q}_{\text{num}}(S^m 0) - \ln Z_R$, whose middle term does not depend on w) has w-gradient bounded by some M. So the marginal is $\geq \text{Dir}(N_\varepsilon) P_{w_0}(D_n) e^{-nM\varepsilon}$, and $\liminf \frac{1}{n} \ln \frac{\text{marginal}}{P^*(D_n)} \geq -q^k \mathcal{W}_{\text{o}} - M\varepsilon$ for every ε; the log-odds of R_k against each prover grow like $n(1-q^k)\mathcal{W}_{\text{o}} - o(n)$. Provers that add σ_φ as a spare component to some R_k decay only polynomially relative to R_k (proof sketch; Proposition 4.14(b)). (b) All members but the memorisers prove $\forall x \varphi$ over Q, which gives the second sentence. By Proposition B.6(ii′) the memoriser class has log-marginal $\geq -C \ln^2 n$ relative to P^*. For each R_k the likelihood ratio to P^* depends only on the empirical frequencies $\hat{p}$ of the $k+1$ categories (head values; tail), since the tail shape cancels; so $\frac{1}{n} \ln \sup_w \frac{P_{R_k, w}(D_n)}{P^*(D_n)} = \sup_w G(\hat{p}, w)$, convex and hence continuous in $\hat{p}$ near p^*, with limit $-q^k \mathcal{W}_{\text{o}} < 0$, and the marginal is below the supremum. With $\mathcal{K}$ finite all non-memorisers lose linearly against the memorisers. (c) (Proof sketch.) Under $\mathsf{L1}^{\text{sel}}$, R_k's filtered law (head v_j/N, tail $A_\rho u \mathcal{Q}_{\text{num}}(S^m 0)/N$, $N = 1 - u + A_\rho u$) is a smooth reparametrisation of a $(k+1)$-category multinomial equal to P^* at one interior point. The Laplace (BIC) expansion (Schwarz, 1978; Clarke and Barron, 1990) (recalled, details not checked) gives log-marginal $-\frac{k}{2} \ln n + O_P(1)$ relative to P^*; memorisers lose $O(\ln^2 n)$; $H_\forall$, H_{open} have log-ratio 0. Not written out: almost-sure control, uniform in k for infinite $\mathcal{K}$. □

likelihood, class	quantity	$n = 0$	10	10^2	10^3	10^4	10^5	10^6
$\mathsf{L1}$, $\{H_\forall, H_{\text{open}}, R_1\}$	$P(\vdash)$, pure	.997	.996	.697	.000	.000	.000	.000
	MAP	H_{open}	H_{open}	H_{open}	R_1	R_1	R_1	R_1
$\mathsf{L1}$, $+R_{2..4}$, mem.	$P(\mathsf{Q} \cup T \vdash)$	.601	1.000	1.000	1.000	1.000	.000	.000
	MAP	H_{open}	H_{open}	H_{open}	R_3	R_4	mem	mem
$\mathsf{L1}$, $+R_{2..40}$, mem.	$P(\vdash)$, pure	.599	.996	.697	.000	.000	.000	.000
	$P(\mathsf{Q} \cup T \vdash)$	.601	1.000	1.000	1.000	1.000	1.000	1.000
	MAP	H_{open}	H_{open}	H_{open}	R_3	R_6	R_9	R_{12}
$\mathsf{L1}^{\text{sel}}$, each class	$P(\vdash)$, pure	.997/.599	.998	.987	1.000	1.000	1.000	1.000

Table 13: Without a guard (c8 Part 3): $P(T \vdash \forall x \varphi \mid D_n)$ in pure logic and over Q, $\varphi = 0 + x = x$, closed numeral data ($q = \frac{1}{2}$), C_{open} at $(c, g, \rho) = (0.3, 0.2, 0.1)$, Dirichlet($\frac{1}{2}$) weights (Proposition 3.12(b) is proved for Laplace-weighted memorisers), prior $2^{-(\text{symbols}+1 \text{ per axiom})}$, memoriser class mass 2^{-6} (lower bracket), mean of 20 seeds; MAP = mode over seeds. Part 4 (multinomial counts, 10 seeds): with $R_{1..40}$ the MAP is $R_{15}, R_{18}, R_{21}, R_{24}, R_{30}$ at $n = 10^7, 10^8, 10^9, 10^{10}, 10^{12}$ and $P(\mathsf{Q} \cup T \vdash) = 1.000$, $P(\vdash)$ pure $= 0$, also with the upper bracket and class mass $\frac{1}{2}$, with $\alpha = 1$, and with $\lambda = 2$ ($R_{14}..R_{30}$) universal Prop N3.

The MAP depth is about $\log_2 n$ minus a slowly growing term (≈ 8 at $n = 10^6$, ≈ 10 at 10^{12}).

B.6 Quantified data and the equational fragment

Proposition B.14 (Quantified data)**.** [(a), (b) proved; computed] universal Prop U12 (a) *If $s \in D$ and $B \cup I_c(\varphi) \nvdash s$, then under any likelihood supported on theorems $B \oplus \sigma_\varphi$ has posterior 0; over Q, one datum $\forall y(0 + Sy = Sy)$ leaves only provers of $\forall x(0 + x = x)$.* (b) *If all theories prove $\forall x\varphi$ at different costs ($\{\forall^L \forall x\varphi, \sigma_\varphi\}$ against $\{\forall x\varphi, \sigma_\varphi\}$ in $\mathsf{C}_{\min}$, L vacuous quantifiers, uniform weight priors, equal prior masses, data $\forall x\varphi$ with probability f_q and instances otherwise), the log Bayes factor tends to $\ln(1 + c + \cdots + c^L)$ when $f_q = 0$ (0.349 at $L = 3$) and grows linearly when $f_q > 0$, at a rate a little above $f_q L \ln \frac{1}{c}$ ([proof sketch] for $f_q > 0$).* (c) [proof sketch] *In $\mathsf{L1}^{\max}$ form a derived sentence becomes an axiom when the derivation steps it saves, at $\geq \ln \frac{1}{c}$ each, exceed its prior cost and an $O(\ln n)$ dilution: frequently used lemmas become axioms.*

Proof of Proposition B.14. (a) If $B \cup I_c \nvdash s$, then $B \oplus \sigma_\varphi \nvdash s$. In AS Ex 4.9's model, $0+Sa = 0+a = b \neq a = Sa$, so $\mathsf{Q}+I_c \nvdash \forall y(0+Sy = Sy)$; and $\mathsf{Q}+\forall y(0+Sy = Sy) \vdash \forall x(0+x = x)$ by Q4 at 0 and Q3 elsewhere. (b) In units of $1 - c$, with $S_B := \sum_{k=0}^{L+1} c^k$: $\{\forall x\varphi, \sigma_\varphi\}$ with weight w on $\forall x\varphi$ gives $P(\forall x\varphi) = \frac{w}{1+wc}$, $P(\varphi(t)) = \frac{1-w+wc}{1+wc}\mathcal{Q}(t)$; $\{\forall^L \forall x\varphi, \sigma_\varphi\}$ gives $P(\forall x\varphi) = wc^L/Z_B$, $P(\varphi(t)) = ((1-w)+wc^{L+1})\mathcal{Q}(t)/Z_B$, $Z_B = wS_B + 1 - w$ (vacuous eliminations succeed with term mass 1). A quantified datum adds $L \ln \frac{1}{c} + \ln \frac{Z_B}{1+wc}$ to the log-ratio; an instance 0.021, 0.094, 0.307, 1.323 nats at $w = 0.05, 0.2, 0.5, 0.9$ ($L = 3$), so the two theories price instances alike only for small w (referee m5). For $f_q = 0$ both Bayes factors against H_{sch} are $\int_0^1 e^{nf_i(w)}dw$ with $f_i(0) = 0$, f_i decreasing, $f'_{\text{both}}(0) = -1$, $f'_B(0) = -(1+c+\cdots+c^L)$; Watson's lemma gives $(n|f'_i(0)|)^{-1}(1+o(1))$ for each. For $f_q > 0$ (sketch) the rate is $\Delta(f_q) := \max_w \mathbb{E}\ell_{\text{both}}(w) - \max_w \mathbb{E}\ell_B(w)$, the Laplace approximation at the interior maxima not written out. (c) (Sketch.) In $\mathsf{L1}^{\max}$ form a derivation's code length is a sum of rule and term costs; adding ψ costs its prior bits and an $O(\ln n)$ dilution, and saves $\ell_T(s) - \ell_{T+\psi}(s)$ steps of cost $\geq \ln \frac{1}{c}$ per datum s. □

c10(c), equal priors, 20 seeds, $n \leq 10^5$: for $f_q = 0$ the log Bayes factor is 0.349 ($L = 3$) and 0.357 ($L = 10$) at every n; for $L = 3$, $\Delta = 0.0396$, 0.1980, 0.7921 at $f_q = 0.01, 0.05, 0.2$ (measured 0.0395, 0.1977, 0.7921; $f_q L \ln \frac{1}{c} = 0.036$, 0.181, 0.722); for $L = 10$, 0.1240, 0.6198, 2.4793 (measured 0.1236, 0.6187, 2.4793). The first version's table (c5 Part B) gave $\{\forall^L \forall x\varphi, \sigma_\varphi\}$ a much smaller prior and so tracked only the refutation in (a): $P(T \vdash \forall x\varphi) = 1 - (1 - f_q)^n$ within Monte Carlo error.

Proposition B.15 (The equational fragment)**.** [(a)–(c) proved; (a) computed] universal Prop U14 (a) *In equational logic over $E = \{Q4, Q5\}$ every derivation of $0 + S^k0 = S^k0$ has at least $k + 1$ axiom leaves.* (b) *If grammar nodes pick rules independently, with at most two premises and mean $\mu < 1$, then $P(\geq L \text{ nodes}) \leq e^{-(L-1)(1-\mu)^2/2}$.* (c) *The full sum favours the schema by $\Omega(k)$ on $0 + S^k0 = S^k0$ when $q > e^{-(1-\mu)^2/2}$; this needs $q > 0.61$, so it does not apply at $q = \frac{1}{2}$; the two-part ($\mathsf{L1}^{\max}$) comparison favours it linearly in k at $q = \frac{1}{2}$ when each leaf pays a citation and a term of point mass $\leq 1 - q$.*

Proof of Proposition B.15. (a) Let $w(t) := \sum_{+\text{-nodes } l+r}(1 + \#S(r))$, $\#S(r)$ the number of S symbols in r. Rewriting $u+0$ to u, or $u+Sv$ to $S(u+v)$, anywhere in a term changes w by exactly -1 (the rewritten node contributes 1, resp. $2 + \#S(v)$ before and 0, resp. $1 + \#S(v)$ after; nodes inside are untouched; nodes above keep the S-count of their right argument), and the reverse by $+1$. $w(0+S^k0) = k+1$, $w(S^k0) = 0$. A derivation (reflexivity, symmetry, transitivity, congruence, substitution, axiom instances) converts by induction into a rewrite chain with at most as many steps as axiom leaves: congruence and transitivity concatenate chains, symmetry reverses one, substitution keeps lengths. (b) Couple the grammar with the Galton–Watson tree of rule labels

and explore it depth-first with premise counts ξ_i; $\geq L$ nodes requires $\sum_{i\leq L-1}(\xi_i-1)\geq 0$; apply Hoeffding's inequality (Hoeffding, 1963) to $\xi_i-1\in[-1,1]$ with mean $\mu-1$ (typed grammars need the maximum of μ over contexts). (c) Full sum: by (a), (b), $P_E(0+S^k0=S^k0)\leq e^{-k(1-\mu)^2/2}$, while the schema gives $\geq p_{\text{cite}}w(1-q)q^k$. Two-part: $\geq k+1$ leaves of cost $\geq \ln\frac{1}{p_{\text{cite}}}+\ln\frac{1}{1-q}>\ln\frac{1}{q}$ (for $q\geq\frac{1}{2}$), against $O(1)+k\ln\frac{1}{q}$ for the schema. □

c6: shortest chains of exactly $k+1$ for $k\leq 5$ by breadth-first search; referee r5: the invariant on 339 250 random rewrites.

B.7 The sentence-only class

Proposition B.16 (Sentence-only class)**.** [(b), (c) proved; computed] universal Prop U16 (a) [proof sketch] *Under* $\mathsf{L0}^{\text{cl}}$*,* $H_\forall$ *has the true law and* Split_k*, Both0 match it only at a point or at the boundary, so* $P(T\vdash\forall x\varphi\mid D_n)\to 1$ *(Laplace expansion uniform in* k *not written out).* (b) *Under* $\mathsf{L1}$*:* $\text{KL}(P^*\|P_{H_\forall})=\ln\frac{1+c}{c}$ *(*1.466*),* $\inf_w \text{KL}(P^*\|P_{\text{Split}_k})=q^k\ln\frac{1+c}{c}$*,* $\inf_w \text{KL}(P^*\|P_{\text{Both0}})=q\ln\frac{1+qc}{qc}$ *(*1.018*).* (c) Split_k *proves* $\forall x\varphi$ *over* Q*, not in pure logic; memorisers in neither.* (d) [computed; limit conjecture] *In pure logic* $P(T\vdash\forall x\varphi\mid D_n)\to 0$ *in every class computed. Over* Q*, with depth* ≤ 4 *memorisers win (limit* 0*); with depth* ≤ 40 *splits win to* $n=10^{12}$ *under four prior variants (Table 14). Conjecture: with all depths the limit is* 1*.*

Proof of Proposition B.16. (a) (Sketch.) Split_k equals the true law at one weight vector ($v_j=\mathcal{Q}_{\text{num}}(S^j0)$, $u=q^k$) and pays $n^{-k/2}$ with learned weights; Both0 only at the boundary; memorisers lose $O(\ln^2 n)$; uniformity in k not written out. (b) $H_\forall$ gives each instance $c/(1+c)$ times its true probability (Theorem 3.3). Split_k: the universal sentence yields itself with mass $(1-c)u$ or $\varphi(S^{k+m}0)$ with mass $(1-c)uc\mathcal{Q}_{\text{num}}(S^m0)$; normalised by $(1-c)(1+cu)$: head $v_j/(1+cu)$, tail $uc\mathcal{Q}_{\text{num}}(S^m0)/(1+cu)$, waste $u/(1+cu)=p_t/c$; Lemma B.11 with $\tau=q^k$, $r_{\text{w}}=1/c$ gives $q^k\ln(1+\frac{1}{c})$, attainable as p_t ranges over $(0,\frac{c}{1+c}]$. Both0 with weight w on $\varphi(0)$: with $M:=(1-w)c/(1+c(1-w))$, the tail $\{j\geq 1\}$ has mass qM with the exact shape, waste M/c, and $\varphi(0)$ mass $1-qM-M/c$; so $\text{KL}=(1-q)\ln\frac{1-q}{1-M(q+1/c)}+q\ln\frac{1}{M}$, convex, minimal at $M=\frac{qc}{1+qc}\leq\frac{c}{1+c}$ with value $q\ln\frac{1+qc}{qc}$. (c) The countermodel and derivation of Proposition 3.12. □

c9 Parts 1–2 confirm the rates to $8.9\cdot10^{-16}$; referee r4 reproduces them.

Lemma B.17 (Survivors prove $\forall x\varphi$ over Q)**.** [proved]
universal Lemma S1 *In* $\mathsf{C}_{\min}$ *with numeral elimination terms and* $\varphi=0+x=x$*, every finite set* T *of sentences with* $P_T(\varphi(S^j0))>0$ *for infinitely many* j *satisfies* $\mathsf{Q}\cup T\vdash\forall x\varphi$*.*

Proof of Lemma B.17. A chain cites some $A\in T$ and eliminates with numerals; T is finite, so one A yields $\varphi(S^j0)$ for infinitely many j. The conclusions are quantifier-free, so $A=\forall\bar{y}\chi$ with χ quantifier-free and $\chi[\bar{t}/\bar{y}]=(0+S^j0=S^j0)$ for numeral tuples with infinitely many j. Numerals contain only 0 and S, so χ is $\tau_0+\tau_1=\tau_2$ with τ_0 either 0 or a variable instantiated by 0 in every such output, and τ_1,τ_2 of the forms S^a0 or S^ay. As j varies, $\tau_1=S^ay_m$ and $\tau_2=S^by_{m'}$, and neither variable is that of τ_0 (which takes only the value 0). If $m=m'$, then $a=b$ and $A\vdash\forall y\varphi(S^ay)$ (instantiate τ_0's variable, if any, by 0); over Q, a uses of Q3 and the closed instances give $\forall x\varphi$. If $m\neq m'$, $A\vdash 0+S^a0=S^b0$ and $A\vdash 0+S^a0=S^{b+1}0$, so $S^b0=S^{b+1}0$, which Q refutes; then $\mathsf{Q}\cup T$ is inconsistent. □

The track's proof treats only $\tau_0=0$; the case of a variable always instantiated by 0 is added here and does not change the conclusion.

Remark B.18 (Kreisel's conjecture). [known (secondary sources only)] universal §9 "All instances have short derivations, so $\forall x\varphi$ is derivable" resembles Kreisel's conjecture, reported open for standard formalisations of PA, settled for some (Parikh, 1973; Baaz and Pudlák, 1993) and false for some theories close to PA (Hrubeš, 2007) (none of these was opened). Split_1 shows that the naive version fails without axioms such as Q3: every numeral instance has a one-step derivation, $\forall x\varphi$ none.

likelihood, class	quantity	$n=0$	20	10^2	10^3	10^4	10^5	10^6
L1, $\{H_\forall, \mathrm{Split}_1, \mathrm{Both0}\}$	$P(\vdash)$, pure	0.996	0.038	0.000	0.000	0.000	0.000	0.000
L1, $+\mathrm{Split}_{1..4}$, mem.	$P(\mathsf{Q}\cup T\vdash)$	0.338	0.926	0.947	0.475	0.000	0.000	0.000
	MAP	mem	Split_1	Split_3	mem	mem	mem	mem
L1, $+\mathrm{Split}_{1..40}$, mem.	$P(\vdash)$, pure	0.336	0.024	0.000	0.000	0.000	0.000	0.000
	$P(\mathsf{Q}\cup T\vdash)$	0.338	0.926	0.947	1.000	1.000	1.000	1.000
	MAP	mem	Split_1	Split_3	Split_6	Split_9	Split_{12}	Split_{15}
$\mathsf{L0}^{\mathrm{cl}}$, $+\mathrm{Split}_{1..40}$, mem.	$P(\vdash)$, pure	0.336	0.997	0.999	1.000	1.000	1.000	1.000

Table 14: Sentence-only class (c9 Part 3): $\varphi = 0 + x = x$, $\mathsf{C}_{\min}$, $c = 0.3$, numerals with $q = \frac{1}{2}$, Dirichlet($\frac{1}{2}$), prior $2^{-(\text{symbols}+1 \text{ per axiom})}$, memoriser class mass 2^{-6} (lower bracket), 20 seeds, MAP = mode; the larger classes also contain $H_\forall$ and Both0. With $\mathrm{Split}_{1..40}$ under L1 the MAP is $\mathrm{Split}_{18}, \mathrm{Split}_{21}, \mathrm{Split}_{24}, \mathrm{Split}_{27}, \mathrm{Split}_{34}$ at $n = 10^7, \ldots, 10^{10}, 10^{12}$ with $P(\mathsf{Q}\cup T\vdash) = 1.000$, also with the upper bracket and class mass $\frac{1}{2}$, with $\alpha = 1$ and with $\lambda = 2$ (c9 Part 4).

The race (c9 Part 5). Log odds of the split family ($k \le 40$) against the memoriser family, mean of 10 seeds: 16.5 nats at $n = 10^2$, 54.7 at 10^4, 105.1 at 10^6, 193.5 at 10^{10} and 286.4 at 10^{12} (minimum over seeds negative only at 10^2), of the order of $L \log_2 L$ with $L = \log_2 n$. *Heuristic (conjecture).* The best memoriser pays about L^2 bits of prior and $(L/2)\ln n$ nats of Dirichlet cost; the best split wastes about $nq^k \ln \frac{1+c}{c}$ nats, balancing the $\Theta(L)$ cost of one more level at $k \approx L - \log_2 L$, and so saves about $\log_2 L$ levels of $\Theta(L)$ nats. *History.* The first version's "$= 1$ over Q" came from $\{H_\forall, \mathrm{Split}_1, \mathrm{Both0}\}$, where all members prove $\forall x\varphi$ over Q; the referee's "$\to 0$ over Q" holds with depth ≤ 4, not with depth ≤ 40 to $n = 10^{12}$.

C Proofs and computations for Section 4

Proofs follow the model track's final notes; steps they leave out are named. The checks are the seeded scripts of `research/tracks/model/checks/` (c2–c16), each with its output file; the referee's independent scripts (r1–r7) are in `referee_code/`. The model track's referee re-did the proofs of Theorem 4.2, Corollaries 4.3 and 4.4, Proposition C.1, Proposition 4.5, Proposition 4.6(a), (b), Theorem 4.16, Examples 4.17 and 4.19 and Lemma 4.18, Theorem 4.7(a), Proposition 4.9, Proposition 4.10, Proposition 4.14(a) and Theorem 4.21 line by line and found no error beyond the corrections recorded in Remark 4.8 and Appendix H. Results added after the referee report were checked by the track and again while writing, not by a referee; the complete list is in Appendix H.1. Nothing is checked in a proof assistant.

C.1 Consistency and rates

Proof of Theorem 4.2. *Step 1 (joint law; Lévy).* Let Π be the law of $(T, X_1, X_2, \ldots)$ on $\mathcal{H} \times \mathcal{S}^\infty$ with $\Pi(T = t, X \in E) = \pi(t)P_t^\infty(E)$, and $M := \sum_t \pi(t)P_t^\infty$ its data marginal. Since

$\mathcal{H}$ is countable, Bayes' rule gives $\pi_n(A) = \Pi(T \in A \mid X_1, \dots, X_n)$ on the M-full event where $M(D_n) > 0$. By Lévy's upward theorem (known), for each fixed $A \subseteq \mathcal{H}$, $\pi_n(A) \to \Pi(T \in A \mid X_\infty)$ Π-a.s.; the convergence event is a data event, so it holds M-a.s. *Step 2 (the data determine the generator).* For $x \in \mathcal{S}^\infty$ let $F(x)(s) := \lim_n n^{-1}\#\{i \le n : x_i = s\}$, and $F(x) := \perp$ if some limit fails to exist or the limits do not sum to 1. By the strong law applied to the countably many s, $P_t^\infty(F = P_t) = 1$ for each t (on that event the limits are the numbers $P_t(s)$, which sum to 1). Hence $\Pi(P_T = F(X)) = 1$. *Step 3 (the limit lives on $\{T : P_T = F(X)\}$).* Choose a version of $\{\Pi(T = t \mid X_\infty)\}_t$ that sums to 1 a.s. Each indicator $1\{P_t = F(X)\}$ is X_∞-measurable, so $\mathbb{E}[1\{P_T = F(X)\} \mid X_\infty] = \sum_t 1\{P_t = F(X)\}\Pi(T = t \mid X_\infty)$ by monotone convergence. The left side is 1 a.s. by Step 2. $F(X)$ ranges Π-a.s. over the countable set $\{P_t : t \in \mathcal{H}\}$; apply Step 1 to the countably many sets $A_\nu := \{t : P_t = \nu\}$ and intersect the full-measure events. Then, M-a.s., $\lim_n \pi_n(A_{F(X)}) = \Pi(T \in A_{F(X)} \mid X_\infty) = 1$. *Step 4 (transfer to the truth).* $M \ge \pi(T^*)P_{T^*}^\infty$, so every M-null event is $P_{T^*}^\infty$-null and Step 3 holds $P_{T^*}^\infty$-a.s. Under $P_{T^*}^\infty$, $F(X) = P_{T^*}$ a.s. by Step 2, so $A_{F(X)} = C^*$, and $\pi_n(C^*) \to 1$. *Pointwise and total variation.* Write $q_T := \pi(T)1[T \in C^*]/\pi(C^*)$. By Corollary 4.3, $\pi_n(T) = \pi_n(C^*)\, q_T$ for $T \in C^*$, so

$$\textstyle\sum_T |\pi_n(T) - q_T| = \sum_{T \in C^*} q_T\,(1 - \pi_n(C^*)) + \pi_n(\mathcal{H} \setminus C^*) = 2\,(1 - \pi_n(C^*)) \to 0.$$

□

The universal track proves the same statement independently (its Lemma U7, total variation via Scheffé's lemma); the exact identity above is ours and gives the same conclusion directly.

Proof of Corollary 4.3. Every $T \in C^*$ has $P_T = P_{T^*}$ as a law on $\mathcal{S}$, so $P_T(D_n) = P_{T^*}(D_n)$ for every sequence, and the posterior ratio is the prior ratio. □

Computed (c9b; every likelihood by matching; three seeds, $n \le 2000$): $\{z+0 = z\}$ and its root split with weights $p_f = (0.5, 0.3, 0.1, 0.1)$ agree to $5.68 \cdot 10^{-14}$ in log-likelihood; an over-general template, a split missing a root and T^* plus the spare $0 = S0$ (weight 0.05, decaying as 0.95^n) go to 0. An earlier check (c9) was tautological for the split (referee m5).

Proof of Corollary 4.4. Put $A := \{T : s \in \mathrm{Th}_d(T)\}$. Then $\pi_n(A) = \pi_n(A \cap C^*) + \pi_n(A \setminus C^*)$. The second term is at most $\pi_n(\mathcal{H} \setminus C^*) \to 0$. By Corollary 4.3 the first equals $\pi_n(C^*)\pi(A \cap C^*)/\pi(C^*)$. Both hold on the single event of Theorem 4.2, so simultaneously for all (s, d). For the two cases named after the corollary: under L0 with full-support $\mathcal{Q}$ and positive weights, $\operatorname{supp} P_T = \bigcup \mathrm{inst}(T)$, so every $T \in C^*$ has T^*'s instance union and hence its Th_d; under L1 with parameters admissible, $\operatorname{supp} P_T = \mathrm{Th}(T)$, so every $T \in C^*$ has $\mathrm{Th}(T) = \mathrm{Th}(T^*)$. □

Proposition C.1 (Rates)**.** [proved]

model Prop 2.4 *Assume W. (a) For $T \notin C^*$, $\pi(T) > 0$: $\frac{1}{n}\ln\frac{\pi_n(T^*)}{\pi_n(T)} \to \mathrm{KL}(P_{T^*} \| P_T) \in (0, \infty]$ a.s. (b) $\sum_{i=1}^n \mathbb{E}\,\mathrm{KL}(P_{T^*} \| M_{i-1}) \le \ln\frac{1}{\pi(C^*)}$. (c) For $A \subseteq \mathcal{H} \setminus C^*$ and $\rho_T := \sum_s (P_{T^*}(s)P_T(s))^{1/2}$, $\mathbb{E}[(\pi_n(A)/\pi_n(T^*))^{1/2}] \le \sum_{T \in A}(\pi(T)/\pi(T^*))^{1/2}\rho_T^n$; so $P[\pi_n(A) \ge \varepsilon] \le C\rho^n\varepsilon^{-1/2}$ if $\sup_A \rho_T \le \rho < 1$ and $C := \sum_{T \in A}(\pi(T)/\pi(T^*))^{1/2} < \infty$ (true for π_λ, $\lambda \ge 2$).*

The rate (c) degenerates for spare templates ($\rho_T \ge (1-\varepsilon)^{1/2}$ for a spare of weight ε), whose decay the weight prior sets (Proposition 4.14).

Proof of Proposition C.1. (a) $\ln\frac{\pi_n(T^*)}{\pi_n(T)} = \ln\frac{\pi(T^*)}{\pi(T)} + \sum_{i \le n} Y_i$ with $Y_i := \ln\frac{P_{T^*}(X_i)}{P_T(X_i)}$ i.i.d. Their negative part is integrable, since $(\ln x)^+ \le x$ gives $\mathbb{E}Y^- = \mathbb{E}(\ln\frac{P_T}{P_{T^*}})^+ \le \mathbb{E}\frac{P_T(X)}{P_{T^*}(X)} \le 1$, and

$\mathbb{E}Y = \mathrm{KL}(P_{T^*}\|P_T) > 0$ by Gibbs' inequality, as $P_T \neq P_{T^*}$. The strong law holds with limit in $(0,\infty]$. If $P_T(X_i) = 0$ for some i, the ratio is $+\infty$ from then on. (b) $M(D_n) \geq \pi(C^*)P_{T^*}(D_n)$. By the chain rule $\ln \frac{P_{T^*}(D_n)}{M(D_n)} = \sum_{i\leq n} \ln \frac{P_{T^*}(X_i)}{M_{i-1}(X_i)}$, and the conditional expectation of the i-th term given D_{i-1} is $\mathrm{KL}(P_{T^*}\|M_{i-1})$. Take expectations. (c) $\frac{\pi_n(A)}{\pi_n(T^*)} = \sum_{T\in A} \frac{\pi(T)}{\pi(T^*)} L_T$ with $L_T := \prod_{i\leq n} \frac{P_T(X_i)}{P_{T^*}(X_i)}$. Use $(\sum a_T)^{1/2} \leq \sum a_T^{1/2}$ and, by independence, $\mathbb{E}L_T^{1/2} = \rho_T^n$. For the probability bound, $\pi_n(A) \leq \pi_n(A)/\pi_n(T^*)$ and Markov's inequality applied to the square root. □

For the degeneration: if $T = T^* \cup \{\tau\}$ with weight ε on τ and T^*'s weights scaled by $1-\varepsilon$, then $P_T \geq (1-\varepsilon)P_{T^*}$, so $\rho_T \geq (1-\varepsilon)^{1/2} \sum_s P_{T^*}(s) = (1-\varepsilon)^{1/2}$.

C.2 Identification of the generator

Proof of Proposition 4.5. $\mathrm{Th}(T_1) = \mathrm{Th}(T_2)$: b follows from a and $a \to b$ by MP, and $a \to b$ follows from b by an A1 instance and MP. Under T_2: b has root $\neg$; every instance of a logical axiom has root $\to$, $\forall$ or $=$, and b is not an axiom of T_2; so no one-node tree concludes b, and $\mu_{T_2}(b) \leq \Pr[\text{the root is a rule node}] = \alpha_{\mathrm{r}}$. With $Z_{T_2} \geq 1 - \alpha_{\mathrm{r}}$ (Lemma 2.10(b)), $P_{T_2}(b) \leq \alpha_{\mathrm{r}}/(1-\alpha_{\mathrm{r}})$. Under T_1 the one-node tree citing b has probability $\alpha_{\mathrm{ax}}/2$, and $Z_{T_1} \leq 1$. The same argument holds under L2. □

Computed (c6; A1–A3 and MP, formulas of size ≤ 3, depth ≤ 2; rule probabilities for (ax, lg, MP) $(0.6, 0.25, 0.15)$ in the proved region, $(0.5, 0.3, 0.2)$ on its boundary, $(0.3, 0.3, 0.4)$ and $(0.2, 0.4, 0.4)$ outside): $P_{T_1}(b) = 0.16$–0.35, $P_{T_2}(b) = 0.007$–0.04, TV $= 0.16$–0.35; the referee's implementation (r6) confirms both bounds. A route to all parameters, not carried out: $P_T(s)$ is a power series in the rule probabilities with nonnegative coefficients, so the difference is real-analytic on the subcritical region and nonzero near $\alpha_{\mathrm{r}} = 0$.

Proof of Proposition 4.6. (a) $\operatorname{supp} P_{T,w} = \bigcup \mathrm{inst}(T)$ by Lemma A.2(c) with positive weights, and Th_d depends only on the instance union (Definition 2.2). (b) *Templates.* Substituting the body into a pattern occurrence $M(\bar{y})$ gives $r(N_1(\bar{y}, \dots), \dots)$; each N_j has a pattern occurrence, because $\bar{y}$ are distinct bound variables and a binder production adds one more. Other occurrences of M receive the same body with metavariable-free arguments, which DT° allows. So $\tau_r \in \mathrm{DT}^\circ$. *Instances.* $\mathrm{inst}(\tau) = \bigsqcup_r \mathrm{inst}(\tau_r)$: at a pattern occurrence of M an instance shows the root production of $\theta(M)$, and distinct productions give distinct subtrees there (for a hole production z_i, the distinct bound variable y_i). *Probabilities.* For $s \in \mathrm{inst}(\tau_r)$ with τ's matcher θ, the matcher of τ_r gives the children's bodies of $\theta(M)$, and the root factorisation gives $\mathcal{Q}(\theta) = p_r\, \mathcal{Q}_{\tau_r}(s)$. (Holes and bound indices are drawn alike, so a child body under a binder has the law of a body with one more argument.) Hence $w_\tau \mathcal{Q}_\tau(s) = \sum_r w_\tau p_r \mathcal{Q}_{\tau_r}(s)$, and the other components are unchanged. *Under* L1, L2. A theory-citation node's conclusion has the same law under (T, w) and (T', w'), by the L0 statement; every other node has the same law, and a tree's validity and conclusion depend only on its nodes' conclusions. So $\mu_{T'} = \mu_T$ and $Z_{T'} = Z_T$. (c) A renaming permutes the factors of $\mathcal{Q}(\theta)$; an argument permutation permutes the holes, which a hole-symmetric $\mathcal{Q}$ ignores. The converse is (a) for single templates; for first-order templates with only 0-ary term metavariables, AS Prop C.1 (under its richness condition) turns equal instance sets into $\tau \equiv \tau'$. That proposition assumes one side has only 0-ary term metavariables (referee m10), so it says nothing about templates such as A1 $= B \to (C \to B)$. □

With *learned* weights the likelihoods differ; that case is Propositions 4.10 and 4.11. Track pa's Prop 2.1 is (b) for T_{Ind} with a shared grammar $\mathcal{Q}(\varphi) = \mathcal{Q}(f)\mathcal{Q}(\cdot \mid f)$: an induction datum with motive root f has probability $w\mathcal{Q}(f)\mathcal{Q}(\varphi \mid f) = w\mathcal{Q}(\varphi)$ under both.

Computed (c2, Part A): $z+0=z$ against its split over $0, S, +, \cdot$ with root law $(0.5, 0.3, 0.1, 0.1)$; maximal difference $1.08\cdot 10^{-19}$ over all 2849 instances with $|t| \le 9$, every instance matched against every template.

Remark C.2 (Chain likelihood, latent citations). [proved; computed] experiments Prop X13, E8 In the chain Ch_J the tie holds for a 0-ary closed-guard term metavariable of a component that is not an MP major premise; in E8, where a datum $\psi(t)$ may be cited or derived from $\varphi(t)$ by MP, the two log-probabilities agree to $1.1 \cdot 10^{-13}$. For E6 see Remark 7.7.

Proof of Remark C.2 (experiments Prop X13). Under $\mathcal{Q}$ a closed term has root f with probability q_f and, given the root, children drawn independently from the same closed-term law (every node of a closed term has the same context). So "cite τ with probability w_τ and draw the body of u from $\mathcal{Q}$" and "cite τ_f with probability $w_\tau q_f$ and draw the bodies of $u_1, \dots, u_r$ from $\mathcal{Q}$" give the same law on the cited sentence, jointly with the bodies of τ's other metavariables. Every later chain step depends only on the current sentence and on the set of MP components, which is the same in T and T' because neither τ nor any τ_f is an MP component. □

C.3 Identification in the limit and Gold

Proof of Theorem 4.7(a). By Theorem 4.2, $\pi_n(C^*) \to 1$, and every $T \in C^*$ has $\operatorname{supp} P_T = \operatorname{supp} P_{T^*}$. At most one L can have posterior mass above $\frac{1}{2}$; once $\pi_n(C^*) > \frac{1}{2}$ it is $\operatorname{supp} P_{T^*}$, which is $\bigcup \operatorname{inst}(T^*)$ under $\mathsf{L0}$ (Lemma A.2(c)) and $\operatorname{Th}(T^*)$ under $\mathsf{L1}$ (Lemma 2.10(c)). □

Proof of Theorem 4.7(b). The proof of Theorem 4.2, with the frequency functional replaced by a support functional. *A countable parameter that the data determine.* Put $U(T,w) := \bigcup \operatorname{inst}(T)$; it takes countably many values. For a sequence x let $F'(x) := \{x_i : i \ge 1\}$. For w in the open simplex, $\operatorname{supp} P_{T,w} = \bigcup \operatorname{inst}(T)$ is a countable set of points of positive probability, each of which occurs a.s., and nothing else occurs; so $P^\infty_{T,w}(F' = U(T,w)) = 1$. Dir_T charges only the open simplex. Hence $\Pi(U = F'(X)) = 1$ for the joint law Π of (T, w, X). *Steps 1 and 3.* The posterior given D_n is $\Pi(\cdot \mid D_n)$ (Bayes' formula with the likelihood as a density with respect to the prior). For each of the countably many sets $A_\nu := \{(T,w) : \bigcup \operatorname{inst}(T) = \nu\}$, Lévy's theorem gives $\pi_n(A_\nu) \to \Pi(A_\nu \mid X_\infty)$ M-a.s. Since $F'(X)$ is X_∞-measurable, $1 = \mathbb{E}[1\{U = F'(X)\} \mid X_\infty] = \sum_\nu 1\{F'(X) = \nu\}\, \Pi(A_\nu \mid X_\infty)$ a.s. So, M-a.s., $\pi_n(A_{F'(X)}) \to 1$. *Step 4, changed.* Let E be the M-null event where this fails. Since $M \ge \pi(T^*) \int P^\infty_{T^*,w}(\cdot)\, \mathrm{Dir}_{T^*}(dw)$, we get $\int P^\infty_{T^*,w}(E)\, \mathrm{Dir}_{T^*}(dw) = 0$, so $P^\infty_{T^*,w}(E) = 0$ for Dir_{T^*}-a.e. w, which is Lebesgue-a.e. w because the Dirichlet density is positive on the open simplex. For such w^*, a.s. $F'(X) = \bigcup \operatorname{inst}(T^*)$ and $\pi_n(A_{\bigcup \operatorname{inst}(T^*)}) \to 1$. □

The pre-referee proof said that the steps of Theorem 4.2 "go through with the countable sum replaced by an integral". That sentence is withdrawn: Step 3 needs a countable family of sets, which the support functional provides and the law itself does not (referee M4, who suggested this route).

Sketch of Theorem 4.7(b′). $P_{T,w}$ lies in the Polish space of probability vectors on $\mathcal{S}$ (ℓ^1 metric), and the frequency functional F recovers it a.s. For a countable convergence-determining family of bounded continuous functions g, Lévy's theorem gives that the posterior mean of $g(P)$ converges to $g(F(X))$ a.s. So the posterior law of $P_{T,w}$ converges weakly to the point mass at P^*, which gives the statement for total-variation balls; transfer to Lebesgue-a.e. w^* as in (b). Not

written out: the choice of the family and the passage from weak convergence to total-variation balls.

Proof of the counter-statement in Remark 4.8. The posterior on w given T is absolutely continuous with respect to Dir_T, so it suffices that each W_T below is Lebesgue-null. Fix T. The set $W_T := \{w : P_{T,w} = P^*\}$ is the intersection of the simplex with an affine subspace, since $w \mapsto P_{T,w}$ is affine. If it had positive Lebesgue measure in the simplex, the affine map would be constant on an open set, hence on the whole simplex, and every component of T would have law P^* (for $|T| = 1$ the condition is $\mathcal{Q}_\tau = P^*$). The posterior on w given T is absolutely continuous with respect to Dir_T. So the posterior mass of the exact class is 0 at every n unless some template σ has $\mathcal{Q}_\sigma = P^* = \sum_\tau w^*_\tau \mathcal{Q}_\tau$. For each σ, linear independence of $\{\mathcal{Q}_\tau : \tau \in T^*\}$ allows at most one such w^*, and there are countably many σ. □

Remark C.3 (Which computed posteriors are covered). [known (Doob's theorem in the (T, w) form); not covered: data-dependent pools] experiments Prop X5 Theorem 4.2 is stated for fixed laws. Run on the parameter (T, w) with prior $\pi(T)\mathrm{Dir}_T(dw)$, as in the proof of Theorem 4.7(b), Doob's argument gives concentration on $\{(T, w) : P_{T,w} = P_{T^*}\}$ for a one-component generator, which is an atom of the prior, in a fixed pool: this covers E1, E4 and E6 with their hand pools. The experiments' data-dependent pool members ($\mathrm{Mem}(D_n)$, causal theories) are covered by no consistency theorem here, only by the pool versions of Theorems 5.6 and 5.7. E2, E3(b), E5(b) and E8 sample at one fixed weight vector, where only Theorem 4.7(b) and conjecture (c) speak: a yardstick, not an instance of a theorem.

Remark C.4 (Memorisation and the bound). [proved; known (prior work, flagged)] model §5.2 A fixed memoriser $\mathrm{Mem}(E)$ dies once a datum outside E appears; before that it pays its prior and, with Dirichlet weights, about $\frac{|E|-1}{2} \ln n$ nats. The memoriser *class* can keep mass $\exp(-O(\ln^2 n))$ (Laplace-weighted memorisers, geometric numerals; Proposition B.6). So the Bayesian needs no bound on the number of templates, unlike the cautious learner (AS Prop 6.4). Identification from stochastic positive data goes back to Horning (1969) and Angluin (1988) (existence checked, content not).

Proof of Proposition 4.9. (a) is Theorem 4.2. (b) Enumerate $\mathcal{L}_\infty = \{e_1, e_2, \dots\}$ and build the text in stages. At stage i, with σ the prefix built so far, pick $k_i > k_{i-1}$ with $\sigma \cup \{e_i\} \subseteq \mathcal{L}_{k_i}$ (possible since the chain increases to $\mathcal{L}_\infty$) and append e_i; the new prefix σ' has positive P_{k_i}-probability. By (a), applied to $P^\infty_{k_i}$ conditioned on the positive-probability prefix σ', almost every continuation reaches $g_n = \mathcal{L}_{k_i}$ at some finite n; pick such a continuation, with elements in $\mathcal{L}_{k_i} \subseteq \mathcal{L}_\infty$, and cut it at a time where $g_n = \mathcal{L}_{k_i}$. The resulting sequence lists every e_i and only elements of $\mathcal{L}_\infty$, and $g_n = \mathcal{L}_{k_i} \neq \mathcal{L}_\infty$ at infinitely many times. (c) By (a) with $i = \infty$, P^∞_∞-a.s. $g_n = \mathcal{L}_\infty$ eventually. □

Computed (E5; Table 37): on data uniform on $\mathcal{L}_5$ the mass of $\mathcal{L}_5$ is $2 \cdot 10^{-6}$ at $n = 64$, 0.661 at 128 and 1.000 from 256 in all 10 seeds ($\mathcal{L}_5$ gains $3 - \log_2 5 \approx 0.68$ bits per datum, repaying 72 prior bits and its Dirichlet cost); on Gold's text against the MAP learner over $\{\mathcal{L}_1, \dots, \mathcal{L}_{60}, \mathcal{L}_\infty\}$ the first 14 stages end (740 data), which shows only that these stages end (experiments' referee m10). *The* $\mathrm{I}\Sigma_n$ *chain* (model §5.2, heuristic). Under L1 with theorem data, PA's generator charges $\mathrm{Th}(\mathsf{PA}) \setminus \mathrm{Th}(\mathrm{I}\Sigma_n)$, so $\mathrm{I}\Sigma_n$ is eventually rejected on PA-data; and PA's generator charges $\mathrm{Th}(\mathsf{PA}) \setminus \mathrm{Th}(\mathrm{I}\Sigma_5)$, so PA is eventually rejected on $\mathrm{I}\Sigma_5$-data. Both masses presumably are tiny, as they need long derivations with complex induction; if so, the number of data needed is of order $1/P_{\mathsf{PA}}(\mathrm{Th}(\mathsf{PA}) \setminus \mathrm{Th}(\mathrm{I}\Sigma_n))$. The masses were not estimated.

C.4 Splits

Proof of Proposition 4.10. (a) The split's instance sets are disjoint (Proposition 4.6(b)), so the labelling sum of Lemma 2.8 has one term, and $\mathcal{Q}_{\tau_f}(s) = \mathcal{Q}_\tau(s)/p_f$ for $s \in \mathrm{inst}(\tau_f)$. So $P^{\mathrm{Dir}}_{T'}(D) = \mathrm{DirMult}(n_f;\alpha)\prod_i \mathcal{Q}_\tau(X_i)/p_{f_i}$, while $P_T(D) = \prod_i \mathcal{Q}_\tau(X_i)$. (b) Stirling: $\ln\Gamma(a+m) = (a+m-\frac{1}{2})\ln m - m + \frac{1}{2}\ln 2\pi + o(1)$ as $m \to \infty$. Applied to $\ln\Gamma(K\alpha+n)$ and the K terms $\ln\Gamma(\alpha+n_f)$,

$$\begin{aligned}\ln \mathrm{DirMult}(n_f;\alpha) = {} & \textstyle\sum_f n_f \ln n_f - n\ln n + (\alpha-\frac{1}{2})\sum_f \ln n_f \\ & - (K\alpha-\tfrac{1}{2})\ln n + \tfrac{K-1}{2}\ln 2\pi + \ln\tfrac{\Gamma(K\alpha)}{\Gamma(\alpha)^K} + o(1).\end{aligned}$$

With $\ln n_f = \ln n + \ln \hat{r}_f$ the first two terms are $n\sum_f \hat{r}_f \ln \hat{r}_f$, and the $\ln n$ terms collect to $(K\alpha - \frac{K}{2} - K\alpha + \frac{1}{2})\ln n = -\frac{K-1}{2}\ln n$. Subtracting $\sum_f n_f \ln p_f$ turns $n\sum_f \hat{r}_f \ln \hat{r}_f$ into $n\mathrm{KL}(\hat{r}\|p)$. (c) Pearson's and Wilks's χ^2 limit (known), with all $n_f \to \infty$ a.s. since p is in the open simplex. (d) The strong law for $\hat{r}$; $\ln \mathrm{DirMult}(n_f;\alpha) = n\sum_f \hat{r}_f \ln \hat{r}_f + O(\ln n)$ also when some $r_f = 0$. (e) The prior. □

With $\alpha = \frac{1}{2}$ the term $(\alpha - \frac{1}{2})\sum \ln \hat{r}_f$ vanishes, and in the well-specified case $\mathbb{E}\Delta_n \approx \frac{K-1}{2} - \frac{K-1}{2}\ln n + c_{K,\alpha}$; for $K = 4$, $c_{4,1/2} = \ln \pi^{-2} + \frac{3}{2}\ln 2\pi = 0.467$, which gives the predictions of Table 15.

n	10^2	10^3	10^4	10^5	10^6
well specified, mean Δ_n	−4.93	−8.34	−11.77	−15.27	−18.85
prediction	−4.94	−8.39	−11.85	−15.30	−18.76
$P(\Delta_n > 0)$	0.007	0	0	0	0
misspecified, mean Δ_n	−0.06	39.7	482.2	4917.5	49356.3
prediction	−0.00	41.0	481.9	4921.9	49353.3

Table 15: Proposition 4.10, computed (model check c2, Part B): Δ_n in nats, $K = 4$, $\alpha = \frac{1}{2}$, $\mathcal{Q}$'s root law $p = (0.5, 0.3, 0.1, 0.1)$, 400 runs per n. Well specified: $r = p$. Misspecified: $r = (0.4, 0.3, 0.2, 0.1)$, $\mathrm{KL}(r\|p) = 0.0494$ nats; predictions from (b) with $\hat{r}$ replaced by r, plus $\frac{K-1}{2}$ (the mean of $n\mathrm{KL}(\hat{r}\|r)$) in both cases. The referee (r4) checked (b) for $\alpha = 0.3, 1, 2$ ($K = 3$): the residual tends to 0 like $1/n$.

Proof of Proposition 4.11. (a) Under (T', w') a theory-citation node chooses τ_r with probability w'_r and draws the children's bodies from $\mathcal{Q}$. By the root factorisation and $\mathrm{inst}(\tau) = \bigsqcup_r \mathrm{inst}(\tau_r)$, its conclusion has the law of a citation of τ whose M has root law w'. All other nodes have the same laws, and validity and conclusions depend only on the nodes' conclusions; so μ and Z coincide, under L0, L1 and L2. (b) By Tonelli,

$$\mathbb{E}[\mathrm{BF}_{n+1} \mid D_n] = \textstyle\int \prod_{i\le n} \frac{P^{(p')}}{P^{(p)}}(X_i) \sum_{x:P^{(p)}(x)>0} P^{(p')}(x)\,\mathrm{Dir}(dp') \le \mathrm{BF}_n.$$

Ville's inequality (known; IL Lemma C.4) and the supermartingale convergence theorem (known). (c) Write

$$\ln \mathrm{BF}_n = \ln \tfrac{P^{\mathrm{Dir}}_{T'}(D_n)}{P^{(r)}(D_n)} + \textstyle\sum_{i\le n} \ln \frac{P^{(r)}(X_i)}{P^{(p)}(X_i)}.$$

The second sum divided by n tends to $\mathrm{KL}(P^{(r)}\|P^{(p)})$ a.s., as in Proposition C.1(a). For the first term: *upper bound:* $P^{\mathrm{Dir}}_{T'}(D_n)/P^{(r)}(D_n)$ is a nonnegative supermartingale under $P^{(r)}$, as in (b);

it converges a.s. to a finite limit, so $\limsup_n n^{-1} \ln$ of it is ≤ 0. *Lower bound:* for $\gamma > 0$ let $U_\gamma := \{p' : \mathrm{KL}(P^{(r)} \| P^{(p')}) < \gamma\}$. By the continuity lemma below it contains a neighbourhood of the interior point r, so $\mathrm{Dir}(U_\gamma) > 0$; let Dir_U be the normalised restriction. By Jensen's inequality,

$$\ln \textstyle\int_{U_\gamma} \prod_i \frac{P^{(p')}}{P^{(r)}}(X_i)\, \mathrm{Dir}(dp') \ge \ln \mathrm{Dir}(U_\gamma) + \sum_i Y_i, \qquad Y_i := \int \ln \frac{P^{(p')}(X_i)}{P^{(r)}(X_i)}\, \mathrm{Dir}_U(dp').$$

The Y_i are i.i.d., and by Tonelli on the positive and negative parts

$$\mathbb{E}Y^+ \le \int \mathbb{E}\tfrac{P^{(p')}(X)}{P^{(r)}(X)}\, \mathrm{Dir}_U(dp') \le 1, \qquad \mathbb{E}Y = -\int \mathrm{KL}(P^{(r)} \| P^{(p')})\, \mathrm{Dir}_U(dp') \ge -\gamma.$$

By the strong law, $\liminf_n n^{-1} \ln[P^{\mathrm{Dir}}_{T'}(D_n)/P^{(r)}(D_n)] \ge -\gamma$ a.s.; let $\gamma \downarrow 0$ along a sequence. *Continuity lemma:* $\mathrm{KL}(P^{(r)} \| P^{(p')}) \to 0$ *as* $p' \to r$. For a tree π, $\Pr_{p'}(\pi)/\Pr_r(\pi) = \prod p'_f/r_f$ over π's theory-citation leaves, f the root production chosen there. Let $L(\pi)$ be the number of these leaves, $\varrho := \min_f p'_f/r_f \le 1$ and $\varrho' := \max_f p'_f/r_f \ge 1$. Then $\mu_{p'}(x) \ge \sum_{\pi \to x} \varrho^{L(\pi)} \Pr_r(\pi) \ge \mu_r(x)\, \varrho^{\mathbb{E}_r[L|x]}$ (Jensen, $\mathbb{E}_r[\cdot \mid x]$ over the valid trees concluding x), and $Z_{p'} \le \mathbb{E}_r[\varrho'^L; \mathrm{valid}]$. So

$$\mathrm{KL}(P^{(r)} \| P^{(p')}) \le \ln \tfrac{1}{\varrho} \cdot \mathbb{E}_r[L \mid \mathrm{valid}] + \ln \frac{\mathbb{E}_r[\varrho'^L; \mathrm{valid}]}{Z_r}.$$

Here $\mathbb{E}_r[L \mid \mathrm{valid}] \le \mathbb{E}_r[N]/Z_r < \infty$, with N the number of derivation nodes ($m < 1$). And $\mathbb{E}_r[\varrho'^L; \mathrm{valid}] \to Z_r$ as $\varrho' \downarrow 1$ by dominated convergence, since $L \le N$ and $\mathbb{E}_r[e^{\zeta N}] < \infty$ for small $\zeta > 0$ (Lemma A.2(d) for the derivation tree). Both terms tend to 0 as $p' \to r$. Under L2, $L \le 2^d$ and the lemma is immediate. □

part	result
(a)	split with weights $(u, 1-u)$ against the schema with root law u, $u \in \{0.1, 0.37, 0.8\}$: maximal difference $1.05 \cdot 10^{-14}$
latency	88 of 5885 conclusions depend on u through derived trees
(b)	$\delta' = 0.5, 0.2, 0.05$: $P(\max_{n \le 3000} \mathrm{BF}_n \ge 1/\delta') = 0.427, 0.133, 0.036$ (1000 runs, every n checked)
(c)	data root law 0.3, schema 0.6: KL = 0.11487 nats; mean $n^{-1} \ln \mathrm{BF}_n =$ 0.11301, 0.11460, 0.11464 at $n = 10^3, 10^4, 10^5$
(d)	mean $\ln \mathrm{BF}_n = -1.765, -6.405$ at $n = 10^2, 10^6$ (400 runs); slope -0.507 (-0.498 on $n \ge 10^4$) against -0.5

Table 16: Proposition 4.11, computed (model check c13): a propositional L2 model, depth 2, atom a, logical axioms A1–A3 instantiated from a grammar truncated to size ≤ 5, MP; $\tau = F \to F$ with F rooted at a or $\neg$, $K = 2$, root law $p_a = 0.6$; 5885 possible conclusions. Citations are latent only weakly here.

C.5 Spare templates

Proof of Proposition 4.14(a). The labellings of the data by templates of $T^* \cup \{\sigma\}$ are those for T^*, with $n_\sigma = 0$. For each labelling, the Dirichlet moment of Lemma 2.8 for $T^* \cup \{\sigma\}$ divided by that for T^* is $\frac{\Gamma(A+\alpha_\sigma)}{\Gamma(A+\alpha_\sigma+n)} \frac{\Gamma(\alpha_\sigma)}{\Gamma(\alpha_\sigma)} \Big/ \frac{\Gamma(A)}{\Gamma(A+n)}$, the same for every labelling; factor it out. Asymptotics: $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$. □

For a symmetric prior ($\alpha_\tau = \alpha_\sigma = \alpha$, $A = K\alpha$) this is experiments' Prop X7(a) and, at $\alpha = \frac{1}{2}$, pa's Prop 5.1: $\log_2$ of the inverse is $\frac{1}{2}\log_2 n + \log_2 \frac{\Gamma(K/2)}{\Gamma((K+1)/2)} + o(1)$ bits. Its prediction for pa's code is compared with pa's computed spare costs in Proposition F.21.

Proof of Proposition 4.14(d1). $P_{T^*\cup\{\sigma\},w} = \sum_{\tau\neq\tau_1} w_\tau \mathcal{Q}_\tau + (w_{\tau_1} + w_\sigma)\mathcal{Q}_{\tau_1}$ for every w. By the aggregation property of Dirichlet laws, $(w_{\tau_1} + w_\sigma, (w_\tau)_{\tau\neq\tau_1}) \sim \mathrm{Dir}(\alpha_1 + \alpha_\sigma, (\alpha_\tau)_{\tau\neq\tau_1})$. So $P^{\mathrm{Dir}}_{T^*\cup\{\sigma\}}$ is the Dirichlet marginal of T^* with α_1 replaced by $\alpha_1 + \alpha_\sigma$, and with disjoint instance sets the one-labelling formula of Lemma 2.8 gives, for every D_n with n_1 data in $\mathrm{inst}(\tau_1)$, exactly

$$\mathrm{BF}^\sigma_n = \frac{\Gamma(A+\alpha_\sigma)\Gamma(A+n)}{\Gamma(A)\Gamma(A+\alpha_\sigma+n)} \cdot \frac{\Gamma(\alpha_1+\alpha_\sigma+n_1)\Gamma(\alpha_1)}{\Gamma(\alpha_1+\alpha_\sigma)\Gamma(\alpha_1+n_1)}.$$

The limit follows from $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$ and $n_1/n \to w^*_1$ a.s. □

We checked the limits in Table 17 against this formula: with $A = 1$, $\alpha_1 = \frac{1}{2}$, $w^*_1 = \frac{1}{2}$ it gives $\ln \mathrm{BF}^\sigma_\infty = \ln(\frac{\pi}{2} \cdot 2^{-1/2}) = 0.1050$ at $\alpha_\sigma = \frac{1}{2}$, 0 at $\alpha_\sigma = 1$ and $\ln \frac{2}{3} = -0.4055$ at $\alpha_\sigma = 2$, as c15 reports.

Sketch of Proposition 4.14(b), (c). Write $w = ((1-u)\tilde{w}, u)$ with $u \sim \mathrm{Beta}(\alpha_\sigma, A)$ independent of $\tilde{w} \sim \mathrm{Dir}(\alpha_\tau)$ (aggregation). The per-datum factor is $1 - u + u\,\mathcal{Q}_\sigma(X)/P_{T^*,\tilde{w}}(X)$. In (b), at $\tilde{w} = w^*$ its mean is $1 - uc$, so the likelihood decays like e^{-ncu} in u, and $\int u^{\alpha_\sigma - 1} e^{-ncu} du \sim \Gamma(\alpha_\sigma)(cn)^{-\alpha_\sigma}$. In (c) the mean is 1 (zero drift) and the log-likelihood in u is locally $-nIu^2/2 + \sqrt{n}\,Zu$ with $I = \sum_s \mathcal{Q}_\sigma(s)^2/P^*(s) - 1$; this quadratic expansion is where the finite-χ^2 hypothesis of (c) is used. The integral against $u^{\alpha_\sigma - 1}$ scales as $n^{-\alpha_\sigma/2}$. Not written out: uniformity in $\tilde{w}$ near w^*, and almost-sure or in-probability control of the fluctuations. This is a boundary Laplace approximation of the kind analysed for overfitted mixtures by Rousseau and Mengersen (2011), whose setting has continuous component parameters (checked by the model track's referee); their result is not applied here.

Sketch of Proposition 4.14(d2). T^* and $T^* \cup \{\sigma\}$ parametrise the same family of laws, so the two marginals integrate the same likelihood against two prior densities on that family, and Laplace's method gives the ratio of the densities at P^*. Not written out: the Laplace step.

case	source	result
(a)	E5(a)	$0 = S0$ never cited, $u \cdot 0 = 0$ never matched: $\log_2$ BF -1.00, -6.83 at $n = 1$, 4096, equal to the formula on every seed; slope -0.500
(a), (b)	c3, r4	formula matched to 4 decimals; slopes -0.499, -0.498; referee: $-\alpha_\sigma$ to 3 decimals for $\alpha_\sigma = \frac{1}{2}, 1, 2$
(c)	E5(a2)	nested $\varphi(Sz)$ in $\{\varphi(z)\}$: mean $\log_2$ BF -2.196 at $n = 10^2$, -7.200 at 10^8 (1000 draws per n); slope -0.251 ± 0.002
(c)	c3b, r4	-0.265 on $10^3 \ldots 10^7$ (s.e. ≈ 0.07 per point); referee -0.230, -0.523, -0.961 for $\alpha_\sigma = \frac{1}{2}, 1, 2$
(d1)	c15	formula equals the numerical integral to 5 decimals; at $n = 10^6$ within 0.003 of the limit
(d2)	c15	$w^*_1 = 0.3$: $\ln \mathrm{BF}^\sigma_n$ changes by -0.031, -0.021, -0.004 per decade; $w^*_1 = \frac{1}{2}$ ($P^* = \mathcal{Q}_\sigma$, density infinite): $+0.22$, $+0.20$, $+0.13$

Table 17: Spare templates, computed. Slopes are of the mean log Bayes factor against the log of n in the same base, so they do not depend on the base. The primary estimate for (c) is E5(a2); its rate remains a proof sketch.

data (true unless stated)	likelihood	posterior lands on	where
same theorems or instances, other axioms			
instances, usage that $\mathcal{Q}$ misfits	L0, L1	root splits; depth-2 nestings	Props. 4.10 and 4.11 and Rem. 4.20
PA instances, connective-rich motives	L0	$T^* + T_\exists$; observed-root fragments	Rem. 7.17
deductively weaker			
closed instances of $\forall x(0 + x = x)$	L0	closed schema $0 + z = z$	Ex. 4.17
instances: numerals only; finite set	L0 (L1 to $n = 512$)	R_m, m growing; $\mathrm{Mem}(D_n)$ (L0, $n = 1024$)	Rem. 4.20
PA instances, atomic motives	L0	$\mathsf{Q} + \{T_=, T_<\}$	Rem. 7.17 and Prop. 7.16
narrow induction practice	two-part code	$\mathsf{Q} + T_E + T_N \subseteq \mathrm{I}\Sigma_1$ (extrapolated)	Ex. 7.14
theorems, no direct induction uses	$\mathsf{L1}^{\mathrm{sch}}$	Q + a finite library	Rem. F.10
unsound			
derived equations $t = 0$, finite class	L0	$T_{\mathrm{esc}} = \{z = 0\}$	Ex. 4.19
the same	$\mathsf{L1}^{\mathrm{eq}}$	T_{esc} or $T_{\mathsf{Q}} \cup \{z = 0\}$ (16/16 at $p_+ = 0.25$)	Ex. C.6
the same, full class	L0, Dir.	memorisation–T_{esc} hybrids (conjecture)	Rem. C.5
prime values of $k^2 + k + 41$	$\mathsf{L1}^{\mathrm{sch}}$	the false schema (counterexample absent)	Ex. F.26
10% false near misses (not true)	L0, L1	verifier accepts falsehoods, 100/100	Rem. 5.10
small n, rarely cited axioms	L0; $\mathsf{L1}^{\mathrm{sch}}$	unsound lumps (E2: unsound MAP in 7/25 seeds at $n = 8$; false acceptance in 5/25)	Rems. 5.11 and F.23

Table 18: Kinds of misspecification and where the posterior lands; statuses are those of the cited items. Computed results hold within their pools or candidate sets. T_f: T_{Ind} with motive root f; T_E, T_N: the two induction templates of narrow practice, on motives $\exists z(s = t)$ and $\neg(s = t)$ (Ex. 7.14).

C.6 Misspecification

Proof of Theorem 4.16. On $\operatorname{supp} P_H$, $\ln(P_T/P_{T_0}) = Y_0 - Y_T$ with $Y_T := \ln(P_H/P_T)$. Y_T has an integrable negative part, $\mathbb{E}(\ln \frac{P_T}{P_H})^+ \le \mathbb{E}\frac{P_T}{P_H} \le 1$, and mean K_T; $Y_0 \in L^1$ because also $K_{T_0} < \infty$. By the strong law, $n^{-1}\sum_{i\le n}(Y_{0,i} - Y_{T,i}) \to K_{T_0} - K_T$ a.s., the value $-\infty$ allowed when $K_T = \infty$. $\mathcal{H}$ is finite. □

Inside $\mathbb{M}$, members with equal KL but different laws trade places like a zero-drift random walk.

Proof of Example 4.17. $P_H = P_\sigma$, so Theorem 4.2 applies with truth σ. Every $T \in C_\sigma$ has $\bigcup \mathrm{inst}(T) = \mathrm{inst}(\sigma) = I_{\mathrm{c}}(0 + x = x)$ by Proposition 4.6(a) (parameters not admissible), hence $\mathrm{Th}(T) = \mathrm{Th}(I_{\mathrm{c}}(0 + x = x))$. Take the structure $\mathbb{N} \cup \{a\}$, standard on $\mathbb{N}$, with $0 + a := 0$ and the other operations extended arbitrarily, for instance $Sa := a$ and $a \circ y := y \circ a := a$ for $\circ \in \{+, \cdot\}$ except $0 + a = 0$. Closed terms denote standard numbers, so every closed instance holds; $\forall x(0 + x = x)$ fails at a. By Corollary 4.4 the posterior probability that $T \vdash \forall x(0 + x = x)$ tends to 0. □

Proof of Lemma 4.18. Let τ cover some $t = 0 \in Z$. *Shape.* The datum has no binders, so τ's rigid skeleton has none. A pattern occurrence of positive arity needs bound variables in scope, so every metavariable is 0-ary. The only formula position in $t = 0$ is the root. *Root metavariable.* Then τ is a formula metavariable, with the false instance $0 = S0$. *Otherwise* $\tau = (u = v)$ with term templates u, v. u matches t, so its rigid nodes are $+$ and 0 and its metavariable occurrences are leaves; hence $\mathrm{val}(u\theta) = \sum_j c_j \,\mathrm{val}(\theta(z_j))$, c_j the multiplicity of z_j in u. v matches 0, so v is 0 or a metavariable z. $v = 0$. Soundness requires $\sum_j c_j \mathrm{val}(\theta(z_j)) = 0$ for all θ; $\theta(z_j) := S0$ for all j shows $c_j = 0$. So τ is ground and covers one sentence. $v = z$, *z not in u.* The value of $u\theta$ does not depend on $\theta(z)$; one of $\theta(z) \in \{0, S0\}$ gives a false instance. $v = z$, *z in u.* Soundness requires $\sum_j c_j \mathrm{val}(\theta(z_j)) = \mathrm{val}(\theta(z))$ for all θ; unit test values give $c_z = 1$ and $c_j = 0$ for $j \neq z$. So u contains z once and is otherwise ground; covering $t = 0$ forces $\theta(z) = 0$, and τ covers exactly one member of Z. □

Proof of Example 4.19. By Lemma 4.18 each T_j covers at most $|T_j|$ members of Z. Since $\operatorname{supp} P_H$ is infinite, $P_H(Z \setminus \bigcup_j \bigcup \mathrm{inst}(T_j)) > 0$, so a.s. some datum falls outside every T_j, which then has likelihood 0 under $\mathsf{L0}$. $P_{T_{\mathrm{esc}}} > 0$ on Z (full-support $\mathcal{Q}$, $z := t$). So $\pi_n(T_{\mathrm{esc}}) = 1$ from that datum on. $T_{\mathrm{esc}} \vdash S0 = 0$ (instance $z := S0$); $\mathsf{Q} \vdash \neg S0 = 0$ (Q1). Q's axioms are sentences outside Z, so a theory containing only them has likelihood 0 at the first datum. □

Computed (c4, Part 2; the run reported in Example 4.19): the twenty sound theories are ground equations for the 20 most likely data, which carry mass 0.956, weighted as the human's conditional law, and share prior 0.99 against T_{esc}; $S0 = 0$ had reached posterior ≥ 0.99 in 90.2% of runs by $n = 50$. Proposition 5.4 uses this example for the verifier.

Remark C.5 (The full class). [conjecture; computed] model Rem 3.5 In the full class with Dirichlet weights the KL-projection picture can fail: theories with finite KL are unsound (Lemma 4.18), with infimum KL 0 approached by hybrids of memorisation and T_{esc}; memorisers are sound but each has infinite KL, yet as a family they code the data with $o(n)$ extra bits. Conjecture: unsound mass $\to 1$. Computed for three hand-picked families: at $n = 10^4$ the hybrid leads memorisation by at least 7000–25000 bits (Table 19).

	$\lambda = 1$			$\lambda = 2$		
seed	11	12	13	11	12	13
distinct data at $n = 10^4$	282	325	330	282	325	330
hybrid − memorisation, $n = 10^3$	≥ 1143	≥ 1207	≥ 1065	≥ 3344	≥ 3505	≥ 3063
hybrid − memorisation, $n = 10^4$	≥ 7370	≥ 8613	≥ 8625	≥ 21703	≥ 24719	≥ 24596
escape alone − memorisation, $n = 10^4$	-4234	-2649	-2719	12871	16289	16120

Table 19: Remark C.5, computed (model check c7): $\log_2$ score (prior plus $\mathsf{L0}$ marginal likelihood, $\mathrm{Dir}(\frac{1}{2})$) relative to memorising all distinct data. Human data $t = 0$, t a $\{0, +\}$-tree (0 with probability 0.7); model $\mathcal{Q}$ root law $(0.5, 0.3, 0.1, 0.1)$ on $(0, S, +, \cdot)$; prior $2^{-\lambda \ell}$ under a simplified code (3 bits per symbol, 2 per metavariable occurrence, Elias γ for the number of templates), not π_λ of Definition 2.4. Hybrid: ground templates for data seen at least twice plus $z = 0$; its score is a lower bound (one labelling). Escape alone: $\{z = 0\}$. Positive favours the unsound theory. At $\lambda = 1$ the escape alone loses to memorisation by $n = 10^4$, but the hybrid does not.

Example C.6 (A derivation likelihood does not reliably rescue the axioms)**.** [computed] model Ex 3.6 $\mathsf{L1}^{\mathrm{eq}}$ derives closed equations over $0, S, +$ (cite, refl, sym, trans, congruences), terms of size $\leq N$,

$P_T = \mu_T/Z_T$ exact. Class: $T_{\mathsf{Q}} := \{z+0 = z,\ z_1+Sz_2 = S(z_1+z_2)\}$ (Q4, Q5) and $T_{\mathsf{Q}} \cup \{0+z = z\}$ (sound); T_{esc} and $T_{\mathsf{Q}} \cup \{z = 0\}$ (unsound). On true equations $t = 0$ over $\{0, +\}$ the cross-entropy minimiser is unsound in all 16 settings in which $\mathcal{Q}$ gives + probability 0.25, and sound in 5 of 8 when + is rare under $\mathcal{Q}$ (Table 20): in grammar 1, citing $z = 0$ pays about 2 nats per + through $\mathcal{Q}$, deriving from Q4, Q5 about 7. Limits: truncated, equational only, three grammars, a finite class.

Conjecture C.7. [conjecture] model Conj 3.7 *In untruncated* $\mathsf{L1}^{\text{eq}}$, $\ln P_{T_{\mathsf{Q}}}(t = 0) = -c_{\text{der}}\#_+(t)+O(1)$ *on balanced* $\{0, +\}$*-terms; so* T_{esc} *beats* T_{Q} *on laws with many* + *iff* $\ln \frac{1}{p_+ p_0} < c_{\text{der}}$*,* p_+, p_0 *the root probabilities under* $\mathcal{Q}$ *(grammar 1:* ≈ 2 *against* ≈ 7*).*

Details of Example C.6. We write $T_{\mathsf{Q}0} := T_{\mathsf{Q}} \cup \{0 + z = z\}$ and $T_{\text{mix}} := T_{\mathsf{Q}} \cup \{z = 0\}$. Each node of the $\mathsf{L1}^{\text{eq}}$ grammar cites a template of T (probability α_{ax}, metavariables from $\mathcal{Q}$), or applies refl ($t \sim \mathcal{Q}$), sym, trans (valid iff the middle terms agree), congruence for S, or congruence for + (a side term $u \sim \mathcal{Q}$, left or right). A tree is valid iff its steps are valid and every term lies in U_N, the closed terms of size $\leq N$. $P_T = \mu_T/Z_T$, with μ_T the least fixpoint of the grammar's equation. These rules are complete for closed equational consequence (Birkhoff; recalled, not checked against the source (Birkhoff, 1935)). A Monte Carlo simulation agrees with the fixpoint ($Z = 0.54909$ against 0.54965 ± 0.00091; $|z| \leq 1.31$ on eight conclusion frequencies). Human law: $t = 0$, t a closed $\{0, +\}$-term from a grammar choosing 0 with probability $1 - h$ and + with h, conditioned on size $\leq N$, $h \in \{0.3, 0.45\}$, with and without the datum $0 = 0$.

grammar	rule probabilities (ax, refl, sym, trans, cong_S, cong_+)	$\mathcal{Q}$ root law $(0, S, +)$	minimiser	settings
1	$(.35, .15, .1, .15, .1, .15)$, $m = 0.65$	$(.55, .2, .25)$	T_{esc} with $0 = 0$; T_{mix} without	8/8 unsound
2	$(.5, .1, .05, .2, .05, .1)$, $m = 0.60$	$(.55, .2, .25)$	as grammar 1	8/8 unsound
3	as grammar 1	$(.8995, .1, .0005)$	$T_{\mathsf{Q}0}$ in 5; T_{mix} in 3 (all with $0 = 0$)	5/8 sound

mean $\ln P_T(t = 0)$ by size of t	1	3	5	7	9
grammar 1: T_{Q}	−1.88	−1.82	−8.45	−15.52	−22.89
grammar 1: T_{esc}	−0.80	−3.23	−5.22	−7.20	−9.19
grammar 3: T_{Q}	−1.62	−1.57	−7.91	−15.41	−23.33
grammar 3: $T_{\mathsf{Q}0}$	−1.63	−1.28	−7.34	−14.55	−22.19
grammar 3: T_{esc}	−0.43	−8.58	−16.30	−24.01	−31.72

Table 20: Example C.6, computed (model check c16). Top: the cross-entropy minimiser over $\{T_{\mathsf{Q}}, T_{\mathsf{Q}0}, T_{\text{esc}}, T_{\text{mix}}\}$ in the eight settings of each grammar ($N \in \{7, 9\}$, $h \in \{0.3, 0.45\}$, with and without $0 = 0$). Bottom: per-datum log-likelihoods at $N = 9$. In grammar 1, T_{Q} pays about 7 nats per extra + and T_{esc} about 2; e.g. at $N = 9$, $h = 0.45$, data $t \neq 0$, the cross-entropies are 7.758 (T_{Q}), 7.124 ($T_{\mathsf{Q}0}$), 4.951 (T_{esc}), 4.904 (T_{mix}) nats, so the odds of T_{Q} against T_{mix} fall by about 2.85 nats per datum. In grammar 3, T_{esc} pays $\ln \frac{1}{p_+ p_0} \approx 7.7$ nats per + through $\mathcal{Q}$; T_{Q}'s increments (6.3, 7.5, 7.9 nats between consecutive sizes) are still rising at size 9, so the large-term limit in grammar 3 is not settled.

C.7 Predictive versus deductive

Proof of Theorem 4.21. (a) is Proposition C.1(b) with the index shifted; (b) is Corollary 4.4 with $d = \infty$, and the two cases are those named after it. (c) $\mathbb{E}_{P_{T_1}} \ln \frac{P_{T_1}(X)}{P_{T_2}(X)} = \mathrm{KL}(P_{T_1} \| P_{T_2})$ per datum, summed over the data; Pinsker's inequality $\mathrm{TV} \le (\mathrm{KL}/2)^{1/2}$ (known). For the misspecified remark: $M(D_n) \ge \pi(T) P_T(D_n)$ for every T and every sequence, so the cumulative log loss of M exceeds that of T by at most $\ln(1/\pi(T))$. □

Proposition C.8 (Data that come with their proofs)**.** [proved]
model Prop 8.2 *Let each datum be a whole valid* L1 *tree from $P_T(\cdot \mid \text{valid})$, showing each node's rule and conclusion and whether a citation leaf cites a theory or a logical axiom, but not which template; weights fixed. (a) $P_T(\pi) = \prod_v P_T^{\mathsf{L0}}(\mathrm{concl}(v))\, G(\pi)/Z_T$ over the theory-citation leaves v, with $P_T^{\mathsf{L0}} := \sum_\tau w_\tau \mathcal{Q}_\tau$, G independent of T, and Z_T a function of P_T^{L0}. (b) $P_T = P_{T'}$ on trees iff $P_T^{\mathsf{L0}} = P_{T'}^{\mathsf{L0}}$: the posterior concentrates on the* L0 *generator class of T^*, with instance union $\bigcup \mathrm{inst}(T^*)$, and splits still tie. (c) So proofs guarantee identification of the instance union $\bigcup \mathrm{inst}(T^*)$; conclusions guarantee only $\mathrm{Th}(T^*)$ (though they often separate more, Proposition 4.5).*

Proof of Proposition C.8. (a) A tree's probability is the product of its nodes' probabilities. A theory-citation leaf whose template is not recorded contributes $\alpha_{\mathrm{ax}} \sum_\tau w_\tau \mathcal{Q}_\tau(\mathrm{concl}(v)) = \alpha_{\mathrm{ax}} P_T^{\mathsf{L0}}(\mathrm{concl}(v))$, the sum over the unrecorded choices factorising over leaves (matching is unique, so $\mathcal{Q}_\tau(\mathrm{concl}(v))$ is the sum over substitutions). No other node depends on T. Validity depends only on conclusions, so $Z_T = \Pr[\text{valid}]$ is a function of the law of theory-leaf conclusions, i.e. of P_T^{L0}. (b) "If" is (a). "Only if": the one-node tree citing s as a theory axiom has probability $\alpha_{\mathrm{ax}} P_T^{\mathsf{L0}}(s)/Z_T$; equality for all s makes P_T^{L0} and $P_{T'}^{\mathsf{L0}}$ proportional, and both sum to 1. The rest is Theorem 4.2, Corollary 4.3 and Proposition 4.6(a), (b). (c) The proof-data generator class of T^* is contained in its conclusion-data class, since equal L0 laws give equal L1 laws; the latter guarantees only $\mathrm{Th}(T^*)$ (Corollary 4.4, L1 with parameters admissible), though conclusions can separate more (Proposition 4.5). □

If the template name is recorded at each citation leaf, the argument is easier.

Remark C.9 (Near misses). [proved (the transfer); open (size of the generator class)]
model Rem 8.3 Hänni's "grade near misses" can be a corruption channel $K(s' \mid s)$ favouring s' near s, $P_T^K(s') := \sum_s P_T(s) K(s' \mid s)$. Each P_T^K is a fixed law with a data-independent normaliser, so Theorem 4.2, Corollary 4.4 and Theorem 5.2 (with C_d^* defined through P^K) hold when the channel is part of the well-specified model. The size principle also holds, but with a full-support channel it is vacuous ($\mathrm{supp}\, P_{T^*}^K = \mathcal{S}$), and theories are separated only through KL. The generator class can grow, since K need not be injective; by how much, for natural metrics, is open. The noise mixtures of Proposition B.1 are related.

Proof of Remark C.9. P_T^K is a probability on $\mathcal{S}$ whose normaliser does not depend on the data. The proofs of Theorem 4.2, Corollary 4.4, Theorem 5.2 and Lemma 2.21 use only that each hypothesis carries a fixed law on a countable set and that the data are i.i.d. from one of them; they therefore hold with P_T^K in place of P_T when the data are i.i.d. from $P_{T^*}^K$. □

D Soundness: proofs and details

Scripts named c*k* are the model track's (`research/tracks/model/checks/`), r*k* its referee's (`research/tracks/model/referee_code/`); the others are in `research/tracks/experiments/`

`checks/`. Each is seeded and writes its output next to itself. Of this section's results the model track's referee checked Theorem 5.2, Propositions 5.5 and D.2 and the trichotomy results Propositions 2.14, 2.18 and 2.19 (now in Section 2.7). Lemma D.4, Theorems 5.6 and 5.7, Remark D.3, the mechanism of Example 5.8 and Proposition 2.17 were added after the referee report and checked by the track and while writing, not by a referee; the complete list of such results is in Appendix H.1. Nothing is checked in a proof assistant.

D.1 Fixed weights

Proof of Theorem 5.2. This is the proof of IL Thm 4.15(b) (IL App. C.9), in the class form of IL Thm 10.9; the role of shadow-measurability there is played by the definition of C_d^*. Write $L_t(T) := \prod_{i \le n(t)} P_T(X_i) \prod_j c_j(T)$.

Step 1: C_d^* keeps the target's likelihood. For $T \in C_d^*$, $P_T = P_{T^*}$, and every constraint has $c_j(T) = c_j(T^*) = 1$ because $\mathrm{Th}_d(T) = \mathrm{Th}_d(T^*)$. So $L_t(T) = L_t(T^*)$, which is positive almost surely. With $Z_t := \sum_T \pi(T) L_t(T)/L_t(T^*)$ we get $\pi_t(C_d^*) = W_d^*/Z_t$.

Step 2: constraints only help. $c_j(T^*) = 1$ and $c_j(T) \le 1$, so pathwise $Z_t \le Z^{\mathcal{H}}_{n(t)}$, where

$$Z_n^{\mathcal{H}} := \sum_T \pi(T) \prod_{i \le n} \frac{P_T(X_i)}{P_{T^*}(X_i)} = \frac{M(X^n)}{P_{T^*}(X^n)}.$$

Step 3: Ville. Under P_{T^*}, $\mathbb{E}[P_T(X)/P_{T^*}(X)] = \sum_{x: P_{T^*}(x)>0} P_T(x) \le 1$. By independence and Tonelli, $Z^{\mathcal{H}}$ is a nonnegative supermartingale for $\sigma(X_1, \dots, X_n)$ with $Z_0^{\mathcal{H}} = 1$. By Ville's inequality (IL Lemma C.4) the event $G := \{\sup_n Z_n^{\mathcal{H}} < 1/\delta'\}$ has probability at least $1 - \delta'$. It depends only on the data.

Step 4: conclusion. On G, in every round and for every history of queries and truthful constraints, $\pi_t(C_d^*) = W_d^*/Z_t \ge W_d^*/Z^{\mathcal{H}}_{n(t)} > W_d^* \delta' \ge \delta$. If $q \notin \mathrm{Th}_d(T^*)$ then $q \notin \mathrm{Th}_d(T)$ for every $T \in C_d^*$, so $\pi_t(\{T : q \notin \mathrm{Th}_d(T)\}) > \delta$ and q is not accepted. The prover's knowledge and randomness enter only through the schedule and the constraints, so the conclusion covers a prover that knows the future data. □

Escalations. Let $Z_t := W_d^*/\pi_t(C_d^*)$, as in Step 1. Escalating a valid query q adds the constraint $1[q \in \mathrm{Th}_d(T)]$, which keeps C_d^* and multiplies Z_t by $\pi_t(\{T : q \in \mathrm{Th}_d(T)\}) < 1 - \delta$, because q was not accepted; data rounds multiply it by factors that form a nonnegative supermartingale from 1 (Step 3). Since $Z_t \ge W_d^*$, on the Ville event of probability $\ge 1 - \delta''$ the number E_t^+ of escalated valid queries satisfies $(1-\delta)^{E_t^+} > W_d^* \delta''$, so $E_t^+ \le (\ln(1/W_d^*) + \ln(1/\delta''))/\delta$ for all t (IL Thm 4.16). An invalid query escalated adds a constraint that removes the theories deriving it; IL bounds these through a reject threshold δ_r, which this protocol does not have, so their number is not bounded.

Computed (Theorem 5.2). `c4`, Part 1: `L0`, fixed weights, $\mathcal{Q}$ root law $(.5, .3, .1, .1)$ on $(0, S, +, \cdot)$, $T^* = \{z + 0 = z\}$ against $\{z + 0 = z,\ S0 + 0 = S0,\ 0 + 0 = S0\}$ (weights $0.49, 0.5, 0.01$), which proves $q := (0 + 0 = S0)$; $\pi(T^*) = 0.3$; 20 000 runs of 300 data. The frequency of $\{\exists n \le 300 : \pi_n(T^*) \le \delta\}$ was 0.044, 0.014, 0.004 against $\delta/\pi(T^*) = 0.10$, 0.033, 0.010. `r7`: nine hypotheses all proving the same invalid query, $\pi(C^*) = 0.385$, $n \le 400$: 0.0365, 0.0125, 0.0027 against $\delta/\pi(C^*) = 0.130$, 0.052, 0.013. So the frequency of invalid acceptance stayed 2–5 times below the bound. The horizons are finite, so these check the bound, not its time-uniformity.

Remark D.1 (The constant is tight). [proved]
model §4.1 (referee m13) IL Prop C.5 shows that $\delta \le W_d^*\delta'$ cannot be improved, with $R^* = \{a, b\}$ against $R' = \{a, c\}$, which generates only a and proves c. That pair is not in the template models (under L0, $\operatorname{supp} P_{T'} \subseteq \operatorname{supp} P_{T^*}$ implies $\mathrm{Th}_d(T') \subseteq \mathrm{Th}_d(T^*)$ for every d; under L1 with parameters admissible it does so for $d = \infty$ only: $T^* = \{a, a \to b\}$ and $T' = T^* \cup \{b\}$ have the same support, but $b \in \mathrm{Th}_{|b|}(T') \setminus \mathrm{Th}_{|b|}(T^*)$). But $T' = \{a, c\}$ with a small fixed weight on c attains IL's acceptance probability $(1-u)^{t^*}$ exactly, and it tends to δ' as $u, W_d^* \to 0$.

Proof of Remark D.1. IL's pair and the template models. Under L0 with full-support $\mathcal{Q}$, $\operatorname{supp} P_T = \bigcup \mathrm{inst}(T)$, and Th_d depends only on the instance union, so $\operatorname{supp} P_{T'} \subseteq \operatorname{supp} P_{T^*}$ gives $\mathrm{Th}_d(T') \subseteq \mathrm{Th}_d(T^*)$ for every d: no T' generates only data of T^* and still proves a sentence T^* does not. Under L1 with parameters admissible $\operatorname{supp} P_T = \mathrm{Th}(T)$ (Lemma 2.10(c)), so the inclusion of supports gives $\mathrm{Th}(T') \subseteq \mathrm{Th}(T^*)$, the case $d = \infty$ only. For finite d it fails: with ground $T^* = \{a, a \to b\}$ and $T' = T^* \cup \{b\}$, $\mathrm{Th}(T') = \mathrm{Th}(T^*)$, so the supports agree, but b is an axiom of T' while every derivation of b from T^* uses a, $a \to b$ and MP, so $b \in \mathrm{Th}_{|b|}(T') \setminus \mathrm{Th}_{|b|}(T^*)$.

The construction. Let a, b, c be ground sentences with $c \notin \mathrm{Th}_d(T^*)$ and $c \in \mathrm{Th}_d(\{c\})$, let $T^* = \{a, b\}$ with fixed weights $(1-u, u)$, $T' = \{a, c\}$ with fixed weights $(1-\varepsilon, \varepsilon)$, prior $(W^*, 1-W^*)$, likelihood L0 and $\delta = W^*\delta'$. Then $W_d^* = W^*$. After t data equal to a, $r_t := P_{T'}(a^t)/P_{T^*}(a^t) = ((1-\varepsilon)/(1-u))^t$, and c is accepted iff $\pi_t(T^*) = W^*/(W^* + (1-W^*)r_t) \le \delta$, i.e. iff $r_t \ge \theta := (1 - W^*\delta')/((1-W^*)\delta')$. Let t^* be the least t with $(1-u)^{-t} > \theta$ (IL Prop C.5). For $t < t^*$, $r_t < (1-u)^{-t} \le \theta$. At t^* the inequality $(1-u)^{-t^*} > \theta$ is strict, so $r_{t^*} > \theta$ for all ε below some $\varepsilon_0 > 0$. A datum b kills T'. So the prover that waits for t^* data and then asks c (an earlier query could be escalated, and the oracle's "no" would kill T') wins iff the first t^* data are a: probability $(1-u)^{t^*}$, exactly IL's value. By IL Prop C.5 it lies in $[(1-u)/\theta, 1/\theta)$ with $1/\theta = (1-W^*)\delta'/(1-W^*\delta')$, so it tends to δ' as $u, W^* \to 0$. □

Truncation (Remark 5.3). On a finite truncation $\mathcal{H}_N \ni T^*$ with the renormalised prior, the proof above runs with $\pi(\cdot)/\pi(\mathcal{H}_N)$ in place of π, so the theorem holds with W_d^* replaced by $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N) \ge \pi(T^*)$.

The computable verifier (Remark 5.3). For a finite class and $d < \infty$, $\mathrm{Th}_d(T)$ is decidable (finitely many derivations of symbol size $\le d$ up to renaming of parameters, axioms checked by matching), the likelihoods are computable reals (L0 by matching; L1 by Proposition 2.24(a)), and the denominator of π_t is positive. So an interval of width $\le \varepsilon$ around the non-deriving mass can be computed; accept iff its upper end is $\le \delta$. Acceptance then implies that the true mass is $\le \delta$ at the same posterior, and Step 4 applies verbatim. With truncated $\mathcal{Q}$ the computation is exact.

The refuted remarks (Remark 5.3). (1) On a truncation $\mathcal{H}_N \ni T^*$ the constant $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N)$ is at least $\pi(T^*)$ but can be below $\pi(C_d^*)$: with $\mathcal{H} = \{T^*, T_2, T_3\}$, $T_2 \in C_d^*$, prior $(0.01, 0.89, 0.10)$ and $\mathcal{H}_N = \{T^*, T_3\}$ it is $0.01/0.11 \approx 0.09$, against 0.90. (2) L2 likelihoods are finite sums only when $\mathcal{Q}$ has finite support: for $T = \{P \to b\}$, P a 0-ary formula metavariable, $P_T(b)$ at depth 1 sums $\alpha_{\mathrm{MP}}\, \mu(A)\, \mu(A \to b)$ over the infinitely many logical-axiom instances A.

Proposition 5.4: computed. `c4`, Part 2, is the run of Example 4.19 (details in Appendix C.6), read with $\delta = 0.01$ and no escalation: the fraction of runs in which $S0 = 0$ had been accepted was 0.341, 0.902, 0.986, 1.000 by $n = 10, 50, 100, 500$. `r3`, Part 2: against a verifier that escalates

every non-accepted query, a prover asking $S0 = 0$ in every round won 11/400 (only when the first datum was uncovered), the waiting prover 400/400.

D.2 The cautious limit

Proposition D.2 (The limit $\delta \to 0$)**.** [proved] model Prop 4.3 *Suppose each P_T has a fixed support* supp_T *(fixed weights; or Dirichlet weights under* L0*, where $P_T^{\mathrm{Dir}}(D) > 0$ iff $D \subseteq \bigcup \mathrm{inst}(T)$).*

(a) $\pi_n(T) > 0$ *iff* $\pi(T) > 0$ *and* $D_n \subseteq \mathrm{supp}_T$*. So $V_{0,d}$ accepts exactly* $\bigcap\{\mathrm{Th}_d(T) : \pi(T) > 0,\ D_n \subseteq \mathrm{supp}_T\}$.

(b) $V_{0,d}$ is sound for every T^* with $\pi(T^*) > 0$ and $D_n \subseteq \mathrm{supp}_{T^*}$, deterministically and without well-specification (IL Thm 4.14). $V_{\delta,d}$ accepts at least as much.

(c) Under L0 *with full-support $\mathcal{Q}$, if π charges every finite theory, $V_{0,d}$ accepts exactly* $\mathrm{Th}_d(D_n)$ *(AS Prop 6.4, read on theorems).*

(d) Under L0*, if* $\mathrm{supp}\,\pi = H_k(\mathrm{DT}^\circ)$*, the theories with at most k templates, the acceptance set of $V_{0,d}$ contains* $\mathrm{Th}_d(\mathrm{Acc}_k(D_n))$*, where* Acc_k *is the cautious k-union verifier of AS Def 2.6. In general classes the inclusion can be strict; for* $H_k(\mathrm{DT}^\circ)$ *itself this is open.*

Proof of Proposition D.2. (a) The likelihood of T (fixed weights, or the Dirichlet marginal under L0) is positive iff every datum lies in supp_T, and $\pi_n(T)$ is proportional to $\pi(T)$ times it. $V_{0,d}$ accepts q iff $\pi_n(\{T : q \notin \mathrm{Th}_d(T)\}) = 0$, i.e. iff $q \in \mathrm{Th}_d(T)$ for every T of positive posterior mass. (b) Such a T^* is one of the theories in the intersection of (a), and truthful constraints keep it; so everything accepted lies in $\mathrm{Th}_d(T^*)$. The accepted set of $V_{\delta,d}$ grows with δ. (c) The memoriser $\mathrm{Mem}(D_n)$, made of ground templates for the distinct data, is charged by π, its support is the set of distinct data, and $\mathrm{Th}_d(\mathrm{Mem}(D_n)) = \mathrm{Th}_d(D_n)$. Every T with $D_n \subseteq \bigcup \mathrm{inst}(T)$ has $\mathrm{Th}_d(D_n) \subseteq \mathrm{Th}_d(T)$, since Th_d is monotone in the axiom set. So the intersection in (a) is $\mathrm{Th}_d(D_n)$; this is AS Prop 6.4 ($\mathrm{Acc}_\infty(D) = D$) read on theorems. (d) A theory of positive posterior mass lies in $H_k(\mathrm{DT}^\circ)$ and covers D_n, so it is in the version space $\mathrm{VS}_k(D_n)$ of AS Def 2.6, and $\bigcup \mathrm{inst}(T) \supseteq \mathrm{Acc}_k(D_n)$. Monotonicity of Th_d gives the inclusion. *Strictness in a general class.* Take the class $\{T_1, T_2\}$ with $T_1 = \{c,\ a \wedge b\}$, $T_2 = \{c,\ b \wedge a\}$ (ground), and $D = \{c\}$. Both theories prove a, so $V_{0,d}$ accepts a for d large enough. The instance-level intersection is $\{c\}$, which does not prove a. Whether the inclusion can be strict for $H_k(\mathrm{DT}^\circ)$ itself is not settled (model §10, problem 10). □

The two verifiers compared. Table 21 contrasts the cautious verifier of AS with the thresholded verifier of Section 5 (model §5.5, with the weight convention stated).
The time for the Bayesian verifier to accept s is set by the competitors that miss it: a split that omits a root of probability p gains a factor $(1-p)^{-n}$ until the first datum with that root.

D.3 Dirichlet weights

Remark D.3 (Vacuity for Dirichlet weights). [proved (L0 and likelihoods affine in w)] model Rem 4.5; experiments §0 (M3) Under L0, or any likelihood affine in w (such as the experiments' chain Ch_J), read Theorem 5.2 with parameter (T, w) and prior $\pi(T)\mathrm{Dir}_T(dw)$. Its class $\{(T, w) : P_{T,w} = P_{T^*,w^*},\ \mathrm{Th}_d(T) = \mathrm{Th}_d(T^*)\}$ has prior mass 0 if $|T^*| \geq 2$, the component laws of T^* are linearly independent, and w^* lies outside a countable set; then $\delta \leq W_d^* \delta'$ forces $\delta = 0$. Under L1, where $P_{T,w}$ is a ratio of power series in w, the same is plausible but not proved.

	cautious k-union verifier (AS)	Bayesian thresholded verifier
assumptions for soundness	realizability ($R^* \in H_k$), bound $k \geq k'$	well-specified data law; $W_d^* \geq \delta/\delta'$; with Dirichlet weights, averaging over w^*, or a threshold shrinking with n for likelihoods linear in w
soundness	deterministic; every prover, every time	probability $\geq 1-\delta'$; every prover, every time (Theorems 5.2, 5.6 and 5.7)
without the assumptions	sound relative to $\mathrm{cl}(R^*)$ (AS Lemma 2.8); without a bound accepts only the data (AS Prop 6.4)	can accept a false sentence with probability 1 (Proposition 5.4); at a fixed w^* a constant threshold fails (Example 5.8)
completeness	exact once an anchor is present (AS Lemma 2.8)	each theorem eventually, a.s. (Proposition 5.5; Dirichlet: a.e. w^*); no anchor, no bound; open under a shrinking threshold
splits	sound splits survive any negatives (AS Thm 6.5)	exact ties at matched fixed weights; with learned weights Occam terms decide; usage statistics can reverse (Propositions 4.6, 4.10 and 4.11)
depends on the target's length	no	yes: $\delta \leq 2^{-\lambda \ell(T^*)}\delta'$; under Theorem 5.7 also on $\lvert T^* \rvert$

Table 21: The cautious verifier of AS against the Bayesian thresholded verifier (model §5.5, with the weight convention stated).

Proof of Remark D.3. The class is the exact class $\{(T,w) : P_{T,w} = P_{T^*,w^*}\}$ of Remark 4.8 intersected with $\{T : \mathrm{Th}_d(T) = \mathrm{Th}_d(T^*)\}$, so its prior mass is at most that of the exact class. The affine-slice argument in the proof of Remark 4.8 (Appendix C.3) uses only that $w \mapsto P_{T,w}$ is affine, which holds under L0 and for the chain Ch_J (whose law is $\sum_i w_i a_i$): it gives $\mathrm{Dir}_T(\{w : P_{T,w} = P_{T^*,w^*}\}) = 0$ for every T unless some one-component theory has law P_{T^*,w^*}, which, by linear independence of T^*'s component laws, happens for at most one w^* per one-component theory, so for countably many w^*. Under L1, $P_{T,w} = \mu_{T,w}/Z_{T,w}$ is a ratio of power series in w, not affine, and the argument does not apply. □

Lemma D.4 (Regret)**.** [(a), (b), (d) proved; (c) known; computed] model Lemma 4.6; experiments Prop X8(c)
Let T have K templates with laws q_τ (under L0, $q_\tau = \mathcal{Q}_\tau$), $P_{T,w} := \sum_\tau w_\tau q_\tau$, weights $\mathrm{Dir}(\alpha)$, $A := \sum_\tau \alpha_\tau$, and

$$R_\alpha(n,K) := \min_{c \in \mathbb{N}^K,\ \sum_\tau c_\tau = n} \frac{\mathbb{E}_{\mathrm{Dir}(\alpha)}\big[\prod_\tau w_\tau^{c_\tau}\big]}{\prod_\tau (c_\tau/n)^{c_\tau}} \qquad (0^0 := 1).$$

(a) $P_T^{\mathrm{Dir}}(x^n) \geq R_\alpha(n,K)\, P_{T,w}(x^n)$ for every w in the closed simplex and every x^n. (b) If $\alpha_\tau \leq 1$ for every τ, then $R_\alpha(n,K) \geq \frac{\Gamma(A)}{e\,(K-1)!\prod_\tau \Gamma(\alpha_\tau)}\,(n+1)^{-(K-1)}$. (c) For $\alpha_\tau \equiv \frac{1}{2}$: $-\ln R_\alpha(n,K) = \frac{K-1}{2}\ln n + O(1)$ (Krichevsky and Trofimov, 1981). (d) For $\alpha_\tau \equiv \alpha$: $R_\alpha(n,K) \leq \frac{\Gamma(K\alpha)\Gamma(\alpha+n)}{\Gamma(\alpha)\Gamma(K\alpha+n)} \sim \frac{\Gamma(K\alpha)}{\Gamma(\alpha)} n^{-(K-1)\alpha}$; so at $\alpha = 1$ the exponent $K-1$ in (b) is tight.

Proof of Lemma D.4. (a) Expand over labellings $\ell : [n] \to T$ with counts $c_\tau(\ell)$:

$$P_{T,w}(x^n) = \sum_\ell \prod_\tau w_\tau^{c_\tau(\ell)} \prod_i q_{\ell_i}(x_i), \qquad P_T^{\mathrm{Dir}}(x^n) = \sum_\ell \mathbb{E}_{\mathrm{Dir}}\Big[\prod_\tau w_\tau^{c_\tau(\ell)}\Big] \prod_i q_{\ell_i}(x_i).$$

For each ℓ the maximum of $\prod_\tau v_\tau^{c_\tau}$ over the simplex is attained at $v = c/n$, so

$$\mathbb{E}_{\mathrm{Dir}}\Big[\prod_\tau w_\tau^{c_\tau}\Big] \;\ge\; R_\alpha(n,K)\prod_\tau (c_\tau/n)^{c_\tau} \;\ge\; R_\alpha(n,K)\prod_\tau w_\tau^{c_\tau}.$$

Sum over ℓ. This termwise argument is experiments Prop X8(c).

(b) Fix c, put $v := c/n$, $\eta := 1/(n+1)$ and $B := \{(1-\eta)v + \eta u : u \text{ in the simplex}\}$. On B, $w_\tau \ge (1-\eta)v_\tau$, so $\prod_\tau w_\tau^{c_\tau} \ge (1-\eta)^n \prod_\tau v_\tau^{c_\tau} \ge e^{-1}\prod_\tau v_\tau^{c_\tau}$, because $(1-\eta)^n = (1+1/n)^{-n} \ge e^{-1}$. The Dirichlet density $\frac{\Gamma(A)}{\prod_\tau \Gamma(\alpha_\tau)}\prod_\tau w_\tau^{\alpha_\tau - 1}$ (with respect to Lebesgue measure on the first $K-1$ coordinates) is at least $\Gamma(A)/\prod_\tau \Gamma(\alpha_\tau)$ on the simplex, since $w_\tau \le 1$ and $\alpha_\tau \le 1$. B is the simplex scaled by η, so its volume is $\eta^{K-1}/(K-1)!$. Hence $\mathbb{E}_{\mathrm{Dir}}[\prod w^c] \ge e^{-1}\prod v^c \cdot \frac{\Gamma(A)}{\prod_\tau \Gamma(\alpha_\tau)} \cdot \frac{\eta^{K-1}}{(K-1)!}$; divide by $\prod v^c$ and minimise over c.

(d) Take $c = (n, 0, \dots, 0)$: $\prod (c/n)^c = 1$ and $\mathbb{E}_{\mathrm{Dir}}[w_1^n] = \Gamma(K\alpha)\Gamma(\alpha+n)/(\Gamma(\alpha)\Gamma(K\alpha+n))$; then $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$.

(c) is the Krichevsky–Trofimov rate (Krichevsky and Trofimov, 1981); it is not re-proved here. □

Computed (Lemma D.4). `c11`, Part A: exact Dirichlet marginals (a Bernstein-form dynamic programme, equal to quadrature to six decimals) for $K = 2, 3$ with overlapping components on four letters, w^* interior and on the boundary, $n \le 3000$. The largest value of regret minus bound (b) was -2.81, -1.00, -6.45 nats ($K = 2$, $\alpha = \frac{1}{2}$; $K = 2$, $\alpha = 1$; $K = 3$, $\alpha = \frac{1}{2}$). Part A2: slopes of the exact $\ln R_\alpha(n, K)$ against $\ln n$ were -0.499, -0.993 ($K = 2, 3$; $\alpha = \frac{1}{2}$) and -0.993, -1.960 ($\alpha = 1$), against $-(K-1)\max(\frac{1}{2}, \alpha)$. The experiments' `kt_regret`: -0.499, -0.998, -1.485 for $K = 2, 3, 4$ at $\alpha = \frac{1}{2}$.

Proof of Theorem 5.6. Steps 1, 2 and 4 of the proof of Theorem 5.2 go through with $L_t(T) := P_T^{\mathrm{Dir}}(X^{n(t)})\prod_j c_j(T)$ and C_d^{Dir} in place of C_d^*. For Step 3 let $M := \sum_T \pi(T) P_T^{\mathrm{Dir}}$ and $Z_n := M(X^n)/P_{T^*}^{\mathrm{Dir}}(X^n)$. Each P_T^{Dir} is a probability on sequences, consistent in n, so M is too, and under $P_{T^*}^{\mathrm{Dir}}$

$$\mathbb{E}[Z_{n+1} \mid X^n] = \sum_{x:\ P_{T^*}^{\mathrm{Dir}}(X^n x) > 0} \frac{M(X^n x)}{P_{T^*}^{\mathrm{Dir}}(X^n)} \le \frac{M(X^n)}{P_{T^*}^{\mathrm{Dir}}(X^n)} = Z_n.$$

Ville's inequality gives $G := \{\sup_n Z_n < 1/\delta'\}$ with $P_{T^*}^{\mathrm{Dir}}(G) \ge 1 - \delta'$. For the consequence, $f(w^*) := P_{T^*,w^*}^{\infty}(G^c)$ has $\int f\, d\mathrm{Dir}_{T^*} = P_{T^*}^{\mathrm{Dir}}(G^c) \le \delta'$, since $P_{T^*}^{\mathrm{Dir}}$ on infinite sequences is the mixture $\int P_{T^*,w}^{\infty}\mathrm{Dir}_{T^*}(dw)$; apply Markov's inequality.

Pool version (experiments Prop X8(a), (b); no escalation). Let F be the countable family, $C := \sum_{T\in F} 2^{-\mathrm{bits}(T)} \le 1$ (Kraft for the experiments' code, Remark A.1), $M_T := P_T^{\mathrm{Dir}}$, and $Z_n := \sum_{T\in F}(2^{-\mathrm{bits}(T)}/C)\, M_T(D_n)/M_{T^*}(D_n)$, a nonnegative supermartingale under M_{T^*} with $Z_0 = 1$ by the computation above. For every $R_n \subseteq F$ with $T^* \in R_n$, however chosen, pathwise

$$\pi_R(T^* \mid D_n) = \frac{2^{-\mathrm{bits}(T^*)} M_{T^*}(D_n)}{\sum_{T\in R_n} 2^{-\mathrm{bits}(T)} M_T(D_n)} \ge \frac{2^{-\mathrm{bits}(T^*)}}{C\, Z_n} \ge \frac{2^{-\mathrm{bits}(T^*)}}{Z_n}.$$

Off the Ville event this exceeds $2^{-\mathrm{bits}(T^*)}\delta' \ge \delta$. Accepting s with $T^* \nvdash_J s$ needs mass $\ge 1 - \delta$ on theories deriving s, hence $\pi_R(T^* \mid D_n) \le \delta$. If F is a fixed pool H plus all memorisers whose codes lie outside the template code, $C \le 1$ may fail; but $C \le \sum_{T\in H} 2^{-\mathrm{bits}(T)} + 1$, because the memorisers' codes form a prefix-free set, and the argument runs with $2^{-\mathrm{bits}(T^*)}/C$ in place of $2^{-\mathrm{bits}(T^*)}$. □

Proof of Theorem 5.7. $Z'_n := M(X^n)/P_{T^*,w^*}(X^n)$ is a nonnegative supermartingale under the i.i.d. law P_{T^*,w^*} with $Z'_0 = 1$, by the computation above with P_{T^*,w^*} as reference law (M is a consistent probability on sequences). Let $G := \{\sup_n Z'_n < 1/\delta'\}$, so $P(G) \geq 1 - \delta'$. Members of C_d^{Dir} have T^*'s marginal likelihood and pass every constraint, and the other constraints are ≤ 1, so

$$\pi_t(C_d^{\mathrm{Dir}}) \;\geq\; \pi(C_d^{\mathrm{Dir}})\frac{P_{T^*}^{\mathrm{Dir}}(X^n)}{M(X^n)} = \pi(C_d^{\mathrm{Dir}})\frac{P_{T^*}^{\mathrm{Dir}}(X^n)/P_{T^*,w^*}(X^n)}{Z'_n} \;\geq\; \frac{\pi(C_d^{\mathrm{Dir}})\,R_\alpha(n,K)}{Z'_n},$$

with $n = n(t)$, by Lemma D.4(a), which needs the likelihood to be linear in w. On G this exceeds $\pi(C_d^{\mathrm{Dir}})R_\alpha(n,K)\delta' \geq \delta_n$. No member of C_d^{Dir} derives an invalid q, so its non-deriving mass exceeds δ_n.

Pool version (experiments Prop X8(c)). With F, $C \leq 1$ and M_T as in the pool version of Theorem 5.6, $Z'_n := \sum_{T\in F}(2^{-\mathrm{bits}(T)}/C)\,M_T(D_n)/P_{T^*,w^*}(D_n)$ is a nonnegative supermartingale under P_{T^*,w^*} with $Z'_0 = 1$, so $P[\sup_n Z'_n \geq 1/\delta'] \leq \delta'$. For every pool $R_n \subseteq F$ with $T^* \in R_n$, pathwise and by Lemma D.4(a),

$$\pi_R(T^* \mid D_n) \;\geq\; \frac{2^{-\mathrm{bits}(T^*)}M_{T^*}(D_n)}{C\,Z'_n\,P_{T^*,w^*}(D_n)} \;\geq\; \frac{2^{-\mathrm{bits}(T^*)}R_\alpha(n,K)}{Z'_n},$$

which off the Ville event exceeds $2^{-\mathrm{bits}(T^*)}R_\alpha(n,K)\delta' \geq \delta_n$. Conclude as in the pool version of Theorem 5.6. □

Example 5.8: details. *Why $T^* \nvdash q$.* In the term model in which + returns its first argument on $(a,0)$ exactly when a has root 0 or S (and returns the formal term $a+b$ otherwise), every instance of $0+0=0$ and $(Sz)+0=Sz$ holds, but $(0\cdot 0)+0$ denotes the term $(0\cdot 0)+0 \neq 0\cdot 0$. *Why $P_{T'} = (1-\varepsilon)P_{T^*,w^*}$ on the data's support.* A datum is $t+0=t$ with $t=0$ or $t=St'$. Under T' it has probability $\mathcal{Q}(t)$, i.e. 0.5 or $(0.5-\varepsilon)\mathcal{Q}(t')$; under P_{T^*,w^*} it has $w_1^* = 0.5/(1-\varepsilon)$ or $w_2^*\mathcal{Q}(t') = (0.5-\varepsilon)\mathcal{Q}(t')/(1-\varepsilon)$. *The decomposition.* T^*'s instance sets are disjoint, so $\ln P_{T^*,w^*}(D) - \ln P_{T^*,\hat{w}}(D) = \sum_f n_f \ln(w_f^*/\hat{w}_f) = -n\mathrm{KL}(\hat{w}\|w^*)$, with $\hat{w}$ the empirical weights. With the previous point,

$$\ln\frac{P_{T'}(D)}{P_{T^*}^{\mathrm{Dir}}(D)} = n\ln(1-\varepsilon) + \Big[\ln P_{T^*,\hat{w}}(D) - \ln P_{T^*}^{\mathrm{Dir}}(D)\Big] - n\,\mathrm{KL}(\hat{w}\,\|\,w^*).$$

The bracket is the regret of Lemma D.4(c) at $K=2$, also Proposition 4.10(b), and $2n\mathrm{KL}(\hat{w}\|w^*) \to \chi_1^2$ in law. *Not written out*: a version of that χ^2 limit uniform over the window $1 \ll n \ll 1/\varepsilon$, which the crossing argument needs. While $\varepsilon n \ll 1$ the log-odds of T' grow like $\frac{1}{2}\ln n$ and cross $\ln 99$ near $n \approx 8000$. *Moderate ε.* The referee's runs (`r3`, geometric grid, $n \leq 8000$ and $n \leq 80000$) found no acceptance at $\varepsilon = 10^{-3}$ and 10^{-4}, at the fixed w^* (median of the largest log-ratio 2.81 and 3.96, below $\ln 99 = 4.60$) and with w^* from the prior. *The shrinking threshold* of the model track's run is $\delta_n = \pi(T^*)\,\delta'\,\frac{1}{e\pi}\,(n+1)^{-1}$ with $\delta' = 0.02$, the bound of Lemma D.4(b) at $K=2$, $\alpha = \frac{1}{2}$ $(1/(e\pi) = 0.1171)$.
`check_fixed_weight` also matched its closed form against the posterior code (ln Bayes factors to $4.1\cdot 10^{-13}$) and checked Lemma D.4(b) at $K=2$ for $n \leq 2000$ (margin ≥ 2.1 nats).

E Time penalties: proofs and details

Scripts cited here are in the folders `checks/` (the model track's) and `referee_code/` (its referee's) of `research/tracks/model/`; each is seeded and writes its output next to itself. Sizes are symbol

implementation	ε	fixed w^*, constant δ	$w^* \sim \mathrm{Dir}(\frac{1}{2}, \frac{1}{2})$, constant δ	fixed w^*, shrinking δ_n
model (`c11`, every $n \le 2 \cdot 10^6$, 300 runs)	10^{-5}	0.997	0.007	0/300
	10^{-6}	1.000	0.003	0/300
experiments (`check_fixed_weight`, same)	10^{-5}	0.977 (median first n 8500)	0.003	0/300
	10^{-6}	1.000 (median first n 7525)	0.007	0/300
model's referee (`r3`, geometric grid of n)	10^{-5}	≥ 0.980	≥ 0.008	–
	10^{-6}	≥ 1.000	≥ 0.010	–

Table 22: Example 5.8: probability that the underivable q is ever accepted; nominal bound $\delta/\pi(T^*) = 0.02$. The referee checked a geometric grid of n, so its numbers are lower bounds. The differences between implementations are Monte Carlo and grid effects.

counts in closure-normal form with de Bruijn indices: a quantifier, a connective, a function or relation symbol, a constant, an index and a parameter each count one.

E.1 The two inductors

Proposition E.1 (Why consistent assigners)**.** [proved] model Prop 6.0 *Let R be a unary predicate, $w_1, w_2, \ldots$ distinct closed terms, and φ_i be $R(w_i)$ or $\neg R(w_i)$ by independent fair coins, all labelled true (a consistent labelling).* $\mathrm{FI}_{\mathrm{all}}$*'s log loss is at most a constant (the program "always accept"); the expected log loss of* AI *and of* $\mathrm{FI}_{\mathrm{cons}}$ *is at least n bits.*

This confirms Hänni's informal separation, and is why the fair comparison is with $\mathrm{FI}_{\mathrm{cons}}$.

Proof of Proposition E.1. $\mathrm{FI}_{\mathrm{all}}$: the program "always accept" has some length c and labels every φ_i true, so $W_{\mathrm{FI}_{\mathrm{all}}}(D_n) \ge 2^{-c}$; with $W_{\mathrm{FI}_{\mathrm{all}}}(\emptyset) \le 1$ (Kraft, for a prefix machine) the log loss $-\log_2 W(D_n) + \log_2 W(\emptyset)$ is at most c. AI and $\mathrm{FI}_{\mathrm{cons}}$: let D be the data so far and $\psi := R(w_i)$ for the next index. A hypothesis compatible with $D + (\psi, 1)$ and with $D + (\neg\psi, 1)$ would prove, or accept, both ψ and $\neg\psi$, against consistency. So the two sets of compatible hypotheses are disjoint subsets of those compatible with D, and the predictive probabilities $a := W(D + (\psi, 1))/W(D)$ and $a' := W(D + (\neg\psi, 1))/W(D)$ satisfy $a + a' \le 1$. The coin shows ψ or $\neg\psi$ with probability $\frac{1}{2}$ each, independently of D, so the expected step loss is $\frac{1}{2}(-\log_2 a - \log_2 a') \ge -\log_2 \frac{a+a'}{2} \ge 1$ bit, by convexity of $-\log_2$. Summing over the n steps gives at least n bits. □

Proof of Theorem 6.3. $W_{\mathrm{FI}} \ge 2^{-c'} W_{\mathrm{AI}}$. For a decider p let f_p be the program: on input φ, dovetail searches for proofs of φ and of $\neg\varphi$ from $B \cup A_p$ (both sets are decidable, so the theorems are recursively enumerable), and output acc or rej for the first proof found. Its code is a fixed wrapper followed by p, so $|f_p| \le |p| + c'$ and $p \mapsto f_p$ is injective. Let p be compatible with D. Then $B \cup A_p$ is consistent, so at most one of $\varphi, \neg\varphi$ is provable, and f_p labels each φ_i with b_i. Also $\Gamma_{f_p} \subseteq \mathrm{Th}(B \cup A_p)$, so $B \cup \Gamma_{f_p}$ is consistent and f_p is a compatible $\mathrm{FI}_{\mathrm{cons}}$ hypothesis. Summing $2^{-|f_p|} \ge 2^{-c'} 2^{-|p|}$ over compatible p gives the claim.

$W_{\mathrm{AI}} \ge 2^{-c''} W_{\mathrm{FI}}$ *(Craig's trick).* For an $\mathrm{FI}_{\mathrm{cons}}$ hypothesis f define A_f^C as in Theorem 6.3. Its decider p_f: given χ, try each of the at most $|\chi|$ decompositions $\chi = \psi^{\wedge m}$; for each, run f for $m - 1$ steps on ψ and check whether it accepts in exactly $m - 1$ steps, and, if $\psi = \neg\varphi$, run f on φ and check whether it rejects in exactly $m - 1$ steps. Every run is bounded, so p_f halts on every input, even when f is partial. Each member $\varphi^{\wedge(k+1)}$ is logically equivalent to $\varphi \in \Gamma_f$

(likewise for $\neg\varphi$), and each member of Γ_f has such a power in A_f^C, with k the halting time. So $\mathrm{Cn}(B \cup A_f^C) = \mathrm{Cn}(B \cup \Gamma_f)$, which is consistent, and $B \cup A_f^C$ decides every φ_i as f does. Hence p_f is compatible whenever f is. Its code is a fixed wrapper followed by f: $|p_f| \leq |f| + c''$, and $f \mapsto p_f$ is injective.

Losses. Put $c := \max(c', c'')$. The semimeasure log loss of an inducer on D_n is $-\log_2 W(D_n) + \log_2 W(\emptyset)$; each of the two terms differs by at most c between AI and $\mathrm{FI}_{\mathrm{cons}}$, and both depend on D_n only as a set. If one weight is 0, so is the other, and both losses are infinite. □

Remark E.2 (The convention, and "what is a sequence"). [proved] model §6.1 Renormalising at each step, $P(b \mid \varphi, D) := W(D + (\varphi, b))/\sum_{b'} W(D + (\varphi, b'))$, the argument of Theorem 6.3 gives only $2c$ bits *per step*. The constant equivalence is proved only in the semimeasure convention, which answers Hänni's "wtf is a sequence here?" (any reveal order, any labelling). His "same spec length as the axiom system plus a const" holds in both directions, and B needs no arithmetic. His route through the schema does need arithmetic; it works with two sorts, or in one sort for sound assigners, and Proposition 6.5 shows that it fails for some consistent unsound ones (Section 6.2).

Per-step renormalisation. With $P_X(b \mid \varphi, D) := W_X(D+(\varphi, b))/(W_X(D+(\varphi, 0)) + W_X(D+(\varphi, 1)))$, the numerator and the denominator each change by a factor in $[2^{-c}, 2^c]$ between the inducers, so each step's ratio lies in $[2^{-2c}, 2^{2c}]$; over n steps the bound is $2cn$ bits, and the argument gives nothing better.

E.2 Hänni's schema

Proof of Proposition 6.4. Consistency. Take the structure whose arithmetic sort is the standard $\mathbb{N}$ and whose L-sort is a model M of Γ_f. In $\mathbb{N}$, $\mathrm{Accept}_f(\ulcorner\varphi\urcorner)$ is true iff f accepts φ, and then $\varphi \in \Gamma_f$ holds in M; likewise for rejections. So every member of A_f holds, and so does Q. $\supseteq$. If $f(\varphi) = \mathrm{acc}$, then $\mathrm{Accept}_f(\ulcorner\varphi\urcorner)$ is a true Σ_1 sentence, hence provable in Q (Σ_1-completeness for $<$-free Σ_1 sentences, which applies as Accept_f is written without $<$; known, e.g. Boolos et al., 2007, chapter on Q, not checked); MP with its member of A_f gives φ. Rejections are similar. So $\Gamma_f \subseteq \mathrm{Cn}(B \cup A_f)$. $\subseteq$. An L-sentence provable from $B \cup A_f$ holds in every structure $\mathbb{N} \oplus M$ with $M \models \Gamma_f$; its truth there depends only on M. So it holds in every model of Γ_f, and by completeness it lies in $\mathrm{Cn}(\Gamma_f)$. □

Proof of Proposition 6.5. Γ_f. Since $\mathrm{Con}(\mathsf{PA})$ holds, the search on $\ulcorner\mathrm{Con}(\mathsf{PA})\urcorner$ never ends, so $\Gamma_f = \{\neg\mathrm{Con}(\mathsf{PA})\}$. $\mathsf{PA}+\neg\mathrm{Con}(\mathsf{PA})$ is consistent by Gödel's second incompleteness theorem. $\neg\mathrm{Con}(\mathsf{PA})$. $\mathrm{Accept}_f(\ulcorner\neg\mathrm{Con}(\mathsf{PA})\urcorner)$ is a true Σ_1 sentence, so PA proves it; with its member of A_f, $\mathsf{PA} \cup A_f \vdash \neg\mathrm{Con}(\mathsf{PA})$. $\mathrm{Con}(\mathsf{PA})$. $\mathsf{PA} \vdash \neg\mathrm{Con}(\mathsf{PA}) \to \mathrm{Accept}_f(\ulcorner\mathrm{Con}(\mathsf{PA})\urcorner)$: if a proof of $\bot$ exists there is a least one, and PA proves that the search halts on it with output acc. This formalisation step is standard and already provable in $\mathrm{I}\Sigma_1$. With the member $\mathrm{Accept}_f(\ulcorner\mathrm{Con}(\mathsf{PA})\urcorner) \to \mathrm{Con}(\mathsf{PA})$ of A_f, $\mathsf{PA} \cup A_f \vdash \neg\mathrm{Con}(\mathsf{PA}) \to \mathrm{Con}(\mathsf{PA})$, so $\mathsf{PA} \cup A_f \vdash \mathrm{Con}(\mathsf{PA})$. Both together are a contradiction. □

E.3 Templates

Proof of Proposition 6.6. Let $\tau \in \mathrm{DT}^\circ$ with $\mathrm{inst}(\tau) \subseteq C_f$, and suppose τ has a metavariable M with an occurrence at position q. Constant bodies exist for every type with two different root symbols: $\lambda\bar{z}.0$ and $\lambda\bar{z}.(0+0)$ for terms, $\lambda\bar{z}.(0=0)$ and $\lambda\bar{z}.\neg(0=0)$ for formulas. By the preservation lemma (AS Lemma 2.4), for a constant body the subtree of $\tau\theta$ at q is that constant, whatever the occurrence's arguments, and the rigid symbols of τ above q are the same in every

instance. Every member of C_f has the form $\mathrm{Accept}_f(N) \to \varphi$, where N is the numeral $\ulcorner\varphi\urcorner$ substituted for the free variable of the fixed formula $\mathrm{Accept}_f(x)$. Since $\mathrm{inst}(\tau) \neq \emptyset$, the path from the root to q is a path of such a member.

Case 1: q is the root or lies in the antecedent. (i) If q lies inside an occurrence of the numeral N (its root included), every member of C_f carries S or 0 at q. Put $\theta(M) := \lambda\bar{z}.(0+0)$ and any bodies elsewhere. Then $\tau\theta$ has $+$ at q. If $\tau\theta$ were in C_f, q would again lie inside a numeral, since the path to q is the same; so $\tau\theta \notin C_f$. (ii) Otherwise q is the root or a position of the fixed skeleton of $\mathrm{Accept}_f(x)$. All members of C_f carry the same symbol there ($\to$ at the root). Choose the constant body of the right type whose root symbol differs from it; then $\tau\theta \notin C_f$. Both contradict $\mathrm{inst}(\tau) \subseteq C_f$.

Case 2: no metavariable occurs at the root or in the antecedent. Then the antecedent of every instance is one fixed sentence $\mathrm{Accept}_f(N_0)$, and the only member of C_f with that antecedent is $\mathrm{Accept}_f(\ulcorner\varphi_0\urcorner) \to \varphi_0$ for the single φ_0 with $\ulcorner\varphi_0\urcorner = N_0$. So τ has at most one instance. But a template with a metavariable has infinitely many instances: each type has infinitely many bodies, and $\theta \mapsto \tau\theta$ is injective (AS Thm 2.5(a)). Contradiction.

So τ is ground. A ground template has one instance; C_f is infinite, so no finite set of DT° templates has instance union C_f, and such a set with instances in C_f covers at most as many members as it has templates.

Reflection schema (pa Prop 4.3). With $\mathrm{Prv}_T(x)$ in place of $\mathrm{Accept}_f(x)$ the members are $\mathrm{Prv}_T(\ulcorner\varphi\urcorner) \to \varphi$, with unary numerals, and the proof goes through word for word. □

Implicit in the notes. Case 1(i) uses that "inside the numeral" is determined by the path to q, which is rigid in τ; we have made this explicit. The notes, and the statement, treat the acceptance part C_f. The rejection part, and templates whose instances mix the two parts, are not treated: at a skeleton position where Accept_f and Reject_f carry different symbols, Case 1(ii) would need a constant body whose root differs from both (Appendix H.6).

Proof of Proposition 6.7. (a) Let $f(\varphi) = \mathrm{acc}$ with $\varphi \in \Sigma_n$. Then $\mathsf{PA} \vdash \mathrm{Accept}_f(\ulcorner\varphi\urcorner)$ (a true Σ_1 sentence) and $\mathsf{PA} \vdash \mathrm{Sent}_{\Sigma_n}(\ulcorner\varphi\urcorner)$ (a true Δ_1 fact). So $\mathsf{PA} + \rho_{f,n} \vdash \mathrm{Tr}_n(\ulcorner\varphi\urcorner)$, and $\mathsf{PA} \vdash \mathrm{Tr}_n(\ulcorner\varphi\urcorner) \to \varphi$. For rejections, $\mathsf{PA} + \rho_{f,n} \vdash \neg\mathrm{Tr}_n(\ulcorner\varphi\urcorner)$ and $\mathsf{PA} \vdash \varphi \to \mathrm{Tr}_n(\ulcorner\varphi\urcorner)$, so $\neg\varphi$ follows. (b) In $\mathbb{N}$, Accept_f and Reject_f hold exactly of f's actual decisions, and Tr_n is Σ_n-truth, so both conjuncts of $\rho_{f,n}$ are true when f is Σ_n-sound. *Length.* f enters $\rho_{f,n}$ only through its index, written as a binary numeral of length $O(|f|)$ inside Accept_f and Reject_f; the rest of the sentence depends on n only. □

E.4 Time penalties

Proof of Proposition 6.8. (a) Parsing χ as an implication with antecedent $\mathrm{Accept}_f(t)$ or $\mathrm{Reject}_f(t)$ is a comparison with a fixed formula, linear in $|\chi|$. Computing the Gödel number of the consequent (or of the formula under its negation) is polynomial in $|\chi|$, and so is writing it as a binary numeral and comparing it with t. With unary numerals, t itself has length $\ulcorner\varphi\urcorner \leq |\chi|$, so writing out the numeral is again polynomial in $|\chi|$. No step runs f. (b) $\chi = \psi^{\wedge m}$ forces $m \leq |\chi|$, and there are at most $|\chi|$ candidate decompositions. For each, f is simulated for $m - 1 < |\chi|$ steps on ψ (and on φ if $\psi = \neg\varphi$), in time polynomial in $|\chi|$ for fixed f. □

Proof of Theorem 6.9. Algorithm. Given w of length m, compute φ_w and the time-constructible bound $\ell(m)$, guess a K-derivation of symbol size at most $\ell(m)$ whose parameter numbers have at most $K+1+\lceil\log_2(\ell(m)+1)\rceil$ bits, $K = O(\ell(m)^e)$ the membership-time bound at size $\ell(m)$ (if any

derivation of size at most $\ell(m)$ exists, such a one does, by Lemma E.28; a parameter counts one symbol whatever its number, so without this bound the guess could be arbitrarily long), and check it: each line is an instance of a logical axiom (A1–A3, A5 and the equality axioms by matching; A4 by first-order matching, which is polynomial), a member of T (time $O(\ell^e)$), or follows by MP or Gen. Accept iff the derivation is valid and ends in φ_w. All checks are polynomial in $\ell(m)$, of a degree that depends only on e and on the degree d_{K} of K's line checks; so the machine runs in time $O(\ell(m)^{c_e})$. *Completeness.* If $w \in X$, such a derivation exists by hypothesis. *Soundness.* If $w \notin X$, then $\neg\varphi_w \in \Gamma_{f_X}$, so $T \vdash \varphi_w$ would make $T \cup \Gamma_{f_X}$ inconsistent. No derivation exists, and the machine rejects. □

Proof of Corollary 6.10. Uniformity. The constant c_e of Theorem 6.9 depends on T only through e. Membership in a finite DT° theory is matching against each template, in time $O(|T| \cdot |\chi|)$ (AS Thm 2.5), so $e = 1$ for all of them, with a T-dependent constant factor that does not change the class. *The hierarchy theorem* (known; exact form not checked against Cook, 1972; Seiferas et al., 1978; Žák, 1983): if t_2 is time-constructible and $t_1(m+1) = o(t_2(m))$, then $\mathrm{NTIME}(t_2) \setminus \mathrm{NTIME}(t_1) \neq \emptyset$. (a) Apply it with $t_1 := s^{c_1}$ and $t_2 := s^{c_1+1}$ to get a decidable $X \in \mathrm{NTIME}(s^{c_1+1}) \setminus \mathrm{NTIME}(s^{c_1})$. Suppose some finite DT° theory T with $T \cup \Gamma_{f_X}$ consistent had, for all but finitely many m, a derivation of size at most $s(m)$ of every φ_w with $w \in X$, $|w| = m$. Put the finitely many words of the exceptional lengths in a table. The machine of Theorem 6.9 with $\ell = s$, plus the table, decides X in $\mathrm{NTIME}(s^{c_1})$: a contradiction. (b) The same with c_e in place of c_1, for each e. *A single X for all e* (proof sketch). Take $X \in \mathrm{NTIME}(s^{\log_2 s}) \setminus \mathrm{NTIME}(s^{(\log_2 s)/2})$, when these bounds are time-constructible and satisfy the gap condition. Every s^{c_e} is eventually below $s^{(\log_2 s)/2}$, so the argument of (a) applies to every e. The time-constructibility and gap conditions for these bounds are not checked. □

Remark E.3 (Upper bounds). [proof sketch (A_f^C, two-sorted A_f); not established for $\rho_{f,n}$] model §6.4 The collapse constructions through A_f^C and the two-sorted A_f pay at most a polynomial of the assigner's *deterministic* running time k in derivation size: cite $\varphi^{\wedge(k+1)}$ and eliminate $\wedge$; or prove the true Σ_1 sentence $\mathrm{Accept}_f(\ulcorner\varphi\urcorner)$ in Q, in size polynomial in the computation (recalled from Pudlák, 1998, not checked), then apply MP. For $\rho_{f,n}$ the same route also needs a PA-proof of the Tarski biconditional for φ, whose size is not bounded here.

Upper bounds (Remark E.3); proof sketch. From A_f^C, a derivation of φ cites $\varphi^{\wedge(k+1)}$, $k = \mathrm{time}_f(\varphi)$, and applies $\wedge$-elimination, in size polynomial in $k|\varphi|$. From A_f in two sorts, a derivation is a Q-proof of the true Σ_1 sentence $\mathrm{Accept}_f(\ulcorner\varphi\urcorner)$ followed by MP; the size of such a Q-proof is polynomial in the length of the computation (recalled from Pudlák, 1998, not checked). From $\rho_{f,n}$, the same plus a PA-proof of the Tarski biconditional $\mathrm{Tr}_n(\ulcorner\varphi\urcorner) \to \varphi$, whose size is not bounded in the notes.

E.5 What the derivation likelihoods charge

The counterexample to the pre-referee claim (Remark 6.11; referee M2). Let $\varphi^{(0)} := \forall x\,(x = x)$, the logical axiom of reflexivity (size 4). Round $j = 1, \ldots, k$: apply $\forall$E with the term $p + p$ (p the first parameter), then Gen on p; this doubles the number of occurrences of the bound variable and keeps one $\forall$ at the root. A final $\forall$E with $p + p$ gives φ_k, an equation whose two sides are balanced sums of 2^{k+1} copies of p; its size is $2(2^{k+2} - 1) + 1 = 2^{k+3} - 1$. The tree has one citation node; each $\forall$E contributes its own node and the 5 grammar nodes of $p + p$ (the $+$ node and two parameters, each with a one-node index chain), each Gen its node and a one-node chain:

$\nu = 1 + 6 + 8k = 7 + 8k$. In the referee's grammar ($\alpha_{\mathrm{ax}} = \alpha_{\mathrm{lg}} = \alpha_{\mathrm{MP}} = 0.2$, $\alpha_{\mathrm{Gen}} = \alpha_{\forall\mathrm{E}} = 0.1$, so $m = 0.6$; seven logical schemas cited uniformly; terms $0{:}.4$, $S{:}.2$, $+{:}.1$, $\cdot{:}.1$, parameter: .2; parameter index geometric with ratio $\frac{1}{2}$) a round costs $-\ln(0.1 \cdot 0.1 \cdot (0.2 \cdot 0.5)^2) - \ln(0.1 \cdot 0.5) = 12.206$ nats and the citation plus final $\forall$E 12.77 nats. Every derivation of φ_k contains φ_k, so $\ell_{\min}(\varphi_k) \geq 2^{k+3} - 1$, and $-\ln P_T(\varphi_k) \leq -\ln \Pr(\text{tree})$ because $Z_T \leq 1$. For every $\kappa > 0$ and C the claimed inequality fails for large k; Table 23 gives the numbers.

One-node example. For $B_m(x) := (x + \cdots + x = 0)$ with m summands and $t := S^m 0$, the A4 instance $\forall x B_m(x) \to B_m(t)$ has size quadratic in m but probability $e^{-O(m)}$: the referee's script gives sizes 145, 1765, 6725 symbols at costs at most 99.4, 369.0, 728.5 nats for $m = 10, 40, 80$ (`r2_l1_size.out`).

k	$\nu = 7 + 8k$	$\lvert\varphi_k\rvert = 2^{k+3} - 1$	$-\ln \Pr(\text{tree})$ (nats)	$\ln \lvert\varphi_k\rvert / \nu$
0	7	7	12.77	0.278
1	15	15	24.97	0.181
3	31	63	49.38	0.134
10	87	8191	134.83	0.104
14	119	131071	183.65	0.099
30	247	$8.6 \cdot 10^9$	378.95	0.093

Table 23: The counterexample to the pre-referee claim of Remark 6.11. Sizes were built and checked by the referee's script for $k \leq 14$ (`r2_l1_size.out`; formula beyond) and independently by `c12_l1_size.out`, Part A, for $k \leq 12$, which also counts $\nu = 7 + 8k$. Costs are the referee's, for the grammar given in the text. $\ln \lvert\varphi_k\rvert / \nu \to \ln 2/8 \approx 0.087$, below the $1/e \approx 0.368$ that Lemma E.4 allows.

Lemma E.4 (Symbol size against grammar choices)**.** [proved; computed] model Lemma 6.10 *Let π be a valid* L1 *tree from a finite T with ν grammar nodes, $a_T \geq 2$ a bound on the sizes of the templates of T and of* K*'s logical templates, and $c_T := 2a_T^2$. Every conclusion in π has symbol size at most $(c_T + 1)\nu^2 e^{\nu/e}$.*

Proof of Lemma E.4. For a node v let $G(v)$ be the number of Gen nodes in its subtree and $\Pi(v) := \prod_u |t_u|$ over the $\forall$E nodes u in its subtree, t_u the substituted term ($\Pi = 1$ if there are none). We show $|\mathrm{concl}(v)| \leq (c_T\nu^2 + G(v))\,\Pi(v)$ by induction on the tree. *Citation of a* DT° *template* (of T, or A1–A3, A5, equality; substitution θ). An occurrence $M(\bar{t})$ becomes $\theta(M)[\bar{t}/\bar{z}]$. The body $\theta(M)$ has at most ν nodes and each argument t_i, a subterm of the template, has size at most a_T, so the piece has size at most νa_T. There are at most a_T occurrences and at most a_T rigid symbols, so $|\tau\theta| \leq a_T + a_T^2\nu \leq c_T\nu \leq c_T\nu^2$. *Citation of A4* ($\forall x B \to B[t/x]$, not a template; Lemma A.5). B and t have at most ν nodes each, so the size is at most $1+(|B|+1)+|B|\,|t| \leq 3\nu^2 \leq c_T\nu^2$ (such a tree has $\nu \geq 3$). *$\forall E$ with term t.* The conclusion $\psi[t/x]$ replaces each occurrence of the bound variable by t, which contains no bound indices, so no index shifting is needed: $|\psi[t/x]| \leq |\psi|\,|t| \leq |\mathrm{premise}|\,|t|$, and Π gains the factor $|t|$. *Gen.* Replacing a parameter by an index keeps the size and the new $\forall$ adds one; G grows by one and $\Pi \geq 1$. *MP.* The conclusion is a proper subformula of the major premise, and G and Π only grow from a child to its parent. Finally $G(v) \leq \nu$, and the substituted terms of different $\forall$E nodes are disjoint sets of grammar nodes, so $\sum_u |t_u| \leq \nu$. For k terms, AM–GM gives $\prod_u |t_u| \leq (\nu/k)^k \leq \sup_{y>0}(\nu/y)^y = e^{\nu/e}$. Hence $|\mathrm{concl}(v)| \leq (c_T\nu^2 + \nu)e^{\nu/e} \leq (c_T + 1)\nu^2 e^{\nu/e}$. □

Computed (`c12_l1_size.out`, Part B). Chains without A4 citations satisfy the stronger bound $(c_T + 1)\nu e^{\nu/e}$ (the citation case gives $c_T\nu$). This is what the script tests: on 9865 conclusions of

3000 random valid ∀E/Gen chains, with terms from a PCFG and cited sentences with up to three quantifiers and repeated variables, neither the inductive bound nor the final bound was violated; the largest $\ln(\text{size})/\nu$ was 0.220. The first version of the lemma had $\nu^{\nu+1}$ in place of $e^{\nu/e}$, and ν in place of ν^2; the A4 case (the referee's one-node example) forced ν^2 (model §12.4).

Proposition E.5 (L1 charges its own code length exponentially)**.** [proved; computed]
model Prop 6.11 *Assume* L1 *with $m < 1$, $\mathcal{Q}$ uniformly subcritical (Definition 2.6), T finite and a geometric parameter law. There are $\zeta, C > 0$, depending on the grammar and on T, with $P_T(\varphi) \le C\, e^{-\zeta\, \nu_{\min}(\varphi)}/Z_T$ for every φ.*

Proof of Proposition E.5. The whole tree (derivation nodes, $\mathcal{Q}$ nodes, parameter-chain nodes) is a multi-type branching process with boundedly many children per node, each node's children drawn independently given its type. Give derivation nodes weight 1, $\mathcal{Q}$ nodes weight $\eta h_{\mathcal{Q}}$ ($h_{\mathcal{Q}}$ from Definition 2.6) and chain nodes weight η. A derivation node has expected weighted offspring at most $m + \eta(v_T h_{\mathcal{Q},\max} + 1)$, with v_T bounding the number of metavariables of a template of T or of K; this is below 1 for small η, since $m < 1$. $\mathcal{Q}$ nodes satisfy the condition with $\rho_{\mathcal{Q}}$, and a chain node continues with probability $g_{\text{par}} < 1$. So the uniform condition holds with $\rho_{\text{br}} := \max(m + \eta(v_T h_{\mathcal{Q},\max} + 1), \rho_{\mathcal{Q}}, g_{\text{par}}) < 1$, and Lemma A.2(b),(d) gives $\mathbb{E}[e^{\zeta\nu}] \le C$ for some $\zeta > 0$. A valid tree concluding φ has $\nu \ge \nu_{\min}(\varphi)$, so $P_T(\varphi) = \mu_T(\varphi)/Z_T \le \Pr[\nu \ge \nu_{\min}(\varphi)]/Z_T \le Ce^{-\zeta\nu_{\min}(\varphi)}/Z_T$ by Markov's inequality. □

Computed (`c12_l1_size.out`, Part C). For the joint process with $m = 0.55$ and the grammar $\mathcal{Q}_A$ of the model appendix, 10^5 simulated trees: mean $\nu = 26.9$; $\Pr(\nu \ge L) = 0.362$, 0.137, 0.0288, 0.00238, $3.0 \cdot 10^{-5}$ for $L = 25, 50, 100, 200, 400$; the log-tail falls by 0.0249 nats per node between $L = 100$ and 200.

Theorem E.6 (Code length bounds the assigner's time, logarithmically)**.** [proved] model Thm 6.12
In the setting of Theorem 6.9, let T be a finite DT° *theory with $T \cup \Gamma_{f_X}$ consistent, and let every φ_w, $w \in X$, have a valid* L1 *tree from T with at most $\ell(|w|)$ grammar nodes, $\ell(m) \ge m$ time-constructible. Then $X \in \text{NTIME}(2^{(d+1)\ell(m)})$, where d, the polynomial degree of the syntactic operations used in checking, depends neither on T nor on ℓ.*

Proof of Theorem E.6. On input w, put $\ell := \ell(|w|)$ and guess a tree with at most ℓ grammar nodes. Each node is a choice from a finite list (rule, template of T, logical schema, production of $\mathcal{Q}$) or one of at most $\ell + O(1)$ variables, so the guess has $O(\ell \ln \ell)$ bits. Compute every node's conclusion explicitly, check validity (MP needs equality of formulas, ∀E a root ∀) and compare the root's conclusion with φ_w. By Lemma E.4 every formula has size at most $S := (c_T + 1)\ell^2 e^{\ell/e}$, and all operations are polynomial of a fixed degree d in that size. So the time is at most $\ell \cdot S^d$, which is at most $2^{(d+1)\ell}$ for $\ell \ge \ell_0(T)$, since $e^{d/e} < 2^d$ and the remaining factor is polynomial in ℓ; smaller inputs go in a table. Completeness and soundness are as in Theorem 6.9: citations name a template of T, so no membership test is needed, and a valid tree is a K-derivation from $\bigcup \text{inst}(T)$ (each ∀E is an A4 instance plus MP). □

Corollary E.7 (Hard assigners make plain L1 pay, logarithmically)**.** [proved, given the hierarchy theorem (known)] model Cor 6.13 *Let $s(m) \ge m$ be time-constructible with $s(m)^2 - (d+1)s(m+1) \to \infty$ (every polynomial; 2^m). There is a decidable X such that for every finite* DT° *theory T with $T \cup \Gamma_{f_X}$ consistent, for infinitely many m some $w \in X$ of length m has $\nu_{\min}(\varphi_w) > s(m)$; under the hypotheses of Proposition E.5, $-\ln P_T(\varphi_w) \ge \zeta\, s(m) - \ln(C/Z_T)$ on those data.*

Proof of Corollary E.7. Apply the hierarchy theorem with $t_1 := 2^{(d+1)s}$ and $t_2 := 2^{s^2}$; the condition $t_1(m+1) = o(t_2(m))$ is $s(m)^2 - (d+1)s(m+1) \to \infty$. This gives a decidable $X \in \mathrm{NTIME}(2^{s^2}) \setminus \mathrm{NTIME}(2^{(d+1)s})$. If, for some T and all but finitely many m, every $w \in X$ of length m had $\nu_{\min}(\varphi_w) \le s(m)$, Theorem E.6 with $\ell = s$ and a table (for the exceptional lengths and for $\ell < \ell_0(T)$) would put X in $\mathrm{NTIME}(2^{(d+1)s})$. Since d does not depend on T, one X serves every T. The likelihood bound is Proposition E.5 with $\nu_{\min}(\varphi_w) > s(m)$. □

Proof of Proposition 6.12. (a) Summing over valid trees π concluding φ,

$$\mu_T^\sigma(\varphi) = \sum_\pi \Pr(\pi) e^{-\kappa|\pi|} \le e^{-\kappa \ell^\sigma_{\min}(\varphi)} \sum_\pi \Pr(\pi) = e^{-\kappa \ell^\sigma_{\min}(\varphi)} \mu_T(\varphi) \le e^{-\kappa \ell^\sigma_{\min}(\varphi)} Z_T;$$

divide by Z_T^σ. (b) List the conclusions of all nodes of the tree in post-order (total size ℓ). For each $\forall$E node v with premise u, insert the A4 instance $\mathrm{concl}(u) \to \mathrm{concl}(v)$ before v, of size $|\mathrm{concl}(u)| + |\mathrm{concl}(v)| + 1$; the substituted term has no bound variables, so it is free for x. Then $\mathrm{concl}(v)$ follows by MP; Gen nodes are K's Gen, and citations are axioms of K or members of $\bigcup \mathrm{inst}(T)$. Each node is the premise of at most one node and there are at most ℓ nodes, so the insertions total at most 3ℓ and the derivation has size at most 4ℓ. If $\ell^\sigma_{\min}(\varphi_w) \le s(m)/4$, there would be a K-derivation of size $\le s(m)$; by Corollary 6.10(a) this fails for some $w \in X$ of length m, for infinitely many m. On those data (a) gives $-\ln P_T^\sigma(\varphi_w) \ge \kappa\, \ell^\sigma_{\min}(\varphi_w) - \ln(Z_T/Z_T^\sigma) > \kappa s(m)/4 - \ln(Z_T/Z_T^\sigma)$. (c) An explicit K-derivation written with at least one bit per symbol has bit length at least its symbol size, which exceeds $s(m)$ on the same data. So $g(\ell_T(\varphi_w)) = 2^{-\kappa \ell_T(\varphi_w)} < 2^{-\kappa s(m)}$, and after normalisation $-\log_2 P_T^g(\varphi_w) = \kappa \ell_T(\varphi_w) + \log_2 Z_T > \kappa s(m) + \log_2 Z_T$. □

Implicit in the notes. Part (c) needs the explicit derivations of Remark 2.23 to be derivations in K (or derivations of a calculus that translates into K with linear overhead, which changes the constant) and the code to spend at least one bit per written symbol; the notes say "bits, which is at least their symbol count". We state both assumptions.

The route to Conjecture 6.13, and what is missing. Represent conclusions with sharing (formula DAGs, grammar-compressed terms), so that $\forall$E need not copy t into every occurrence; for the doubling family above a DAG of size $O(k)$ suffices. That every tree with ν grammar nodes has such representations of size polynomial in ν is part of what has to be shown. MP needs equality of two compressed formulas; equality of compressed strings and of grammar-compressed trees is decidable in polynomial time (recalled by the track, not checked: Plandowski 1994; Busatto, Lohrey and Maneth 2008; Schmidt-Schauß 2005). The missing step is Gen and de Bruijn indices under sharing: abstracting a parameter turns it into an index whose value depends on the binder depth at each occurrence, so a shared subterm can need different indices in different contexts. With that step, the proof of Theorem E.6 would give $\mathrm{NTIME}(\ell^c)$. Model §10 lists this as problem 12, together with the derivation-penalised equivalence (problem 13), a single-sorted repair of Hänni's schema for consistent but unsound assigners (problem 14), and for which f the set A_f^C, or a set with its consequences and polynomial membership, is a finite union of DT° templates (problem 15). The follow-ups answer problem 13 for hard bounds (Proposition E.21), problem 14 in the form of Hänni's schema (Proposition E.17), and problem 15 for literal consequences on literal data (Proposition E.24; conservativity open).

Remark E.8 (Prior variants on the E2 data). [computed] experiments E7 On the E2 data (25 seeds), a membership-time factor $(1+|T|)^{-\tau}$ shifts the log-odds of T^* against a hand competitor by at most $\log_2(78/2) \approx 5.3$ bits at $\tau = 1$; it changes nothing visible from $n = 64$ on and at $n \le 32$

favours small unsound lumps, slightly at $\tau = 1$ (unsound MAP at $n = 8$ in 9 instead of 7 seeds) and markedly at $\tau = 4$ (mean unsound mass at $n = 8$ 0.372 against 0.256; a false probe accepted in 5 instead of 2 seeds). A steeper simplicity penalty $\lambda = 2$ makes the MAP at $n = 8$ unsound in 22 seeds ($\lambda = 1$: 7) but removes the false spare slot faster ($2 \cdot 10^{-4}$ against 0.004 unsound mass at $n = 512$); $\lambda = \frac{1}{2}$ (proper only on finite pools) the reverse: neither removes both failures (Table 39).

Remark E.9 (Links to the other sections). [(i) proved; (ii), (iv): proof sketch; numbers of (iv): computed] universal Prop U2t, §8; experiments E7; pa §3.8, F9 (i) No penalty monotone in derivation length or time reverses Theorem 3.4: in $\mathsf{C}_{\min}$ a chain weight non-increasing in the number of eliminations keeps $P^f_{H_\forall}(s) \leq P^f_{H_{\mathrm{sch}}}(s)$ on every datum $s \neq \forall x\varphi$ (Proposition B.1, U2t). This counts steps, not symbols. (ii) In the equational fragment $\{\mathrm{Q4}, \mathrm{Q5}\}$, a Kt-style penalty on generating $0+S^k0 = S^k0$ charges $O(\ln k)$, less than the derivation charge of Proposition B.15 (at least $k+1$ axiom leaves). (iii) The experiments' membership-time factor is a small extra simplicity penalty by construction (Remark E.8). (iv) A time-bounded evaluation of the likelihood sums over the derivations found within the bound, so it lower-bounds $\mu_T(s)$ and favours adopting theorems with long proofs as axioms; a two-part code is the extreme case. Under $\mathsf{L1}^{\mathrm{sch}}$ with its default axiom prior, memorising already beats deriving at the first occurrence for 11-line proofs (Proposition 7.9, Table 3). (ii) and (iv) agree: a derivation likelihood favours the theory that makes the data cheap to state, not the logically minimal one. A PA comparison under a derivation-time penalty ($\mathrm{I}\Sigma_n$ as Q plus one partial-truth sentence, against the schema) was not quantified.

E.6 Which computation is free

The results of Section 6.6 come from two follow-up tracks, `research/tracks/time-followup/` and `research/tracks/short-derivations/`, each refereed once by two referees (Appendix H.4). The proofs follow their final notes. Scripts cited below are in those folders (`checks/` for the tracks, `referee_*/` for their referees); each is seeded or deterministic and writes its output next to itself.

Machines. Hypotheses are programs of the universal prefix machine U: words of a prefix-free code, read to their end before the input, so $\mathrm{time}_p(x) \geq |p|$; for a fixed wrapper W, $W^\frown q$ is again a program that finds the end of q. *Assumption (S)* (efficient self-simulation; standard for multitape machines, not checked for a particular U): there are constants $\alpha, a \geq 1$ such that, for the fixed wrappers used below, reading q, copying it and computing $|q|$ takes at most $\alpha(|q|+1)$ steps; parsing a sentence, or comparing or copying strings of total length L, at most $\alpha(L+1)^2$ steps; and $W^\frown q$ can run the *dovetailing schedule* on q (at stage s, start q on the s-th sentence and advance every unfinished run by one simulated step) while keeping a *work counter* T (the simulated steps, plus the lengths of the sentences started, plus $|q|$), within $\alpha(T+1)^a$ steps up to work T, not counting output, writing a string of length L costing at most αL steps. The counter starts at $|q|$ and grows with every simulated step, so distinct detections happen at distinct counter values. Put $a' := \max(a, 2)$ and $a^* := \max(a, 3)$. *Clocked strings.* U_0 is a fixed plain universal machine. For a time-constructible τ and a string f, $f^\tau(\varphi) := U_0(f, \varphi)$ if U_0 halts within $\tau(|\varphi|)$ steps with output acc or rej, and f^τ abstains otherwise. A string has weight $w(f) := 2^{-|E(f)|}$, where $E(f)$ is the Elias-δ code of $|f|+1$ followed by f; $\sum_f w(f) = 1$, and the tails of this sum are computable. *Overheads* (standard polynomial-overhead simulations; assumed, like (S)): q is a fixed nondecreasing polynomial with nonnegative coefficients such that, on literal sequences (Definition E.18, $|\varphi_w| = |w| + 2$), (i) a string labelling every φ_w correctly within $T(|\varphi_w|)$ steps of U_0 (or a U-program within T steps of U) yields a multitape machine deciding X in time

$q(C_f T(m+2)+m)$, $m=|w|$, C_f depending on the string; (ii) a multitape machine deciding X within $T(|w|)$ steps yields a string labelling each φ_w correctly within $q(T(|w|)+|\varphi_w|)$ steps of U_0; (iii) U_0 runs a U-program p for T steps within $q(C_pT)$ steps. Since $q(Cx)\le C^{\deg q}q(x)$ for $C\ge 1$, a growing factor m absorbs every constant from some length on. DTIME and NTIME refer to multitape machines.

Definition E.10 (Penalised inductors)**.**

time-followup Defs 1.1, 1.2 Let t be time-constructible, nondecreasing, superadditive, with $t(m)\ge m$. Compatibility is as in Definition 6.2.

- $\mathrm{AI}_t^{\mathrm{chk}}$: deciders p with $\mathrm{time}_p(\chi)=O(t(|\chi|))$, weight $2^{-|p|}$. $\mathrm{AI}_t^{\mathrm{Kt}}$: all deciders, weight $2^{-|p|}/c_p$, $c_p:=\sup_\chi \mathrm{time}_p(\chi)/t(|\chi|)$ (weight 0 if $c_p=\infty$).
- $\mathrm{AI}_t^{\mathrm{gen}}$: enumerators g, programs without input writing $a_1,a_2,\dots$, with $A_g:=\{a_i\}$; e_i is the step at which a_i is completely written, $e_0:=0$. Weight $2^{-|g|}$ if $e_i-e_{i-1}=O(t(|a_i|))$ (the *delay form*; polynomial delay, Johnson et al., 1988), 0 otherwise. Variants: the *strict form* $e_i=O(t(|a_i|))$ and the *cumulative form* $e_i=O(t(|a_1|+\cdots+|a_i|))$. Strict implies delay, and delay implies cumulative by superadditivity.
- $\mathrm{FI}_{\mathrm{cons}}^{\tau}$: clocked strings f with $B\cup\Gamma_{f^\tau}$ consistent, weight $w(f)$, compatible with D if f^τ labels every datum correctly; $\mathrm{FI}_{\mathrm{all}}^{\tau}$ drops consistency. $\mathrm{FI}_{\mathrm{cons}}^{\mathrm{poly}}$ mixes over pairs (f,k), f clocked at $(|\varphi|+2)^k$, with weight $w(f)2^{-2\lceil\log_2(k+1)\rceil-1}$.
- A *predictor* maps a history D and a sentence φ to (q_0,q_1) with $q_b\ge 0$, $q_0+q_1\le 1$; its loss is $\ell_P(D_n):=\sum_{j\le n}-\log_2 q_{b_j}(D_{j-1},\varphi_j)$, which for a mixture in the semimeasure convention equals $\ell(D_n)$, by telescoping. It is *computable* if a total computable function maps (D,φ,k) to rationals within 2^{-k} of q_0,q_1. Hänni's S is read on labelled sentences: its components see the labelled history and the next sentence. Only this is used: S is a computable predictor, and it is the Bayes mixture of its components as counted (each predicting $\frac12$ before its start and when over budget), one of which, "label 1 with probability 1", runs within every budget, has prior weight $\pi_1>0$ and is tracked from a finite step ℓ_1 on.

Lemma E.11 (Bounded loss on an infinite sequence)**.** [proved] time-followup Lemma 1.6 *Let $W(D)=\sum_h v(h)[h \text{ compatible with } D]$ with $\sum_h v(h)<\infty$ and compatibility inherited by prefixes, and let D_n be the prefixes of an infinite labelled sequence. Then $\sup_n \ell(D_n)<\infty$ iff a single h with $v(h)>0$ is compatible with every D_n.*

Proof. If h is, $W(D_n)\ge v(h)$ and $\ell(D_n)\le\log_2(W(\emptyset)/v(h))$. If none is, each indicator is eventually 0, so $W(D_n)\to 0$ by dominated convergence and $\ell(D_n)\to\infty$. □

Every inductor of this subsection is such a mixture. The *fit class* of an inductor is the set of X with $\sup_n \ell(D_n^X)<\infty$ (literal sequences: Definition E.18).

Lemma E.12 (Padded Craig sets)**.** [proved, under (S); computed]

time-followup Lemma 2.1; scripts: c2_padding, referee_penalty r2_enumeration *Let q be a program and S_q the set of sentences on which it halts with output "yes". The enumerator $e_q:=W_{\mathrm{enum}}{}^\frown q$ runs the dovetailing schedule on q and, when it first detects that q says yes on χ, at work counter T, writes $\chi^{\wedge(T+1)}$; $\mathrm{Pad}(q)$ is the set written. Then: (a) $\mathrm{Cn}(B\cup\mathrm{Pad}(q))=\mathrm{Cn}(B\cup S_q)$; (b) $\mathrm{Pad}(q)$ has a decider $d_q:=W_{\mathrm{dec}}{}^\frown q$ with $\mathrm{time}_{d_q}(\chi)\le\alpha(|q|+1)+3\alpha(|\chi|+1)^{a'}$; (c) e_q writes each member once; the member a_i, written at counter T_i, has $|a_i|\ge T_i+1$, and $e_i-e_{i-1}\le 2\alpha|a_i|^a$, $e_i\le 5\alpha|a_i|^{a^*}$ and $e_i\le 2\alpha(|a_1|+\cdots+|a_i|)^a$; (d) $|d_q|,|e_q|\le|q|+c$, and $q\mapsto d_q$, $q\mapsto e_q$ are injective.*

Proof. (a) $\chi^{\wedge m}$ is equivalent to χ; each $\chi \in S_q$ is detected once, so it has exactly one padded copy, and every member is a copy of a member of S_q. (b) d_q reads and copies q and computes $|q|$, in $\alpha(|q|+1)$ steps. If $\chi = \psi^{\wedge m}$ with $m \geq 2$, then ψ is the left conjunct of χ, and m is fixed by $|\chi|$, since $|\psi^{\wedge m}| = m|\psi| + 3(m-1)$ in primitive syntax; the only other candidate is $m = 1$, $\psi = \chi$. Both are found within $\alpha(|\chi|+1)^2$ steps. A candidate with $m - 1 < |q|$ is rejected, as the counter starts at $|q|$. Otherwise d_q runs the schedule without writing until the counter exceeds $m-1$, and accepts iff q is detected saying yes on ψ exactly at counter $m-1$; as $m \leq |\chi|$, this costs at most $\alpha(|\chi|+1)^a$ steps by (S). The runs are bounded by the counter, so d_q halts on every input, also when q is partial. (c) Detections are of distinct sentences, so each member is written once, and $|a_i| = |\chi_i^{\wedge(T_i+1)}| \geq T_i + 1$. Between the completions of a_{i-1} and a_i the schedule costs at most $\alpha(T_i+1)^a$ and writing a_i at most $\alpha|a_i|$: the delay bound. Up to a_i at most T_i+1 members were written, at distinct counters $T_j \leq T_i$, each of length at most $(T_j+1)(T_j+3)$ since $|\chi_j| \leq T_j$ (the length of χ_j is part of the counter); so the total written length is at most $4(T_i+1)^3 \leq 4|a_i|^3$, and $e_i \leq \alpha|a_i|^a + 4\alpha|a_i|^3$. Finally $e_i \leq \alpha(T_i+1)^a + \alpha\sum_{j\leq i}|a_j|$. (d) d_q and e_q are fixed wrappers followed by q. □

Proof of Theorem 6.14. (a) *Lower bounds.* For an $\mathrm{FI}_{\mathrm{cons}}$ hypothesis f let q_f say yes on φ if $f(\varphi) = \mathrm{acc}$ and on $\neg\varphi$ if $f(\varphi) = \mathrm{rej}$; then $S_{q_f} = \Gamma_f$ and $|q_f| \leq |f| + c$. By Lemma E.12(a), $\mathrm{Cn}(B \cup \mathrm{Pad}(q_f)) = \mathrm{Cn}(B \cup \Gamma_f)$, which is consistent and decides each datum as f does, so d_{q_f} and e_{q_f} are compatible whenever f is. By (b) and (c) they meet the membership bound and every form of the generation bound for $t(m) \geq m^{a^*}$, and by (d) the maps are injective and add a constant to the length. *Upper bounds.* $\mathrm{AI}_t^{\mathrm{chk}}$ has a subset of AI's hypotheses with the same weights; apply Theorem 6.3. An enumerator g goes to $f_g :=$ "search for proofs of φ and of $\neg\varphi$ from $B \cup A_g$" (A_g is r.e.), with $|f_g| \leq |g| + c$; if g is compatible, f_g labels each datum correctly and $\Gamma_{f_g} \subseteq \mathrm{Cn}(B \cup A_g)$ is consistent with B. (b) Let X be AI, $\mathrm{AI}_t^{\mathrm{chk}}$ or $\mathrm{AI}_t^{\mathrm{gen}}$. By (a) and Theorem 6.3, $W_X(D) \geq 2^{-c} W_{\mathrm{FI}}(D)$, and $W_X(\emptyset) \leq 1$ by Kraft. For a clocked string h let g_h be a fixed wrapper that runs h^τ and diverges when it abstains, followed by $E(h)$. It computes the same partial map as h^τ, so it is an $\mathrm{FI}_{\mathrm{cons}}$ hypothesis, compatible with D iff h is; $|g_h| = |E(h)| + c_1$, and $h \mapsto g_h$ is injective. (Equal sets Γ would not suffice: an assigner rejecting ψ and one accepting $\neg\psi$ and abstaining on ψ have the same Γ but differ on the datum $(\psi, 0)$.) So $W_{\mathrm{FI}}(D) \geq 2^{-c_1} W_{\mathrm{FI}^\tau_{\mathrm{cons}}}(D)$ and $\ell_X(D) \leq -\log_2 W_{\mathrm{FI}^\tau_{\mathrm{cons}}}(D) + c + c_1 = \ell_{\mathrm{FI}^\tau_{\mathrm{cons}}}(D) - \log_2 W_{\mathrm{FI}^\tau_{\mathrm{cons}}}(\emptyset) + c + c_1$, where $W_{\mathrm{FI}^\tau_{\mathrm{cons}}}(\emptyset)$ is at least the weight $2^{-c_\perp}$ of the always-abstaining string (B consistent). Put $c_\tau := c + c_1 + c_\perp$. □

Proposition E.13 (A Kt penalty costs a logarithm)**.** [proved, under (S); computed]
time-followup Prop 2.3; scripts: referee_penalty r1_kt_prior *Let $t(m) \geq m^{a^*}$ be time-constructible, m_0 the least length of a sentence, and $W_{\mathrm{FI}^1}(D) := \sum_f 2^{-|f|}/(|f|+1)$ over the $\mathrm{FI}_{\mathrm{cons}}$ hypotheses compatible with D. Then $W_{\mathrm{AI}_t^{\mathrm{Kt}}}(\emptyset) \leq t(m_0)$, and there is a constant c with $2^{-c} W_{\mathrm{FI}^1}(D) \leq W_{\mathrm{AI}_t^{\mathrm{Kt}}}(D) \leq 2^c t(m_0) W_{\mathrm{FI}^1}(D)$ for every D; so the two losses differ by at most $2c + 2\log_2 t(m_0)$, and an assigner f pays $\log_2(|f|+1) + O(1)$ bits beyond its unpenalised weight.*

Proof. $\mathrm{time}_p(\chi) \geq |p|$, so $c_p \geq |p|/t(m_0)$ and p has weight at most $t(m_0)2^{-|p|}/|p|$; by Kraft the total is at most $t(m_0)$. *Lower bound.* For an $\mathrm{FI}_{\mathrm{cons}}$ hypothesis f let $d := d_{q_f}$. By Lemma E.12(b), as $(|\chi|+1)^{a'} \leq 2^{a'}|\chi|^{a^*} \leq 2^{a'} t(|\chi|)$ and $t \geq 1$, $c_d \leq \alpha(|q_f|+1) + 3\alpha 2^{a'} \leq \beta(|f|+1)$ with β depending only on U and the coding; so d has weight at least $2^{-|f|-2c}/(\beta(|f|+1))$, is compatible when f is, and $f \mapsto d$ is injective. *Upper bound.* A compatible p goes to the proof-search assigner f_p of Theorem 6.3, compatible, with $|p| \leq |f_p| \leq |p| + c$; p's weight is at most $t(m_0)2^{-|p|}/|p| \leq t(m_0)(c+2)2^c\, 2^{-|f_p|}/(|f_p|+1)$, and $p \mapsto f_p$ is injective. Apply both bounds to D and to $\emptyset$. □

Lemma E.14 (What Q proves about computations)**.** time-followup Lemma 1.4 *Write $\underline{k}$ for the binary numeral of k ($\underline{0} := 0$, $\underline{1} := S0$, $\underline{2k} := SS0 \cdot \underline{k}$, $\underline{2k+1} := SS0 \cdot \underline{k} + S0$ for $k \geq 1$) and S^k0 for the unary one. (a)* [known: the MRDP theorem (Matiyasevich, 1970; Davis, 1973); not checked] *There are polynomials E, E' with natural-number coefficients such that program e halts on input x with output y iff $\mathbb{N} \models \exists \bar{c}\,(E(e,x,y,\bar{c}) = E'(e,x,y,\bar{c}))$. Put $\mathrm{Out}(e,x,y) := \exists \bar{c}\,(E = E')$, a Σ_1 formula without $<$ and without bounded quantifiers. (b)* [proved] *For every closed L_A-term t of value k, $\mathsf{Q} \vdash t = S^k0$. (c)* [proved] *For $a \neq b$, $\mathsf{Q} \vdash \neg S^a0 = S^b0$; so Q proves every true and refutes every false closed equation, and proves every true sentence $\exists \bar{y}\,(s = s')$. (d)* [proved; computed: referee script `r2_q_models`] *For every k, $\mathsf{Q} \vdash \forall y\,(\exists z\,(z + y = S^k0) \to y = 0 \vee y = S0 \vee \cdots \vee y = S^k0)$.*

Proof. (b) Induction on t: St' by the equality axioms; $\mathsf{Q} \vdash S^a0 + S^b0 = S^{a+b}0$ by b uses of Q5 and one of Q4; $\mathsf{Q} \vdash S^a0 \cdot S^b0 = S^{ab}0$ by induction on b with Q6, Q7 and the + case. (c) For $a < b$, a uses of Q2 turn $S^a0 = S^b0$ into $0 = S^{b-a}0$, against Q1. A closed equation reduces by (b) to one between unary numerals. For a true $\exists \bar{y}(s = s')$, instantiate $\bar{y}$ by the numerals of a witness. (d) Induction on k. For $k = 0$: if $y = Sy'$ (Q3), then $z + Sy' = S(z + y') \neq 0$ by Q5 and Q1, so $y = 0$. From k to $k+1$: if $y \neq 0$, then $y = Sy'$ and $S(z + y') = S(S^k0)$, so $z + y' = S^k0$ by Q2, and the induction hypothesis places y', hence y. □

Proof of Theorem 6.15.

time-followup Thm 3.2, Cor 3.3, Prop 3.4; scripts: c1_diagonal, c4_revision Part A, referee_logic r1_quine_diagonal Let $\mathrm{DET} := \forall e \forall x \forall y \forall y' \forall \bar{c} \forall \bar{c}'\,(E(e,x,y,\bar{c}) = E'(e,x,y,\bar{c}) \wedge E(e,x,y',\bar{c}') = E'(e,x,y',\bar{c}') \to y = y')$, a true Π_1 sentence, and $H_0 := \mathsf{Q} \cup \{\mathrm{DET}\}$. *Construction.* Let $G(e', n)$ be the program that, for $j = 1, \ldots, n$, forms $\sigma_j := \mathrm{Out}(\underline{e'}, \underline{j}, S0)$, asks P for approximations (a_0, a_1) within 2^{-j-3} of its pair on the history $((\sigma_1, \beta_1), \ldots, (\sigma_{j-1}, \beta_{j-1}))$ and the sentence σ_j, and sets $\beta_j := 0$ if $a_0 \leq a_1$, $\beta_j := 1$ otherwise; it outputs β_n. G is total, as P is. By Kleene's recursion theorem (Rogers, 1967, §11.2; section not checked) there is e with $e(n) = G(e, n)$ for all n. The values β_j computed inside $G(e, n)$ do not depend on n, so $e(j) = \beta_j \in \{0, 1\}$ and $\sigma_j = s_j$; by Lemma E.14(a), $\mathbb{N} \models s_n$ iff $e(n) = 1$, so the label b_n is β_n, and P is queried exactly on the true histories D_{j-1}. *H_0 decides each s_n.* If $b_n = 1$, s_n is a true existential equation, provable in Q (Lemma E.14(c)). If $b_n = 0$, $\mathrm{Out}(\underline{e}, \underline{n}, 0)$ is one, so Q proves it, and with s_n and DET it gives $S0 = 0$, against Q1; so $H_0 \vdash \neg s_n$. (H_0 is not complete; only the truth of DET is used below.) *The loss of P.* With $q_b := q_b(D_{j-1}, s_j)$ and $\varepsilon := 2^{-j-3}$, the tie rule gives $q_{b_j} \leq a_{b_j} + \varepsilon \leq a_{1-b_j} + \varepsilon \leq q_{1-b_j} + 2\varepsilon$; with $q_0 + q_1 \leq 1$ this gives $q_{b_j} \leq \frac{1}{2} + 2^{-j-3}$, so $-\log_2 q_{b_j} \geq 1 - \log_2(1 + 2^{-j-2})$, and $\sum_j \log_2(1 + 2^{-j-2}) \leq \sum_j 2^{-j-2}/\ln 2 = 1/(4 \ln 2) < 0.361$. (With exact outputs the loss is at least n; the bound is attained up to rounding, script `c4_revision`, Part A.) *Axiom induction.* H_0 is true in $\mathbb{N}$ and B in an expansion of $\mathbb{N}$, so $B \cup H_0$ is consistent, and H_0 is compatible with every D_n. It is finite, so it has a decider running in linear time and an enumerator writing finitely many axioms; let $\omega > 0$ be the least of their weights in AI, $\mathrm{AI}^{\mathrm{chk}}_t$ and $\mathrm{AI}^{\mathrm{gen}}_t$. Then $W(D_n) \geq \omega$, $W(\emptyset) \leq 1$, and $\ell(D_n) \leq \log_2(1/\omega)$, whatever P is. *S, $\mathrm{FI}^{\tau}_{\mathrm{all}}$ and $\mathrm{FI}^{\tau}_{\mathrm{cons}}$.* S is a computable predictor (Definition E.10). For $\mathrm{FI}^{\tau}_{\mathrm{all}}$ put $P(D) := (W_{\mathrm{FI}^{\tau}_{\mathrm{all}}}(D) + 2^{-|D|})/2$ and $q_b(D, \varphi) := P(D + (\varphi, b))/P(D)$. Then $q_0 + q_1 \leq 1$, as $\mathrm{FI}^{\tau}_{\mathrm{all}}$ is a semimeasure and the uniform part halves at each step; P is computable, as clocked compatibility is decidable and the weights have computable tails, and $P(D) \geq 2^{-|D|-1} > 0$; and $W_{\mathrm{FI}^{\tau}_{\mathrm{all}}}(\emptyset) = 1$, so $P(\emptyset) = 1$ and $\ell_P(D_n) = -\log_2 P(D_n)$. On P's sequence $W_{\mathrm{FI}^{\tau}_{\mathrm{cons}}}(D_n) \leq W_{\mathrm{FI}^{\tau}_{\mathrm{all}}}(D_n) \leq 2P(D_n)$, so $\ell_{\mathrm{FI}^{\tau}_{\mathrm{all}}}(D_n) \geq n - 1.37$ and $\ell_{\mathrm{FI}^{\tau}_{\mathrm{cons}}}(D_n) \geq \ell_P(D_n) - 1 + \log_2 W_{\mathrm{FI}^{\tau}_{\mathrm{cons}}}(\emptyset) \geq n - 1.37 - c_\perp$, with $c_\perp := -\log_2$ of the weight of the always-abstaining string. *The converse.* On the fair-coin literals of Proposition E.1, all labelled true, S gives the label sequence at least the probability its constant component gives it, times π_1: at least $\pi_1 2^{-\ell_1}$. So $\ell_S \leq \log_2(1/\pi_1) + \ell_1$, and $\mathrm{FI}^{\tau}_{\mathrm{all}}$ has loss at most $-\log_2 w(\text{always accept})$ once τ covers the constant time of "always accept". By Proposition E.1, AI (and $\mathrm{FI}_{\mathrm{cons}}$) lose at least n bits in expectation. □

"Axiom induction becomes strictly stronger than every time-bounded function inductor", in the first version of the track's answer, is refuted by the converse part: S and $\mathrm{FI}^{\tau}_{\mathrm{all}}$ are time-bounded and incomparable with AI. The statement proved is strict superiority over time-penalised *consistent* function induction, the fair comparison of Section 6.1.

Proposition E.15 (The evaluator; a time-hierarchy effect)**.** [(a), (b) proved; (c) proof sketch, resting on Hänni's construction of S and his theorem for it] time-followup Prop 4.1 *(a) Let u be the assigner that, on an input $\mathrm{Out}(\underline{e},\underline{n},S0)$, runs e on n and outputs acc if the result is 1 and rej if it is anything else (looping if e loops), and abstains on all other inputs. Then $\Gamma_u \subseteq \mathrm{Th}(\mathbb{N})$, so u is consistent with every B true in an expansion of $\mathbb{N}$, and $|u|$ is a constant. (b) On every sequence of Theorem 6.15, u labels each datum correctly, so unpenalised $\mathrm{FI}_{\mathrm{cons}}$ (and $\mathrm{FI}_{\mathrm{all}}$) has $\ell(D_n) \le |u|$ there. The running time of u on s_n is at least that of e on n, which includes the time P spends on its first n predictions. (c) Let S_p be Hänni's predictor with per-step time $p(n)\log n$ and state reuse. The labels that defeat S_p are produced by a predictor with per-step time $O(p(n)\log n)$ and state reuse that predicts its own labels with certainty; so $S_{p'}$, for a slightly larger p' (e.g. $np(n)$), has bounded loss on the sequence that defeats S_p.*

Proof. (a) $\mathrm{Out}(\underline{e},\underline{n},S0)$ is true iff e outputs 1 on n, so u's decisions are true. (b) In Theorem 6.15 e is total with values in $\{0,1\}$; then $\ell(D_n) = -\log_2 W(D_n) + \log_2 W(\emptyset) \le |u|$. (c) Not written out: Hänni's version of his theorem that gives the compared predictors the same state reuse, and the machine-model caveat he names. □

Proposition E.16 (Disguised assigners keep their weight)**.** [proved, under (S)] time-followup Prop 5.1(a) *Let X be $\mathrm{AI}^{\mathrm{chk}}_t$ or $\mathrm{AI}^{\mathrm{gen}}_t$ (any form), $t(m) \ge m^{a^*}$. For a class Θ of deductively closed theories let $W(\Theta,D)$ be the weight of the compatible hypotheses p with $\mathrm{Cn}(B \cup A_p) \in \Theta$. There is a constant c with $2^{-c}W_{\mathrm{AI}}(\Theta,D) \le W_X(\Theta,D) \le 2^c W_{\mathrm{AI}}(\Theta,D)$ for all Θ and D. So $W_X(\Theta,D) = 0$ iff $W_{\mathrm{AI}}(\Theta,D) = 0$; otherwise the posterior shares $W(\Theta,D)/W(D)$ under X and AI differ by a factor at most 2^{2c}, and so do the posterior odds of any two classes of positive weight, "genuine" and "disguised" alike.*

Proof. Lower bound. An AI hypothesis p semi-decides A_p via $q_p :=$ "yes iff p accepts"; Lemma E.12 gives d_{q_p} and e_{q_p}, of length at most $|p|+c$, meeting the penalty, with $\mathrm{Cn}(B \cup \mathrm{Pad}(q_p)) = \mathrm{Cn}(B \cup A_p)$, and compatibility depends only on the theory. *Upper bound.* For $\mathrm{AI}^{\mathrm{chk}}_t$ the hypotheses are AI's with the same weights. For $\mathrm{AI}^{\mathrm{gen}}_t$ map g to d_{q_g}, $q_g :=$ "yes iff g writes the input", which preserves the theory and adds at most c to the length. Divide, using $W(D) = W(\text{all theories}, D)$. □

So under these penalties a slow consistent assigner f keeps prior weight at least $2^{-|f|-c}$, as a padded Craig set, and its posterior share stays within a constant factor of its unpenalised one.

Proposition E.17 (Hänni's schema with a binary witness bound)**.** [proved, given the MRDP theorem (known); computed] time-followup Prop 2.5; scripts: referee_logic r2_q_models *One sort, $L \supseteq L_A$, $B \supseteq \mathsf{Q}$ decidable, true or not. With* Out *from Lemma E.14(a) write $\mathrm{Out}(\underline{\ulcorner f \urcorner}, x, \underline{\ulcorner \mathrm{acc} \urcorner}) = \exists \bar{y}\,(P_f(x,\bar{y}) = P'_f(x,\bar{y}))$, $\bar{y} = (y_1,\dots,y_r)$, and put $\mathrm{Acc}^{\le k}_f(x) := \exists \bar{y}\,(y_1 \le \underline{k} \wedge \dots \wedge y_r \le \underline{k} \wedge P_f(x,\bar{y}) = P'_f(x,\bar{y}))$, with $y \le s$ written $\exists z\,(z + y = s)$; likewise $\mathrm{Rej}^{\le k}_f$. Let $A^{\mathrm{bd}}_f := \{\mathrm{Acc}^{\le k}_f(\underline{\ulcorner \varphi \urcorner}) \to \varphi,\ \mathrm{Rej}^{\le k}_f(\underline{\ulcorner \varphi \urcorner}) \to \neg\varphi : \varphi \in \mathrm{Sent}_L,\ k \in \mathbb{N}\}$. Then $\mathrm{Cn}(B \cup A^{\mathrm{bd}}_f) = \mathrm{Cn}(B \cup \Gamma_f)$, for sound and unsound assigners alike, and membership in A^{bd}_f is decidable in time polynomial in $|\chi|$ without running f.*

Proof. Fix φ and k. If f accepts φ and the acceptance equation has a solution in $\{0,\dots,k\}^r$, Q proves the antecedent (instantiate $\bar{y}$, and each z_i by the numeral of $k - y_i$; every conjunct is

a true closed equation, Lemma E.14(c)), so the axiom is Q-equivalent to $\varphi \in \Gamma_f$. Otherwise Q refutes the antecedent: by Lemma E.14(b), (d) each y_i is one of $0, S0, \dots, S^k0$, and each of the $(k+1)^r$ cases makes $P_f = P'_f$ a false closed equation, which Q refutes; so the axiom is a theorem of Q. Rejections are the same, and every member of Γ_f has its axiom (take k large). Membership: parse χ against the fixed formula with two numeral slots and compare one slot with $\underline{\ulcorner\varphi\urcorner}$. □

In the counterexample of Proposition 6.5 every antecedent $\mathrm{Acc}_f^{\leq k}(\underline{\ulcorner\mathrm{Con}(\mathsf{PA})\urcorner})$ has no solution, so $\mathsf{PA} \cup A_f^{\mathrm{bd}}$ is equivalent to $\mathsf{PA} + \neg\mathrm{Con}(\mathsf{PA})$, which is consistent: the numeral bound keeps out the nonstandard witnesses. This is a repair in the form of Hänni's schema, for every consistent assigner, in one sort; in substance it is Craig's idea, the bound k playing the role of the padding.

Deduction charged: syntax. Formulas are as in Definition 2.2: closure-normal form with de Bruijn indices, parameters $p_0, p_1, \dots$ read under universal closure, connectives $\neg, \to$, quantifier $\forall$, a finite signature with equality. A *line* has no dangling index; a sentence has no parameter either. $|\chi|$ counts nodes (Appendix E); for the formula nodes v of χ, $F(\chi) := \sum_v |\chi_v|$, χ_v the subtree at v. $\mathrm{Gen}_p(\chi) := \forall\,\mathrm{abs}_p(\chi)$, where abs_p replaces each occurrence of p at binder depth d by the index d (a vacuous $\forall$ if p does not occur). K has A1–A3, A4 $\forall B \to B[t]$ (t index-free), A5 $\forall(B \to C) \to (B \to \forall C)$ (B closed), reflexivity $\forall(\#0 = \#0)$ and substitutivity $p = q \to (B \to B')$, and the rules MP and Gen. A derivation from Ax is a sequence of lines, each justified as an axiom (logical, or in Ax), by MP from two earlier lines, or by Gen on a parameter from an earlier line. $\mathrm{size}(\pi)$ is the sum of the line sizes; $I(\pi)$ is the set of distinct formulas of axiom lines; the *material* is $M(\pi) := \sum_{\alpha \in I(\pi)} |\alpha|$, and $M_{\mathrm{nl}}(\pi)$ the same over the nonlogical ones. A line λ *sits at* a formula node v of α *with* k *abstractions* if the k nodes above v are quantifier nodes, the topmost being u, and $\alpha_u = \mathrm{Gen}_{q_k}(\cdots \mathrm{Gen}_{q_1}(\lambda)\cdots)$ for some parameters q_i; with $k = 0$, $\lambda = \alpha_v$. A derivation is *normal* if its lines are distinct and each line but the last is a premise of a later line.

Definition E.18 (Readings of "short axioms"; certificate function induction)**.** short-derivations Defs 1.2, 1.3, 1.6; time-followup Def 1.3 Let L be finite and B decidable, budgets nondecreasing. A hypothesis is a program p deciding A_p in time $O(t(|\chi|))$ such that $Ax_p := B \cup A_p$ is closed under permutations of the parameters, condition (F1). It is compatible with D if Ax_p is consistent and each $\varphi_i^{b_i}$ ($\varphi^1 := \varphi$, $\varphi^0 := \neg\varphi$) has a K-derivation π_i from Ax_p in which, for $\mathrm{AI}^{\mathrm{a1}}[t,h]$, every axiom line has size at most $h(|\varphi_i|)$; for $\mathrm{AI}^{\mathrm{a2}}[t,h]$, every line justified as a member of Ax_p does; for $\mathrm{AI}^{\mathrm{b}}[t,g]$, $M(\pi_i) \leq g(|\varphi_i|)$; for $\mathrm{AI}^{\mathrm{b2}}[t,g]$, $M_{\mathrm{nl}}(\pi_i) \leq g(|\varphi_i|)$; for $\mathrm{AI}[t,d]$, $\mathrm{size}(\pi_i) \leq d(|\varphi_i|)$. The weight is $2^{-|p|}$. $\mathrm{FI}_{\mathrm{cert}}[t',d']$: a hypothesis is a program V with outputs in $\{\mathrm{acc}, \mathrm{rej}, \perp\}$ and $\mathrm{time}_V(\varphi, c) = O(t'(|\varphi| + |c|))$ on all inputs; $f_V(\varphi)$ is acc (rej) if some c with $|c| \leq d'(|\varphi|)$ has $V(\varphi, c) = \mathrm{acc}$ (rej), and undefined otherwise; $B \cup \Gamma_{f_V}$ must be consistent; V is compatible if f_V labels every datum correctly; the weight is $2^{-|V|}$. *Polynomial budgets.* $\mathrm{AI}^{\mathrm{b}}[\mathrm{poly}, \mathrm{poly}]$ mixes over triples (p,i,j), with $A_{p,i} := \{\chi : U_0(p,\chi) = \text{yes within } (|\chi|+2)^i \text{ steps}\}$ satisfying (F1) with B, derivations of material at most $(|\varphi_i|+2)^j$, and weight $w(p)2^{-2\lceil\log_2(i+1)\rceil - 2\lceil\log_2(j+1)\rceil - 2}$; $\mathrm{AI}^{\mathrm{a1}}[\mathrm{poly}, \mathrm{poly}]$ likewise with every axiom line of size at most $(|\varphi_i|+2)^j$; $\mathrm{FI}_{\mathrm{cons}}^{\mathrm{poly}}$ is as in Definition E.10. Constant factors in running times are absorbed by raising the exponent, as $|\chi| + 2 \geq 2$. *Literal sequences.* L contains a constant e, unary function symbols s_0, s_1 and a unary relation symbol R; $t_w := s_{w_1}(\cdots s_{w_k}(e)\cdots)$ and $\varphi_w := R(t_w)$, so $|\varphi_w| = |w| + 2$; $B = \emptyset$; D^X lists the data $(\varphi_w, [w \in X])$ over all words w in length-lexicographic order; Γ_{f_X} is consistent (the term algebra with $R := \{t_w : w \in X\}$). Results whose proofs compute φ_w from w hold for every injective encoding computable and invertible in polynomial time, the paper's $R(\underline{w})$ included; the template constructions read w off t_w.

Every axiom set constructed below is a set of sentences, so it satisfies (F1); so do template instance sets. A finite injective renaming extends to a permutation.

Lemma E.19 (Subformula lemma)**.** [proved; computed] short-derivations Lemmas 1.1, 2.1, 2.2, Cor 2.3, Prop 2.4; scripts: c1_subformula, referee_logic r1_subformula_independent, r3_sharp_constant *(a) Every derivation π from Ax containing a line φ becomes, by deleting lines, a normal derivation from Ax with last line φ; so $I(\pi)$ and the size do not grow. (b) Let π be normal with last line φ and $J := I(\pi) \cup \{\varphi\}$. There is an injective map o from the lines of π to the formula nodes of the members of J (nodes of different members counted separately) such that every axiom or MP line sits at $o(\lambda)$ with 0 abstractions, and every Gen line γ sits at $o(\gamma)$ with $k \geq 0$ abstractions, $q_1, \ldots, q_k$ being the parameters abstracted along a chain of Gen steps that starts at γ. Hence every line has size at most $\max_{\alpha\in J} |\alpha|$, and $\mathrm{size}(\pi) \leq \sum_{\alpha\in J} F(\alpha) \leq (M(\pi) + |\varphi|)(M(\pi) + |\varphi| + 1)/2$. (c) For the least material $M_{\min}(\varphi)$ and least size $\ell_{\min}(\varphi)$ of derivations of φ from Ax: $M_{\min}(\varphi) \leq \ell_{\min}(\varphi) \leq (M_{\min}(\varphi) + |\varphi|)(M_{\min}(\varphi) + |\varphi| + 1)/2$; more precisely, every derivation π of φ can be replaced by one with instances among $I(\pi)$ and size at most $\sum_{\alpha\in I(\pi)\cup\{\varphi\}} F(\alpha)$. (d) The constant $\frac{1}{2}$ cannot be lowered: for sentences $r \neq b$, the root of b not $\forall$, $\beta := \forall^k r \to b$ and $Ax := \{r, \beta\}$, the derivation r, $\forall r, \ldots, \forall^k r$ (vacuous Gen), β, b is normal, and its size and $(M + |b|)(M + |b| + 1)/2$ are both $k^2/2 + O(k)$.*

Proof. (a) Delete every line whose formula occurred earlier, redirecting citations to the first occurrence; cut after φ and keep only its ancestors, in order. Justifications stay valid and nothing is added. (b) *Axiom and MP lines.* Let $o(\alpha)$ be the root of $\alpha \in I(\pi)$, and for an MP line B with major premise $A \to B$ let $o(B)$ be the right child of $o(A \to B)$. The major premise has root $\to$, so it is not a Gen line (root $\forall$); by induction on the line number $o(A \to B)$ is defined and carries $A \to B$, so the subtree at $o(\lambda)$ is λ. *Gen lines.* For a Gen line γ choose a chain $\gamma = \gamma_0, \ldots, \gamma_k =: e$ in which γ_{i+1} is justified by Gen on γ_i, abstracting q_{i+1}, and e is the minor premise of an MP line or the last line. It exists and ends: by normality each γ_i other than the last line is a later premise, not a major one (root $\forall$ against $\to$), so a minor or a Gen premise, and line numbers increase. If e is the minor premise A of an MP line with major premise $A \to B$, let o_e be the left child of $o(A \to B)$; if $e = \varphi$, let o_e be the root of $\varphi \in J$ (then φ is a Gen line, so $\varphi \notin I(\pi)$). As $e = \mathrm{Gen}_{q_k}(\cdots \mathrm{Gen}_{q_1}(\gamma) \cdots)$, one descends from o_e through k quantifier nodes; $o(\gamma)$ is the node reached, and γ sits there with k abstractions. *Injectivity.* If λ sits at v and the subtree at v has no dangling index, λ is that subtree: an abstracted parameter occurring in λ would leave an index pointing above v. Let $\lambda \neq \lambda'$ with $o(\lambda) = o(\lambda') = v$. If both are axiom or MP lines, both equal the subtree at v, against distinctness. If λ is one and λ' a Gen line, the subtree at v is λ, which has no dangling index, so $\lambda' = \lambda$. If both are Gen lines, with chains of lengths $k \leq k'$, the node k levels above v is o_e, with subtree $e = \gamma_k$, and γ'_k sits there, so $\gamma'_k = \gamma_k$; as each line has one justification, $\gamma'_{k-1} = \gamma_{k-1}$, and so on down to $\gamma' = \gamma$: a contradiction. *Bounds.* $|\lambda|$ is the size of the subtree at $o(\lambda)$, so injectivity gives $\mathrm{size}(\pi) \leq \sum_{\alpha\in J} F(\alpha)$ and the bound on each line. Each node of α lies in the subtree of each of its formula-node ancestors, so $F(\alpha) \leq \sum_u (\mathrm{depth}(u) + 1)$; in breadth-first order the i-th node has depth at most $i - 1$, so $F(\alpha) \leq |\alpha|(|\alpha| + 1)/2$. As $x \mapsto x(x + 1)/2$ is superadditive and $\sum_{\alpha\in J} |\alpha| \leq M(\pi) + |\varphi|$, the last bound follows. (c) $M(\pi) \leq \mathrm{size}(\pi)$, as every member of $I(\pi)$ is a line; conversely normalise by (a) and apply (b). (d) The lines are distinct and each is used by the next. $\mathrm{size} = (k + 1)|r| + k(k + 1)/2 + |\beta| + |b|$ with $|\beta| = |r| + k + 1 + |b|$, and $M = |r| + |\beta|$; also $J = \{r, \beta, b\}$ and $F(\beta) = |\beta| + \sum_{j=1}^{k}(|r| + j) + F(r) + F(b)$, so $\sum_J F - \mathrm{size}$ is a constant. □

Normality is needed: unused lines, such as vacuous Gen chains, and repeated lines can be added to any derivation at will. Material counts the logical instances too; the case where they are free is short-derivations Prop 2.6, not reproduced here.

Lemma E.20 (Renaming; canonical closure)**.** [proved; computed]
short-derivations Lemma 4.1; scripts: c3_short_axioms *(a) For an injective renaming ρ of parameters and a derivation π, $\rho(\pi)$, applied to every line and to the parameter of every Gen step (vacuous ones included), is a derivation with the same sizes whose nonlogical axiom lines are the ρ-images of those of π; under (F1) it is a derivation from Ax. (b) Let Ax satisfy (F1), let D_H be the set of formulas with a derivation from Ax all of whose lines have size at most H, and let* $\mathrm{can}(\chi)$ *rename the parameters of χ to $p_0, p_1, \ldots$ in order of first occurrence. Then* $\mathrm{can}(D_H)$ *is the least set S of canonical formulas of size at most H that contains* $\mathrm{can}(\alpha)$ *for every axiom instance α of size at most H (logical or in Ax), contains* $\mathrm{can}(B)$ *whenever it contains some $A \to B$ and* $\mathrm{can}(A)$*, and contains* $\mathrm{can}(\mathrm{Gen}_p(X))$ *whenever it contains X, p occurs in X or is one fixed parameter not in X, and $|\mathrm{Gen}_p(X)| \le H$.*

Proof. (a) $\rho(A \to B) = \rho A \to \rho B$; $\rho(\mathrm{Gen}_p\chi) = \mathrm{Gen}_{\rho p}(\rho\chi)$, as ρ is injective; the logical schemas are closed under injective renaming (A4: $\rho(\forall B \to B[t]) = \forall \rho B \to (\rho B)[\rho t]$; A5's side condition concerns indices; reflexivity has no parameter; in substitutivity $\rho B'$ arises from ρB by replacing occurrences of ρp by ρq). (b) By (a), D_H is closed under injective renamings, so $\chi \in D_H$ iff $\mathrm{can}(\chi) \in D_H$. $S \subseteq \mathrm{can}(D_H)$ by induction on the construction of S (an axiom is one line; for MP concatenate the derivations of $A \to B$ and A and add B, which is smaller than $A \to B$; Gen adds one line). Conversely, by induction on a derivation with lines of size at most H: for an MP line B, $\mathrm{can}(A \to B) = \sigma A \to \sigma B$ for the canonicalising σ and $\mathrm{can}(\sigma A) = \mathrm{can}(A) \in S$, so $\mathrm{can}(B) \in S$; for a Gen line, $\mathrm{can}(\lambda) = \sigma\lambda$ and $\mathrm{Gen}_{\sigma p}(\sigma\lambda) = \sigma\mathrm{Gen}_p(\lambda)$ has the canonical form of $\mathrm{Gen}_p(\lambda)$, and all vacuous Gens give the same formula. □

The derivation code. Code writes each line as a 2-bit tag (axiom, MP, Gen), then the Elias-γ codes (Elias, 1975) of its premise line numbers plus one and, for Gen, of its parameter number plus one, then its formula in preorder with $b_s := \lceil \log_2(\sigma_L + 6) \rceil$ bits per token, an index or parameter being an escape token followed by the γ code of its number plus one; a 2-bit end tag closes the code. $|\chi|_{\mathrm{bit}}$ is the length of a formula's code, so $|\chi| \le |\chi|_{\mathrm{bit}}$ and $|\neg\chi|_{\mathrm{bit}} = |\chi|_{\mathrm{bit}} + b_s$.

Proposition E.21 (Material, size and certificates: exact budgets)**.** [(a) proved; (b), (c) proved, under (F1); a form of the correspondence between proof systems and NP (Cook and Reckhow, 1979) (known); computed] short-derivations Thm 3.1; time-followup Thm 4.2; scripts: c4_cert_embedding, time-followup c3_certificate *Let L be finite, B decidable in polynomial time and satisfying (F1), t, g, d, d' time-constructible and nondecreasing, and $G(m) := (g(m)+m+1)(g(m)+m+2)/2$. There are constants c, β and a fixed polynomial m^{k_0} such that for every finite D, with $t^+(m) := m^{k_0}(t(m)+1)$ and $t'^{\#}(m) := m^{k_0} + t'(m)$: (a) $W_{\mathrm{AI}[t,g]}(D) \le W_{\mathrm{AI}^{\mathrm{b}}[t,g]}(D) \le W_{\mathrm{AI}[t,G]}(D)$, hypothesis by hypothesis; (b) $W_{\mathrm{AI}^{\mathrm{b}}[t,g]}(D) \le 2^c W_{\mathrm{FI}_{\mathrm{cert}}[t^+,\beta G\log_2(G+2)]}(D)$ and $W_{\mathrm{AI}[t,d]}(D) \le 2^c W_{\mathrm{FI}_{\mathrm{cert}}[t^+,\beta d\log_2(d+2)]}(D)$; (c) $W_{\mathrm{FI}_{\mathrm{cert}}[t',d']}(D)$ is at most 2^c times each of $W_{\mathrm{AI}^{\mathrm{b}}[t'^{\#},2d'+m+31]}(D)$, $W_{\mathrm{AI}^{\mathrm{a1}}[t'^{\#},d'+m+16]}(D)$ and $W_{\mathrm{AI}[t'^{\#},2d'+2m+31]}(D)$.*

Proof. (a) A derivation has material at most its size, and by Lemma E.19(c) a derivation of material at most g can be replaced by one with no new instances and size at most $(g + |\varphi^b|)(g + |\varphi^b| + 1)/2 \le G(|\varphi|)$; the compatibility relations are nested, with the same weights. (b) For a hypothesis p let $V_p(\varphi, c)$ be acc if c decodes to a K-derivation of φ from Ax_p of material at most $g(|\varphi|)$ (of size at most $d(|\varphi|)$, for $\mathrm{AI}[t,d]$), rej if it decodes to one of $\neg\varphi$, and $\bot$ otherwise. *Short certificates exist.* If $\varphi_i^{b_i}$ has a derivation within the budget, it has one, π, of size $s \le G(|\varphi_i|)$ (by (a); $s \le d(|\varphi_i|)$ directly for $\mathrm{AI}[t,d]$) and no new instances. π has at most s parameters, Gen parameters included; an injective renaming maps them to $p_0, \ldots, p_{s-1}$ and gives, by Lemma E.20(a) and (F1), a derivation from Ax_p with the same sizes. Its indices and line numbers are below s, so

each token costs at most $b_s + 2\log_2 s + 1$ bits and each line at most $3(2\log_2 s + 1) + 2$ more: $|\mathrm{Code}(\pi)| \le \beta s \log_2(s+2)$, β depending on b_s. *Time.* Checking a decoded line is polynomial for the logical axioms (A4 by matching), MP, Gen and B, and takes $O(t(|c|))$ for membership via p; so V_p runs in time $O(t^+(|\varphi| + |c|))$. *Consistency and weights.* Ax_p is consistent, so no φ has both an acc and a rej certificate, and $\Gamma_{f_{V_p}} \subseteq \mathrm{Cn}(Ax_p)$ is consistent with B; every datum has a certificate within the bound, so f_{V_p} labels it correctly; $|V_p| \le |p| + c$, and $p \mapsto V_p$ is injective. (c) Let $Z := \forall(\#0 = \#0)$, $C_c := \nu_{c_1}(\cdots \nu_{c_k}(Z)\cdots)$ with $\nu_0 := \neg$ and $\nu_1 := \forall$ (vacuous), and $\theta_c := Z \to (C_c \to Z)$, an instance of A1 of size $|c| + 14$ from which c is read off uniquely (strip $\neg$ and $\forall$ until Z remains). For a hypothesis V let $A_V := \{\theta_c \to \varphi : V(\varphi, c) = \text{acc},\ |c| \le d'(|\varphi|)\} \cup \{\theta_c \to \neg\varphi : V(\varphi, c) = \text{rej},\ |c| \le d'(|\varphi|)\}$, a set of sentences. Membership: parse $\chi = \theta \to \psi$, read c off θ, and accept iff $|c| \le d'(|\psi|)$ and $V(\psi, c) = \text{acc}$, or $\psi = \neg\varphi$, $|c| \le d'(|\varphi|)$ and $V(\varphi, c) = \text{rej}$; as d' is time-constructible the length test stops after $|c| + 1$ steps; the time is $O(t'^{\#}(|\chi|))$, and the decider p_V has $|p_V| \le |V| + c$. As θ_c is valid, each member is equivalent to its member of Γ_{f_V}, and each member of Γ_{f_V} has one, so $\mathrm{Cn}(B \cup A_V) = \mathrm{Cn}(B \cup \Gamma_{f_V})$, which is consistent. The derivation θ_c (A1), $\theta_c \to \varphi^b$ (axiom), φ^b (MP) has material $2|c| + 29 + |\varphi^b|$, largest instance $|c| + 15 + |\varphi^b|$ and size $2|c| + 29 + 2|\varphi^b|$, with $|\varphi^b| \le |\varphi| + 1$. □

Proof of Theorem 6.16. (a) is Lemma E.19(b), (d); (b) is Proposition E.21: (a) gives the first chain, (b) and (c) the equivalence with $\mathrm{FI}_{\mathrm{cert}}$ and the embedding into reading (a1). □

Lemma E.22 (Clocked Craig sets)**.** [proved, under (S); computed]

time-followup Cor 4.3(a); scripts: referee_logic r3_enum_parse_derive, Part D *For a time-constructible τ and a string f, the Craig set $A^C_{f^\tau}$ of Theorem 6.3 is a set of sentences with $\mathrm{Cn}(A^C_{f^\tau}) = \mathrm{Cn}(\Gamma_{f^\tau})$; its membership is decidable in time polynomial in $|\chi|$, of a degree independent of f and τ; and each φ that f^τ labels b has a derivation of φ^b from it of size at most $\gamma(\tau(|\varphi|) + 1)(|\varphi| + 4)$, γ a constant.*

Proof. Membership: there are at most two candidates $\chi = \psi^{\wedge m}$ (proof of Lemma E.12(b)); check $m - 1 \le \tau(|\varphi|)$ by running τ's constructor for at most m steps, and simulate f on U_0 for $m - 1$ steps. Consequences: as in Theorem 6.3. Derivations: cite $(\varphi^b)^{\wedge(k+1)}$, $k \le \tau(|\varphi|)$; if $k \ge 1$, add the substitution instance $B := \varphi^b$, $C := (\varphi^b)^{\wedge k}$ of a fixed derivation of $(B \wedge C) \to B$ from A1–A3 and MP (one exists, as the propositional fragment of K is complete; the referee's script builds one of 287 lines), and apply MP. Substitution instances of derivations are derivations, of size linear in $|B| + |C|$, and $|(\varphi^b)^{\wedge(k+1)}| \le (k+1)(|\varphi| + 1) + 3k \le (\tau(|\varphi|) + 1)(|\varphi| + 4)$. □

Proof of Corollary 6.17(a). short-derivations Cor 3.2, Prop 3.4; time-followup Cor 4.3(c)–(e) By Lemma E.11, $\mathrm{AI}^{\mathrm{b}}[\mathrm{poly}, \mathrm{poly}]$ fits D^X iff one triple (p, i, j) is compatible with every D^X_n, and likewise for the other mixtures. *Fit class $\subseteq$ NP $\cap$ coNP.* For such a triple, V_p of Proposition E.21(b) runs in polynomial time with certificates of polynomial length, and $f_{V_p}(\varphi_w) = [w \in X]$ for every w; so $w \in X$ iff some short certificate makes V_p accept φ_w, and $w \notin X$ iff some short certificate makes it reject. $\supseteq$. Combine polynomial-time verifiers for X and for its complement into one V with a tag bit; $f_V = f_X$ is consistent. Proposition E.21(c) gives a decider p_V of a set of sentences whose membership time on U_0 is at most $C(|\chi| + 1)^{i_0}$, so $A_{p_V,i} = A_V$ for $i \ge i_0 + \log_2 C$, with material at most $2d'(m) + m + 31 \le (m+2)^j$ for some j; (p_V, i, j) is compatible with every D^X_n. *If* NP $\cap$ coNP $\neq$ P, take X in the difference: $\mathrm{AI}^{\mathrm{b}}[\mathrm{poly}, \mathrm{poly}]$ fits D^X, while a pair (f, k) of $\mathrm{FI}^{\mathrm{poly}}_{\mathrm{cons}}$ fitting it would put X in $\mathrm{DTIME}(q(C_f(m+4)^k + m)) \subseteq$ P (overhead (i)); so $\ell_{\mathrm{FI}^{\mathrm{poly}}_{\mathrm{cons}}}(D^X_n) \to \infty$. *If* P = NP, the two inductors have bounded loss on the same infinite sequences of sentences. If (p, i, j) fits D, acceptance, $\exists c\,(|c| \le \beta G \log_2(G+2) \wedge V_p(\varphi, c) = \text{acc})$ with G polynomial, and rejection are NP predicates of φ, so they are decided in polynomial time; this gives (f, k) with f, clocked at $(|\varphi|+2)^k$, equal to f_{V_p}, whose Γ lies in $\mathrm{Cn}(B \cup A_{p,i})$ and which labels every datum of D. If (f, k) fits

D, the Craig decider of f clocked at $(|\varphi|+2)^k$ fits D (Lemma E.22): its set consists of sentences, its membership degree is fixed up to a constant that the exponent absorbs, and its derivations have material at most their size, at most $\gamma((|\varphi|+2)^k+1)(|\varphi|+4) \le (|\varphi|+2)^{k+O(1)}$. Apply Lemma E.11 both ways. *Against clocked function induction.* We use the deterministic time hierarchy theorem (Hartmanis and Stearns, 1965; Hennie and Stearns, 1966) (known; exact form not checked): if T_2 is time-constructible and $T_1 \log T_1 = o(T_2)$, then $\mathrm{DTIME}(T_2) \setminus \mathrm{DTIME}(T_1) \neq \emptyset$. Let τ be time-constructible, $T_1(m) := m\, q(m\tau(m+2)+m)$, T_2 time-constructible and nondecreasing with $T_1 \log T_1 = o(T_2)$, and $X \in \mathrm{DTIME}(T_2) \setminus \mathrm{DTIME}(T_1)$. If a τ-clocked f labelled every φ_w correctly, then $X \in \mathrm{DTIME}(q(C_f\tau(m+2)+m))$ by overhead (i), and $q(C_f\tau(m+2)+m) \le T_1(m)$ for $m \ge C_f^{\deg q}$; with a table for the shorter words $X \in \mathrm{DTIME}(T_1)$, a contradiction; so $\ell_{\mathrm{FI}^{\tau}_{\mathrm{cons}}}(D_n^X) \to \infty$. Let τ_3 be time-constructible with $\tau_3(m) \ge q(C_X T_2(m)+m)$, C_X the constant of a machine for X. By overhead (ii) some τ_3-clocked string labels every φ_w correctly, and by Lemma E.22 its Craig decider, with membership in a fixed polynomial time t_C, is compatible with all of D^X in $\mathrm{AI}^{\mathrm{b}}[t_C, d_{\tau_3}]$, $d_{\tau_3}(m) := \gamma(\tau_3(m)+1)(m+4)$. *Uncharged degrees.* Let $\mathrm{AI}^{\mathrm{b}}_{\mathrm{unch}}$ mix over pairs (p,j), p a decider satisfying (F1) with membership time $O(|\chi|^i)$ for some i, weight $2^{-|p|-2\lceil \log_2(j+1)\rceil - 1}$, and material at most $(|\varphi_i|+2)^j$. Then $\sup_D(\ell_{\mathrm{FI}^{\mathrm{poly}}_{\mathrm{cons}}}(D) - \ell_{\mathrm{AI}^{\mathrm{b}}_{\mathrm{unch}}}(D)) = \infty$, whether or not P = NP. For $r \ge 1$ let $N := 2^{2^r}$ and, by the hierarchy theorem, $X_N \in \mathrm{DTIME}(m^N) \setminus \mathrm{DTIME}(m^{N/2})$, decided by a program of length $c_0 + O(\log r)$ (the diagonal construction, with N computed from r). The decider p_N of $\{\varphi_w : w \in X_N\} \cup \{\neg\varphi_w : w \notin X_N\}$ has polynomial membership time (of degree about N, uncharged) and length $c_0' + O(\log r)$; each datum is an axiom, with material at most $|\varphi_w|+1 \le (|\varphi_w|+2)^1$, so $(p_N, 1)$, of weight $2^{-|p_N|-3}$, fits D^{X_N} and $\ell_{\mathrm{AI}^{\mathrm{b}}_{\mathrm{unch}}}(D_n^{X_N}) \le |p_N| + 3$. Fix b, depending only on q, with $q(C(m+4)^k + m) \le m^{bk}$ for all $k \ge 1$ and $m \ge m(C)$. If (f,k) fitted D^{X_N}, then $X_N \in \mathrm{DTIME}(m^{bk})$ after a table, so $bk > N/2$. By the argument of Lemma E.11, $W_{\mathrm{FI}^{\mathrm{poly}}_{\mathrm{cons}}}(D_n^{X_N})$ tends to at most $\sum_{k > N/(2b)} 2^{-2\lceil \log_2(k+1)\rceil - 1} \le b/N$: the 2^{j-1} values of k with $\lceil \log_2(k+1)\rceil = j$ contribute 2^{-j-2}, so the sum over $k \ge K$ is at most $1/(2K)$ (script `c4_revision`, Part C). As $W_{\mathrm{FI}^{\mathrm{poly}}_{\mathrm{cons}}}(\emptyset)$ is at least a constant, the regret on D^{X_N} is at least $2^r - O(\log r)$ for large n. □

Proposition E.23 (Reading (a1): exponential time, both ways)**.** [proved, under (F1); computed] short-derivations Thm 4.2, Prop 4.3; scripts: c3_short_axioms *(a) Let $Ax := B \cup A_p$ satisfy (F1), with membership in time t_{Ax}, and put $H(\varphi) := \max(h(|\varphi|), |\varphi|+1)$. The assigner g_p that accepts φ if $\varphi \in D_{H(\varphi)}$, rejects it if $\neg\varphi \in D_{H(\varphi)}$ and $\varphi \notin D_{H(\varphi)}$, and abstains otherwise has $|g_p| \le |p| + c$ and $\Gamma_{g_p} \subseteq \mathrm{Cn}(Ax)$, runs in time $2^{O(H \log H)}(1 + t_{Ax}(H))^{O(1)}$, and labels correctly every datum of every D with which p is (a1)-compatible for the budget h. So the fit class of $\mathrm{AI}^{\mathrm{a1}}[t,h]$ is contained in $\bigcup_C \mathrm{DTIME}(2^{CH\log H}(t(H)+1)^C)$, $H := h(m+2)+m+3$. (b) Let $L \supseteq \{e, s_0, s_1, R, C, C'\}$, C, C' binary, $B = \emptyset$, and let M be an alternating machine with branching at most 2 that decides X in space $s(m) \ge m$ and halts on every path. Code the configurations of M on w as words of length at most $c_M(s(|w|)+1)$, written as terms t_c. A_M consists of the ground sentences $C(t_w, t_c)$ for accepting halting c and $C'(t_w, t_c)$ for rejecting halting c; $C(t_w, t_{c'}) \to C(t_w, t_c)$ for existential c with successor c'; $C(t_w, t_{c_1}) \to (C(t_w, t_{c_2}) \to C(t_w, t_c))$ for universal c with successors c_1, c_2; the duals for C', with existential and universal exchanged; and $C(t_w, t_{\mathrm{init}(w)}) \to R(t_w)$, $C'(t_w, t_{\mathrm{init}(w)}) \to \neg R(t_w)$. Then A_M is a set of sentences with membership decidable in polynomial time by a decider of length $|M| + c$, $A_M \cup \Gamma_{f_X}$ is consistent, and every datum $\varphi_w^{[w \in X]}$ has a derivation from A_M by MP alone with at most $2^{c_M'(s(|w|)+1)}$ lines, each of size at most $c_M''(s(|w|)+1)$.*

Proof. (a) A canonical formula of size $n \le H$ is a sequence of n tokens, each a symbol of L, $\neg$, $\to$, $\forall$, $=$, one of fewer than H indices or one of fewer than H parameters; so there are at most

$N := (\sigma_L + 2H + 4)^{H+1} = 2^{O(H \log H)}$ of them. Enumerate them, test each against the logical schemas (A4 by matching) and against Ax (by (F1), $\chi \in Ax$ iff $\mathrm{can}(\chi) \in Ax$), and close under the rules of Lemma E.20(b): at most N rounds of $O(N \,\mathrm{poly}(H))$ steps, $N^{O(1)}(1+t_{Ax}(H))$ in all, and a polynomial more for running p inside a wrapper. If p is (a1)-compatible, normalise the derivation of $\varphi_i^{b_i}$ (Lemma E.19(a), which does not enlarge I); by Lemma E.19(b) every line then has size at most $\max(h(|\varphi_i|), |\varphi_i^{b_i}|) \le H(\varphi_i)$, so $\varphi_i^{b_i} \in D_{H(\varphi_i)}$, and as Ax is consistent $\varphi_i^{1-b_i} \notin D_{H(\varphi_i)}$; so g_p labels φ_i with b_i. The fit class follows by Lemma E.11, with $|\varphi_w| = m+2$. (b) *Membership.* Parse the shape, decode the configurations and check M's local transition relation on w. *Consistency.* In the free term algebra let C hold of (t_w, t_c) iff c is accepting for M on w (the least fixed point of the alternating acceptance condition), C' likewise for rejecting, and $R := \{t_w : w \in X\}$. The axioms are the closure conditions of these fixed points and the two output axioms, which hold as M decides X; when all paths halt, every configuration is accepting or rejecting and not both, by induction on rank. *Derivations.* For $w \in X$ list the configurations of an accepting computation tree from $\mathrm{init}(w)$ in order of rank and derive $C(t_w, t_c)$ for each: cite a halting fact, or cite the step axiom and apply MP once or twice; finish with the output axiom and MP. The case $w \notin X$ is dual. Every line is an atom or a subformula of an axiom, of size $O(s(|w|))$, and there are $2^{O(s)}$ configurations. □

Proof of Corollary 6.17(b). short-derivations Cor 4.4, Prop 4.5 *Fit class.* $\subseteq$: Proposition E.23(a) with polynomial h, t. $\supseteq$: EXPTIME = APSPACE = $\bigcup_k$ ASPACE(m^k) (Chandra et al., 1981) (known; not checked); with a clock all paths halt, and Proposition E.23(b) with $s = m^k$ gives a set of sentences with membership clocked at $(|\chi| + 2)^i$ for some i and instances of size at most $(|\varphi_w| + 2)^j$ for some j. *A time limit.* For computable h, t put $H := h(m + 2) + m + 3$ and $T(m) := 2^{H^2}(t(H) + 1)^H$. T is computable and eventually exceeds $2^{CH \log H}(t(H) + 1)^C$ for each C, and diagonalising against T-clocked machines gives a decidable $X \notin \mathrm{DTIME}(T)$; by Proposition E.23(a) and Lemma E.11, $\ell_{\mathrm{AI}^{\mathrm{a1}}[t,h]}(D_n^X) \to \infty$, while unpenalised $\mathrm{FI}_{\mathrm{cons}}$ fits D^X through f_X. With $X \in \mathrm{EXPTIME} \setminus \mathrm{P}$ (hierarchy theorem), $\mathrm{AI}^{\mathrm{a1}}[\mathrm{poly}, \mathrm{poly}]$ fits D^X and, by overhead (i), $\mathrm{FI}_{\mathrm{cons}}^{\mathrm{poly}}$ does not. *Predictors.* Let P be a predictor whose pair, on a history of j literal data and a next datum, can be approximated within 2^{-k} in time polynomial in $j + k$. Let w_j be the j-th word and $\beta_j := 0$ iff $a_0 \le a_1$, where (a_0, a_1) approximate P's pair within 2^{-j-3} on the history $((\varphi_{w_1}, \beta_1), \ldots, (\varphi_{w_{j-1}}, \beta_{j-1}))$ and φ_{w_j}; put $X := \{w_j : \beta_j = 1\}$. No self-reference is needed, as the data are literals, and the loss computation in the proof of Theorem 6.15 gives $\ell_P(D_n^X) \ge n - 1/(4 \ln 2)$. Computing $\beta_1, \ldots, \beta_n$ takes time polynomial in n, and $n < 2^{m+1}$ for $|w_n| = m$, so $X \in \mathrm{DTIME}(2^{am}) \subseteq \mathrm{ASPACE}(O(m))$ for some a (Chandra et al., 1981); with a clock, Proposition E.23(b) gives one set of sentences, with polynomial membership and every instance of size $O(m)$, that fits D^X under (a1) at a linear budget. Hänni's S has per-step time $p(n) \log n$ given the previous step's state (Hänni, 2026), so replaying its first j steps costs $\sum_{i \le j} p(i) \log i$, polynomial in j; this premise is his claim, not checked here. □

The converse fails, as for Theorem 6.15: on the fair-coin literals of Proposition E.1, S and $\mathrm{FI}_{\mathrm{all}}^{\tau}$ have bounded loss, while every reading of axiom induction, whose hypotheses are consistent, loses n bits in expectation.

Proposition E.24 (Horn templates simulate a verifier)**.** [proved; computed] short-derivations Prop 6.1, Rem 6.4(3); scripts: c5_templates, referee_complexity rc2_templates_left *Let M be a deterministic three-tape machine (input, certificate, work; tape alphabet $\Gamma \cup \{\mathit{blank}\}$, $\{0,1\} \subseteq \Gamma$) that computes $V(w, c) \in \{\mathrm{acc}, \mathrm{rej}, \perp\}$, halts on every input, outputs $\perp$ unless $w, c \in \{0,1\}^*$, and has f_V consistent; let X_{acc} (X_{rej}) be the set of words with an accepting (rejecting) binary certificate. Assume the normal form: no transition writes a blank, and none moves a head right from a cell that is blank after the step (at no*

cost: add a symbol that M treats as blank and write it instead). Let $L \supseteq \{e, \mathrm{pr}, R, C\} \cup \{s_a : a \in \Gamma\}$ with pr *a binary function symbol and C a binary relation symbol, and $B = \emptyset$. A tape with its head is coded by two terms over the s_a ending in e: the cells left of the head, nearest first, and the cells from the head on; a configuration in state q is $\mathrm{conf}(q, \dots) := \mathrm{pr}(t_q, \mathrm{pr}(l_0, \mathrm{pr}(r_0, \dots)))$, t_q a word term for q's index. T_M consists of first-order-pattern templates, all of whose metavariables are 0-ary term metavariables: Start, $C(x, \mathrm{conf}(q_0, (e, x), (e, z), (e, e)))$; one template $C(x, pattern) \to C(x, result)$ per transition (state, three read symbols and, for each head that moves left, its left neighbour or the left end); Accept, $C(x, \mathrm{pr}(t_{q_{\mathrm{acc}}}, y)) \to R(x)$; and Reject, $C(x, \mathrm{pr}(t_{q_{\mathrm{rej}}}, y)) \to \neg R(x)$. Then: (i) T_M has $O(|Q_M|)$ templates, each of size $O(\log |Q_M|)$, for fixed Γ; (ii) $T_M \cup \Gamma_{f_V}$ is consistent; for words, $T_M \vdash R(t_w)$ iff $w \in X_{\mathrm{acc}}$ and $T_M \vdash \neg R(t_w)$ iff $w \in X_{\mathrm{rej}}$; (iii) if $V(w, c) \in \{\mathrm{acc}, \mathrm{rej}\}$ after τ steps, the corresponding literal has a derivation from $\bigcup \mathrm{inst}(T_M)$ by MP alone, with $2\tau + 3$ lines, each of size at most $c_M(\tau + |w| + |c| + 1)$. Every consistent computable assigner f that abstains outside the φ_w is f_V for such a V ("c is a halting computation of f on φ_w; output its output"), so a finite Horn template theory consistent with Γ_f proves exactly f's decisions among the word literals.*

Proof. (iii) Cite the Start instance with $x := t_w$, $z := t_c$; τ times, cite the transition instance at the current configuration and apply MP; cite Accept or Reject and apply MP. Configuration terms have size $O(\tau + |w| + |c| + \log |Q_M|)$. (ii) Take the free term algebra $\mathcal{H}$ with equality the identity, and let $C^{\mathcal{H}}$ be the closure of the Start instances under the transition templates. Distinct templates have disjoint patterns, so the facts derived from one Start instance form a single run. For $x = t_w$, $z = t_c$ it is M's run on (w, c), by the normal form. If x or z has a node other than an s_a or e, the run follows M on the longest word prefixes $(\mathrm{wp}(x), \mathrm{wp}(z))$ and gets stuck when a head reaches that node, where M would read a blank; if it halts first, M makes the same run on the prefixes, since the node is never read. *Model 1*: $R := \{t_w : w \in X_{\mathrm{acc}}\} \cup \{\text{non-word } x : \text{some run from } x \text{ accepts}\}$. Accept holds by construction. Reject holds on words as $X_{\mathrm{acc}} \cap X_{\mathrm{rej}} = \emptyset$, and on a non-word x because an accepting and a rejecting run from x would give $V(\mathrm{wp}(x), \cdot) = \mathrm{acc}$ and $= \mathrm{rej}$ on binary certificates, so $\mathrm{wp}(x)$ would be a binary word (as V outputs $\perp$ otherwise), against the consistency of f_V. Γ_{f_V} holds. This gives consistency and $T_M \vdash R(t_w) \Rightarrow w \in X_{\mathrm{acc}}$. *Model 2*: R on words is the complement of X_{rej}, the same as Model 1 on non-words; it gives $T_M \vdash \neg R(t_w) \Rightarrow w \in X_{\mathrm{rej}}$. The converses are (iii). Instances with parameters are read under closure; in $\mathcal{H}$ a closure holds iff all its closed instances do, and these are instances again. (i) Count. The last claim: checking a computation is decidable, so V halts on every input, and $f_V = f$. □

Proof of Corollary 6.17(c).

short-derivations Prop 4.12; scripts: referee_logic r4_a2_finite_theory Let X be decidable and M a machine in the normal form of Proposition E.24 computing $V(w, c) :=$ acc if $w \in X$ and rej if $w \notin X$, for binary w, c, and $\perp$ otherwise. Let $T_M^{\forall}$ be the set of universal closures of the templates of T_M: a finite set of sentences, of largest size c_M and total size Σ_M. Model 1 of Proposition E.24(ii) satisfies $T_M^{\forall} \cup \Gamma_{f_X}$, as its elements are closed terms, so a universal closure holds iff all its closed instances do. In the derivation of Proposition E.24(iii) with c empty, replace each cited instance by its closure in $T_M^{\forall}$ followed by A4 instances and MP, which are logical; the nonlogical instances used are members of $T_M^{\forall}$. For an unbounded nondecreasing h with $h(m) \geq m + 1$, the decider of $T_M^{\forall} \cup \{\varphi_w^{[w \in X]} : h(|\varphi_w|) < c_M\}$, a finite set of sentences, is (a2)-compatible with all of D^X, the finitely many short data being cited as themselves; for (b2) the same holds with Σ_M in place of c_M. So every decidable X is in the fit class, for every $t \geq$ linear. □

Corollary E.25 (Corollary 6.10 is tight on literal data)**.** [proved, given the transcript method of Book et al. (1970) (known; not checked)] short-derivations Cor 6.2(iii), (iv) *On literal sequences: (i) if $X \in$*

$\mathrm{NTIME}(\tau)$, $\tau(m) \geq m$ *time-constructible, then a finite first-order-pattern theory* T *with* $T \cup \Gamma_{f_X}$ *consistent derives every* φ_w, $w \in X$, *within size* $c_X(\tau(|w|)+|w|)^2$*; (ii) so for the* X *of Corollary 6.10(a) (with word terms), which lies in* $\mathrm{NTIME}(s^{c_1+1})$, *a finite template theory consistent with* Γ_{f_X} *derives each* φ_w, $w \in X$, *within size* $c(s(m)^{c_1+1}+m)^2$, *while every finite* DT° *theory consistent with* Γ_{f_X} *needs size above* $s(m)$ *infinitely often.*

Proof. (i) A verifier reads a guessed transcript of a run of a nondeterministic machine for X (the transitions taken, with the symbols read and written) and checks it tape by tape on its work tape, in time $O(\tau+|w|)$, with a binary certificate of length $O(\tau)$ (Book et al., 1970). It only accepts, so $X_{\mathrm{rej}}=\emptyset$ and Model 1 of Proposition E.24 sets R to X on words, so the negative literals of Γ_{f_X} hold as well. Apply Proposition E.24(iii). (ii) The proof of Corollary 6.10(a), guess and check, works for word terms as for $R(\underline{w})$. □

Proposition E.26 (Soft charges)**.** [proved; computed] short-derivations Def 5.1, Thm 5.2(a), (b), Prop 5.3; scripts: c4_cert_embedding, referee_complexity rc1_rate_one *For a hypothesis* p *let* $\ell_p^{\mathrm{bit}}(\psi)$ *be the least code length of a derivation of* ψ *from* Ax_p *(*∞ *if none).* Soft axiom induction $\mathrm{AI}^\sigma_\kappa[t]$ *has* $W(D) := \sum_p 2^{-|p|}\prod_i 2^{-\kappa\ell_p^{\mathrm{bit}}(\varphi_i^{b_i})}$ *over deciders with time* $O(t)$ *and* Ax_p *consistent (Hänni's graded score with* $g(\ell)=2^{-\kappa\ell}$*, both polarities, not normalised; (F1) is not required).* Soft certificate function induction $\mathrm{FI}^\sigma_{\mathrm{cert}\,\kappa,\mu}[t']$ *has* $W(D) := \sum_V 2^{-|V|}\prod_i 2^{-\kappa\lambda_V(\varphi_i^{b_i})-\mu|\varphi_i|_{\mathrm{bit}}}$ *over verifiers with time* $O(t')$ *and* $B\cup\Gamma_{f_V}$ *consistent,* f_V *using certificates of any length, where* $\lambda_V(\varphi^1)$ *(*$\lambda_V(\varphi^0)$*) is the least length of an accepting (rejecting) certificate. Both are semimeasures, and* $W(\emptyset)$*, the weight of the consistent hypotheses, depends on neither rate. Let* $e_1 := 31b_s+24$, $r := 31b_s+26$, c' *(*c''*) a constant plus* $\log_2$ *of the inverse empty-data weight of soft axiom (certificate) induction. For every* D*: (a)* $\ell_{\mathrm{FI}^\sigma_{\mathrm{cert}\,\kappa,\kappa}[t^+]}(D) \leq \ell_{\mathrm{AI}^\sigma_\kappa[t]}(D) - \kappa\sum_i(2+b_s[b_i=0]) + c'$*; (b)* $\ell_{\mathrm{AI}^\sigma_\kappa[t'^\#]}(D) \leq \ell_{\mathrm{FI}^\sigma_{\mathrm{cert}\,2b_s\kappa,2\kappa}[t']}(D) + \kappa e_1|D| + c''$*, and* $\ell_{\mathrm{AI}^\sigma_\kappa[t'^\#]}(D) \leq \ell_{\mathrm{FI}^\sigma_{\mathrm{cert}\,r\kappa,r\kappa}[t']}(D) + c''$*; (c) on the all-negative literal sequence,* $\ell_{\mathrm{AI}^\sigma_\kappa}(D^\emptyset_n) - \ell_{\mathrm{FI}^\sigma_{\mathrm{cert}\,\kappa_c,\kappa_s}}(D^\emptyset_n) = (\kappa-\kappa_s)\sum_{i\leq n}|\varphi_{w_i}|_{\mathrm{bit}} + \kappa(b_s+4)n + O(1)$*; so no pair of rates* (κ_c,κ_s) *gives two-sided constant regret with* $\mathrm{AI}^\sigma_\kappa$*, and the factor* r *in (b) cannot be* 1.

Proof. Semimeasures. For a consistent hypothesis at most one of $\varphi, \neg\varphi$ has finite cost, and every factor is at most 1. (a) Let $V_p(\varphi, c)$ be acc if c, followed by $\mathrm{Code}(\varphi)$ and the end tag, decodes to a derivation of φ from Ax_p, rej if this holds for $\neg\varphi$, and $\perp$ otherwise. The code of a shortest derivation of φ^b without its last formula and end tag is a certificate, so $\lambda_{V_p}(\varphi^b) \leq \ell_p^{\mathrm{bit}}(\varphi^b) - |\varphi^b|_{\mathrm{bit}} - 2$, and V_p's factor $2^{-\kappa\lambda-\kappa|\varphi|_{\mathrm{bit}}}$ is at least $2^{-\kappa\ell_p^{\mathrm{bit}}(\varphi^b)+\kappa(2+b_s[b=0])}$. The certificate contains the codes of all parameter numbers, so no renaming hypothesis is needed. V_p checks in time $O(t^+)$, $\Gamma_{f_{V_p}} \subseteq \mathrm{Cn}(Ax_p)$ is consistent with B, $|V_p| \leq |p|+c$, and $p \mapsto V_p$ is injective; sum over p and use $W_{\mathrm{FI}^\sigma_{\mathrm{cert}}}(\emptyset) \leq 1$. (b) With θ_c as in the proof of Proposition E.21(c) and no bound on $|c|$, A_V is decided in time $O(t'^\#)$ and $\mathrm{Cn}(B\cup A_V) = \mathrm{Cn}(B\cup\Gamma_{f_V})$. The derivation θ_c, $\theta_c \to \varphi^b$, φ^b has code length $2b_s|c|+2|\varphi^b|_{\mathrm{bit}}+29b_s+24$ (θ_c has $|c|+8$ ordinary nodes and six index nodes of $1+b_s$ bits; four 2-bit tags; the γ codes of the two premises, $1+3$ bits), which is at most $2b_s|c|+2|\varphi|_{\mathrm{bit}}+e_1$ and, as $|\varphi|_{\mathrm{bit}} \geq 1$, at most $r(|c|+|\varphi|_{\mathrm{bit}})$. Sum over V and use $W_{\mathrm{AI}^\sigma}(\emptyset) \leq 1$. (c) A code lists at least one line, whose 2-bit tag and formula precede the 2-bit end tag, so $\ell_p^{\mathrm{bit}}(\psi) \geq |\psi|_{\mathrm{bit}}+4$ and $\ell_{\mathrm{AI}^\sigma_\kappa}(D) \geq \kappa\sum_i(|\varphi_i^{b_i}|_{\mathrm{bit}}+4)$. On $D^\emptyset$ the decider $p_\emptyset$ of $\{\neg\varphi_w\}$ derives each datum in one line, of code length $|\varphi_w|_{\mathrm{bit}}+b_s+4$, so $\ell_{\mathrm{AI}^\sigma_\kappa}(D^\emptyset_n)$ is $\kappa\sum_{i\leq n}(|\varphi_{w_i}|_{\mathrm{bit}}+b_s+4)$ up to an additive term in $[0,|p_\emptyset|]$. Every $\mathrm{FI}^\sigma_{\mathrm{cert}}$ factor is at most $2^{-\kappa_s|\varphi_i|_{\mathrm{bit}}}$, which $V_\emptyset(\varphi_w,\varepsilon) :=$ rej attains with $\lambda=0$; so $\ell_{\mathrm{FI}^\sigma_{\mathrm{cert}}}(D^\emptyset_n)$ is $\kappa_s\sum_{i\leq n}|\varphi_{w_i}|_{\mathrm{bit}}$ up to a term in $[0,|V_\emptyset|]$. If $\kappa_s \leq \kappa$ the difference tends to $+\infty$; if $\kappa_s > \kappa$ it tends to $-\infty$, since $\sum_{i\leq n}|\varphi_{w_i}|_{\mathrm{bit}} \geq \sum_{i\leq n}|w_i| \geq (n/2)(\log_2 n-3)$ in length-lexicographic order. □

So at a common rate the certificate side gains at least 2κ bits per datum on every sequence. No soft fit classes or soft separations from deterministic function induction are claimed: every soft loss diverges on every infinite sequence, by the bound in (c).

Proposition E.27 (The parameter channel)**.** [proved; computed]
short-derivations Prop 7.1; scripts: referee_logic r2_parameter_channel *Let V be a verifier with time t_V and a time-constructible certificate bound L_V, and let* $\mathrm{cert}(N)$ *be the binary expansion of $N+1$ without its leading* 1*. Let A_V^{par} consist of $\neg\varphi \to \neg(p_N = p_N)$ for $V(\varphi, \mathrm{cert}(N)) = \mathrm{acc}$ and of $\neg\neg\varphi \to \neg(p_N = p_N)$ for $V(\varphi, \mathrm{cert}(N)) = \mathrm{rej}$, over sentences φ with $|\mathrm{cert}(N)| \le L_V(|\varphi|)$. Then* $\mathrm{Cn}(B \cup A_V^{\mathrm{par}}) = \mathrm{Cn}(B \cup \Gamma_{f_V})$*; A_V^{par} violates (F1); with time measured in symbols its membership is decidable in time $O(|\chi|^2 + L_V(|\chi|) + t_V(|\chi| + L_V(|\chi|)))$; and every φ^b with $f_V(\varphi) = b$ has a 9-line derivation from it of size $9|\varphi^b| + 54$, material $5|\varphi^b| + 40$ and largest instance $3|\varphi^b| + 13$, whatever the certificate. So for $X \in \mathrm{NP} \cap \mathrm{coNP}$ this set fits D^X under (a1) and (b) at linear budgets, with polynomial membership, and Proposition E.23(a) without (F1) would put* $\mathrm{NP} \cap \mathrm{coNP}$ *inside* $\bigcup_c \mathrm{DTIME}(2^{cm \log m})$*, an inclusion that is open.*

Proof. The closure of a member is $\forall x(\neg\varphi \to \neg x = x)$, equivalent to φ (for rejections to $\neg\varphi$), and every member of Γ_{f_V} has a member. Renaming p_N to p_0 gives a member only if V decides φ with the empty certificate, so (F1) fails once some φ needs a non-empty one. Membership: parse χ, reject if φ contains a parameter, read at most $L_V(|\varphi|) + 2$ bits of $N+1$, and run V; a sentence has only indices bounded by its depth. Derivation, with $\psi := \varphi^b$ and $E := (p_N = p_N)$: $\forall(\#0 = \#0)$ (reflexivity); $\forall(\#0 = \#0) \to E$ (A4); E; $E \to (\neg\psi \to E)$ (A1); $\neg\psi \to E$; $\neg\psi \to \neg E$ (the member); $(\neg\psi \to \neg E) \to ((\neg\psi \to E) \to \psi)$ (A3); $(\neg\psi \to E) \to \psi$; ψ, of sizes 4, 8, 3, $|\psi| + 9$, $|\psi| + 5$, $|\psi| + 6$, $3|\psi| + 13$, $2|\psi| + 6$ and $|\psi|$. For $X \in \mathrm{NP} \cap \mathrm{coNP}$ take V combining verifiers for X and its complement; L_V and t_V are polynomial, and with $|\varphi^b| \le |\varphi| + 1$ the budgets $h(m) = 3m + 16$ and $g(m) = 5m + 45$ suffice. Proposition E.23(a), if it held for this set, would give the inclusion stated. □

Lemma E.28 (Truncating parameter indices)**.** [proved; computed] short-derivations Lemma 7.2, Thm 7.4(a), (b); scripts: c6_parameters, Part A *Write formulas as strings of cells, an index or parameter number being a marker cell, its binary expansion plus one bit per cell, and an end cell. Let a machine reading its input sequentially decide membership in Ax within $T(|\chi|)$ steps, T nondecreasing, and let π be a derivation from Ax whose nonlogical axiom lines have size at most H. Put $K := T(H)$, let $q_0, \ldots, q_{r-1}$ be the parameters of π (Gen parameters included) whose numbers N have more than K bits in the expansion of $N+1$, and $w := \lceil \log_2(r+1) \rceil$. Let ρ fix the other parameters and map q_j to the parameter whose number plus one has as expansion the first K bits of q_j's, then* 1*, then j in w bits. Then $\rho(\pi)$ is a derivation from Ax with the same line sizes, its parameter numbers have at most $K + 1 + w$ bits, and $r \le \mathrm{size}(\pi)$. Consequently Theorem 6.9 holds as stated for axiom sets that violate (F1), with membership time measured in symbols, and so does the inclusion of the fit class of reading (b) in* $\mathrm{NP} \cap \mathrm{coNP}$*.*

Proof. ρ is injective: the images have exactly $K + 1 + w$ bits and distinct serials, while the fixed parameters have at most K. So logical axioms, MP and Gen stay valid (Lemma E.20(a) uses only injectivity). A nonlogical axiom line χ and $\rho\chi$ agree on their first $K + 1$ cells: before the first parameter with more than K bits the cells are equal, that parameter's bits start at some cell $s \ge 2$, after its marker, and its first K bits are kept. The machine halts within $T(|\chi|) \le K$ steps on both inputs, as $|\rho\chi| = |\chi| \le H$, and reads only cells $1, \ldots, K + 1$, so $\rho\chi \in Ax$. Every such parameter occurs in a line or names a Gen step, so $r \le \mathrm{size}(\pi)$. For Theorem 6.9: if φ_w has a derivation of size at most ℓ, it has one whose parameter numbers have $O(\ell^e)$ bits ($K = O(\ell^e)$,

$w \le \log_2(\ell+1)+1$), so of bit length $O(\ell^{e+1})$; the machine of Theorem 6.9 guesses and checks such derivations. For reading (b), Lemma E.19(c) turns material at most g into size at most G. □

Without (F1), the finer statements (the clock $2^{O(h \log h)}$ of Proposition E.23(a) and the certificate length of Proposition E.21(b)) are not claimed; membership time measured in bits rather than symbols is treated in short-derivations Prop 7.1(e) and Thm 7.4.

Not reproduced here. The tracks also treat a content-relative generation penalty, enumerators of Craig sets and the strict generation form (time-followup Props 2.4, 2.7, Rem 2.6), history-free predictors (time-followup Rem 4.6), the bounded schema, Hänni's two-sorted schema, a trace schema and $\rho_{f,n}$ under the readings (a2), (b2) (short-derivations Props 4.6–4.8, Cor 4.9, Rem 4.10), and least-prime-factor certificates (short-derivations Ex 5.6); statements, status and proofs are in their notes.

F Proofs and computations for Section 7

Proofs follow the pa track's final notes (and the experiments track's for Props X9–X11); steps they leave out are named. The pa track's referee re-checked every derivation of Propositions 7.3, 7.8 and F.3 and of Remark F.4 with an independent checker, recomputed the cost tables, the toy tower and the Euler example by independent code, and read the proofs of Propositions 7.11, 7.12, 7.19, F.5, F.12, F.21 and F.22, Remark F.13 and Proposition 7.13(a), finding them correct for the stated models after the corrections recorded in Appendix H. Results added in the revision were checked only by the tracks and while writing, not by a referee; the complete list is in Appendix H.1. Nothing is checked in a proof assistant.

F.1 The code, the checker and the reproduction

The code $\mathsf{L1}^{\mathrm{sch}}$ in detail (Definition 7.1). Sizes: in the derivation computations $|\cdot|$ is `nd.size`, in which a quantifier and its variable count 2, and $\beta = \log_2 23 \approx 4.52$; in the computations built on AS's code `u7` (Section 7.3, Examples 7.14 and F.18, Section 7.6) it is the `dtlib` size, in which a de Bruijn $\forall$ counts 1, at 5 bits per symbol. The *decodable* variant also codes the number of lines, the premise counts of tc lines (Elias γ) and the variables of subst and $\exists$I lines. The decodable variant adds 4.8–7.3% to derivation costs; with at least $\log_2$ of the alphabet size bits per written symbol (about $\log_2 25$ with the variable names used) it is a prefix code, so Kraft holds [proved for $\beta \ge \log_2$(alphabet size)] and $\mathsf{L1}^{\mathrm{sch}}$ is the two-part form of a sub-probability likelihood. The reported $\beta = \log_2 23$ under-charges symbols by about 3%, which changes no conclusion (pa §0). $\mathsf{L1}^{\mathrm{naive}}$ writes every formula out at the instance, each axiom leaf instantiated independently; a derivation that writes P K times then costs about $\beta(K-1)(|\varphi|-1)$ bits more than citing. The 9 formula symbols of the grammar are $=, \neg, \wedge, \vee, \to, \forall, \exists, \top, \bot$.

The checker. `checks/nd.py` checks natural-deduction derivations line by line. Rules: hyp (open a hypothesis, to be discharged), ax (an instance of an axiom of the theory), tc (tautological consequence of earlier lines, by truth tables over their prime subformulas), $\to$I, $\forall$E (with a capture check), $\forall$I and $\exists$E (with eigenvariable conditions), $\exists$I, refl and subst (equality). A derivation must end with no open hypotheses. Schematic derivations are written in a unary predicate symbol P; substituting a formula for P, after renaming the derivation's bound and eigenvariables away from it, maps derivations to derivations (the substitution theorem for predicate symbols; Shoenfield,

1967, location not verified). `test_nd.py` replays the four derivations of Proposition 7.3 on 200 random concrete motives with a parameter (seed 20261008): 200/200 check. Negative tests: the checker rejects all of 10 unsound steps (a ∀I eigenvariable free in a hypothesis, a tc step that is not a tautology, an ∃E eigenvariable that is not fresh in the conclusion or in a hypothesis, a mismatched subst, capture in ∀E, a wrong goal, an open hypothesis, tc through a quantifier, a wrong ∃I witness). The referee's independent re-checker (own substitution, α-normal form, truth tables, rule semantics and statements of all axioms and schemas) accepts all 20 derivations of the first version and rejects all 173 corrupted variants (a negated conclusion on some line, or a bad eigenvariable).

A script matched every number quoted in track pa's notes against the outputs (87 quotes, 0 failures; three planted errors were caught). All twelve outputs were byte-identical on a second run. Seeds: `c1` 7, `c2` `random.Random(n)` per n (as in u7), `c3` 1, `c4` 11 and 12, `c5` 5 and 6, `c6` 41 and 42, `c8` 31, 33, 11–13.

F.2 Equivalent axiomatisations

Proof of Proposition 7.3. Write "φ is progressive" for $\forall x(\theta_\varphi \to \varphi)$. (A) Assume $\varphi(0)$ and $\forall x(\varphi \to \varphi(Sx))$; we show progressiveness. Let $\theta_\varphi(x)$. By Q3, $x = 0$, and then $\varphi(x)$; or $x = Sw$. In the second case $w + S0 = S(w + 0) = Sw$ by Q5 and Q4, so $w < x$ by Dlt; hence $\varphi(w)$, hence $\varphi(Sw)$, i.e. $\varphi(x)$. CVI(φ) gives $\forall x\varphi$. (B) Assume φ progressive and apply Ind to θ_φ. Base: $y < 0$ gives $y + Sz = 0$ for some z (Dlt), so $S(y+z) = 0$ by Q5, contradicting Q1; so $\theta_\varphi(0)$. Step: assume $\theta_\varphi(x)$ and $y < Sx$. Then $y + Sz = Sx$ for some z, so $S(y + z) = Sx$ (Q5) and $y + z = x$ (Q2). By Q3, $z = 0$ or $z = Sv$. If $z = 0$, then $y = x$ (Q4) and $\varphi(x)$ follows from $\theta_\varphi(x)$ by progressiveness. If $z = Sv$, then $y + Sv = x$, so $y < x$ and $\varphi(y)$ by $\theta_\varphi(x)$. So $\theta_\varphi(Sx)$, and $\forall x\theta_\varphi(x)$; progressiveness gives $\forall x\varphi$. (C1) If LNP(φ) fails, $\exists x\varphi$ holds and every x with $\varphi(x)$ has some $y < x$ with $\varphi(y)$. Then $\neg\varphi$ is progressive, so CVI($\neg\varphi$) gives $\forall x\neg\varphi$, a contradiction. (C2) Let φ be progressive and suppose $\exists x\neg\varphi$. LNP($\neg\varphi$) gives an x with $\neg\varphi(x)$ and $\forall y < x\, \varphi(y)$; progressiveness gives $\varphi(x)$, a contradiction. The base axioms cited by the checked derivations are exactly those named: (A) CVI, Dlt, Q3–Q5; (B) Dlt, Ind, Q1–Q5; (C1), (C2) only the schema. □

Remark F.1 (Patterns). [proved] pa Rem 1.2 $\mathrm{CVI}(P), \mathrm{LNP}(P) \in \mathrm{PAT}$, while $T_{\mathrm{Ind}} \in \mathrm{DT}^\circ \setminus \mathrm{PAT}$ ($P(0)$, $P(Sx)$). So PA is Q, Dlt and one higher-order pattern (compare AS Prop 5.18: Q plus the first-order pattern IndEq).

For restricted motive classes the classical equivalences ($\mathrm{I}\Sigma_n \Leftrightarrow \mathrm{I}\Pi_n \Leftrightarrow \mathrm{L}\Sigma_n \Leftrightarrow \mathrm{L}\Pi_n$; Kaye, 1991; Hájek and Pudlák, 1993, theorem numbers not checked) hold over the usual base theory; their proofs use bounded collection ($\mathrm{B}\Sigma_n$, provable in $\mathrm{I}\Sigma_n$; known) because θ_φ adds a bounded quantifier. Some fail for parameter-free schemes (Kaye et al., 1988); DT° templates cannot say "motive in Σ_n", so only the full-schema equivalence matters here.

theory	forms derived	lines	symbols	overhead	decodable
$\mathsf{PA}_{\mathrm{Ind}}$	CVI, LNP	63, 82	204, 332	1468.0, 2234.1	1551.2, 2341.8
$\mathsf{PA}_{\mathrm{CVI}}$	Ind, LNP	39, 20	127, 117	880.4, 646.4	945.1, 676.9
$\mathsf{PA}_{\mathrm{LNP}}$	Ind, CVI	65, 27	253, 128	1692.1, 757.5	1792.3, 799.1

Table 24: Overhead in bits per use of deriving a non-primitive form $f(P)$ over citing it ($\mathsf{L1}^{\mathrm{sch}}$, $\beta = \log_2 23$); a citation costs 15.5–15.8 bits; template priors Ind 72, CVI 81, LNP 86 bits. Computed; upper bounds (pa §1.2).

Proof of Proposition 7.4. $\mathrm{CL}(\mathsf{PA}_S; D_n) = \mathrm{prior}_S + \sum_{i \le n}(c_S(f_i) + L_{\mathcal{Q}}(\varphi_i))$. The prior is a constant, and the $L_{\mathcal{Q}}$ terms are the same under every S. The differences $c_S(f_i) - c_{S'}(f_i)$ are i.i.d. and bounded, so the strong law gives the limit. The posterior odds are $2^{-\Delta \mathrm{CL}}$, which tend to 0 or ∞ exponentially when the limit is nonzero. □

Proposition F.2 (Lower bound)**.** [proved] pa Prop 1.4 *Every* $\mathsf{L1}^{\mathrm{sch}}$*-derivation of* $\mathrm{CVI}(P)$ *from* $\mathsf{PA}_{\mathrm{Ind}}$ *costs at least* $\log_2 10 + 11\beta \approx 53$ *bits more than citing it in* $\mathsf{PA}_{\mathrm{CVI}}$*; every derivation of* $\mathrm{Ind}(P)$ *from* $\mathsf{PA}_{\mathrm{CVI}}$ *at least* $\log_2 10 + 9\beta \approx 44$ *bits more than citing it in* $\mathsf{PA}_{\mathrm{Ind}}$*.*

Proof of Proposition F.2. For $\mathrm{CVI}(P)$; $\mathrm{Ind}(P)$ is the same with its 11-symbol antecedent ($\mathrm{Ind}(P)$ fails in the structure $\{0, a\}$ with $S0 = 0$, $Sa = a$, $P = \{0\}$, and it is not an instance of CVI, whose antecedent has root $\forall$). Both theories have 9 axioms, so a citation costs $\log_2 10 + \log_2 9 + 2\beta$ (it writes $P(x)$). The term positions of both forms lie under binders or are 0, so no $\forall$E step can have produced them from a shorter variable without capture; formulas on the chain below have full size. Let π derive $\mathrm{CVI}(P)$ from $\mathsf{PA}_{\mathrm{Ind}}$. 1. π *cites an axiom.* $\mathrm{CVI}(P)$ is not valid: read $<$ as the full relation and P as empty; then $\forall y(y < x \to P(y))$ fails for every x (take $y = x$), so the antecedent holds, while $\forall x P(x)$ fails. So some line is a citation, costing at least $\log_2 10 + \log_2 9$. 2. *Another line writes at least 13 symbols.* Follow the premises back from the last line through $\forall$E (premise $\forall v A$ with $A[t/v]$ the current formula), $\forall$I (premise the matrix) and $\exists$E (premise with the same conclusion). Every formula on this chain is $\mathrm{CVI}(P)$ with a quantifier prefix and terms in place of variables, which keeps the connective skeleton. The chain ends at a line of another rule. It is not ax: the Q axioms and Dlt do not contain P; an Ind instance, even after removing a $\forall$-prefix, is an implication whose antecedent is a conjunction, while every chain formula, prefix removed, has an antecedent with root $\forall$. It is not refl or $\exists$I (wrong root). A hyp, tc or subst line writes a formula with $\mathrm{CVI}(P)$'s skeleton, at least 18 symbols (13 skeleton symbols and one at each of 5 term positions); a hyp line still writes it, even if discharged later. A $\to$I line writes its discharged antecedent, which has the skeleton of $\mathrm{CVI}(P)$'s antecedent: at least 13 symbols. So some non-citation line costs at least $\log_2 10 + 13\beta$. 3. Premise references cost ≥ 0. So $\mathrm{bits}(\pi) \ge 2\log_2 10 + \log_2 9 + 13\beta$, which exceeds the citation by $\log_2 10 + 11\beta$. □

Proposition F.3 (Recursion conventions)**.** [proved; computed] pa Prop 1.5 $\mathsf{Q} + T_{\mathrm{Ind}}$ *and the theory with Q4, Q5 replaced by* $0 + x = x$, $Sx + y = S(x + y)$ *prove each other's recursion laws with one induction each, so they are equivalent; the per-use overheads are 213.8 and 593.4 bits one way, 213.8 and 545.1 the other. The posterior adopts the convention the data use.*

Proof of Proposition F.3. Each law is proved by one induction: in T_{left}, $x + 0 = x$ on x ($0 + 0 = 0$; $Sx + 0 = S(x + 0) = Sx$) and $\forall y(x + Sy = S(x + y))$ on x; in T_{right}, $0 + x = x$ on x and $Sp + x = S(p + x)$ on x with parameter p. The four derivations (11, 22, 11 and 21 lines) are checked; they cite only T_{Ind} and the other convention's two laws. The last claim is Proposition 7.4 with these overheads. □

Remark F.4 (ZF). [proved; computed; known; conjecture (reverse overheads)] pa §1.4 Separation from Replacement, Replacement from Collection and Foundation from $\in$-induction are checked, at 574–1432 bits per use against template priors of 72–185 bits (Table 25). Conversely, Collection from Replacement holds over ZF but fails without Power Set (Zarach, 1996; Gitman et al., 2016) and is open here without Foundation; EInd from Found needs transitive closures (Kaye and Wong, 2007); these reverse overheads were not formalised (conjecture: much larger). So, given the conjectured reverse overheads, the textbook ZF axioms win when the data are direct uses of them and of nothing they derive only at a cost.

derivation (checked)	lines	symbols	writes of P	overhead	template prior
Sep from $\mathrm{Repl}_{\mathrm{J}}(\lambda xy.\,\varphi(x) \wedge y = x)$	36	207	19	1209.3	72 (Sep)
$\mathrm{Repl}_{\mathrm{sp}}$ from Coll	15	108	12	574.1	185 ($\mathrm{Repl}_{\mathrm{sp}}$)
$\mathrm{Repl}_{\mathrm{J}}$ from Coll and two Sep instances	43	237	21	1432.0	167 ($\mathrm{Repl}_{\mathrm{J}}$)
Found from EInd($\neg x \in S$)	21	138	0	758.1	109 (Found)

Table 25: ZF: overheads in bits per use ($\mathsf{L1}^{\mathrm{sch}}$, $\beta = \log_2 23$). Sep is $\forall X \exists Y \forall u (u \in Y \leftrightarrow u \in X \wedge \varphi)$; $\mathrm{Repl}_{\mathrm{J}}$ is the image form $\forall x \forall y \forall z (\psi \wedge \psi[z/y] \to y = z) \to \forall X \exists Y \forall y (y \in Y \leftrightarrow \exists x (x \in X \wedge \psi))$; $\mathrm{Repl}_{\mathrm{sp}}$ the bounding form with $\exists!$ spelled out; Coll, EInd, Found as in AS Table 6. The checker caught a capture error in a first version of the Sep constructor. Computed; upper bounds (pa §1.4).

Further numbers for Remark 7.5. The ratios of (i) are 0.609 and 0.765 under the decodable code, and the first is 0.671–1.052 under $\mathsf{L1}^{\mathrm{naive}}$. On a 0.1 grid of $(p_{\mathrm{Ind}}, p_{\mathrm{CVI}})$, $\mathsf{PA}_{\mathrm{Ind}}$ is the best minimal theory for $p_{\mathrm{Ind}} \geq 0.7$, $\mathsf{PA}_{\mathrm{CVI}}$ in a broad middle region and $\mathsf{PA}_{\mathrm{LNP}}$ for large p_{LNP}; $\mathsf{PA}_{\mathrm{all}}$, if allowed, wins every mixed cell. It pays $81 + 86$ prior bits and $\log_2(11/9) \approx 0.29$ bits per citation more than $\mathsf{PA}_{\mathrm{Ind}}$, and is ahead in expectation after 12, 2, 1, 8 schema uses for $(p_{\mathrm{CVI}}, p_{\mathrm{LNP}}) = (0.01, 0)$, $(0.05, 0)$, $(0.10, 0.05)$, $(0, 0.01)$; one CVI use alone saves 1468 bits. Under $\mathsf{L1}^{\mathrm{naive}}$ the overhead grows by about $\beta(K-1)$ bits per motive symbol, K the number of written copies of P: CVI from $\mathsf{PA}_{\mathrm{Ind}}$ ($K = 14$) costs 1621, 1915, 2515, 3832 bits at $|\varphi| = 5, 10, 20, 40$ (schematic: 1468), and Ind from $\mathsf{PA}_{\mathrm{CVI}}$ ($K = 18$) 1088, 1487, 2264, 4030 (schematic: 880); mean of 5 random motives per size, seed 7.

F.3 The MDL finding

Grammars and usage laws. AS's code `u7` compares T^* with its splits under three body codes (AS Table 12): NAIVE, a fixed code of $\log_2 23$ bits per symbol; PC, sequential KT per template with context (parent symbol, child index); DPC, the same with the depth below the motive root added to the context. We added SDPC, one learned grammar shared by all templates with context (depth, parent, child index), the split templates' bodies being coded at the positions they occupy in T_{Ind}, so that a split changes only how the motive's root symbol is coded; RDPC, in which T^*'s grammar may condition on the motive's root while the splits keep the per-template DPC; and CF, a probabilistic grammar with one nonterminal per sort, shared by all templates. Usage laws: G1, AS's natural law (random motives of depth 2 with quantifiers only over atoms and scope-dependent terms; all seven roots occur, so $H_{\mathrm{used}} = H_{\mathrm{root}}$); G2 (roots $=, \neg, \wedge, \to$; symbols depend only on depth, parent and index, so DPC is well specified); G3 (roots $=, \to, \forall$). Reading of Table 26: SDPC keeps T_{Ind} whole by about the prior difference, falling by 1, 2.5 and 3 bits per doubling (Proposition F.6(c)); RDPC keeps it whole with a margin growing like $\ln n$; under NAIVE, PC, DPC and CF the split wins linearly at large n (CF: -0.067 bits per datum on G1 and -0.177 on G2 for H_{used} at $n = 256000$), since CF codes the motive's root with the pooled law of all formula symbols, and moving the root into the template index buys the divergence of the roots' law from it on every induction datum.

Proposition F.5 (Chain rule)**.** [proved] pa Prop 2.1 *With fixed parameters, if T^* has weight w on T_{Ind} and body law $\mathcal{Q}(\varphi) = \mathcal{Q}(f)\mathcal{Q}(\varphi \mid f)$, f the root, then H_F ($F \supseteq$ the roots in the support) with weights $w\mathcal{Q}(f)$ on T_f and body laws $\mathcal{Q}(\cdot \mid f)$ has $P_{H_F} = P_{T^*}$.*

Proof of Proposition F.5. For an induction datum $\mathrm{Ind}(\varphi)$ whose motive has root f, $P_{H_F} = w\mathcal{Q}(f)\mathcal{Q}(\varphi \mid f) = w\mathcal{Q}(\varphi) = P_{T^*}$; ground data have the same weight under both. □

Proposition F.6 (Occam identity)**.** [(a) proved; (b), (c) proved under the growth condition of (b); computed]

pa Prop 2.3 *Use* L0 *with a Dirichlet($\frac{1}{2}$) citation index and a learned positional grammar shared by all templates (SDPC: KT per context c, alphabet A_c);* $\mathrm{KT}(k; A)$ *is* $-\log_2$ *of the Dirichlet($\frac{1}{2}$) marginal of counts k over A symbols. Suppose every datum is covered by exactly one template of H and one of H', and, for a set C of contexts, H' codes datum by datum exactly the (context, symbol) pairs that H codes outside C. (a)* $\mathrm{CL}(H') - \mathrm{CL}(H) = \Delta\mathrm{prior} + \mathrm{KT}(n'; |H'|) - \mathrm{KT}(n; |H|) - \sum_{c \in C} \mathrm{KT}(m_c; |A_c|)$ *(n, n' citation counts, m_c symbol counts in c). (b) If all nonzero counts grow linearly,* $\mathrm{KT}(k; A) = N\hat{H}(k) + \frac{A-1}{2}\log_2 N + O(1)$, $N = \sum k$. *(c) If $C = \{c\}$, the data visiting c are those of one template of H, and H''s template and the pair (H's template, symbol in c) determine each other, then* $\mathrm{CL}(H') - \mathrm{CL}(H) = \Delta\mathrm{prior} + \frac{(|H'|-|H|)-(|A_c|-1)}{2}\log_2 n + O(1)$*; for a split of T_{Ind} into roots F under SDPC,* $\Delta\mathrm{prior} + \frac{|F|-9}{2}\log_2 n + O(1)$.

Proof of Proposition F.6. (a) The total code is the prior, plus the KT code of the citation index, plus a sum over contexts of the KT code of the symbols coded there. Sequential KT in a context equals the Dirichlet($\frac{1}{2}$) marginal of that context's counts, whatever their order (known; it is the Pólya-urn form of the Dirichlet-multinomial; Krichevsky and Trofimov, 1981). By hypothesis the two hypotheses have the same count vectors in every context outside C, so those terms cancel; H' has no counts in C. The index and prior terms remain. (b) $\mathrm{KT}(k; A)\ln 2 = \ln\Gamma(N + \frac{A}{2}) - \ln\Gamma(\frac{A}{2}) - \sum_{a:k_a>0}[\ln\Gamma(k_a + \frac{1}{2}) - \ln\Gamma(\frac{1}{2})]$. Stirling, $\ln\Gamma(m+a) = (m+a-\frac{1}{2})\ln m - m + \frac{1}{2}\ln 2\pi + o(1)$, applied to the $j+1$ growing arguments (j the number of nonzero counts) gives $N\ln N - \sum_a k_a \ln k_a + \frac{A-1}{2}\ln N + \frac{1-j}{2}\ln 2\pi + j\ln\Gamma(\frac{1}{2}) - \ln\Gamma(\frac{A}{2}) + o(1)$; the first two terms are $N\hat{H}(k)$ in nats. (c) Let τ_0 be the template of H whose data visit c. H''s counts are those of H for the other templates and $m_{c,a}$ for the templates that correspond to (τ_0, a). Hence $n\hat{H}(n') = n\hat{H}(n) + m_c\hat{H}(m_c)$ (chain rule for empirical entropies), and the entropy terms of (a) cancel exactly. The $\log_2$ terms give $\frac{|H'|-1}{2} - \frac{|H|-1}{2} - \frac{|A_c|-1}{2}$ times $\log_2 n$, since m_c is linear in n; the rest is $O(1)$. For a split of T_{Ind} into the roots F, $|H'| - |H| = |F| - 1$ and $|A_c| = 9$. □

Remark F.7 (Refuted: a constant margin). [refuted] pa §2.2; referee m2 The first version called the SDPC margin of H_{root} constant ($+770 \to +762$ bits for $n = 10^3 \to 2.56 \cdot 10^5$). It falls by 1 bit per doubling (Proposition F.6(c), $|F| = 7$); extrapolated crossover $n \approx 2^{780}$.

Corrections and reproduction. The first version, like u7 itself, charged prior bits only for templates that some datum uses, while the KT index ranged over all seven (referee m1). Under G1 nothing changes; under G2 (no $\vee, \forall, \exists$ at the root) 355 prior bits were missing and under G3 470. Every template is now charged; H_{used} is AS's split H_F with the index restricted to F. With u7's own bookkeeping the script reproduces AS Table 12 exactly at $n = 1000$ (G1: -223, $+2325$, $+2701$; G2: -915, $+1141$, $+1428$); the G2 rows of AS Table 12 carry the same undercount, which changes no conclusion there. A first SDPC run ($+803$ to $+12270$ bits) was an encoding artefact and is refuted.

Proof of Proposition 7.8. (a) Each wrapper $w_f(P)$ ($\neg\neg P$, $P \wedge P$, $P \vee P$, $(0 = 0) \to P$, $\forall z P$, $\exists z P$, with z not free in P) is logically equivalent to P and has root f, so $\mathrm{Ind}(w_f(P)) \in \mathrm{inst}(T_f)$, and $\mathrm{Ind}(P)$ follows by Lemma F.15 (a special case of Proposition F.16(a)). The six checked

law, split	n	NAIVE	PC	DPC	SDPC	RDPC	CF
G1, H_{root}	1000	−223.0	+2325.3	+2700.6	+770.0	+1676.6	+709.8
	16000	−16466.1	−1198.5	−3035.6	+765.8	+3724.6	−301.8
	256000	−274655.9	−94088.5	−163027.5	+761.9	+5985.9	−17241.7
G2, H_{used}	1000	−926.0	+1129.8	+1417.1	+403.7	+870.1	+236.5
	16000	−21484.6	+1035.8	+2805.3	+393.7	+1566.1	−2341.2
	256000	−350773.1	−12391.6	+4289.5	+383.7	+2346.1	−45211.8
G3, H_{used}	1000	−1300.8	+772.5	+543.0	+284.6	+931.3	+270.0
	16000	−24985.6	−3058.9	−9258.6	+272.8	+1736.6	−246.7
	256000	−404445.7	−75061.9	−186306.5	+260.8	+2564.9	−7814.9

Table 26: CL(split) − CL(T^*) in bits ($T^* = \mathsf{Q} + T_{\text{Ind}}$; positive: T_{Ind} stays whole). Prior parts +780 (G1), +425 (G2), +310 (G3). SDPC for H_{root} under G2 and G3: +769.9 → +761.8 and +769.7 → +761.9. CF on G1 at $n = 256000$: −0.067 bits per datum. Computed (pa §2.2; `c2_mdl.out`, `c2b_cf.out`).

derivations have 17–19 lines; overheads over citing Ind(P): ¬ 409.9, ∧ 441.6, ∨ 441.6, → 488.9, ∀ 369.2, ∃ 425.6 bits. No wrapper exists for the root =. (b) The instances of $T_=$ are induction for equations, so $\mathsf{Q} + T_= \subseteq$ IOpen; IOpen does not prove that $\sqrt{2}$ is irrational (Shepherdson, 1964), and PA does. □

F.4 Theorem data

Proposition F.8 (Gibbs)**.** [proved]
pa Prop 3.2 *If $\sum_s 2^{-\ell(s)} \le 1$ (e.g. $\ell(s) = \beta|s|$ for a Polish-notation code with $\beta \ge \log_2$ of the alphabet size) and P has finite entropy, then $\mathbb{E}_P[-\log_2 P] \le \mathbb{E}_P[\ell]$. So on data from P_{T^*} the generator's code length for a new datum is in expectation at most the prior cost of memorising it (take $\ell(s) = \beta_{\text{ax}}|s|$, $\beta_{\text{ax}} \ge \log_2$ of the alphabet size).*

Proofs of Propositions 7.9 and F.8. Proposition 7.9: adding s adds $\beta_{\text{ax}}|s|$ prior bits; each occurrence of s costs ℓ_{cite} instead of $D_T(s)$; the other data change only through the index. Proposition F.8: let $Z := \sum_s 2^{-\ell(s)} \le 1$ and $q := 2^{-\ell}/Z$. Then $\mathbb{E}_P[\ell + \log_2 P] = \sum_s P(s) \log_2 \frac{P(s)}{Zq(s)} = \mathrm{KL}(P\|q) - \log_2 Z \ge 0$, with equality iff $P = q$ and $Z = 1$. Adopting a new datum costs at least $\ell(s)$ prior bits when the axiom prior is itself such a code, so memorisation can gain only on repeated data. □

Proof of Proposition 7.10. Th(Q∪S) ⊆ Th(PA), and equality would make PA finitely axiomatisable, which it is not (Ryll-Nardzewski, 1952). So some $\sigma \in \mathrm{Th}(\mathsf{PA}) \setminus \mathrm{Th}(\mathsf{Q} \cup S)$ has $P_{\mathsf{PA}}(\sigma) > 0$ (Lemma 2.10, parameters admissible) and $P_{\mathsf{Q}\cup S}(\sigma) = 0$; it occurs a.s. after finitely many data. There are countably many such S. The second claim is Theorem 4.2. For $\mathsf{PA} \cup S$ under W: for generic fixed weights its law differs from P_{PA}, so $\mathrm{KL}(P_{\mathsf{PA}} \| P_{\mathsf{PA}\cup S}) > 0$ and Proposition C.1(a) gives exponential decay at that rate; whether some special weight gives $P_{\mathsf{PA}\cup S} = P_{\mathsf{PA}}$ was not checked. With Dirichlet weights the model is outside W: $\mathsf{PA} \cup S$ then contains the generator with extra components, as a spare slot does, and polynomial decay as in Proposition 4.14 is a proof sketch. □

Proposition F.9 ("Must prove" with refutation)**.** [proved] pa Prop 3.4 *Under S_{prove} with bounded derivability and R_d: (a) the posterior is the prior restricted to the score-1 theories; (b) $\mathsf{Q} + T_{\text{Ind}}$*

beats $\mathsf{Q} \cup \{s_1, \ldots, s_m\}$ *once* $\beta_{\mathrm{ax}} \sum_i |s_i| > \beta_{\mathrm{ax}}|T_{\mathrm{Ind}}|$ *(72 bits at the default* $\beta_{\mathrm{ax}} = \beta$*; one or two library theorems suffice, associativity alone:* $17 \log_2 23 = 76.9 > 72$*), if* d *covers its derivations; (c) any unrefuted theory with a smaller prior that derives every datum within* d *beats both, e.g. an inconsistent theory whose shortest refutation of a datum (and shortest derivation of a negative datum) is longer than* d*. (The bare formula metavariable* F_0 *is refuted by* R_d *as soon as* d *covers a one-line citation of a negated datum, which (b) requires.)*

Proof of Proposition F.9. (a) is the definition of the score. (b) and (c) compare prior bits among score-1 theories. In (c), an inconsistent theory derives every sentence, so it has score 1 unless R_d refutes it, which needs a refutation of a datum or a derivation of a negative datum within d. The example in (b), at $\beta_{\mathrm{ax}} = \beta$: associativity alone, $17\beta = 76.9 > 72$ bits; or $\beta(7 + 13) = 31.7 + 58.8 = 90.5$. □

The theorems in other axiomatisations. The six theorems of Table 3 cost 1148.4–4163.7 bits more in $\mathsf{PA}_{\mathrm{CVI}}$ and 2241.2–8117.4 bits more in $\mathsf{PA}_{\mathrm{LNP}}$ than in $\mathsf{Q}+T_{\mathrm{Ind}}$, with $\beta^* = 143$–501 and 240–861 bits per symbol (upper bounds: Ind is re-derived at each use; a direct $\mathsf{PA}_{\mathrm{CVI}}$ derivation of $\forall x\, 0 + x = x$ by the referee costs 652 bits more instead of 1161). The four recursion-convention theorems of Proposition F.3 give $\beta^* = 30.5$–45.6. Decodable-code costs are 3.4–7.5% higher. All 19 derivations are checked.

Remark F.10 (Lemmas, compressible data, theorem streams). [computed; known] pa §§3.3–3.5 (i) Lemmas used often become axioms: commutativity costs 1358.6 bits with its two lemmas derived in line and 504.7 with them adopted (prior 90.5; compare Proposition B.14(c)). (ii) A schema compresses its direct instances about fourfold; adopting a non-primitive form $f(P)$ beats deriving and memorising after 1–2 uses. (iii) On streams of fresh induction uses (probability u) and theorems of a library fixed in advance, for $u = 0$ the MAP is Q plus the library from the first datum, a finite subtheory of PA and so strictly weaker (known; Ryll-Nardzewski, 1952); for $u = 0.1, 0.5$ it is $\mathsf{Q} + T_{\mathrm{Ind}}$ plus the library, with PA's theorems, from $n = 10$ (the first checkpoint after $n = 1$).

Compressible data and streams (Remark F.10). Per datum $f(\varphi)$ (φ a random motive, seed 41) the options cost: derive in $\mathsf{PA}_{\mathrm{Ind}}$, memorise $(\beta|f(\varphi)| + \ell_{\mathrm{cite}})$, or adopt $f(P)$ $(\ell_{\mathrm{cite}} + \beta|\varphi|$, plus $\beta|f(P)|$ once). Ind, $|\varphi| = 21$: 101.5 (a citation) against 431.7. CVI, $|\varphi| = 20$: derive 1565.0, memorise 332.2, adopt 97.0 (+81); $|\varphi| = 418$: 3365.4, 5733.3, 1897.3. LNP, $|\varphi| = 21$: 2335.5, 350.3, 101.5 (+86). Streams (seed 42; bits relative to $\mathsf{Q} + T_{\mathrm{Ind}}$, $n = 1000$): $u = 0$: $\mathsf{Q} + T_{\mathrm{Ind}}$ plus library -561857, Q plus library -562036 (MAP); $u = 0.1$: -513380 (MAP) against -489898; $u = 0.5$: -285950 (MAP) against -147800.

F.5 Data from $\mathrm{Th}(\mathbb{N})$, the chain and narrow practice

Proof of Proposition 7.11. If T proves a false φ, then $\neg\varphi \in \mathrm{Th}(\mathbb{N})$ and $T \nvdash \neg\varphi$ by consistency. Otherwise T's theorems are true, and since they are r.e. while $\mathrm{Th}(\mathbb{N})$ is not, some true σ has $T \nvdash \sigma$. In both cases some s has $\mu(s) > 0 = P_T(s)$, and the probability that it is absent from n draws is $(1 - \mu(s))^n \to 0$. □

Proof of Proposition 7.12. An inconsistent theory derives every sentence, so its 0/1 likelihood is never 0 without a rule. Under R_∞ it derives the negation of the first datum. Under R_{d_n} its shortest refutation has some finite size d, and it is refuted once $d_n \geq d$. Consistency of r.e.

theories is Π_1-complete (known), so no computable rule refutes exactly the inconsistent theories. (AS Thm 6.15 is a related but different statement about DTRC-type accept sets.) □

Proposition F.11 (Reflection is not a template)**.** [proved] pa Prop 4.3 *Every* DT$^\circ$ *template whose instances lie in* $\mathrm{Refl}_T := \{\mathrm{Prv}_T(\ulcorner\varphi\urcorner) \to \varphi\}$ *(unary numerals) is ground.*

Proof of Proposition F.11. As for Proposition 6.6 (Appendix E.3), with Prv_T for Accept_f. □

Proposition F.12 (Regret against a theory plus exceptions)**.** [proved] pa Prop 4.4 *Under* L_ε*, for every* T *and* D_n*, with* E *the data* T *neither derives nor refutes:* $-\log_2 P_{\mathrm{mix}}(D_n) \le -\log_2 \pi(T) + \sum_{s\notin E} -\log_2((1-\varepsilon)P_T(s)) + \sum_{s\in E} -\log_2(\varepsilon\mu_0(s))$*. With 0/1 support the same holds for* $T \cup E$*, the exceptions memorised at* $\beta_{\mathrm{ax}}|e|$ *prior bits each.*

Remark F.13 (Pointwise limits are trivial). [proved; open]
pa Rem 4.5 Under 0/1 support every surviving consistent theory eventually decides each sentence correctly (memorisation alone suffices). With inconsistent theories in the class, each is eventually refuted under R_{d_n}, but whether the posterior mass of not-yet-refuted inconsistent theories deriving a false sentence tends to 0 is open.

Proofs of Proposition F.12 and Remark F.13. $P_{\mathrm{mix}}(D_n) \ge \pi(T)\prod_s \mathsf{L}_\varepsilon(T;s)$; for $s \notin E$ that T derives, $\mathsf{L}_\varepsilon(T;s) \ge (1-\varepsilon)P_T(s)$ (if T refutes s the bound is $+\infty$); for $s \in E$, $\mathsf{L}_\varepsilon(T;s) \ge \varepsilon\mu_0(s)$. Take $-\log_2$. With $\varepsilon = 0$, apply the same to $T \cup E$, whose likelihood is positive on all of D_n. For Remark F.13: a true s occurs eventually, after which every surviving theory derives it; for a false s, $\neg s$ occurs eventually, after which no surviving consistent theory derives s. Each inconsistent theory is refuted under R_{d_n} after finitely many steps (Proposition 7.12), but never all of them at once in an infinite class; their remaining mass can be large (Proposition 7.19), and whether it tends to 0 is open. □

Remark F.14 (Toy drift). [computed] pa §4 Along $T_0 = \mathsf{PA}$, $T_{k+1} = T_k + \mathrm{Con}(T_k)$, with Geometric($\frac{1}{2}$) data levels and prior $2^{-50(k+1)}$, the MAP is the largest level seen (16 at $n = 10^5$) and the regret grows like 50 bits per level (798.9 bits at $n = 10^5$; linear, 0.2637 bits per datum, if the theories' level law is misspecified). The toy covers only the provable part of the data: under a full-support law a positive fraction of the data is unprovable in every T_k. Each step adds a ground sentence, since one reflection sentence suffices (Proposition 6.7) and the reflection schema is not a template (Proposition F.11).

Details (seed 1; data levels Geometric(ρ), each T_k generating its theorems with a truncated geometric level law of rate s): for $\rho = s = \frac{1}{2}$ the MAP level and regret are 2, 98.1 bits at $n = 10$, 9, 448.6 at 10^3 and 16, 798.9 at 10^5, and the probability that the next datum is unprovable is $7.6 \cdot 10^{-6}$ at 10^5; with L_ε, $\varepsilon = 0.01$, the MAP lags the record (11 at 10^5) and the regret is 674.0; for $\rho = s = 0.8$ the regret is 2598.9 bits at 10^5 (MAP 52); for $\rho = \frac{1}{2}$, $s = 0.3$ it is 0.2637 bits per datum at 10^5. The referee's independent re-implementation gives identical numbers.

Proof of Proposition 7.13. (a) $\mathrm{I}\Sigma_n$ is finitely axiomatisable for $n \ge 1$ (known). $\mathrm{I}\Sigma_{n+1} \vdash \mathrm{Con}(\mathrm{I}\Sigma_n)$ (known; Hájek and Pudlák, 1993, traced by the referee to Cor. I.4.34 through a secondary source; not verified), so $\mathsf{PA} \vdash \mathrm{Con}(\mathrm{I}\Sigma_n)$, while $\mathrm{I}\Sigma_n \nvdash \mathrm{Con}(\mathrm{I}\Sigma_n)$ by Gödel's second theorem, $\mathrm{I}\Sigma_n$ being consistent (true). So $P_{\mathsf{PA}}(\mathrm{Con}(\mathrm{I}\Sigma_n)) > 0 = P_{\mathrm{I}\Sigma_n}(\mathrm{Con}(\mathrm{I}\Sigma_n))$, and $\mathrm{Con}(\mathrm{I}\Sigma_n)$ occurs a.s. Eliminating each $\mathrm{I}\Sigma_n$ is not concentration, since the tail $\{\mathrm{I}\Sigma_m : m > N\}$ could hold mass; Theorem 4.2 gives $\pi_n(C^*_{\mathsf{PA}}) \to 1$, and no $\mathrm{I}\Sigma_n$ is in C^*_{PA} since its law has a different support. The waiting

time is about $1/P_{\mathsf{PA}}(\mathrm{Con}(\mathrm{I}\Sigma_n))$. (b) *Sketch.* Deriving $\mathrm{Ind}(\varphi)$, $\varphi \in \Sigma_n$, from σ_n goes through the Tarski biconditional $\varphi \leftrightarrow \mathrm{Tr}_n(\ulcorner\varphi\urcorner)$ (the predicate of Proposition 6.7), whose derivation grows with $|\varphi|$, while citing T_{Ind} costs $L_{\mathcal{Q}}(\varphi)$ plus a constant. Not formalised; the growth rate was not checked. (c) Refuted by Examples 7.14 and F.18. The step that fails: AS Thm 5.16 concerns a single covering template (it does hold with parameters, contrary to a remark of the referee), and the wrappers of Proposition 7.8 need a formula metavariable. □

Lemma F.15 (Equivalent motives)**.** [proved]
pa Lemma 4.7 *If* $B \vdash \forall\bar{p}\forall x(\chi \leftrightarrow \psi)$ *then* $B + \mathrm{Ind}(\chi) \vdash \mathrm{Ind}(\psi)$.

Proof of Lemma F.15. From $\forall x(\chi \leftrightarrow \psi)$ at 0, x and Sx: $\psi(0)$ gives $\chi(0)$, and $\psi(x) \to \psi(Sx)$ gives $\chi(x) \to \chi(Sx)$. So the antecedent of $\mathrm{Ind}(\psi)$ gives that of $\mathrm{Ind}(\chi)$, which gives $\forall x\chi$, hence $\forall x\psi$. Generalise over the parameters. □

Proposition F.16 (Read-once induction templates)**.** [proved] pa Prop 4.8 *Let* $\tau = T_{\mathrm{Ind}}[P := \lambda x.C]$ *with* C *read-once: built by connectives and quantifiers from atoms* $A(x,\bar{z}) = B(x,\bar{z})$ *(*A, B *term metavariables) and* $F(x,\bar{z})$ *(*F *a formula metavariable), each metavariable occurring once. (a) If* C *has a formula metavariable,* $\mathrm{Q1} + \mathrm{inst}(\tau) \vdash \mathrm{Ind}(\psi)$ *for every* ψ*. (b) Otherwise every instance has* C*'s formula skeleton, so* $\mathsf{Q} + \mathrm{inst}(\tau) \subseteq \mathrm{I}\Sigma_k$ *for some* k*, and* $\mathrm{I}\Sigma_k \subsetneq \mathsf{PA}$.

Proof of Proposition F.16. (a) Fix the path from C's root to one occurrence $F(x,\bar{z})$. Instantiate every other atom as a constant: $\top := (0 = 0)$ $(A, B := \lambda.0)$, $\bot := (S0 = 0)$ $(A := \lambda.S0$, $B := \lambda.0)$, other formula metavariables $\lambda.(0 = 0)$ or $\lambda.(S0 = 0)$; Q1 refutes $S0 = 0$. Since each metavariable occurs once, these choices are independent. By induction on its structure, every read-once subformula can be made logically equivalent (over Q1) to $\top$ and, by another choice, to $\bot$ (for $\leftrightarrow$ use $(\top,\top)$ and $(\top,\bot)$; quantifiers over a constant become vacuous). So each sibling on the path can be made $\top$ or $\bot$ as needed: $\top$ beside $\wedge$ and $\leftrightarrow$, $\bot$ beside $\vee$, $\top$ as the antecedent of $\to$ when the path goes to the consequent, and $\bot$ as the consequent when it goes to the antecedent (that step then acts as a negation, as $\neg$ does). Set $F := \lambda x\bar{z}.\psi'(x)$, which ignores $\bar{z}$, so the quantifiers on the path become vacuous. By propositional logic, Q1 and the vacuous-quantifier laws, $\mathrm{Q1} \vdash \forall x(C\theta \leftrightarrow \psi')$ or $\mathrm{Q1} \vdash \forall x(C\theta \leftrightarrow \neg\psi')$, depending only on the path. Take $\psi' := \psi$ in the first case and $\psi' := \neg\psi$ in the second. Then $\mathrm{Q1} \vdash \forall x(C\theta \leftrightarrow \psi)$, and Lemma F.15 applied to the instance $\tau\theta = \mathrm{Ind}(C\theta)$ gives $\mathrm{Ind}(\psi)$. (b) Term metavariables are replaced by terms, which contain no logical symbols, so every instance has C's skeleton. A formula with a fixed skeleton is logically equivalent to a prenex formula of fixed complexity Σ_k, and Lemma F.15 turns the instances into Σ_k induction. $\mathrm{I}\Sigma_k \subsetneq \mathsf{PA}$ because $\mathsf{PA} \vdash \mathrm{Con}(\mathrm{I}\Sigma_k)$ (known, as in Proposition 7.13(a)) while $\mathrm{I}\Sigma_k \nvdash \mathrm{Con}(\mathrm{I}\Sigma_k)$. □

Remark F.17 (Read-once is not necessary)*.* [proved; computed; open]
pa Rem 4.9 $T_{\mathrm{Ind}}[P := \lambda x.\, x = 0 \vee F(x)]$ is not read-once, and every instance's motive holds at 0; yet $\mathrm{Q1}, \mathrm{Q2} + \mathrm{Ind}(x = 0 \vee \exists y(x = Sy \wedge P(y))) \vdash \mathrm{Ind}(P)$ (checked, 47 lines). Which non-read-once templates give full induction is open.

The checked derivation has 227 written symbols, costs 1403.3 bits against a 15.4-bit citation, and cites only Ind, Q1, Q2. The motive χ is not equivalent to P, so Lemma F.15 does not apply.

Example F.18 (G1 keeps PA)**.** [computed; proof sketch (slope)]
pa §4.3 (2) G1 has 157 motive skeletons, each equivalent to a Σ_2 formula, so a term-only skeleton theory covers G1 inside $\mathrm{I}\Sigma_2$; coverage does not force PA. The theory of the 112 skeletons seen by $n = 3000$ (119 templates) covers those data and is 42838.5 bits behind T^* (prior part 43040),

but it gets likelihood 0 at the first unseen skeleton. The covering theory of all 157 skeletons (164 templates), inside $\mathrm{I}\Sigma_2$, has asymptotic slope $\frac{164-8}{2} - 19 \cdot \frac{9-1}{2} = +2.0$ bits per doubling against T^* (19 contexts; Wilks's theorem, from memory; proof sketch); its code length was not computed.

Details of Examples 7.14 and F.18. Narrow practice (the referee's counterexample r8, rechecked by the track): $\mathrm{CL}(H_{\text{narrow}}) - \mathrm{CL}(T^*) = +230.8, +212.5, +192.5, +174.6$ bits at $n = 100, 300, 1000, 3000$; $|T_{\text{Ind}}| = 14$, $|T_E| = 39$, $|T_N| = 31$ (`dtlib`), prior difference 280 bits; coverage checked on 3000 data in seeds 31 and 33; the code, the identity (three contexts: motive root and the two "=" positions) and its expansion give $+230.8/+230.8/+232.0$ at $n = 100$ and $+174.6/+174.6/+174.6$ at 3000. G1: motives of depth 2 quantifying only atoms give $1+7+3\cdot 49+2 = 157$ skeletons, all found among 200000 motives (seed 12); 112 occur in the first 3000 data. Each motive is a Boolean combination of atoms and single quantifiers over atoms, so it is equivalent to a Σ_2 formula. The theory of the 112 skeletons seen in the first 3000 data (119 templates) covers those data and is +42905.6 and +42838.5 bits behind at $n = 1000, 3000$ (prior part +43040); while it covers the data its asymptotic slope is -20.5 bits per doubling, so it closes in, but it gets likelihood 0 at the first unseen skeleton. T^* uses 19 formula contexts. The theory of all 157 skeletons (164 templates) covers G1, and its slope is +2.0 bits per doubling; its code length was not computed. The entropy terms cancel only asymptotically: G1 draws each connective independently given its context, so their difference is $O_P(1)$ by Wilks's theorem for nested multinomial models (cited from memory; proof sketch).

F.6 Root fragments and the Q axioms

The experiments' $T^* = \mathrm{Q1..Q7} + T_{\text{Ind}}$ has one-hole motives without parameters; its language has $<$ with no Q axiom about it. T_f for $f \in \{=, <\}$ has motive $a(x)\, f\, b(x)$ with 1-ary term metavariables; for $\neg$, $\neg P_1(x)$; for $\wedge, \vee, \to, \leftrightarrow$, $P_1(x)\, f\, P_2(x)$; for $\forall, \exists$, $f\, y\, P_1(x, y)$.

Proposition F.19 (Root fragments)**.** [proved] experiments Prop X9 $\mathrm{inst}(\{T_f : \textit{9 roots}\}) = \mathrm{inst}(T_{\text{Ind}})$*; for each connective root f and any set Q^- of axioms among Q1–Q7, $\mathsf{Q}^- \cup \{T_f\}$ and $\mathsf{Q}^- \cup \{T_{\text{Ind}}\}$ have the same consequences; the argument fails for $=$ and $<$.*

Proof of Proposition F.19. (a) Every motive has one of the 9 roots, and an instance of T_{Ind} with root f is the instance of T_f whose sub-bodies are read off the motive (for $\forall, \exists$ the bound variable becomes P_1's second argument); conversely $\mathrm{inst}(T_f) \subseteq \mathrm{inst}(T_{\text{Ind}})$. (b) One direction is (a). For the other, given φ choose: $P_1 = P_2 = \varphi$ for $\wedge, \vee$; $P_1 = \neg\varphi$ for $\neg$; $P_1 = (0 = 0)$, $P_2 = \varphi$ for $\to, \leftrightarrow$; $P_1(x, y) = \varphi(x)$ for $\forall, \exists$ (nonempty domains). The chosen motive ψ satisfies $\vdash \forall x(\psi \leftrightarrow \varphi)$, so $\mathrm{Ind}(\psi) \vdash \mathrm{Ind}(\varphi)$. (c) No such body exists for an atomic root; Proposition 7.16 shows the atomic fragments are weaker. □

Proof of Proposition 7.16. *T^* proves every induction instance with parameters.* For $\varphi(x, \bar{w})$ take the parameter-free motive $\psi(x) := \forall \bar{w}[(\varphi(0, \bar{w}) \wedge \forall y(\varphi(y, \bar{w}) \to \varphi(Sy, \bar{w}))) \to \varphi(x, \bar{w})]$. $\psi(0)$ is valid; if $\psi(x)$ and the base and step hold for $\bar{w}$, then $\varphi(x, \bar{w})$ and so $\varphi(Sx, \bar{w})$: $\psi(Sx)$. T_{Ind} gives $\forall x\psi$, which is induction for φ. So $T^* \supseteq \mathsf{PA}$ and $T^* \vdash \sigma$, as $\mathsf{PA} \vdash \sigma$ (known). $\mathsf{Q} + \{T_=, T_<\} \nvdash \sigma$. By Shepherdson (1964) (secondary sources only) there is a model M of Q plus open induction with parameters in which $x \cdot x = 2 \cdot y \cdot y$ has a solution with $y \neq 0$; M is the non-negative part of a discretely ordered ring, with $Sx := x + 1$, so Q1–Q7 hold (Q3 by discreteness). Instances of $T_=$ are open induction, so they hold. Read $<$ as the empty relation: every instance of $T_<$ has the false

conjunct $a(0) < b(0)$ in its antecedent, so it holds; σ does not mention $<$. So $M \models \mathsf{Q} + \{T_=, T_<\}$ and $M \not\models \sigma$. □

The same holds for any theory of Q axioms and term-only induction-type templates: by the argument of Proposition F.16(b), which needs only a fixed formula skeleton, it lies in some $\mathrm{I}\Sigma_k \subsetneq \mathsf{PA}$.

Proposition F.20 (Redundant Q axioms)**.** [proved; computed]
experiments Prop X11 $\mathrm{Q}1, \mathrm{Q}2, \mathrm{Q}4..\mathrm{Q}7 + T_{\mathrm{Ind}} \vdash \mathrm{Q}3$*; no other* Q_i *is redundant given full induction.*

Proof of Proposition F.20. T_{Ind} cites $\mathrm{Ind}(\psi)$ for $\psi(x) := x = 0 \vee \exists y(x = Sy)$; $\psi(0)$ holds and $\psi(Sx)$ holds with $y := x$, so $\forall x\psi$, which gives Q3. For the other axioms: in a structure in which every element is the value of a numeral, every induction instance holds (induction on k in the metatheory). Each structure below has this property, satisfies the Q_j, $j \neq i$, and falsifies Q_i: $i = 1$: $\{0\}$, all operations 0; $i = 2$: $\{0, 1\}$, $S0 = S1 = 1$, $x + 0 = x$, $x + 1 = 1$, $x \cdot 0 = 0$, $x \cdot 1 = x$; $i = 4$: $\mathbb{N}$, $x \oplus y := x + y + 1$, $x \otimes y := y(x + 1)$; $i = 5$: $\mathbb{N}$, $x \oplus y := x$, $x \otimes y := 0$; $i = 6$: $\mathbb{N}$, standard $+$, $x \otimes y := xy + 1$; $i = 7$: $\mathbb{N}$, standard $+$, $x \otimes y := 0$. Each verification is a line of arithmetic (e.g. $i = 4$: $x \oplus Sy = x + y + 2 = S(x \oplus y)$, $x \otimes Sy = (y + 1)(x + 1) = (x \otimes y) \oplus x$); `check_models.out` evaluates all seven axioms in each structure on a finite range. □

F.7 The Bayesian DTRC, the ∀-merge and Euler's polynomial

Proposition F.21 (Spare slot)**.** [proved; computed] pa Prop 5.1 *A never-used template* τ *added to* K *templates costs* $\beta|\tau| + \frac{1}{2}\log_2 n + \log_2 \frac{\Gamma(K/2)}{\Gamma((K+1)/2)} + o(1)$ *bits (Proposition 4.14(a));* 19.03 *and* 19.82 *bits at* $n = 1000, 3000$ *for* $K = 8$*,* $\tau = F \to F$*, against the computed* 19.0*,* 19.8*.*

Proof of Proposition F.21. The Dirichlet($\frac{1}{2}$) marginal of counts $(n_1, \ldots, n_K, 0)$ over $K + 1$ categories divided by that of $(n_1, \ldots, n_K)$ over K is $\frac{\Gamma((K+1)/2)\Gamma(n+K/2)}{\Gamma(K/2)\Gamma(n+(K+1)/2)}$, and $\Gamma(n+a+\frac{1}{2})/\Gamma(n+a) \sim n^{1/2}$. Add $\beta|\tau|$ prior bits. □

Proposition F.22 ("Does not contradict" learns nothing)**.** [proved] pa Prop 5.2 *If the likelihood is* $\mu_0(s)[T \nvdash \neg s]$ *with* μ_0 *independent of* T*, the posterior is the prior restricted to compatible theories.*

Proofs of Propositions 7.19 and F.22. Proposition F.22: the likelihood is the constant $\mu_0(D)$ for every compatible theory and 0 otherwise. Proposition 7.19: every T-derivation is a T'-derivation, and each of its citations loses a factor $1 - w$ under the rescaled weights; the best T'-derivation is at least as probable. With Dirichlet weights and σ unused, Proposition F.21 applies. □

The noisy likelihood L_ε is a simple model that credits derivations without killing gaps, as Hänni asks; the plain "does not contradict" score has no size principle (Proposition 2.22(b)).

Remark F.23 (Unsound lumps)*.* [computed] pa §5.4; experiments E2 At $n = 10, 30$ the unsound lump $F_0 + T_{\mathrm{Ind}}$ leads by 174.7 and 42.0 bits (F_0 codes the rarely seen Q axioms more cheaply than seven ground templates cost), and $V_{\delta,0}$ accepts the false $0 = S0$ for every $\delta \geq 3 \cdot 10^{-13}$; the negative $0 = S0$ refutes the lump. This is AS's "errs towards unsoundness for rarely used ground axioms" (AS App. E.7) in Bayesian form; it does not contradict Theorems 5.6 and 5.7 ($\pi(\mathsf{Q} + T_{\mathrm{Ind}}) = 2^{-330}$ here, and the data are not from a model generator). One run; frequencies over seeds: Remark 5.11.

Remark F.24 (Overlapping templates)*.* [proof sketch] pa Rem 5.4 For the lump and F_0 the code assigns each datum to its cheapest covering template, a two-part bound; if the greedy assignment

is optimal (not proved), the exact marginal is at most about n bits lower, and the lump stays ≥ 219 bits behind at $n = 100$.

Sketch for Remark F.24. The exact marginal sums over all assignments of data to covering templates. Its code length is at least the body code length under the best assignment, because the index probabilities of all assignments sum to at most 1. If the best assignment is the greedy one (induction data on T_{Ind}, whose motive is coded once, rather than on F_0, which codes the whole sentence), the exact code is lower than the two-part code by at most the greedy index cost, about n bits for two covering templates. At $n = 100$ the lump would then still be at least $319.3 - 100 \approx 219$ bits behind. Optimality of the greedy assignment is not proved.

candidate	$n = 10$	30	100	3000
root split (7 T_f)	+776.4	+773.6	+772.7	+768.3
spare $+ (F \to F)$	+15.9	+16.6	+17.4	+19.8
memorise instances	+1792.3	+5392.7	+15901.1	+399693.6
over-general $\mathsf{Q} + T_{\mathcal{L}_\infty}$ (AS Prop 5.12)	+549.5	+1500.6	+3832.5	+82717.0
lump $F_0 + T_{\text{Ind}}$	−174.7	−42.0	+319.3	+8929.8
bare F_0	+584.0	+2148.6	+6067.9	+137387.5

Table 27: Bayesian DTRC: CL(candidate) − CL($\mathsf{Q} + T_{\text{Ind}}$) in bits on G1 practice; one seed; hand-picked candidates; $T_{\mathcal{L}_\infty}$ is AS's over-general induction template. Computed (pa §5.2; $n = 300, 1000$ in the text below).

Full dtrc run. Table 27 at $n = 300$, 1000: root split +771.4, +769.9; spare +18.2, +19.0; memorise +41418.2, +133845.4; over-general +8964.3, +27876.8; lump +1048.9, +3205.5; bare F_0 +14712.6, +46251.0 bits (the first version had eight entries that did not match the output; referee M1). Without negatives $\mathsf{Q} + T_{\text{Ind}}$ has mass $2.6 \cdot 10^{-53}$ and $2.2 \cdot 10^{-13}$ at $n = 10, 30$, the lump 1.000 at $n = 10$ and $7.679 \cdot 10^{-97}$ at $n = 100$; from $n = 100$ the remainder beside $\mathsf{Q} + T_{\text{Ind}}$ is almost all on the spare ($5.808 \cdot 10^{-6}$). The second negative is $(0 = 0) \wedge \forall x(0 = S0 \to 0 = S0) \to \forall x(0 = S0)$. The theories without $0 = S0$ as an instance have mass $2.6 \cdot 10^{-53}$ and $2.2 \cdot 10^{-13}$ at $n = 10, 30$, hence the acceptance by $V_{\delta,0}$ for $\delta \geq 3 \cdot 10^{-13}$.

Remark F.25 ($\forall x\varphi$ as a sound merge). [computed; identity proved] pa §5.5; referee M5 On data $t + 0 = t$ (t random closed terms) the schema $z + 0 = z$ wins (AS Prop 6.13; Table 28). The sound split by the head of t covers every closed instance but does not entail $\forall x(x + 0 = x)$; it closes in at 5 bits per doubling (Proposition F.6(c), $\frac{3-13}{2}$), crossover near $n = 10^8$: a Bayesian ω-gap (Theorem 3.9), slower than in Proposition 3.10 because the parameter probability is learned. $\forall x(x + 0 = x)$ itself loses at a rate set by the rule code (Table 29): $\log_2 10$ bits per datum under the fixed code, linearly (2 bits per datum at $n = 10^6$) under a learned rule law shared across depths, $\frac{R-1}{2} \log_2 n + O(1)$ under a learned depth-indexed code, and $h(f)$ bits per datum under the latter in a practice with a fraction f of φ-instances (h the binary entropy). The direction holds under every code (Theorem 3.4; a learned shared c: Proposition B.1).

Example F.26 (Euler's polynomial (F4))**.** [computed; proof sketch (covering, eventual behaviour)] pa §6 F4; referee m9 Data: Prime($k \cdot k + k + 41$), k geometric(0.9) conditioned on primality (false k: $40, 41, 44, \dots$). The false schema Prime($z \cdot z + z + 41$) beats memorisation by 51445 bits at $n = 100$ and 314225 at $n = 10^6$ (flat grammar); no finite unguarded covering union is true without memorising. Under a depth-indexed grammar the predictive probability of $k = 40$ falls

candidate	$n = 10$	30	100	300	1000	3000
split by head of t (sound)	+115.5	+109.1	+101.0	+93.3	+84.6	+76.7
memorise instances	+519.5	+1263.8	+3749.9	+12204.1	+34023.6	+83197.3
over-general $z_1 + 0 = z_2$	+145.5	+351.5	+927.2	+2618.8	+7975.5	+22702.2
$\forall x(x + 0 = x)$, fixed rule code	+38.2	+104.7	+337.2	+1001.6	+3326.9	+9970.8

Table 28: $\forall x\varphi$ as a sound merge: $\mathrm{CL} - \mathrm{CL}(z + 0 = z)$ in bits on data $t + 0 = t$ (t random closed terms, seed 12). The over-general template is refuted by the negative $0 + 0 = S0$. With Q3 in the theory the head split would give $\forall x(x + 0 = x)$ by cases; here the theories are the templates alone. A first run in which the head split overtook the schema at $n = 1000$ was an artefact of coding split bodies from a root context (refuted). Computed (pa §5.5).

	$R = 10$ (rules of ND)			$R = 2$ ($\mathsf{C}_{\min}$: cite, $\forall$E)		
n	fixed	learned, shared	learned, by depth	fixed	shared	by depth
10	33.2	23.4	12.5	10.0	20.0	2.5
10^3	3321.9	2004.0	41.1	1000.0	2000.0	5.8
10^6	3321928.1	2000004.0	85.9	1000000.0	2000000.0	10.8

Table 29: $\mathrm{CL}(H_\forall) - \mathrm{CL}(H_{\mathrm{sch}})$ in bits on pure instance data under three rule codes (term costs cancel). The depth-indexed code gives $\frac{R-1}{2}\log_2 n + O(1)$ ($4.31 \log_2 n$ at $n = 10^6$ for $R = 10$, tending to 4.5). In a mixed practice with a fraction f of φ-instances the depth-indexed difference is $h(f)$ bits per datum at $n = 10^6$ (0.4691 at $f = 0.9$ and 0.1; 1.0001 at $f = \frac{1}{2}$), the shared one 1.896, 1.377, 0.483 bits per datum at $f = 0.9, 0.5, 0.1$. Computed in closed form; matches the referee's r3 (pa §5.5).

like $1/n$, yet the schema entails a falsehood; under the flat grammar memorisation should win beyond $n = 10^6$.

Euler's polynomial (Example F.26). $\mathrm{Prime}(m) := \neg m = 0 \wedge \neg m = S0 \wedge \forall a \forall b(a \cdot b = m \to a = S0 \vee b = S0)$ has a 27-symbol frame and three occurrences of $m = k \cdot k + k + 41$ (the first version counted two; referee m9), so $|\mathrm{datum}(k)| = 27 + 3(3(k + 1) + 45)$ with unary numerals; the template has 171 symbols. Under the unconditioned geometric law the false instances have mass 0.0049 ($\rho = 0.9$) and 0.0727 ($\rho = 0.97$), summed over false $k < 200$. Code length of memorisation minus the false schema (bits), flat / depth-indexed grammar, and the schema's predictive probability of $k = 40$: $\rho = 0.9$, $n = 100$: 51445/51373, $1.9 \cdot 10^{-3}/4.1 \cdot 10^{-4}$; $n = 10^4$: 161030/160816, $1.4 \cdot 10^{-3}/4.8 \cdot 10^{-5}$; $n = 10^6$: 314225/320382, $1.4 \cdot 10^{-3}/5.0 \cdot 10^{-7}$; $\rho = 0.97$, $n = 10^6$: 1820112/1920893, $8.6 \cdot 10^{-3}/5.0 \cdot 10^{-7}$ (seeds 5, 6; primality by trial division; the referee's closed-form recomputation is identical). *Covering (sketch).* A template that covers infinitely many data must have a metavariable at or above a position of the numeral k (otherwise k is fixed), so some instance puts there a term that makes the sentence false: the numeral $k = 41j$ at the positions of k ($41 \mid k^2 + k + 41$), or a composite such as $41 \cdot 41$ at a position above them. So every finite unguarded covering union is false unless it memorises. *Eventual behaviour (sketch).* Under the flat grammar the schema pays a linear KL term (about 0.007 bits per datum at $\rho = 0.9$) for the false k it predicts, while memorisation's prior grows like $(\ln n)^2$, so memorisation eventually wins, beyond $n = 10^6$; under the depth-indexed grammar both costs grow like $(\ln n)^2$ and the false schema stays ahead. Constants estimated, not computed. A refutation datum ($40^2 + 40 + 41 = 41^2$) removes the schema.

G Experiments: proofs and details

G.1 Grammar, prior code and pools

$\mathcal{Q}$. A term node is 0, S, $+$, $\cdot$ with weights .45, .35, .10, .10, a variable with total weight .3 split uniformly over the variables in scope (holes and bound variables alike), or p with weight .15 where the guard allows it. A formula node is $=$, $<$, $\neg$, $\wedge$, $\vee$, $\to$, $\leftrightarrow$, $\forall$, $\exists$ with weights .35, .20, .10, .10, .05, .10, .02, .05, .03. Weights are renormalised over the options available. The mean number of same-sort children is 0.75 for closed terms, at most 0.75 for other terms and 0.72 for formulas, so bodies are finite almost surely; the referee's exact size distributions agree.

Prior code. $\mathrm{bits}(\tau)$ is $-\log_2$ of a subcritical stochastic grammar over templates (formula and term nodes, metavariable occurrences, arities and guards); a theory of m distinct components costs $|\gamma(m)| + \sum_i \mathrm{bits}(\tau_i) - \log_2 m!$ with γ the Elias code. The referee's independent implementation gives the same lengths to four decimals; Kraft's inequality holds (Remark A.1).

Pool construction. The pool at n is built by four rules. (1) Hand theories, listed per experiment in Appendix G.4. (2) For every build point $b \le n$: Min of random subsets of D_b (AS Thm 3.3), skeleton clusters of D_b (DTRC without refutation, stopped at k clusters; AS §7.1) and, in PA runs, DTRC with the PA refuter on D_b and the observed-root split $\mathsf{frag\text{-}observed@}b$ (Q plus the fragments of the motive roots seen in D_b); a theory built at b stays in the pool for all $n \ge b$. Build points: E1 $b \in \{1, 2, \ldots, 64\}$ (powers of 2); E3(a) $\{8, 32, 64\}$; E2, E3(b), E7 $\{8, 16, 32, 64\}$; E4 and E6 have none. (3) In PA runs (E2, E3(b), E7), for every pool theory T with at least two components, $\mathrm{Trim}(T, D_n)$: the components i of T with $a_i(d) > 0$ for some datum d of D_n, i.e. those that could have been the first citation of a datum seen (under $\mathsf{L0}$: those that match one). (4) $\mathrm{Mem}(D_n)$. The first version built the data-derived theories once, from the first 16, 40 or 64 data, and used them at every n, and had no trimmed theories (Remark G.1).

Pools. Theories are deduplicated at each n by their components (canonical template plus guards); if $\mathrm{Mem}(D_n)$ equals a pool theory, that theory keeps its mass and Mem becomes an alias without mass (the first version counted it twice, which affected E6 at $n = 1, 2$ only). Under $\mathsf{L0}$ $\mathrm{Mem}(D_n)$ is the best memoriser covering D_n: a further ground sentence costs prior bits and the factor of Proposition 4.14(a) and cannot raise the likelihood. In E2 a theory of Q axioms and induction-type templates is tagged equivalent to T^* iff it contains Q1, Q2, Q4–Q7 and a full-induction template (T_{Ind}, $\mathsf{IndSwap}$ or a connective fragment); other tags come from a budgeted refuter, and an unrefuted theory is "unknown". The false probes are $0 = S0$, $\forall x\,(x + 0 = 0)$, induction with a wrong base, $\forall x\,(Sx = x)$ and $\forall x \forall y\,(y = x)$; held-out induction instances are drawn outside the training stream.

$\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$. The class probabilities c_i are exact for components $\forall^a \chi$ with χ quantifier-free, term metavariables only, every leading variable used and no MP components (a finite Markov chain on the number of leading $\forall$ and the presence of p); other components are left out of $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ pools. Monte Carlo agrees on the eight E6 forms (40000 chains each; largest $|z| = 2.17$).

G.2 Proof of Proposition 8.1

Proof idea. (a) is the matching theorem, AS Thm 2.5. (b) expands the product and takes Dirichlet moments (Lemma 2.8); the upper bound uses concavity of the log-likelihood on the simplex. (c) Every step chooses among finitely many outcomes of total probability 1, and no run fails. (d) Each predecessor set is in bijection with the matchers of a derived DT° template, which are unique. The details follow.

(a) Bodies from $\mathcal{Q}$ contain no parameter other than p, so $\tau\theta$ is already in closure-normal form

and the event "the datum is s" is the event $\tau\theta = s$. By AS Thm 2.5 at most one θ satisfies $\tau\theta = s$, so $P(\tau\theta = s) = \sum_{\theta:\tau\theta=s} \prod_M \mathcal{Q}(\theta(M))$ has at most one term, $\prod_M \mathcal{Q}(\theta_s(M))$. A body containing p has probability 0 under a closed guard. □

(b) Expanding, $\prod_j \sum_i w_i a_{ij} = \sum_z \prod_i w_i^{n_i(z)} \prod_j a_{z_j j}$, and the Dirichlet moments $\mathbb{E} \prod_i w_i^{n_i} = \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_i \frac{\Gamma(\alpha+n_i)}{\Gamma(\alpha)}$ give the formula (Lemma 2.8). *The programme.* The summand depends on z only through $\prod_j a_{z_j j}$ and the counts. A datum covered by one component contributes a fixed factor and one count. For the others, with $S_j := \{i : a_{ij} > 0\}$, the sum over their assignments is the polynomial $\prod_j \sum_{i \in S_j} a_{ij} x_i$ evaluated by the linear functional $x^c \mapsto \prod_i \Gamma(\alpha + c_i + c_i^0)/\Gamma(\alpha)$ (c^0 the fixed counts), times $\Gamma(K\alpha)/\Gamma(K\alpha+n)$. A dense array (at most three overlapping components) or variable elimination over count vectors computes the coefficients exactly. In the latter a component's factor is applied, and its count dropped, once no later group contains it; this is valid because the functional is a product over components. *The bounds.* All terms are nonnegative, so any single term is a lower bound. For the upper bound, $\mathbb{E}_w L(w) \le \max_w L(w)$ with $L(w) = \prod_j \sum_i w_i a_{ij}$, and $f := \ln L$ is concave on the simplex. For an interior w with $L(w) > 0$ and any w° in the simplex, $f(w^\circ) \le f(w) + \nabla f(w) \cdot (w^\circ - w) \le f(w) + \max_i \partial_i f(w) - n$, since $\nabla f(w) \cdot w^\circ \le \max_i \partial_i f(w)$ and $w \cdot \nabla f(w) = \sum_j \sum_i w_i a_{ij} / \sum_k w_k a_{kj} = n$. □

(c) At most J rules are applied. Each step chooses among finitely many outcomes (stop, or a rule that applies) with probabilities summing to 1; each rule's input (a term from $\mathcal{Q}_\mathrm{e}$, bodies from $\mathcal{Q}$, a uniform choice among the applicable MP components) has a probability distribution ($\mathcal{Q}$ is proper). A rule is applied only when it applies, so no run fails; the runs have total probability 1, and the datum is a function of the run. □

(d) Let $G_k(s)_i$ be the probability that the chain reaches step k at s after citing component i, divided by w_i; so $G_0(s)_i$ is the `L0` coefficient of (a). For $1 \le k \le J$ the chain leaves s' only if a rule applies to it, then takes rule r with probability $(1 - c_\mathrm{stop}) c_r / \sum_{r' \text{ applies to } s'} c_{r'}$ and input ξ with probability $P(\xi)$. Hence

$$G_k(s) = \sum_{(s',r,\xi):\, r(s',\xi)=s} G_{k-1}(s')\,(1 - c_\mathrm{stop}) \frac{c_r}{\sum_{r' \text{ applies to } s'} c_{r'}} P(\xi), \qquad a_i(s) = \sum_{k \le J} G_k(s)_i \,\mathrm{stop}_k(s),$$

with $\mathrm{stop}_k(s) = 1$ if $k = J$ or no rule applies to s, and c_stop otherwise. The recursion is exact iff every predecessor s' with $G_{k-1}(s') > 0$ is enumerated. The following give complete predecessor sets.

(d1) $\forall E$ *guided by templates.* Let $U = \forall x\, B$ be a component whose derived template $D := B[u/x]$ (u a fresh 0-ary term metavariable) is in DT°. Then x is no argument of a metavariable occurrence (else an argument would contain u), so x occurs only at rigid positions, bodies never contain x, and $(B\theta)[t/x] = D(\theta \cup \{u \mapsto t\})$. So the pairs (θ, t) with $\forall\mathrm{E}(U\theta, t) = s$ correspond to the matchers of D against s, of which there is at most one (AS Thm 2.5). If x does not occur in B, the predecessor is $\forall x\, s$, with total weight $\sum_t \mathcal{Q}_\mathrm{e}(t) = 1$.

(d2) $\forall E$ *from* P *or* $\forall x\, P(x)$ *at step* 0. For a parameter-free s the predecessors are $\forall x\, \psi_S$, with ψ_S the sentence s with a nonempty set S of occurrences of one closed term t replaced by x. Since the PCFG probability is a product over nodes and every node of ψ_S lies in a context with a variable, $\mathcal{Q}(\forall x\, \psi_S) = \mathcal{Q}(\forall x\, s) \prod_{o \in S} r_o$ with $r_o := (p_\mathrm{var}/\nu_o)/\mathcal{Q}_\mathrm{v}(t)$; here p_var is the normalised variable weight, ν_o is 1 plus the binder depth at o, and $\mathcal{Q}_\mathrm{v}(t)$ is the probability of t in a context with a variable. So $\sum_{S \ne \emptyset} \mathcal{Q}(\forall x\, \psi_S) = \mathcal{Q}(\forall x\, s)\big(\prod_o (1 + r_o) - 1\big)$; for $\forall x\, P(x)$ the same holds up to the factor of the root $\forall$. The code uses this closed form only in theories without MP components, where these parameter-free predecessors admit only $\forall$E. If s contains p and the guard is closed, t contains p and S is all occurrences of t.

(d3) *Gen, then* $\forall E$ ($J = 2$). A step-1 Gen output is $\mathrm{gen}(s_0) = \forall x\, s_0[x/p]$ for a cited $s_0 = T\theta_0$ containing p, and $\forall\mathrm{E}(\mathrm{gen}(s_0), t) = s_0[t/p]$. Let D be the component with its rigid occurrences of p replaced by a fresh 0-ary metavariable u, and suppose $D \in \mathrm{DT}^\circ$ (no rigid p inside an argument; the track's notes leave this hypothesis implicit). Then $s_0[t/p] = D\theta$ with $\theta(M) = \theta_0(M)[t/p]$ and $\theta(u) = t$, so θ is the unique matcher of D against s. Closed-guard bodies satisfy $\theta_0(M) = \theta(M)$; open-guard bodies $\theta_0(M)$ arise from $\theta(M)$ by replacing some occurrences of t by p; without a rigid p, t occurs in some open-guard body. The code enumerates exactly these candidates and keeps those with $\forall\mathrm{E}(\mathrm{gen}(s_0), t) = s$.

(d4) *Gen and MP.* $\mathrm{gen}^{-1}(\forall x\, \chi) = \chi[p/x]$ if χ uses x and s has no parameter. For an MP component $A \to B$, $B \in \mathrm{DT}^\circ$ contains A's metavariables, so the matcher θ of B against s is unique and the predecessor is $A\theta$.

(d5) *Pruning.* A candidate is skipped only if its ($\forall$-spine, parameter flag) is outside the abstract set of step $k-1$, computed forward from the components; by induction on k every step-k sentence lies in that set. □

The code admits into L1 pools only theories whose $\forall$-rooted components have derived templates in DT° (twice derived at $J = 2$) and whose formula-metavariable components are P or $\forall x\, P(x)$ at $J = 1$ (none at $J = 2$). A term with more than 12 occurrences in a datum stops the enumeration and raises a counter; it was 0 in every run. The support check does not test the hypothesis of (d3). It holds for every component without a rigid p, which includes every hand theory of E1 and E6; for data-derived theories at $J = 2$ (E1 *allq2*) exactness rests on the computed checks: the guided recursion equals a brute-force recursion over all subsets of all occurrences to 10^{-9} on samples from 10 theories at $J = 1, 2$ (same class, so not independent), and the referee's own recursion agrees on 904 triples at $J = 1, 2$ to $2.1 \cdot 10^{-14}$.

G.3 Proof of Proposition 7.6

(a) A_{xy} and A_{yx} differ by the order of two universal quantifiers. M_x^{open} cites $\forall x\,(x + p = p + x)$ (body $z := p$), which under the closure reading is $\forall w \forall x\,(x + w = w + x)$, i.e. A_{yx}; and each of its instances $\forall x\,(x + b = b + x)$, b a term with at most the parameter p, follows from A_{xy} by $\forall$E (and Gen on p under the closure reading). Likewise M_y^{open} cites $\forall y\,(p + y = y + p)$, i.e. A_{xy}. Part (a) corrects a remark of the experiments track's referee that open guards cannot reach $\forall x \forall y$ under the single-parameter convention: one bound variable and the parameter already give two variables.

(b) Take the domain $\mathbb{N} \sqcup \{\alpha, \beta\}$, standard on $\mathbb{N}$, with $<$ arbitrary, $S\alpha = \alpha$, $S\beta = \beta$, and for $a \in \{\alpha, \beta\}$, $n \in \mathbb{N}$: $a + n = n + a = a$; $a \cdot n = n \cdot a = a$ if $n \neq 0$ and 0 if $n = 0$; $\alpha + \alpha = \alpha \cdot \alpha = \alpha$, $\beta + \beta = \beta \cdot \beta = \beta$, $\alpha + \beta = \alpha \cdot \beta = \alpha$, $\beta + \alpha = \beta \cdot \alpha = \beta$. Each $\mathbb{N} \cup \{a\}$ is closed under $S, +, \cdot$ and $+$ is commutative on it. Closed terms denote naturals, so the closed instances of S_{ab} hold. An instance $\forall x\,(x + t = t + x)$ of M_x, t closed of value n, holds for $x \in \mathbb{N}$ by arithmetic and for $x = a$ since $a + n = a = n + a$; likewise for M_y. An instance $\forall w\,(f(w) + g(w) = g(w) + f(w))$ of S_{ab}^{open} (one parameter) holds because for $w \in \mathbb{N}$ both values lie in $\mathbb{N}$ and for $w = a$ both lie in $\mathbb{N} \cup \{a\}$. But $\alpha + \beta = \alpha \neq \beta = \beta + \alpha$, so A_{xy} fails and, by soundness, is not derivable. With two parameters, $p_1 + p_2 = p_2 + p_1$ would be an instance of S_{ab}^{open}; the convention excludes it. □
Computed (`checks/check_models.out`): the 285 closed terms with at most 7 nodes denote naturals; the instances of S_{ab}, M_x, M_y hold for $x \in \{0, \ldots, 14, \alpha, \beta\}$, and those of S_{ab}^{open} for 154 terms with at most 5 nodes and $w \in \{0, \ldots, 3, \alpha, \beta\}$; A_{xy} and the instance $\forall x\,(x + p = p + x)$ of M_x^{open} fail at (α, β).

G.4 The experiments in detail

This subsection gives each experiment's setup, its tables and the findings not stated where Table 5 points. Pool names are explained in the caption of Table 5.

E1: $\forall x\varphi$ from data about φ. For $\varphi \in \{x+0=x,\ 0+x=x,\ \neg Sx=0\}$, seeds 0–4 and $n = 1, 2, 4, \ldots, 256$, five generators are used: *sch* ($\mathsf{L0}$ citations of $\sigma_\varphi = \varphi(z)$, closed terms); *all* ($\mathsf{Ch}_1$ from $H_\forall$, closed $\forall$E terms, so about half of the data are $\forall x\varphi$ itself); *allq* (the same with open terms, e.g. $\varphi(Sp)$); *open* (Ch_1 from H_{open}, whose Gen steps emit sentences such as $\forall y\,\varphi(Sy)$); and *allq2* (Ch_2 from $H_\forall$: $\forall$E with an open term, then Gen). The pool (15–28 theories at $n = 256$) holds $H_\forall$, H_{sch}, H_{open}, H_{both}, $H_\forall \cup H_{\text{open}}$, the over-specific $\{\varphi(0), \varphi(Sz)\}$, two root splits, three spare-slot theories, over-general generalisations of $\varphi(z)$ and F_0, causal theories and $\text{Mem}(D_n)$. The likelihoods are $\mathsf{L0}$, $\mathsf{L1}$ with the generator's own chain, and $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ on *sch* data. Every generator has one component and is an atom of the prior in the fixed hand pool, so Doob's argument in the (T, w) form covers the hand theories; the causal theories and $\text{Mem}(D_n)$ are covered by no consistency theorem (Remark C.3). Results in Table 30.

data	lik.	mass of the generating theory at $n = 4, 8, 32, 256$	MAP at 256	$P(\vdash_1 \forall x\varphi)$	first n
sch	$\mathsf{L0}$	0.87, 0.97, 1.000, 1.000	H_{sch}	$2 \cdot 10^{-7}$	16
sch	$\mathsf{L1}$	0.84, 0.96, 1.000, 1.000	H_{sch}	$3 \cdot 10^{-7}$	16
sch	$\mathsf{L1}^{\text{sel}}_{\text{cit}}$	0.43, 0.42, 0.53, 0.59	H_{sch}	0.484; 0.269	never
all	$\mathsf{L1}$	0.87, 0.99, 1.000, 1.000	$H_\forall$	1.000	16
allq	$\mathsf{L1}$	0.73, 0.90, 1.000, 1.000	$H_\forall$	1.000	16
open	$\mathsf{L1}$	0.89, 1.000, 1.000, 1.000	H_{open}	1.000	8
allq2	$\mathsf{L1}$	0.88, 0.997, 1.000, 1.000	$H_\forall$	1.000	16

Table 30: E1. Means over 5 seeds and over the three formulas (per-formula values in `code/results/e1_universal.md`; under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ the generator's mass at $n = 256$ is 0.516 for $x+0=x$ and $0+x=x$ and 0.731 for $\neg Sx=0$). $P(\vdash_1 \forall x\varphi)$ at $n = 256$; under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ it is 0.484 for $x+0=x$ and $0+x=x$ and 0.269 for $\neg Sx=0$. Last column: first n from which the generator keeps mass ≥ 0.99, worst run. Under $\mathsf{L0}$ (rows omitted) $H_\forall$ dies at its first instance; on *all* data H_{both} has mass ≥ 0.995 from $n = 32$, on *allq* data $H_\forall \cup H_{\text{open}}$ has 1.000 from $n = 16$, and on *open* and *allq2* data $\text{Mem}(D_n)$ has 1.000 from $n = 16$ and 64; $P(\vdash_1 \forall x\varphi) = 1.000$ at $n = 256$ in all four.

(1) The factor of Theorem 3.3, exactly. In Ch_J with closed $\forall$E terms, $H_\forall$ emits $\forall x\varphi$ with probability c_{stop} and otherwise $\varphi(t)$ with probability $c_{\text{ch}}\mathcal{Q}(t)$, after which no rule applies; H_{sch} emits $\varphi(t)$ with probability $\mathcal{Q}(t)$. So on closed instances the $\log_2$ odds $H_{\text{sch}} : H_\forall$ gain $\log_2(1/c_{\text{ch}}) = 1$ bit per datum under $\mathsf{L1}$ and stay at the prior odds under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ ($c_\forall = c_{\text{ch}}$, $c_{\text{sch}} = 1$); this is experiments Prop X6 (proved), the Ch_J case of the unnormalised μ_T formula c^n of Theorem 3.3 with $c = c_{\text{ch}}$. Computed: $n + 0.090$ bits ($x+0=x$, $0+x=x$) and $n + 1.440$ ($\neg Sx=0$) at every n and seed, within $1.1 \cdot 10^{-11}$ bits of n plus the prior difference from the referee's code; the $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ tie holds to 10^{-13}. *(2) The ω-gap.* Under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ $P(T \vdash \forall x\varphi \mid D_n)$ tends to the prior share, in this code, of the provers among the theories tied with H_{sch} (Theorem 3.9(b), Table 11), while held-out closed instances get mass ≥ 0.999 from $n = 8$ in every run except *open* data under $\mathsf{L0}$ (Theorem 3.8). *(3) The size principle.* At $n = 8$ on *sch* data, $\text{Mem}(D_n)$ has mass $\leq 2.5 \cdot 10^{-12}$ and the over-general theories $\leq 7 \cdot 10^{-5}$, both falling exponentially (Theorem B.4). This is one memoriser; the memoriser class can decay more slowly (Proposition B.6). Under $\mathsf{L1}$ $\text{Mem}(D_n)$ is not provably the best memoriser, since a ground $\forall$-sentence can raise its likelihood (the referee

found $\mathrm{Mem}(D_n) \cup \{\forall x\,(x+0=x)\}$ 11–16 bits worse, the margin shrinking). *(4) Quantified data.* From a prover of $\forall x\varphi$, L1 finds the generator by $n = 16$ and $P(T \vdash \forall x\varphi \mid D_n) \to 1$ (Proposition 3.10(a), Proposition B.14(a)). Under L0 derived sentences can only be axioms, so theories that list them win, and with Gen outputs of unbounded variety L0 ends at $\mathrm{Mem}(D_n)$.

E2: PA without labels. Data are L0 citations from $T^* = \mathrm{Q1..Q7} + T_{\mathrm{Ind}}$ (numbering of Section 2) at fixed weights 0.05 per Q axiom and 0.65 for T_{Ind}, with one-hole motives from $\mathcal{Q}$; seeds 0–24; $n = 8, \ldots, 512$; the likelihood is L0 with the same $\mathcal{Q}$. Besides the causal theories, $\mathrm{Trim}(T, D_n)$ and $\mathrm{Mem}(D_n)$, the pool holds T^*, $T^* - \mathrm{Q}_i$, frag-complete, frag-atoms, the spares $T^* + T_\wedge$, $T^* + \{\forall x\,(0+x=x)\}$ and $T^* + \{0 = S0\}$, IndSwap, two over-general induction templates, Q-lumped $= \{\forall x \forall y\, P(x,y),\ \forall x\, P(x),\ T_{\mathrm{Ind}}\}$ and F_0. Tags come from Propositions 7.16, F.19 and F.20 and a budgeted refuter ("unsound" is certain). The weights are fixed, so the applicable soundness statement is the pool version of Theorem 5.7, not Theorem 5.6. Results in Tables 31 and 32; identification is discussed in Remark 7.18, false acceptances in Remark 5.11.

n	T^*	equiv.	weaker	unsound	acc.	MAP [tag] (count of 25)
8	$2 \cdot 10^{-21}$	$4 \cdot 10^{-13}$	0.744	0.256	2	skel4@8 [w] 12, DTRC@8 [w] 6, Trim(Q-lumped) [u] 4, …
16	$9 \cdot 10^{-7}$	$5 \cdot 10^{-6}$	0.803	0.197	3	skel6@16 [w] 9, skel4@16 [w] 5, Q-lumped [u] 3, …
32	0.197	0.197	0.727	0.077	0	$T^* - \mathrm{Q2}$ [w] 7, skel6@32 [w] 6, T^* 5, …
64	0.751	0.791	0.200	0.009	0	T^* 19, $T^* - \mathrm{Q3}$ [e] 1, …
128	0.991	0.991	0	0.009	0	T^* 25
512	0.996	0.996	0	0.004	0	T^* 25

Table 31: E2: posterior mass by tag (means over 25 seeds). "Acc.": seeds in which $V_{0.05,0}$ accepts a false probe. Tags: [e] equivalent to T^*, [w] sound and strictly weaker, [u] unsound. At $n = 512$ the unsound mass is the false spare $T^* + \{0 = S0\}$. Pool names: skel$k$@b, a skeleton clustering of D_b into k clusters; DTRC@b, DTRC with a PA refuter on D_b; frag-$*$, Q plus root fragments T_f; IndSwap, Q plus a swapped equivalent induction template (Table 5).

(1) The last of Q1, Q2, Q4–Q7 is first cited at datum 20–97 (Q3 is redundant by Proposition F.20). Before that the sound but weaker $\mathrm{SeenQ}(D_n)$ leads, since an uncited axiom costs its prior bits and the factor of Proposition 4.14(a). *(2)* The false acceptances of Remark 5.11 occur at $n \le 16$, when an unsound lump beats SeenQ: Q-lumped (by up to 12.7 bits) or its trim, or, in one seed (seed 1, $n = 16$), a data-derived skeleton cluster, skel4@16, with mass 0.999. With the first version's pool they occurred in 19 of 25 seeds (Table 32). *(3)* frag-complete has T^*'s instance set and its laws include T^*'s (Proposition 4.6(b)). It trails T^* by about the prior difference (698.9 bits; 696.9 at $n = 8$), and the gap grows by 4.2 bits per doubling of n from $n = 64$ to 512 (asymptotically $(16-8)/2 = 4$, the Occam term of Proposition 4.10(c)). *(4)* Unused spare sentences lose 0.49 bits per doubling and the nested $T_\wedge$ 0.25 (Proposition 4.14(a), (c)); so the inconsistent $T^* + \{0 = S0\}$ loses its mass only polynomially (Remark 4.15). *(5)* The equivalent but uncited IndSwap has likelihood 0 from the first induction datum: citation identifies the cited axioms. Over-general induction templates lose linearly.

E3: misspecified usage. All data are theorems of the target, but their law is no pool theory's (Theorem 4.16). (a) Instances of $\varphi = (x + 0 = x)$ under other term laws: *heavy* (a zeta(1.5) numeral with probability 0.7, else $S^k(t_1 \circ t_2)$), *numerals* ($S^k 0$, $k \sim \mathrm{Geom}(0.2)$), *small* ($\mathcal{Q}$-terms with at most 4 symbols: 12 distinct sentences) and *skewQ* (another PCFG); L0 ($n \le 1024$) and L1 (Ch_1, $n \le 512$); seeds 0–4. The pool adds to E1's hand pool the closed-guard numeral

	SeenQ(D_n)			CL(T) − CL(T^*), bits					acc. seeds	
n	Q seen	mass	vs Q-l.	frag-c.	$+T_\wedge$	$+\forall x(0{+}x{=}x)$	$+0{=}S0$	Q-lumped	legacy	causal
8	2.5	0.744	−16.8	696.9	87.6	13.8	5.1	−100.0	18	2
16	4.0	0.803	−38.1	698.2	87.9	14.2	5.4	−41.0	3	3
32	5.7	0.923	−87.6	700.0	88.2	14.6	5.9	51.8	2	0
64	6.7	0.991	−255.0	703.6	88.5	15.1	6.3	248.2	0	0
512	7.0	0.996	−2928.5	716.3	89.2	16.6	7.8	2928.5	–	0

Table 32: E2, 25 seeds. Left: the sound competitor SeenQ(D_n) = Trim(T^*, D_n); its code length minus that of Q-lumped ranges over seeds from −49.1 to +8.9 at $n = 8$ and from −92.8 to +12.7 at 16. At $n \le 32$ it is usually in the pool under the name of a data-derived theory; from $n = 64$ it is usually T^* or $T^* - \mathrm{Q}_i$. Middle: code lengths relative to T^* (means; positive means worse); the over-general induction templates are 147 and 538 bits behind at $n = 8$, 8303 and 32624 at 512. Right: seeds in which $V_{0.05,0}$ accepts a false probe, with the first version's legacy pool (data-derived theories from the first 16, 40, 64 data, used at every n; no trimmed theories) and with the causal pool; at some $n \le 64$, 19 against 5 seeds. Mean unsound mass, legacy against causal: 0.720/0.256, 0.147/0.197, 0.081/0.077, 0.009/0.009 at $n = 8, 16, 32, 64$. The legacy counts reproduce the referee's. The referee's variant, legacy plus SeenQ, accepts in seeds 1, 4, 11 ($n = 16$) and 17 ($n = 8$); the causal pool accepts in these and in seed 14 at $n = 8$, where Trim(Q-lumped) is the MAP with mass 1.000, so it is not uniformly kinder to soundness.

splits $R_m = \{\varphi(0), \ldots, \varphi(S^{m-1}0), \varphi(S^m z)\}$, $m \le 16$, and the nested $C_1 = \{\varphi(z), \varphi(Sz)\}$ and $C_2 = C_1 \cup \{\varphi(SSz)\}$. (b) E2's T^* and weights with other motive laws: *dtrc-motive* (AS's motive generator), *root-skew* (roots $\to, \wedge, =, \exists$ with .5, .2, .2, .1), *deep* (more connectives) and *atomic* ($a(x) = b(x)$ with probability 0.6, $a(x) < b(x)$ with 0.4); L0; $n \le 2048$; E2's pool plus $T^* + T_f$ for every root f. Results in Tables 33 to 35, discussed in Remark 4.20 for (a) and Remark 7.17 for (b).

data	MAP at the largest n (5 seeds; L0 and L1 alike)	inst. set	gain	$P(\vdash$ held-out)	superset
heavy	C_2 (3 seeds), $\{\varphi(z), \varphi(SSz)\}$ (2)	same	212 → 455	1.000	$3 \cdot 10^{-63}$
numerals	R_9 or R_{10} at $n = 512$; R_{11} or R_{12} at 1024	subset	1271 → 2897	0.333	0
small	H_{sch} up to $n = 512$; Mem(D_n) at 1024 (L0)	same→finite	0 → 156	1.000 → 0.000	$7 \cdot 10^{-5}$
skewQ	frag1, a root split (from $n = 512$)	same	70 → 212	1.000	$2 \cdot 10^{-20}$

Table 33: E3(a). Gain: code length of H_{sch} minus that of the MAP under L0, bits, $n = 512 \to 1024$ (means). $P(\vdash$ held-out): mean mass deriving $\varphi(9)$, $\varphi(2+3)$, $\varphi(1 \cdot 4)$. Superset: largest mass on over-general theories at the largest n over seeds and likelihoods. Families: Table 34.

(a) The gain of the MAP over H_{sch} grows linearly in n. On numerals the winner keeps changing as n grows: the best split deepens (R_4/R_5 at $n = 128$, R_9/R_{10} at 512, R_{11}/R_{12} at 1024, interior to the family), like the drift of Proposition 3.12(d) in the calculus of Section 3; $n = 1024$ is too small to test its part (b). On heavy tails a depth-2 nesting (C_2, or $\{\varphi(z), \varphi(SSz)\}$) wins, and a deeper one would probably win if the exact programme could hold it (conjecture). With numerals only the held-out $\varphi(2+3)$, $\varphi(1 \cdot 4)$ are not derived, and with a finite selected support the axioms become the 12 observed sentences: as in Example 4.17, all data are theorems of the target and the posterior goes to a strictly weaker theory. L1 gives the same MAPs as L0 for $n \le 512$. *(b)* Under connective-rich motive laws the winners contain T_{Ind} or a connective fragment and Q1, Q2,

	heavy data		*numerals* data				
n	C_1	C_2	R_4	R_9	R_{11}	R_{12}	R_{16}
128	−0.2	−11.5	−162.0	−86.6	−3.0	+46.6	+300.2
512	−76.3	−204.4	−978.8	−1270.0	−1248.5	−1220.7	−1015.0
1024	−165.5	−447.0	−2063.1	−2835.1	−2894.6	−2895.1	−2751.7

Table 34: E3(a): code length relative to H_{sch} on the extended families (L0, mean over 5 seeds, bits; negative means preferred). The best R_m moves to larger m as n grows and is interior; C_2 pulls away from C_1 linearly. The first version's pool stopped at R_4 and C_1, which it named as the winners.

motives	MAP at $n = 256$ / 1024 / 2048 (5 seeds)	equiv.	weaker	T^*	unsound at 16
dtrc-motive	T^* / $T^* + T_\exists$ / $T^* + T_\exists$	1.000	0	$5 \cdot 10^{-70}$	0.25; 1.0
root-skew	$T^* + T_\to$ / frag-observed@16, @32 / same	1.000	0	0	0.52; 1.0
deep	T^* / T^* / T^*	0.998	0	0.998	$2 \cdot 10^{-11}$; $7 \cdot 10^{-11}$
atomic	frag-atoms (4), T^* (1) / frag-atoms / frag-atoms	$2 \cdot 10^{-178}$	1.000	$4 \cdot 10^{-298}$	0.21; 0.999

Table 35: E3(b). Masses at $n = 2048$; unsound mass at $n = 16$ as mean; max over seeds. "Equiv.": mass on T^*-equivalents within the pool; see Remark 7.17. Code lengths relative to T^* at $n = 2048$: *root-skew* frag-observed@b −1387 to −1509 bits; *dtrc-motive* $T^* + T_\exists$ −228 to −318; *deep* frag-complete +665 to +674; *atomic* frag-atoms −986 to −1028.

Q4–Q7, so they are equivalent to T^* (Proposition F.19); yet under *dtrc-motive* and *root-skew* the MAP leaves T^*: the theorems survive, the axioms do not. Under *root-skew* the winners lack the fragments of the roots not seen, so they cite held-out induction instances with mass only 0.40 at $n = 2048$, although they prove them all. Unsound lumps recur at $n = 16$.

E4: an adaptive prover. The prover wins if $V_{\delta,1}$ ever accepts some s with $T^* \nvdash_1 s$. T^* always has one component, so Theorem 5.6 bounds the win probability by δ' for a suitable δ. (A) The tight construction of IL Prop C.5 (Hänni and Claude (Anthropic), 2026b) with a Dirichlet decoy: $T^* = \{\varphi(z)\}$ on numerals with $P(t = 0) = 1 - u$, decoy $T' = \{\varphi(0), 0 = S0\}$, prior mass $\pi(T^*)$ on T^*, $\delta = \pi(T^*)\delta'$; the prover asks $0 = S0$ every round. It wins iff the first t^* data are $\varphi(0)$, with t^* the first n at which $\pi_n(T^*) \le \delta$ given $\varphi(0)^n$, so $P(\text{win}) = (1-u)^{t^*}$ exactly. The decoy's Dirichlet factor delays t^* compared with IL's fixed-weight decoy (Remark D.1). (B, C) $\varphi = (x + 0 = x)$, $T^* = H_{\text{sch}}$, E1's hand pool H plus $\text{Mem}(D_n)$ and, per setting, one extra theory; $\delta = W^*_{\text{pool}}\delta'$ with the pool-version constant $W^*_{\text{pool}} := 2^{-\text{bits}(T^*)}/(\sum_{T \in H} 2^{-\text{bits}(T)} + 1) = 7 \cdot 10^{-6}$ of Theorem 5.6, and $\delta' = 0.05$. Each round the prover asks the eight sentences $\forall x\,(x + 0 = x)$, $1 + 0 = 0$, $0 + 0 = 1$, $0 = S0$, $2 + 0 = 3$, $7 + 0 = 8$, $(1 + 1) + 0 = S(1 + 1)$, $0 = 0$, which T^* does not derive (two are true; the code's list has nine entries because it writes $1 + 0 = 0$ twice); 100 streams, every $n \le 256$, L0 and L1. Results in Table 36, discussed in Remark 5.10.

E5: spare slots, $\mathcal{L}_\infty$ against $\mathcal{L}_5$, and Gold's text. (a) Exact Bayes factors of $\{\varphi(z)\}$ plus a spare against $\{\varphi(z)\}$ on its own L0 data, $\varphi = (x + 0 = x)$: the unused spares ($0 = S0$; $z \cdot 0 = 0$) match Proposition 4.14(a) to all printed digits; the nested $\varphi(Sz)$ decays more slowly, supporting the $n^{-1/4}$ of Proposition 4.14(c), itself a proof sketch. A false spare sentence is thus removed by positive data only like $n^{-1/2}$. (a2) For the nested spare the Bayes factor depends only on n and the number n_S of S-rooted data: under $\{\varphi(z), \varphi(Sz)\}$ a datum $\varphi(t)$ has

(A) tight construction

u	$\pi(T^*)$	δ'	t^*	exact P(win)	MC (10^5)	pipeline	selected	bound
0.05	0.01	0.01	150	0.00046	0.00046	0/400	20/20	0.01
0.01	0.001	0.05	680	0.00108	0.00123	not run	20/20	0.05
0.5	0.01	0.01	10	0.00098	0.00095	2/400	20/20	0.01
0.1	0.1	0.1	47	0.00707	0.00694	5/400	20/20	0.1
0.2	0.05	0.2	17	0.02252	0.02298	6/400	20/20	0.2

(B, C) pools; L0 / L1; bound 0.05

setting	prover wins	largest invalid mass	MAP at $n = 256$
B well specified	0/100, 0/100	0.452, 0.533	H_{sch} ×100
C1 10% near misses $t + 0 = St$	100/100, 100/100	1.000, 1.000	$H_{\text{sch}} \cup \{z + 0 = Sz\}$ ×100
C2 heavy-tailed terms	0/100, 0/100	0.452, 0.536	C_1 ×96, H_{sch} ×4
C3 only $\varphi(0)$; decoy in pool	0/100, 0/100	0.107, 0.112	Mem ×100

Table 36: E4. (A) $\pi(T^*)$: prior mass of T^* against the decoy. "Pipeline": wins of the full posterior–derivability–threshold pipeline, which equals the closed-form event on all 400 streams in the two settings checked; "selected data": only $\varphi(0)$ shown. (B) The largest invalid mass sits on $\forall x\,(x + 0 = x)$ at $n \le 6$ ($n \le 3$ in 184 of 200 streams), carried by H_{open} (mean 0.23 under L0, 0.21 under L1), by $H_\forall$ under L1 (0.15) and by a generalisation of $\varphi(z)$. (C1) First acceptance ($0 + 0 = 1$ or $2 + 0 = 3$) at the first mistaken datum in 90 of 100 streams per likelihood, 2–5 data later in the rest.

probability $\mathcal{Q}(t)(1 - w_2 + w_2/q_S)$ if t is S-rooted and $\mathcal{Q}(t)(1 - w_2)$ otherwise ($q_S = 0.35$), so $\text{BF} = \mathbb{E}[(1 - w_2 + w_2/q_S)^{n_S}(1 - w_2)^{n-n_S}]$, $w_2 \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$. Quadrature: substitution $w_2 = u^2$, $2 \cdot 10^5$-point trapezoid rule; at $n = 500$ it agrees with the exact programme to 10^{-4} bits, and the referee's exact Beta-moment sum agrees with it to $3 \cdot 10^{-4}$ bits up to $n = 10^6$. (b) With $\mathcal{L}_k$ the k ground sentences $\varphi(S^j 0)$, $j < k$, and $\mathcal{L}_\infty = \{\varphi(z)\}$ under $\mathcal{Q}_{\text{num}}$, the posterior is over $\{\mathcal{L}_1, \ldots, \mathcal{L}_{40}, \mathcal{L}_\infty\}$. On data uniform on the five sentences of $\mathcal{L}_5$ it switches from $\mathcal{L}_\infty$ to $\mathcal{L}_5$, and on data from $\mathcal{L}_\infty$ it keeps $\mathcal{L}_\infty$. Proposition 4.9(a) proves this for fixed generators; here $\mathcal{L}_5$ has Dirichlet weights and is sampled at one weight vector, so the run is a yardstick (Remark C.3). (c) Against the MAP learner over $\{\mathcal{L}_1, \ldots, \mathcal{L}_{60}, \mathcal{L}_\infty\}$, stage i presents $\varphi(0), \ldots, \varphi(S^{i-1}0)$ round-robin until the MAP is $\mathcal{L}_i$. The first 14 stages all end: the start of a text on which, if every stage ends, the MAP changes infinitely often (Proposition 4.9(b), (c)). All numbers are in Table 37.

E6: generating axiomatisations. The eight forms of commutativity of Proposition 7.6 are compared. The fixed pool is the eight forms, $A_{xy} \cup A_{yx}$, $A_{xy} \cup S_{ab}$ and $\text{Mem}(D_n)$. Data come from Ch_2 (closed ∀E terms) on A_{xy} and on M_x, and from L0 citations of S_{ab}; seeds 0–4; $n \le 256$; likelihoods L1 (that chain) and $\mathsf{L1}^{\text{sel}}_{\text{cit}}$. Results in Table 38, discussed in Remark 7.7. On A_{xy} data the strongest early competitors are M_y (0.39 at $n = 2$, 0 from $n = 4$) and M_y^{open}, which also proves A_{xy}; on M_x data the competitor is M_x^{open}, and once M_x wins the mass on provers of A_{xy} goes to 0, correctly, since the generator does not prove it. Under $\mathsf{L1}^{\text{sel}}_{\text{cit}}$ the open-guard forms are not tied with the closed ones: their bodies may contain p, so their law on closed outputs differs, and they lose.

E7: time factor and steeper penalty. E2's data and pools, 25 seeds; priors with $(\lambda, \tau) \in \{(1,0), (1,1), (1,4), (2,0), (\frac{1}{2}, 0)\}$. The time factor costs $\tau \log_2(1 + |T|)$ bits (T^* has 77 symbols), so at $\tau = 1$ it moves the log-odds between T^* and any hand competitor by at most $\log_2(78/2) \approx 5.3$ bits. At $n \le 32$ it favours small unsound lumps, slightly at $\tau = 1$ and markedly at $\tau = 4$; the

spare added to $\{\varphi(z)\}$	mean $\log_2$ BF at $n = 1, 16, 256, 4096$	slope	prior bits
false sentence $0 = S0$ (never cited)	$-1.00,\ -2.84,\ -4.83,\ -6.83$	-0.500	8.4
disjoint schema $z \cdot 0 = 0$ (never matched)	$-1.00,\ -2.84,\ -4.83,\ -6.83$	-0.500	14.5
nested $\varphi(Sz)$	$+0.17,\ -1.05,\ -2.85,\ -3.97$	-0.281	21.3
nested $\varphi(Sz)$, quadrature	$n = 10^2 \dots 10^8$: see caption	-0.251 ± 0.002	21.3

Table 37: E5 (10 seeds; Bayes factors without the prior; slopes in $\log_2$ BF per doubling of n, on $n = 256 \dots 4096$ except for the quadrature row, $10^2 \dots 10^8$). (a2) The nested spare by quadrature: mean $\log_2$ BF -2.196, -3.028, -3.867, -4.658, -5.575, -6.332, -7.200 at $n = 10^2, \dots, 10^8$ (s.e. 0.028–0.031); slope -0.251 ± 0.002 on $10^2 \dots 10^8$ and -0.251 ± 0.003 on $10^4 \dots 10^8$; per decade -0.250, -0.253, -0.238, -0.276, -0.228, -0.261; 1000 draws of $n_S \sim \mathrm{Bin}(n, 0.35)$ per n, numpy seed 0; the first version reported -0.259 from 200 draws, and the referee's independent sum -0.233 on $10^2 \dots 10^7$ with 40 draws. (b) $\mathcal{L}_5$ data: $\mathcal{L}_5$ has mass $2 \cdot 10^{-6}$ at $n = 64$, 0.66 at 128, 1.000 from 256 (10/10 seeds); $\mathcal{L}_\infty$ data: $\mathcal{L}_\infty$ has 1.000 from $n = 16$. $\mathcal{Q}_{\mathrm{num}}(S^j 0) = 2^{-(j+1)}$; prior bits $\mathcal{L}_1$ 10.9, $\mathcal{L}_5$ 89.4, $\mathcal{L}_8$ 187.9, $\mathcal{L}_\infty$ 17.1; per datum $\mathcal{L}_5$ gains $3 - \log_2 5 \approx 0.68$ bits, which repays the 72-bit prior difference and $\mathcal{L}_5$'s Dirichlet cost of about $2 \log_2 n$ bits between $n = 64$ and 128. (c) Stage lengths $1, 24, 81, 115, 104, 83, 68, 55, 45, 39, 33, 33, 32, 27$ (740 data); they shorten after $i = 4$ because $\mathcal{L}_\infty$'s mean cost on round-robin data, $1 + (i-1)/2$ bits, outgrows $\mathcal{L}_i$'s $\log_2 i$.

data from	generator's mass at $n = 2, 4, 8, 16, 256$	first n (≥ 0.99), per seed	P(proves A_{xy}) at $n = 4$; 256
A_{xy}	0.34, 0.90, 1.000, 1.000, 1.000	8, 8, 8, 4, 2	1.000; 1.000
M_x	0.60, 0.80, 0.90, 0.99, 1.000	32, 16, 8, 16, 16	0.205; $7 \cdot 10^{-49}$
S_{ab}	0.28, 0.56, 0.95, 1.000, 1.000	16 in every seed	0.060; $1 \cdot 10^{-11}$

Table 38: E6 under $\mathsf{L1}$ (means over 5 seeds). Under $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ at $n = 256$, for all three sources: A_{xy} 0.044, A_{yx} 0.044, S_{ab} 0.128, M_x 0.392, M_y 0.392, open forms $\leq 6 \cdot 10^{-9}$, so P(proves A_{xy}) = 0.088. The redundant pairs stay below 10^{-9} under $\mathsf{L1}$: $A_{xy} \cup A_{yx}$ at most $9 \cdot 10^{-11}$ and $A_{xy} \cup S_{ab}$ at most $4 \cdot 10^{-10}$ (means, every n and source). Prior bits: A_{xy}, A_{yx} 31.6; $S_{ab}, S^{\mathrm{open}}_{ab}$ 30.0; the four M forms 28.4. Provability of A_{xy} is read from Proposition 7.6, not from $\vdash_J$.

largest change of a reported mass (of T^*, of its equivalents or of the unsound theories, over seeds and n) is 0.181 at $\tau = 1$ and 0.563 at $\tau = 4$. This follows from the factor's definition and does not test whether a time penalty prices Hänni's collapse (Proposition 6.8 and Theorem 6.9). Results in Table 39, discussed in Remark E.8.

E8: splits under a derivation likelihood. Let $\varphi(t) := (t{+}0 = t)$, $\psi(t) := (0{+}t = t)$, whole := $\{\varphi(z), \psi(z), \varphi(z) \to \psi(z)\}$, and split := whole with $\varphi(z)$ replaced by $\varphi(0), \varphi(Sz), \varphi(z{+}z'), \varphi(z{\cdot}z')$ ($K = 4$ root productions). Under Ch_1 a citation of $\varphi(t)$ stops or applies MP with the cited conditional, giving $\psi(t)$, which can also be cited: the component behind a ψ-datum is latent. Data come from Ch_1 on whole at fixed weights $(0.5, 0.25, 0.25)$, with bodies from $\mathcal{Q}$ (*wellspec*) or from another PCFG (*skew*: roots $0, S, +, \cdot$ with .5, .1, .2, .2); seeds 0–9; $n \leq 4096$. Table 40 gives Bayes factors without the prior, which adds 76.4 bits in favour of whole; the findings are in Remark 4.12. At matched weights ($w_\varphi p_f$) the two theories give every datum the same $\mathsf{L1}$ probability, to $1.1 \cdot 10^{-13}$ (experiments Prop X13, Remark C.2). With well-specified usage the split loses 1.60 bits per doubling of n under $\mathsf{L1}$ and 1.76 under $\mathsf{L0}$ on $n = 1024 \dots 4096$ (on $256 \dots 4096$: Remark 4.12). With misspecified usage its mean gain per datum at $n = 4096$ is 0.1995 bits under $\mathsf{L1}$ and 0.0731 under $\mathsf{L0}$; the gain per datum between $n = 1024$ and 4096 is in

(λ, τ)	$(1,0)$	$(1,1)$	$(1,4)$	$(2,0)$	$(\frac{1}{2},0)$
T^* at $n=32$	0.197	0.197	0.197	0.040	0.186
T^* at $n=64$	0.751	0.751	0.752	0.594	0.719
T^* at $n=128$	0.991	0.992	0.993	1.000	0.961
T^* at $n=512$	0.996	0.996	0.996	1.000	0.980
unsound at $n=8$	0.256	0.293	0.372	0.860	$3 \cdot 10^{-4}$
unsound at $n=16$	0.197	0.217	0.262	0.792	$7 \cdot 10^{-5}$
unsound at $n=32$	0.077	0.084	0.103	0.737	0.014
unsound at $n=64$	0.009	0.009	0.008	0.206	0.039
unsound at $n=512$	0.004	0.004	0.004	$2 \cdot 10^{-4}$	0.019
acc. at $n=8,16,32,64$	2, 3, 0, 0	2, 3, 0, 0	5, 4, 1, 0	18, 19, 18, 5	0, 0, 0, 0
MAP at $n=8$: e/w/u	0/18/7	0/16/9	0/16/9	0/3/22	0/25/0

Table 39: E7 (E2's data and pools; means over 25 seeds). "Acc.": seeds in which $V_{0.05,0}$ accepts a false probe; no seed accepts from $n = 128$. Last column: MAP tags equivalent / weaker / unsound at $n = 8$. T^* has mass below 10^{-3} at $n \leq 16$ in every row. $\lambda = \frac{1}{2}$ is proper only on finite pools. The first version (5 seeds, legacy pool) reported a largest time-factor effect of 0.002 and 0.016; the referee's re-run of that pool with SeenQ gave an unsound MAP at $n = 8$ in 19 of 25 seeds at $\lambda = 2$, the same direction as here.

Remark 4.12.

n	*wellspec*, L1	*wellspec*, L0	*skew*, L1	*skew*, L0
16	−1.7 (−3.4, 0.8)	−2.2 (−4.5, 1.2)	1.7 (−1.5, 8.3)	−0.2 (−2.3, 5.6)
64	−3.5 (−5.5, 1.2)	−2.8 (−5.2, 0.1)	7.7 (−0.4, 14.3)	1.2 (−2.9, 5.7)
256	−7.3 (−8.3, −6.1)	−6.8 (−7.8, −5.6)	48.8 (37.9, 72.6)	17.4 (6.4, 34.2)
1024	−9.8 (−11.6, −6.5)	−9.3 (−11.4, −6.2)	196.5 (165.4, 232.8)	72.1 (41.9, 98.0)
4096	−13.0 (−14.6, −11.3)	−12.9 (−14.5, −11.4)	817.0 (738.9, 912.3)	299.6 (210.7, 351.5)

Table 40: E8: $\log_2$ Bayes factor split : whole without the prior (mean over 10 seeds; range).

Remark G.1 (Claims of the first version). [refuted] experiments §16.4; referee M1–M4 Each claim is kept with its counterexample. (1) E2: "in 3 of 5 seeds an unsound lump has mass ≥ 0.997 at some $n \leq 32$ and a $\delta = 0.05$ verifier accepts a false sentence". True of a pool that used unseen data and lacked SeenQ(D_n); with causal pools, 2 of seeds 0–4 and 5 of 25. Counterexample to its representativeness: seed 2 at $n = 32$, where the old skel4 (built from 40 data) had mass 1.0 and the causal skel4@32 has 0.85. (2) E2: "MAP at $n = 8$: DTRC($n = 16$) three times": built from data 9–16, unseen at $n = 8$. (3) E6: "the five closed-guard forms are logically equivalent": Proposition 7.6(b). (4) "Soundness at a constant threshold for any well-specified data": fails for multi-component T^* at fixed weights (Example 5.8). (5) PA: "robust at the level of theorems, not of axioms": E3(b) atomic motives (Proposition 7.16). (6) E1: "$P(T \vdash \varphi(t^*) \mid D_n) = 1.000$ from $n = 8$ in every configuration": *open* data under L0 give 0 at $n = 4, 8$. (7) E2: "no false probe is accepted": the mean over seeds hid seeds 1, 2 and 4.

G.5 Bounds and counters

Exact L1 *pools.* E1's support check left out 65 (*open*) and 43 (*allq2*) data-derived theories for $\varphi = \neg Sx = 0$, summed over seeds, mostly skeleton clusters, and nothing else. *Dirichlet bounds* replaced the exact sum, with the largest posterior mass the theory could have had: in E1 for some skeleton theories and Mem(D_n) under L1, at most $8 \cdot 10^{-14}$; in E2 for skel6@32 and skel6@64 in 14

cases over 25 seeds and both pools, 0 to double precision; in E3 and E4(B, C), where the largest possible mass is recorded per run, 0 in every run; in E6, checked in seed 0 only, for $\mathrm{Mem}(D_n)$ in two cases, 0. *Fallbacks.* The likelihood and derivability-oracle fallback counters were 0 in every run of E1, E3(a) and E4; the referee re-ran the 75 E1 jobs with the oracle instrumented (5896 queries, no fallback).

G.6 Reproduction and checks

All commands run from `code/` (Python 3.11, numpy 2.4, scipy 1.17). Results are in `results/*.md` and `*.json`, the first version's in `results/v1/`. AS's `dtrc` code is read, not modified. `python3 -m pytest -q tests` runs the 21 unit tests (18 s); `experiments/e`k`_*.py` runs Ek; `sh run_all.sh` runs everything (about 50 minutes on 4 shared cores; E3 10 min, E4 32 min, the rest at most 3 min each). Seeds: E1 0–4, E2 and E7 0–24, E3 and E6 0–4, E4 streams 0–399 (A) and 0–99 (B, C), E5 and E8 0–9; the random subsets for Min have seed $(c + \text{seed}) \cdot 1000 + b$ with $c = 1000$ (E1), 500 (E2), 77 (E3). After the batch run E1–E3 were re-run with their final reporting code; all posteriors are identical, and only derived labels changed. Two `L0` entries of the first version's E1 table were wrong on its own results and are corrected in Table 30. The unit tests compare each core computation with brute force or Monte Carlo, and with the referee's independent `ref_core.py` (within 10^{-9}). Nothing is checked in a proof assistant. The check scripts (`research/tracks/experiments/checks/`, outputs next to them) give: `check_x6`, E1's `L1` odds within $1.1 \cdot 10^{-11}$ bits of prior $+\,n$ with prior differences (0.0899, 0.0899, 1.4404 bits) from the referee's code, and the $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ tie to $1.0 \cdot 10^{-13}$; `check_fixed_weight`, Example 5.8 in this code (every $n \le 2 \cdot 10^6$, 300 runs: 0.977, 1.000 at fixed w^*; 0.003, 0.007 with w^* from the prior; 0 with the shrinking threshold) and $R_\alpha(n, 2) \ge 1/(e\pi(n + 1))$ for $n \le 2000$; `kt_regret`, slopes -0.499, -0.998, -1.485 of $\log_2 R_\alpha(n, K)$ for $K = 2, 3, 4$; `check_models`, the structures of Propositions 7.6 and F.20; `check_e4_tight` and `check_e4_where`, E4 as in Table 36; `check_e1_v1`, E1's generator masses unchanged against the first version to three decimals; `check_bounded` and `check_classprob_open`, as in Appendices G.1 and G.5. The referee's own code reproduced the first version's E1–E7 bit for bit, and the final code reproduces the referee's legacy E2 counts, E3 extensions and atomic-motive run.

H How the results were established

H.1 Process

Roles. A research brief (`research/00-brief.md`) quoted the question and set out hypotheses H1–H7 to be confirmed, refined or refuted; Table 41 gives their verdicts. Four research tracks then worked in parallel: *model* (the general theory), *universal* (the case $\forall x\varphi$), *pa* (PA and ZF) and *experiments* (the package `code/bai` and the experiments E1–E8). Each track wrote notes (`notes.md`) with seeded check scripts. An adversarial referee for each track read the notes, re-derived the proofs, wrote independent code (`referee_code/`) and filed a report (`referee.md`). The track revised its notes into `notes-final.md`, which ends with a verification log mapping every referee issue to its resolution. The referees found no fatal error. They raised, by track: model, 5 major and 16 minor issues and 6 questions; universal, 3 major and 11 minor issues and 7 questions; pa, 5 major and 11 minor issues and 5 questions the track had missed; experiments, 5 major and 14 minor issues. A planning editor then wrote an outline, a notation file and a ledger of every claim with its status and source, and reconciled 43 cross-track conflicts (Appendix H.5). Section writers wrote each section from the final notes only, re-reading the proofs as they went and recording what they found (Table 43). A framing editor wrote the introduction, the discussion and this appendix. Five AI reviewers then read the complete draft: three checked the mathematics and the numbers of the sections against the research notes and their outputs, one the consistency of the whole paper, one its readability. They reported 10 fatal and 66 major issues; the fatal ones were statements that claimed more than their proofs or sources support (among them parts of Theorem 3.4 and Propositions 7.10 and F.9 and Definition 2.12). A lead editor checked every fatal and major issue against the research notes and decided the fixes, and four section editors applied them (Table 43, second part; the reviews, decisions and edit logs are in `research/paper-review/`). Two follow-up tracks on the time penalty came later (Appendix H.4).

What this means for the reader. All roles were separate Claude instances (Anthropic). No human has checked the mathematics. Nothing is checked in a proof assistant. Track pa's natural-deduction derivations are checked line by line by a small checker, `nd.py`, which rejects ten planted unsound steps and the referee's 173 mutants (Appendix F.1); its referee re-checked the derivations with an independent checker. The referees checked the first versions of the four research notes. The following results were added or re-proved after the referee reports; they were checked by their track, re-read by the section writers and by the reviewers of the paper, but not by a referee (numbered as in the tracks' notes, which the grey source notes give):

- *model*: Lemma 1.6, re-proved (Lemma A.2); Props 2.7, 2.8 (Propositions 2.24 and A.8); Ex 3.6 and Conj 3.7 (Example C.6 and Conjecture C.7); the exact tightness remark after Thm 4.1 (Remark D.1); Rem 4.5 (Remark D.3); Lemma 4.6 and Thms 4.7, 4.8 (Lemma D.4 and Theorems 5.6 and 5.7); the mechanism of Ex 4.9 (Example 5.8); the new proof of Thm 5.1(b) (Theorem 4.7); Prop 5.5(d1), (d2) (Proposition 4.14); Prop 5.6 (Proposition 4.11); Lemma 6.10, Prop 6.11, Thm 6.12, Cor 6.13, Prop 6.14 and Conj 6.15 (Lemma E.4, Propositions 6.12 and E.5, Theorem E.6, Corollary E.7, and Conjecture 6.13); Prop 7.6 (Proposition 2.17); Prop 8.2 and Rem 8.3 (Proposition C.8 and Remark C.9);
- *universal*: Props U2n, U2t (Proposition B.1); Prop B2 (Proposition 3.6); the new proof of U10(c3), U10(d2) for the pair, and U10(e) (Theorem 3.9); Lemma N0 and Props N1–N3 (Lemma B.11 and Propositions 3.12, B.12 and B.13); U12(b) for $f_q = 0$ (Proposition B.14); the two-part part of U14(c) (Proposition B.15); the exact rates of U16(b) (Proposition B.16);

	the brief's hypothesis	verdict	where
H1	template prior; derivation grammar; size principle	confirmed with qualification ($\mathcal{Q}$ uniformly subcritical; Hänni's scores lack the size principle)	§ 2
H2	concentration on $\{T : P_T = P_{T^*}\}$; misspecified limits at KL-minimisers	confirmed for fixed weights; with Dirichlet weights the exact class has mass 0 (Rem. 4.8) and the instance union is identified for a.e. weight vector under L0 (Thm. 4.7(b)); KL-minimisers for finite classes only	Thms. 4.2, 4.7 and 4.16 and Rem. 4.8
H3	verifier sound with probability $\geq 1 - \delta/\pi(T^*)$	confirmed for fixed weights; refuted as stated for Dirichlet weights, then repaired: on average over the weights, and with a shrinking threshold for likelihoods linear in the weights	Thms. 5.2, 5.6 and 5.7
H4(a)	memorisation and over-general templates lose exponentially	over-general: confirmed; memorisation: refuted (Laplace weights, geometric numerals)	Thm. B.4 and Prop. B.6
H4(b)	instances do not favour $H_\forall$ over H_{sch}	confirmed in the comparisons analysed, with exact factors; non-logical exceptions under $\mathsf{L1}^{\mathrm{sel}}$ with a background; open with $\wedge$-rules	Thm. 3.4 and Prop. 3.6
H4(c), (d)	"all future data are instances" $\to 1$; "the axioms prove $\forall x\varphi$" need not	confirmed: the Bayesian ω-gap	Thm. 3.9
H4(e)	open instances and Gen make $H_{\mathrm{open}} \vdash \forall x\varphi$	confirmed for parameter data; without a guard the ω-step need not be taken under L1	Prop. 3.10
H4(f)	quantified data favour provers of $\forall x\varphi$	confirmed	Prop. B.14
H5	positive data and the size principle get around Gold; spares cost $\approx \frac{1}{2}\log_2 n$ bits with Dirichlet($\frac{1}{2}$) weights	confirmed with qualification: some spares never vanish	Props. 4.9 and 4.14
H6	a membership-time penalty prices the collapse	refuted: it leaves the direction from function to axiom induction intact; written derivation size prices it	Prop. 6.8, Rem. 6.1, and Thm. 6.14
H7	PA and ZF: equivalent forms, Th($\mathbb{N}$), mixtures, failures	usage decides; no consistent r.e. theory survives Th($\mathbb{N}$), and inconsistent ones are refuted only at growing depth; narrow practice can end weaker; ten robust failures	Tab. 4

Table 41: The hypotheses of the research brief (`research/00-brief.md`), which were to be checked, not assumed, and their verdicts (model §9.1, universal §0, pa Summary, experiments §0).

Lemma S1 (Lemma B.17);

- *pa*: Prop 1.4 (Proposition F.2); Prop 2.3 (Proposition F.6); Props 3.1–3.4 (Propositions 7.9, 7.10, F.8 and F.9) and the theorem-data computations of §3 (Table 3 and Remark F.10); the full proof of Prop 4.3 (Proposition F.11); Lemma 4.7, Prop 4.8, Rem 4.9 and Conj 4.10 (Lemma F.15, Proposition F.16, Remark F.17, and Conjecture 7.15); the G1 skeleton theory (Example F.18); Rem 5.4 (Remark F.24);
- *experiments*: the full proof of Prop X8(c) (Theorem 5.7); Props X10–X13 (Propositions 7.6, 7.16 and F.20 and Remark C.2); E8 (Remark 4.12);
- *the follow-ups on the time penalty* (written after the review of the paper, so not read by its reviewers): the items added or changed in their revision, marked in Table 44;
- *the paper*: the reconciliations marked "editor" in their source notes (Remarks 3.7 and 5.9) and every change made in the review of the paper (Table 43, second part).

The appendices with the proofs point to this list instead of keeping their own.

H.2 Corrections made during verification

Table 42 lists every claim of the first versions that was refuted or weakened, with the reason and where the paper states what holds instead. "Found by" gives the referee's issue number (M major, m minor) or "track" when the track found it in its own revision; pa items carry their final numbers, with the first version's in parentheses. Four refuted claims that change what the reader should believe are also kept in the main text (Remarks 3.5, 3.11, 4.8 and 6.11); the others are stated only here and in the appendix of their section.

Table 42: Claims refuted or weakened during verification.

item; found by	claim of the first version	reason; what holds now	where
model track			
Def 1.5; M5	"every type has mean offspring below 1" makes $\mathcal{Q}$ subcritical	no formula type of L_A satisfies it; with infinitely many types a chain is infinite with probability $\prod_k (1-1/(k+2)^2) = \frac{1}{2}$; the domination step fails. Refuted; replaced by uniform weighted subcriticality	Remark A.3 and Definition 2.6
Rem 1.10; m3	a stronger theory has a larger Z_T, "hence a smaller $P_T(s)$ on each datum"	$T = \{a, a \to b\}$, $T' = T \cup \{b\}$: $P_T(b) = 0.032$, $P_{T'}(b) = 0.346$, for ℓ the symbol size of the least MP tree, normalised over a finite universe of 570 formulas, where the normaliser need not be ≤ 1 (`r6`); in general a datum whose shortest derivation gets shorter can gain. Refuted; true for data whose shortest derivation is unchanged	Remark 2.23, Appendix A.5
Thm 5.1(b); M4	the exact generator class gets posterior mass $\to 1$ for a.e. w^*	its mass is 0 at every n ($\lvert T^*\rvert \geq 2$, independent components, w^* off a countable set). Refuted; replaced by identification of the instance union	Remark 4.8 and Theorem 4.7

item; found by	claim of the first version	reason; what holds now	where
Prop 6.9; M2	plain L1 charges derivation symbol size	a conclusion of size $2^{k+3}-1$ from a tree with $7+8k$ grammar nodes. Refuted; replaced by Lemma 6.10–Prop 6.14 and Conj 6.15	Remark 6.11
§7 (c5); m4	renormalisation is coherent essentially when every theory is complete	coherent in 199 of 300 posteriors, 2 of them with all theories complete. Refuted; the 50/50 rule characterised instead	Remark A.9 and Proposition 2.17
§4; M1	Thm 4.1 applies to the Dirichlet-weight model	vacuous there; a constant threshold accepts with probability 0.977–1.000 against nominal 0.02. Refuted; averaged and shrinking-threshold versions proved	Remark D.3, Example 5.8, and Theorems 5.6 and 5.7
§5.3; M3	"a derivation likelihood does not change the MDL finding" [proved]	proved only for matched fixed weights. Withdrawn; Prop 5.6 with (d) a conjecture	Proposition 4.11 and Remark 4.13
§4.1; m8	"W^* only grows under truncation"; "L2 likelihoods are finite sums"	the truncated constant can fall to 0.09 against $\pi(C_d^*)=0.90$; with full-support $\mathcal{Q}$ the sums are infinite. Refuted	Remark 5.3, Appendix D.1
brief H6; track	a collapse schema "requires running T to check membership", so a Kt penalty prices it	membership is polynomial without running T. Refuted for the direction from function to axiom induction; the other direction is Theorem 6.14(b)	Proposition 6.8
Hänni's note; track	his schema is consistent for every consistent assigner	single-sorted over PA it can be inconsistent. Refuted in that reading; his conclusion holds via Craig	Proposition 6.5 and Theorem 6.3
weakened; m1, m2, m5, m6, m9–m12, m14–m16	—	Cor 6.8's constant uniform only for template theories (m1); Prop 4.2's prover replaced by a waiting prover, as an every-round prover wins only 11/400 against escalation (m2); check c9 was tautological for the split and is replaced by c9b, which computes every likelihood by matching (m5); Prop 2.5's numbers corrected to 0.16–0.35 (m6); consistency hypothesis added to Prop 1.9(b) (m9); Prop 2.6(c) restricted to first-order templates with 0-ary term metavariables (m10); $\alpha_{\mathrm{ax}}+\alpha_{\mathrm{lg}}>0$ (m11) and parameters admissible (m12) added to Lemma 1.7; split ties only at matched fixed weights (m14); Prop 2.5's limit 0 or ∞ only if $T^*\in\{T_1,T_2\}$ (m15); Ex 3.4, Rem 3.5 are L0 only, and L1 was computed in Ex 3.6 (m16)	the corresponding items of Sections 2 and 4 to 6
universal track			
U11(e); M1	without a guard H_{open} wins on closed data and the posterior takes the ω-step	$R_1=\{\varphi(0),\varphi(Sz)\}$ beats H_{open} by $(1-q)\mathcal{W}_{\mathrm{o}}$ per datum; $P(T\vdash\forall x\varphi\mid D_n)=0.000$ from $n=1000$. Refuted under L1; holds for $\{H_\forall,H_{\mathrm{open}}\}$ and under $\mathsf{L1}^{\mathrm{sel}}$	Remark 3.11 and Proposition 3.12
remark after U3; m8	$\wedge$-detours cannot break the factor c	in $\mathsf{C}_\wedge$ the ratio is $0.15553>c=0.15$. Refuted; whether the direction can reverse is open	Remark 3.5

item; found by	claim of the first version	reason; what holds now	where
U16(b), (c); M2	"$= 1$ over Q" in the sentence-only class	true only in a three-hypothesis class; with depth ≤ 4 memorisers win, with depth ≤ 40 splits win to $n = 10^{12}$; the referee's "$\to 0$ over Q" holds only for bounded depth. Restated, limit a conjecture	Proposition B.16, Appendix B.7
U10(c); m3	$\vdash_{\mathsf{C}}$ and $\vdash$ conflated in the faithful regime	split into (c1), (c2) with the extra hypothesis	Theorem 3.9
weakened; M3, m1, m2, m4–m7, m9–m11	—	U10(c3) restricted to $\mathsf{C}_{\min}$-like calculi (M3); memoriser universe named, as the prior was improper (m1); Σ_1-completeness of Q restricted to $<$-free sentences (m2); learned weights are outside U7, Prop B2 added (m4); U12(b) recomputed with equal priors (m5); U14(b) does not apply at $q = \frac{1}{2}$, U14(c) is a sufficient condition (m6); U10(d) for quantified φ a sketch in general (m7); inconsistent theories reported as Inc and the verifier moved to $\mathrm{Bel}_{\mathrm{cons}}$ (m9); wording (m10); the probability qualifier of U6 restored (m11)	the corresponding items of Section 3
pa track			
Prop 4.6(c) (old 3.5(c)), F8; M4	with Σ_n motives every covering DT$^\circ$ theory has PA-strength induction, so the posterior lands on the $\mathcal{L}_\infty$ side	a theory inside $\mathrm{I}\Sigma_1$ covers a Σ_1/open practice and overtakes $\mathsf{Q} + T_{\mathrm{Ind}}$ near $n \approx 2^{26.7}$; a term-only theory inside $\mathrm{I}\Sigma_2$ covers G1. Refuted	Proposition 7.13 and Examples 7.14 and F.18
§2.2; m2	the SDPC margin of the root split is constant	it falls by 1 bit per doubling. Refuted	Remark F.7 and Proposition F.6
§2.2 first run; track	under a shared grammar the root split loses by a growing margin ($+803 \cdots + 12270$ bits)	an encoding artefact (bound variables coded as holes). Refuted	Section 7.3
§5.5 (old §4.5) first run; track	the head-symbol split of $x + 0 = x$ overtakes the schema at $n = 1000$	an artefact of root-context coding; the split closes in at 5 bits per doubling. Refuted	Remark F.25 and Table 28
§5.5 (old §4.5); M5	under $\mathsf{L1}$ the odds of $\forall x(x + 0 = x)$ fall like $2^{-3.3n}$, "as H4 predicts"	true for a fixed rule code only; $\frac{R-1}{2} \log_2 n$ under a learned depth-indexed code. Refuted as a general rate	Remark F.25
§7; M2–M4	"it robustly gets a theory deductively equivalent to PA"	true only within the hand-picked candidate sets and for axiom-instance data. Refuted as stated	Section 7.7.1
§1.2; m5	a derivation of CVI(P) costs more "because it has two lines"	a citation also writes $P(x)$. Argument wrong; conclusion re-proved	Proposition F.2
numbers; M1, m1, m9	eight entries of the DTRC table; the G2/G3 margins; the Euler lead	table regenerated from the output and every quote checked by script (M1); every template charged (m1); lead $2.1 \cdot 10^5 \to 3.1 \cdot 10^5$ bits at $n = 10^6$ (m9)	Tables 26 and 27 and Example F.26

item; found by	claim of the first version	reason; what holds now	where
weakened; m3, m4, m6–m8, m10, m11	—	break-even ratios are code-specific (m3); a decodable variant of the code added (m4); one refutation rule R_d (m6); eliminating each $\mathrm{I}\Sigma_n$ is not concentration, Doob's theorem added (m7); a depth theorem of AS cited too loosely, replaced by Π_1-completeness of consistency (m8); the toy tower covers only the provable part of the data (m10); two-part bounds for overlapping templates added as Rem 5.4, a proof sketch (m11, accepted in part)	the corresponding items of Section 7, Remarks F.14 and F.24
experiments track			
E2; M1	in 3 of 5 seeds an unsound lump has mass ≥ 0.997 at some $n \leq 32$ and a $\delta = 0.05$ verifier accepts a false sentence	the pool used unseen data and lacked SeenQ(D_n); with causal pools 5 of 25 seeds. Refuted as a frequency	Remarks 5.11 and G.1
E2; M1	the MAP list at $n = 8$	built from data 9–16, unseen at $n = 8$. Refuted	Remark G.1
E6; M2	the five closed-guard forms of commutativity are logically equivalent	S_{ab}, M_x, M_y are strictly weaker. Refuted	Proposition 7.6
§0; M3	soundness at a constant threshold for any well-specified data	fails for multi-component T^* at fixed weights. Refuted	Example 5.8
§5; M4	PA: "robust at the level of theorems, not of axioms"	atomic motives lead to $\mathsf{Q} + \{T_=, T_<\}$, strictly weaker. Refuted	Proposition 7.16 and Remark 7.17
E1, E2; track	$P(T \vdash \varphi(t^*) \mid D) = 1.000$ from $n = 8$ in every E1 configuration; no false probe accepted on E2 data	open data under L0 give 0 at $n = 4, 8$; the mean over seeds hid seeds 1, 2 and 4. Refuted	Remark G.1
weakened; m1–m14; track	—	the "brute force" tests shared code with the implementation; a test against the referee's independent implementation added (m1); Prop X5's almost-everywhere qualification (m2, M3); citations of superseded drafts replaced (m3); Mem(D_n) is the best memoriser only under L0 (m4); held-out PA instances drawn outside the stream (m5); every n checked in E4(B, C) (m6); E7's time-factor result holds by construction (m7); $\mathsf{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ is per-citation rejection (m8); summary wording on the prior share of tied provers (m9); Gold's text only if every stage ends (m10); E3(a) gains compared within L0 (m11); E3(a) families extended, so the MAP is interior (m12); fallback counters reported, 0 in every run (m13); the E5(a2) slope, first -0.259 from 200 draws, now -0.251 ± 0.002 from 1000 draws per n (m14); Mem(D_n) double-counted in E6 at $n = 1, 2$, $T^* - \mathrm{Q3}$ mis-tagged, two wrong L0 entries in the first E1 table (track)	Section 8, Appendix G.4

item; found by	claim of the first version	reason; what holds now	where
follow-ups on the time penalty (Appendix H.4)			
Claude's summary to Hänni; Hänni	"The time penalty doesn't block your collapse argument. The collapse into function induction (with Craig's trick) survives a penalty on checking or generating axioms."	conflated the two directions of Theorem 6.3. Penalties on checking or generating axioms leave the direction from function to axiom induction intact; the other direction runs proof search, which they leave free, so once consistent function induction is penalised the equivalence is one-sided and axiom induction strictly stronger. Corrected in Remark 6.1(1), §6.4, the abstract, the introduction, the discussion and the README	Theorems 6.14 and 6.15
§6.4; time-followup RL-m1, RP-M2	a generation penalty fails because "$\varphi^{\wedge(k+1)} \in A_f^C$ comes from k steps of f and is longer than k"	covers single axioms, not an enumeration of the set (time-followup Rem 2.6 treats enumerators of A_f^C). The conclusion holds through Craig sets padded by the enumerator's own work	Lemma E.12 and Theorem 6.14
proof of model Thm 6.7; short-derivations RLg-M1	"guess a derivation of size at most $\ell(\lvert w\rvert)$ and check it"	a parameter counts one symbol whatever its number, so the guessed derivation had no bound on its bit length. The proof now truncates parameter numbers; the statement stands, with membership time measured in symbols	Theorem 6.9 and Lemma E.28
time-followup §0; RL-M1, RP-M5	axiom induction "becomes strictly stronger than every time-bounded function inductor"	Hänni's S and $\mathrm{FI}^{\tau}_{\mathrm{all}}$ are time-bounded and incomparable with it. Refuted; true against time-penalised consistent function induction	Theorem 6.15
short-derivations Prop 5.3; RC-M1	whether the soft correspondence holds with rate factor $r = 1$ is open	no pair of rates gives two-sided constant regret. Refuted	Proposition E.26
weakened; both follow-ups	—	at polynomial budgets the comparisons with polynomial-time function induction hold per sequence, and need the membership degree charged; uncharged, function induction fails to dominate unconditionally (RP-M4, RC-M2). The total weight of the Kt-penalised prior is at most $t(m_0)$, not 1 (RP-m1). Closure of the axiom set under renaming parameters added as a hypothesis (RLg-M1, RC-M3). The diagonal loss bound sharpened to $n - 1/(4\ln 2)$ (RL). Further items in the tracks' logs (time-followup §10, short-derivations §12)	Section 6.6, Appendix E.6

Track against referee. In five places the final notes disagree with the referee, with reasons, and the paper follows the final notes. (1) Pa, m11: AS Thm 5.16 holds in closure-normal form with parameters; the real gap is that it concerns a single covering template (Proposition 7.13). (2) Pa, M4: the law G1 does not force a formula metavariable; its 157 skeletons are covered by a term-only theory inside $\mathrm{I}\Sigma_2$, and the posterior keeps PA through the Occam balance (Example F.18). (3) Experiments, M2: open guards do reach $\forall x \forall y$ under the single-parameter convention (Propo-

sition 7.6(a)). (4) Universal, M2: the referee's "$\to 0$ over Q" holds only with bounded split depth (Proposition B.16). (5) Model, m13: the tightness of the soundness constant holds exactly, not only as a supremum (Remark D.1).

H.3 Corrections made while writing and reviewing the paper

The section writers re-read the final notes against their proofs and outputs; the first part of Table 43 lists what they changed or qualified. None of these changes a headline result; several narrow a scope or lower a status. The second part lists the corrections made after the review of the complete draft, one line each, with the reviewers' issue ids (MA, MB, C without hyphen: the three mathematical reviews; C-: consistency; R-: readability). They narrow several headline statements to what the sections prove and correct parts of four statements (Theorem 3.4, Definition 2.12, and Propositions 7.10 and F.9).

Table 43: Corrections and qualifications made while writing (first part) and after the review of the paper (second part).

section	what was found, and what the paper does	where
while writing		
model	Kraft's inequality for the prior needs theories taken up to renaming of metavariables; added to the definition. The number of templates is coded as $\gamma(k+1)$, not $\gamma(k)$, so that the empty theory has a code.	Definition 2.1, Appendix A.1
model	Lemma 1.7(b), $Z_T \geq \alpha_{\mathrm{ax}} + \alpha_{\mathrm{lg}}$, fails for $T = \emptyset$; $Z_\emptyset \geq \alpha_{\mathrm{lg}}$ suffices where it is used. Prop 1.9(c) uses only $\operatorname{supp} P_T \subseteq \mathrm{Th}(T)$, which needs no hypothesis.	Appendices A.4 and A.5
model	The notation file's "$\leq c$" for the C_{U3} factor bounds only derivations that cite the added axiom. The direction (factor at most 1) is not proved for $\mathsf{C}_\wedge$; the table caption says for which calculi it is. Some "size principle" marks in Table 7 are the writer's applications of Lemma 2.21 and Theorem B.4, flagged in its caption. Prop 2.8 needs $\mathcal{Q}$ symmetric in the holes.	Tables 6 and 7 and Proposition A.8
universal	The proof of Lemma S1 omitted a case (a summand that is a variable always instantiated by 0); added, conclusion unchanged. Thm U2's equality under Hänni's scores needs $\forall x\varphi$ consistent. In c5 Part A, H_{both} carries a uniform prior on its weight. U5's "survival 0.503" is a per-datum estimate of $p_S = \frac{1}{2}$, so only the gain per datum is reported. The notes' citations of the superseded first E3 result and of "only derivation length prices computation" are replaced (conflicts C9, C11).	Lemma B.17 and Tables 9 and 10
ident	"A false spare sentence makes $T^* \cup \{\sigma\}$ inconsistent" holds only when T^* refutes σ. "Well specified: Occam loss" under $\mathsf{L1}$ is a conjecture; only a bounded gain is proved. Spares in the span of T^*'s components never vanish, contrary to a summary sentence of the model notes. Check c7: the escape template alone loses to memorisation at $\lambda = 1$ by $n = 10^4$ (-2649 to -4234 bits); only the hybrid beats memorisation. (A note that the misspecified prediction of check c2 Part B is off at $n = 100$ was itself wrong; see MA-09 below.)	Remarks 4.13 and 4.15, Proposition 4.14, and Table 19
sound	The laws of Bel, Dis, Indep (model Props 7.1(a), (c), (d), 7.2) need unbounded derivability, since $\mathrm{Th}_d(T)$ is not closed under consequence; stated for $d = \infty$. Check c5 records only 2- and 3-monotonicity; 4-monotonicity was checked by the referee on two atoms. The computed support of Thm 4.1 ($n \leq 300$, 400) checks the bound, not time-uniformity.	Section 2.7, Appendices A.7 and D.1

section	what was found, and what the paper does	where
time	The closing statement's upper bound rests on a proof sketch, so its status is lowered. Prop 6.14(b), (c) depend on the hierarchy theorem. Prop 6.14(c) also needs the explicit derivations to be K-derivations (or to translate at linear cost) and a code with at least one bit per written symbol. The size of the PA-proof of the Tarski biconditional in the upper bound for the reflection sentence is not bounded in the notes. Prop 6.4 is proved for the acceptance part of Hänni's schema only.	Remarks 6.1 and E.3 and Propositions 6.6 and 6.12
pa	Rem 4.9's "no instance is logically equivalent to $\mathrm{Ind}(P)$" is not proved; a weaker true statement is used. Prop 5.3 is proved for the two-part likelihood, not for the normalised $\mathsf{L1}$, and is stated under $\mathsf{L1}^{\max}$. The "astronomically large" waiting times of Prop 4.6(a) are a heuristic. Props 3.3 and 4.6(a) need fixed weights (setting W); that $\mathsf{PA} \cup S$ decays polynomially is a proof sketch for Dirichlet weights. "The redundant union wins after 1–12 uses" counts schema uses in expectation; one use of CVI alone saves 1468 bits. "E3(b) keeps the theorems, not the axioms" holds for two of three connective-rich laws.	Remarks 7.5, 7.17 and F.17 and Propositions 7.10, 7.13 and 7.19
experiments	The proof of Prop X4(c) (Gen, then $\forall$E, at $J = 2$) treats a derived component as a DT° template without saying so, and the code's support check does not test it; the hypothesis is added, and for data-derived theories at $J = 2$ exactness rests on the computed checks. At $\lambda = 2$ the verifier also accepts a false probe in 5 of 25 seeds at $n = 64$ (unsound mass 0.206), which the notes do not report.	Proposition 8.1, Appendix G.2 and Table 39
after the review of the paper		
abstract, intro	C-01, MB-01: "closed instances never favour $\forall x\varphi$" restricted to the comparisons analysed, with the non-logical exception under $\mathsf{L1}^{\mathrm{sel}}$ with a background and the open $\wedge$-detour case.	Theorem 3.4 and Proposition 3.6
intro, Table 41	C-02: H2's verdict said that concentration holds for a.e. weight vector with Dirichlet weights; the exact class has mass 0, and only the instance union is identified a.e. under $\mathsf{L0}$. C-14, C-19, C-20: H7, H4(a), H5 cells.	Remark 4.8 and Theorem 4.7
intro	C-03: "plain $\mathsf{L1}$ is proved to charge only logarithmically" read as an upper bound; it is a lower bound, and a polynomial charge is conjectured. C-11, C-12: the collapse result needs PA and Σ_n-soundness; the charge is a polynomial root of the nondeterministic time.	Corollary E.7, Proposition 6.7, and Remark 6.1
abstract, intro, discussion	C-04, C-05, C-06, C-08: identification needs fixed weights; KL-minimisers only in finite classes; the referees checked first versions only; misspecified, there is no guarantee for the verifier, and a waiting prover wins in one example.	Corollary 4.4, Theorem 4.16, and Proposition 5.4
intro, discussion	C-09, C-10: the shrinking threshold needs a likelihood linear in the weights; with a selection model only the reported part of the theorem set is identified. C-13: the MDL finding with the misfit clause. C-15: a false spare sentence is removed polynomially with learned weights, not "only by refutation". R-20: exact computation only for citation, its Dirichlet sum and the two-step chain.	Theorems 3.9 and 5.7, Remarks 4.13 and 4.15, and Proposition 8.1
model	MA-02, MA-26: L_ε at $\varepsilon = 1$ is a per-datum relaxation of S_{nc}, equal to it only in a special case, and tends to P_T as $\varepsilon \to 0$; $S_{\mathrm{nc},\beta}$ defined. MA-03: the refuted part of the graded-score remark argued by an inequality; the referee's numbers with their setting. MA-27: $Z_T \geq 1 - \alpha_{\mathrm{r}}$ for $T \neq \emptyset$.	Definition 2.12, Remark 2.23, and Lemma 2.10
ident	MA-04, MA-16: Proposition 4.14(c) needs $\sum_s \mathcal{Q}_\sigma(s)^2 / P^*(s) < \infty$; a spare in the span but outside the convex hull is not covered. MA-07: which experiments a consistency theorem covers. MA-09: the misspecified predictions of the c2 table lacked $\frac{K-1}{2}$.	Proposition 4.14, Remark C.3, and Table 15
sound	MA-01: in E4(C1) the accepted queries are theorems of the data's generator; the verifier is unsound only relative to the intended T^*, and Theorem 5.7 does not cover the constant threshold used. MA-05: vacuity proved only for $\mathsf{L0}$ and likelihoods affine in w. MA-06: under $\mathsf{L1}$ the support inclusion gives the theorem inclusion for $d = \infty$ only.	Remarks 5.10, D.1 and D.3

section	what was found, and what the paper does	where
sound	MA-08, C-25: pool version of the shrinking-threshold theorem. MA-12: the pool threshold when the prior weights sum to more than 1. MA-19: IL's escalation bound transfers to valid queries only. MA-15: the odds of $T^* \cup \{\psi\}$ stay at the prior under the scores; a sketch for schema-generated ψ.	Theorems 5.6 and 5.7, Appendix D.1, and Remark 2.20
universal	MB-01, MB-22, MB-24: the opening, the title of Theorem 3.4 and its "Not covered" paragraph name the $\mathsf{L1}^{\mathrm{sel}}$ background effects. MB-02, MB-14: (B1) and (B3) restricted; the score case of (B3) by Herbrand's theorem. R-22: Theorem 3.3 displays its formulas.	Theorems 3.3 and 3.4
universal	MB-03: under S_{nc} and S_g the provers share one factor; under S_{nc} the limit set also contains non-provers. MB-08: Proposition 3.10(d) for finite classes only. MB-10: Theorem 3.8 assumes setting W. MB-25: filters with $P_T(S) > 0$ in Theorem 3.9(e).	Propositions 3.10 and B.8 and Theorems 3.8 and 3.9
time	MB-04–MB-07: Remark 6.1 rewritten (no time-penalised prior is defined; Σ_n-sound assigners over PA; a consistency hypothesis added; the upper bound a proof sketch and not established for $\rho_{f,n}$). MB-09: the counterexample costs exactly $12.77 + 12.21k$ nats. MB-18: Proposition 6.6 for the acceptance part only. C-16: Accept_f, Reject_f written without $<$. MB-12, MB-20: the constant equivalence is proved only in the semimeasure convention, and Hänni's one-sort argument only for sound assigners.	Remarks 6.1, 6.11, E.2 and E.3 and Propositions 6.4 and 6.6
pa	C1: in Proposition F.9(c) the bare F_0 is refuted by R_d; the example is an inconsistent theory with long refutations. C2: under fixed weights $\mathsf{PA} \cup S$ loses exponentially; polynomially only with Dirichlet weights (sketch). C3: the axiom prior rate β_{ax} is separated from the proof rate β, and the memorisation threshold restated.	Propositions 7.9, 7.10 and F.9 and Definition 7.1
pa	C4: the theory of the seen skeletons dies at the first unseen one; the covering theory's slope is a sketch. C5: E3(b)'s $2 \cdot 10^{-178}$ is within the pool. C6: F3 is harmless only for non-atomic formula roots. C7: the ZF conclusion is conditional on the reverse overheads. C8: the answer for theorem data under the full-sum $\mathsf{L1}$.	Example F.18, Remarks 7.17 and F.4, Table 4, and Section 7.4
pa, experiments	C9, C37: Kraft for $\beta \geq \log_2$ of the alphabet size. C10: the limit for inconsistent theories is open. C11: the overall answer scoped to broad usage laws and cited axioms. C12: the hypothesis of (d3) is not checked by the code. C13: Proposition F.6(b), (c) under the growth condition. C14, C15: decodable costs 3.4–7.5%; the Euler masses summed over $k < 200$. C21: which theorem covers each experiment.	Definition 7.1, Remark F.13, Sections 7.7.1 and 8.2, Propositions 8.1 and F.6, and Example F.26
all	Further precision and wording fixes: MA-10–MA-25, MB-11–MB-26, C16–C36, C-18–C-42, R-01–R-45 (moves to the appendices, glosses, notation, typography). The issue lists, the decisions and the edit logs are in `research/paper-review/`.	—

H.4 The follow-ups on the time penalty

The correction. After the first version of this paper, Claude's summary to Hänni said: "The time penalty doesn't block your collapse argument. The collapse into function induction (with Craig's trick) survives a penalty on checking or generating axioms." He objected (verbatim): "if we penalize time on the function induction side and in the axiom generator but crucially NOT in the proof search from axioms, then intuitively i disagree with this: i think the time penalty should help? like the function induction guy that runs the proof search from axioms will be extremely slow; there's nothing comparably slow on the axiom induction side". He is right: the sentence conflated the two directions of Theorem 6.3. Penalties on checking or generating axioms leave the direction from function to axiom induction intact (Theorem 6.14(a)); the direction from axiom

to function induction runs proof search, which they leave free, and once function induction is penalised it fails (Theorem 6.14(b), Theorem 6.15). The paper's summary remark said "do not block the collapse" without naming the direction, and the abstract, the short answer, A5, the discussion and the README repeated it; all are corrected. Hänni then proposed (verbatim): "hmm interesting point. but perhaps we can make AI actually interestingly time-limited by requiring that given statements have derivations from short axioms?"

The tracks and their checking. Two follow-up tracks took up the objection and the proposal: `research/tracks/time-followup/` and `research/tracks/short-derivations/` (the second builds on the final version of the first). Each wrote notes with seeded check scripts, was refereed once, by two adversarial referees who wrote independent code, and was revised into `notes-final.md`, which ends with a verification log mapping every referee issue to its resolution. Neither was refereed again after its revision. No referee found a fatal issue. *time-followup*: the logic referee raised 2 major and 18 minor issues, the penalty referee 5 major and 20 minor. *short-derivations*: the logic referee raised 1 major and 12 minor issues, the complexity referee 3 major (one shared with the logic referee) and 14 minor. Each track reran its own and its referees' scripts from scratch copies, with byte-identical outputs, except one script changed on purpose (short-derivations `c2`, a corrected trial-division count that the paper does not use). The writer of Section 6.6 and Appendix E.6 worked from the final notes only. As elsewhere, nothing was checked by a human or in a proof assistant. Table 44 lists what the paper takes from the follow-ups.

Table 44: Results taken from the follow-ups, with their verification. "Refereed" means checked by both referees of the track in its first version; "revision" marks what was added or changed after the reports, which no referee checked. TF: time-followup; SD: short-derivations.

result (track item)	paper	verification	checks
padded Craig sets; checking and generation penalties vacuous from FI to AI (TF Lemma 2.1, Thm 2.2)	Lemma E.12, Theorem 6.14(a)	refereed; delay form and the strict-form analysis added in revision (RP-M1, RL-m1)	`c2`; RP `r2`
Kt charge (TF Prop 2.3)	Proposition E.13	first an upper bound only; two-sided bound proved in revision (RP-m1, m2)	RP `r1`
domination of time-penalised consistent FI (TF Thm 3.1)	Theorem 6.14(b)	refereed; hypothesis corrected in revision (RL-m5)	—
diagonal separation; S, $\mathrm{FI}^{\tau}_{\mathrm{all}}$, $\mathrm{FI}^{\tau}_{\mathrm{cons}}$; incomparability (TF Thm 3.2, Cor 3.3, Prop 3.4)	Theorem 6.15	refereed; bound sharpened to $n - 1/(4 \ln 2)$, Diophantine formula, and Prop 3.4 proved without Hänni's theorem in revision (RL, RL-m14, m16)	`c1`, `c4` A; RL `r1`
arithmetic facts (TF Lemma 1.4)	Lemma E.14	added in revision	RL `r2`
bounded loss on an infinite sequence (TF Lemma 1.6)	Lemma E.11	added in revision	—
evaluator; hierarchy effect (TF Prop 4.1)	Proposition E.15	refereed; (c) a proof sketch	—
disguised assigners (TF Prop 5.1(a))	Proposition E.16	refereed; converse written in revision (RL-m9)	—

result (track item)	paper	verification	checks
bounded schema (TF Prop 2.5)	Proposition E.17	refereed; Diophantine form and reframing in revision (RL-m10, m16)	RL `r2`
subformula lemma (SD Lemmas 1.1, 2.1, 2.2, Cor 2.3, Prop 2.4)	Lemma E.19	refereed; constant $\frac{1}{2}$ and (d) added in revision (RLg-m3, m4)	`c1`; RLg `r1`, `r3`
renaming and canonical closure (SD Lemma 4.1)	Lemma E.20	refereed	`c3`
material, size and certificates (SD Thm 3.1; TF Thm 4.2)	Proposition E.21, Theorem 6.16	refereed; renaming hypothesis (F1), the bit-code step and the linear budgets of (c) added in revision (RLg-M1, RC-M3)	`c4`; TF `c3`; RL `r3`
clocked Craig sets (TF Cor 4.3(a))	Lemma E.22	refereed	RL `r3` D
reading (b) at polynomial budgets; uncharged degrees; clocked FI (SD Cor 3.2, Prop 3.4; TF Cor 4.3)	Corollary 6.17(a)	refereed; degree charge and per-sequence form in revision (RP-M4, RC-M2); uncharged non-domination added (RC-M2)	RP `r3`; `c4` C
reading (a1): closure, alternation, EXPTIME, predictors (SD Thm 4.2, Prop 4.3, Cor 4.4, Prop 4.5)	Proposition E.23, Corollary 6.17(b)	refereed; (F1) added (RLg-M1), and the loss bound updated to $n - 1/(4 \ln 2)$, in revision	`c3`
readings (a2), (b2) (SD Prop 4.12)	Corollary 6.17(c)	added in revision (RLg-m2)	RLg `r4`
Horn templates; tightness (SD Prop 6.1, Rem 6.4(3), Cor 6.2)	Proposition E.24, Corollary E.25	refereed; normal form, $2\tau + 3$, the one-sided case and all literal assigners added in revision (RLg-m8, RC-m4, m5, m9, m13)	`c5`; RC `rc2`
soft charges (SD Def 5.1, Thm 5.2, Prop 5.3)	Proposition E.26	refereed; restated in revision ($r = 1$ refuted, RC-M1)	`c4`; RC `rc1`
parameter channel; truncation (SD Prop 7.1, Lemma 7.2, Thm 7.4)	Proposition E.27, Lemma E.28	channel found by RLg-M1; truncation added in revision	RLg `r2`; `c6`

Referees: RL, RP are time-followup's logic and penalty referees; RLg, RC short-derivations' logic and complexity referees. Scripts are in each track's `checks/` and `referee_*/` folders.

H.5 Cross-track conflicts

The tracks used different calculi, priors and names, and some cited each other's superseded results. The ledger records 43 conflicts, and the review of the paper found one more; Table 45 gives each with its resolution.

Table 45: Cross-track conflicts and their resolutions.

	conflict	resolution
C1	the per-datum ∀E factor differs between calculi	each factor stated with its calculus and normalisation (Table 6); the direction is shared
C2	two selection-aware likelihoods	$\mathtt{L1}^{\mathrm{sel}}$ (stream) for Prop B2; $\mathtt{L1}^{\mathrm{sel}}_{\mathrm{cit}}$ (per citation) for E1, E6 (Remarks 3.7 and A.6)
C3	prior codes differ, so prior shares differ	every share names its code (Table 11)
C4	three verifier constants	one theorem with classes C^*_d, C^{Dir}_d, $C^*_\forall$
C5	Thm 4.1 used for Dirichlet weights	never; Theorems 5.6 and 5.7 instead

	conflict	resolution
C6	Ex 4.9 numbers differ between implementations	ranges with all sources (Table 22)
C7	pa cites E2's superseded "3 of 5"	5 of 25 (causal pools); pa §5.4 is one run
C8	E6 cited as identification among deductively equivalent forms	logically and merely instance-equivalent forms stated separately
C9	universal cites the superseded first E3(a)	final E3(a) only
C10	nested spare-slot slopes differ	E5(a2) -0.251 ± 0.002 primary; the rate stays a proof sketch
C11	"only derivation length prices computation"	"derivation length in written symbols"
C12, C31	pa's two-part code called $\mathsf{L1}$	named $\mathsf{L1}^{\mathrm{sch}}$; its numbers are not $\mathsf{L1}$ results
C13	sign of the Occam drift (fixed vs learned grammar)	both with scopes; Proposition F.6 reconciles
C14	rate of the ∀E penalty	depends on the rule law and normalisation; the direction does not
C15	guard learning rate (exponential vs logarithmic)	both, in their models (ρ fixed or learned)
C16	memorisation verdicts	each with its setting; H4(a) refuted for memorisation in general
C17	misspecification gives weaker or unsound theories	classified by kind of data (Table 18)
C18–C23	clashing names and symbols	the notation of Section 2
C24	numbering of Robinson's axioms	the same in all tracks
C25, C26	symbol counts; Dirichlet α	bit counts name their code; tables state α
C27	parameter admissibility	stated wherever $\operatorname{supp} P_T = \mathrm{Th}(T)$ is used
C28	fixed weight vectors in the experiments	a yardstick, not an instance of a theorem (Remark C.3)
C29	E8 per-doubling rates	quoted with their windows
C30	"PA-equivalents from $n = 100$" vs "SeenQ first"	different data laws and candidate sets; both with scopes
C32	Hänni's equivalence: argument vs conclusion	argument fails single-sorted over PA; conclusion holds (Proposition 6.5 and Theorem 6.3)
C33	instance-only generators vs E1	no conflict; Ch_1 is $\mathsf{C}_{\min}$-like
C34	three spare-slot formulas	one statement, three sources (Proposition 4.14)
C35	model Prop 5.6(b) vs E8	E8 not cited for (b)
C36	U11(e) vs AS Prop 4.5	the Bayesian counterpart, with the refuted claim labelled
C37	over-Q limit for splits	class-dependent (Propositions 3.12 and B.16)
C38	track-vs-referee disagreements	the final notes, with reasons (Appendix H.2)
C39	three verifier definitions	the $\mathrm{Bel}_{\mathrm{cons}}$ verifier inherits soundness (Remark 5.9)
C40	units of d in Th_d	stated at each use
C41	"statements given without proof are easily derivable" in two tracks	the same message, different pairs; no conflict
C42	cost of computing $\mathsf{L1}$	the wording of Proposition 2.24
C43	Thm B under $\mathsf{L0}^{\mathrm{cl}}$ vs E1 under $\mathsf{L0}$	different likelihoods
review C-16	the model notes use Σ_1-completeness of Q for $\mathrm{Accept}_f(\ulcorner\varphi\urcorner)$ (Props 6.2, 6.3); the universal notes restrict it to $<$-free sentences, as Q has no axiom about $<$	Accept_f and Reject_f are written without $<$ (bounded quantifiers as $\exists z\,(z + x = y)$); Q is Σ_1-complete for $<$-free Σ_1 sentences (Sections 2.1 and 6.2 and Proposition 6.4)

H.6 Conjectures, proof sketches and open items

Conjectures. Theorem 4.7(c) (every w^*); Proposition 4.11(d) (Occam rate under $\mathsf{L1}$); Remark C.5 (unsound mass in the full class); Conjecture C.7; Proposition B.6(iii) (memoriser rate); the limits in Proposition 3.12(d) and Proposition B.16(d); Conjecture 6.13; Conjecture 6.18 (the

follow-ups neither prove nor refute it; they concern labelled data); Conjecture 7.15; the reverse overheads in Remark F.4; that a deeper nesting would win on the heavy-tailed data of E3(a) (Appendix G.4).

Proof sketches. Lemma A.5(b); Theorem 4.7(b′); Proposition 4.14(b), (c), (d2), and with them the waiting time in Remark 4.15; Remark 2.20 when ψ is generated by a schema; the mechanism of Example 5.8; Theorem 3.9(d2) in general; Proposition 3.12(c); Proposition B.14(b) for $f_q > 0$ and (c); Proposition B.16(a); the spine formula of Remark 3.5; Corollary 6.10(b) for a single X; Remark E.3 for A_f^C and the two-sorted A_f (for $\rho_{f,n}$ no upper bound is established), and the upper bound in Remark 6.1(3); Remark E.9(ii) and the argument of (iv); Proposition E.15(c), the time-hierarchy effect under a time bound; Proposition 7.13(b); the Dirichlet-weight part of Proposition 7.10; the slope of the covering theory in Example F.18; the covering argument and the eventual behaviour in Example F.26; Remark F.24. *Heuristics*: the waiting times in Proposition 7.13(a) and for the $\mathrm{I}\Sigma_n$ chain after Proposition 4.9. *Computed, not proved*: exactness of the $\mathsf{L1}$ coefficients for data-derived theories at $J = 2$ (Proposition 8.1(d)).

Open items found while writing and reviewing. These are not resolved in the paper. (1) Model Lemma 1.2(b) has only a proof sketch; the missing step is named in Appendix A.3. (2) The direction of Theorem 3.4 is not proved in $\mathsf{C}_\wedge$. (3) In Proposition B.15 the full-sum comparison at $q = \frac{1}{2}$ is open; in Proposition 3.10(d) the limit 0 is not proved for countable classes with learned-weight provers. (4) Proposition 4.14(c) needs a finite χ^2-type sum, and a spare in the span of T^*'s components but outside their convex hull is not covered; Remark D.3 is proved for $\mathsf{L0}$ and likelihoods affine in w only; the size of the generator class under near-miss channels is open (Remark C.9). (5) In Example 5.8 a χ^2 limit uniform over the window $1 \ll n \ll 1/\varepsilon$ is not written out; completeness under the shrinking threshold, and the strictness of Proposition D.2(d) for $H_k(\mathrm{DT}^\circ)$, are open. (6) In Remark E.3 the Q-proof size bound is recalled from Pudlák (1998) and the size of the Tarski biconditional's proof is not bounded; for Corollary 6.10 with a single X, the time-constructibility and gap conditions are not checked; the route to Conjecture 6.13 lacks Gen with de Bruijn indices under sharing, and a polynomial bound on the size of shared representations; Proposition 6.6 is not proved for the rejection part of the schema, nor for templates whose instances mix the two parts; whether templates block the collapse for merely consistent assigners is open (Proposition 6.7); the $\mathrm{I}\Sigma_n$ comparison under a derivation-time penalty proposed by pa's referee was not quantified. (7) Pa: the stronger form of Rem 4.9 (Remark F.17); Proposition 7.19 under the normalised $\mathsf{L1}$; Proposition 7.10 and Proposition 7.13(a) with Dirichlet weights; whether the mass of not-yet-refuted inconsistent theories deriving a false sentence tends to 0 (Remark F.13). (8) Experiments: exactness of $\mathsf{L1}$ for data-derived theories at $J = 2$ rests on computed checks (brute force at $J = 1, 2$ and the referee's recursion on 904 triples), not on Proposition 8.1(d), whose hypothesis the code does not check. (9) Further open problems of the research notes: whether the two-part maximum can change a deductive limit (Proposition A.8); whether $\mathsf{L1}$ separates equivalent theories for all grammar parameters (Proposition 4.5); partial splits that only $\mathsf{L1}$ keeps alive (Remark 4.13); soundness bounds in terms of the distance from the data law to the pool, and robust posteriors; the strictness in Proposition D.2(d); an unknown selection learned jointly with the templates (Theorem 3.9(e)); formulas with several variables; a single-sorted repair of Hänni's schema short of Craig's idea (Proposition E.17 gives one in the form of his schema, with Craig's padding as a numeral bound); the $\mathrm{I}\Sigma_n$ comparison under a derivation-time penalty; Conjecture 6.18; theorem data under plain $\mathsf{L1}$ or the full sum; Collection from Replacement without Foundation; a learned instantiation grammar and cheap remedies for early unsound

lumps (Remark 8.2); the generator class under natural corruption channels (Remark C.9). (10) From the follow-ups on the time penalty (Section 6.6; time-followup §7, short-derivations §9): the soft certificate correspondence with an additive overhead per datum, and soft separations from deterministic function induction (Proposition E.26); at polynomial budgets, the case $\mathrm{P} \neq \mathrm{NP}$ with $\mathrm{NP} \cap \mathrm{coNP} = \mathrm{P}$, and uniform domination in either direction (Corollary 6.17); the gap between $\mathrm{DTIME}(2^h)$ and $\mathrm{DTIME}(2^{O(h \log h)})$ for reading (a1); history-using time-bounded predictors such as S against reading (b) at polynomial budgets; whether a single hypothesis proves all diagonal sentences of Theorem 6.15 within budgets polynomial in their length; finite template theories on non-literal data, an additive description length, and conservativity of the Horn theories of Proposition E.24; whether the regret of the Kt-penalised inducer against $\mathrm{FI}^{\tau}_{\mathrm{cons}}$ is bounded; a computable consistency filter in place of "$B \cup A_p$ consistent". The open problem of the discussion of the first version, whether a symbol-size penalty makes axiom induction equal to function induction over assigners with short certificates, is answered for hard bounds by Proposition E.21; for soft charges only a correspondence up to a constant factor in the rate holds (Proposition E.26).

H.7 References not verified

The bibliography files record, for each entry, what was checked. The following were not verified against their sources, and the text hedges each claim that rests on them. *Content or theorem not checked:* Hutter (2007) (bibliographic data and abstract only); Leike and Hutter (2015) (abstract); Gaifman (1964) (condition cited from memory; secondary sources use constants); Doob (1949) (bibliographic data confirmed only through a bibliography of Doob's publications; the theorem is cited, not read); Krichevsky and Trofimov (1981) (bibliographic data checked by the model track's referee; the KT rate is cited as known and was not re-checked); Berk (1966) and Kleijn and van der Vaart (2006) (abstracts; conditions not read); Ghosal and van der Vaart (2017) (theorem number only via citing papers); Horning (1969) and Angluin (1988) (existence only); Rousseau and Mengersen (2011) (issue number not re-confirmed; not applied); Craig (1953), Seiferas et al. (1978) and Žák (1983) (full text not read); Cook (1972) (exact form of the theorem not checked; pages omitted); Pudlák (1998) (the Q-proof bound not checked in the chapter); Boolos et al. (2007) (chapter not checked); Shepherdson (1964) (secondary sources only); Zarach (1996) (editor and publisher of the volume not checked); Gitman et al. (2016) and Kaye and Wong (2007) (abstracts); Kaye et al. (1988) (existence). *Locations or numbering not checked:* Mendelson (1997) (system K, numbering); Athreya and Ney (1972) (chapter); Hájek and Pudlák (1993) (Cor. I.4.34 traced by a referee through a secondary source) and Kaye (1991) (chapter numbers); Shoenfield (1967) (location; publisher recalled). *Recalled from memory or copied:* Church (1936), Turing (1936), Birkhoff (1935), Schwarz (1978), Clarke and Barron (1990), Hoeffding (1963), Shafer (1976) (cited only for the framework), Ville (1939) (data copied from IL's bibliography), Ryll-Nardzewski (1952) (data copied from AS's bibliography) and Schwartz (1965) (data checked by the model track's referee only). *Not opened:* Parikh (1973), Baaz and Pudlák (1993) (editors and pages not recorded) and Hrubeš (2007). *Added with the follow-ups on the time penalty, as recorded in their notes and not re-checked here:* Chandra et al. (1981) and Book et al. (1970) (not checked); Hartmanis and Stearns (1965) and Hennie and Stearns (1966) (exact form of the hierarchy theorem not checked); Matiyasevich (1970) and Davis (1973) (the MRDP theorem, not checked); Rogers (1967) (section not checked); Cook and Reckhow (1979), Johnson et al. (1988) and Legg (2006) (bibliographic data confirmed by a follow-up referee; content cited as known); Elias (1975) (cited as known). *Cited without an entry, from memory:* Wilks's theorem, the Paris–Kirby results on fragments, the sequential form of the KT estimator, Levin's Kt and

the compressed-string algorithms behind the route to Conjecture 6.13.

H.8 Reproduction

Every computed number comes from a seeded script whose output is saved next to it.

- *Track checks.* `research/tracks/<track>/checks/` holds each track's scripts and their `.out` files (model c1–c16; universal c1–c10; pa `test_nd`, c1–c10; experiments' check scripts), and `referee_code/` the referees' independent scripts and outputs. Universal re-ran all ten of its scripts and the referee's seven, and pa all twelve of its scripts, in a scratch directory, with byte-identical outputs; pa's `c10_quotes.py` checks 87 quoted numbers of its notes against the outputs.
- *Code.* From `code/`: `python3 -m pytest -q tests` (21 unit tests, including a comparison with the referee's independent implementation); `sh run_all.sh` (tests and E1–E8, about 50 minutes on four shared cores); seeds and per-experiment commands in Appendix G.6. Results are in `code/results/*.md` and `*.json`, the first version's in `code/results/v1/`.
- *Paper.* From `paper/`: `./build.sh` merges the bibliographies and runs pdfLaTeX and BibTeX; `./test-section.sh <section> <appendix>` builds one section alone.